Statistics: Pure and Applied

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Preface

These notes started as two separate things. The first half grew out of teaching introductory statistics, where the goal is to take somebody who has seen a bit of probability and get them to the point where they can build a confidence interval or run a test and actually say what the numbers mean. The second half grew out of a graduate course in mathematical statistics, where we go back and ask why any of that works in the first place. So the book is split the same way.

Part I, Applied Statistics, is about doing statistics. We start from the central limit theorem, because almost everything in an introductory course leans on it, and then we build estimation, confidence intervals and hypothesis tests for one and two samples. After that we compare many groups at once with ANOVA, fit lines with linear regression, analyze counts with contingency tables, and finally model probabilities with logistic regression. Each chapter tries to follow the same recipe, which is to find an estimator, find its sampling distribution, and then use that distribution to say something about the population.

Part II, Mathematical Statistics, is about why. We build the probability machinery first (moment generating functions and the different ways a sequence of random variables can converge, along with the inequalities that make those limits work), and only then turn to estimation, so that every theorem we need is already on the table when we use it. Maximum likelihood, Fisher information, sufficiency and the Neyman–Pearson lemma all come after that. If you read Part I first, a lot of Part II will feel like finally getting the receipts for things we waved our hands at. If you go straight to Part II, you should be comfortable with calculus, a bit of real analysis, and probability at the level of a first course.

Problems sit right after the idea they are about, instead of being piled up at the end of a chapter. A lot of them are short calculations or proofs we skipped on purpose, and others ask you to simulate something in Python, which is honestly the fastest way to believe a limit theorem. A few of the applied problems are adapted from Larsen and Marx [4], which is a

lovely book and the one we learned a lot of this from. For the theory, Casella and Berger [2] and Wasserman [6] are the standard references, and Agresti [1] is the place to go for categorical data.

Notation

Throughout the book, $X_1, \ldots, X_n$ is a random sample of size n, meaning independent and identically distributed (i.i.d.) random variables. The sample mean is $\bar{X}$ (or $\bar{X}_n$ when we want to stress the sample size), and the sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

A generic parameter is θ, with parameter space Θ, and an estimator of it is $\hat{\theta}$ (or $\hat{\theta}_n$). Densities and mass functions are written $f(x)$ or $f(x;\theta)$, distribution functions are $F(x)$, and Φ and ϕ are the standard normal CDF and density. The likelihood is $L(\theta)$ and the log-likelihood is $\ell(\theta)$. Critical values are always *upper-tail* values, so z_α is the number with $\mathbb{P}(Z > z_\alpha) = \alpha$, and likewise $t_{\alpha,\nu}$ and $\chi^2_{\alpha,\nu}$ for the t and χ^2 distributions with ν degrees of freedom. A 95% interval for a mean therefore uses $z_{0.025} = 1.96$.

Hieu

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Part I

Applied Statistics

1

Samples, Data, And The CLT

"Aw shit here we go again"

– CJ, GTA San Andreas

There is this major misconception that probability and statistics are one and the same. They're not. There's also this misconception that statistics is applied probability, but this is about as incorrect as saying that physics is applied mathematics. The truth is that statistics and probability try to answer different questions. It just so happens that statistics uses probability, but that does not necessarily mean that they are the same thing. But that gets us asking: what is the point of statistics? What questions are we trying to answer? How can we answer such questions?

1.1 The Issue Of Data

The purpose of statistics is inconsistent depending on whom you ask. If you were to ask me, I would say that the point of statistics is to describe emergent phenomena, where we can take simple microscopic systems and extrapolate them to reveal truths about very complicated macroscopic systems. On the other hand, if you were to ask a classical statistician, they would tell you that statistics is a collection of tools to be able to describe data, more specifically large amounts of data. On the one hand, you have this very grand interpretation of what statistics is, and on the other, it boils down to just a toolbox. But if you sit and meditate on these ideas, you'll realize that perhaps they are doing the same thing. In both points of view, statistics is ultimately the thing that connects microscopic models to

macroscopic reality. But why do we need statistics to do this? The fundamental problem comes from the fact that **data is random.**

Well, not quite. Data is not necessarily random. If we knew all the initial conditions, if we knew every little interplay between variables, and we knew exactly how these variables change with time, then we can describe the future exactly. The issue isn't so much about the data, the issue is the *complexity* of the data. We have a complete(ish) theory of probability. Even if the data was random, if we knew *how* it is was random, (i.e. what the random variable is) then we would know everything there is to know about the data. For example, suppose you have a collection of data points that you know are Poisson distributed

$$\{X_1, X_2, ..., X_n\} \sim \text{Poisson}(\lambda)$$

Then you would know that the distribution of the data (assuming i.i.d.) would follow

$$\begin{aligned} \mathbb{P}(X_1, X_2, ..., X_n) &= \prod_{i=1}^{n} \mathbb{P}(X_i = x_i) \\ &= \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \end{aligned} \tag{1.1}$$

We could collect averages, find the variance, blah blah blah. The issue is that we *don't know* where the data is coming from. That is the fundamental problem of statistics! We have data, but we don't know any probabilistic properties of the data. What kind of information can we extract from it? And how can we do so?

1.2 The Emergent Distribution

Let us do a very simple thought experiment. Suppose there is a drunk man who is so drunk, he can't even properly stand still. Every second, he either takes a step to the right (with probability p) or a step to the left (with probability $1 - p$). On the microscopic scale, this is just the Bernoulli random variable: the simple coin toss distribution. We know what the probability distribution for the Bernoulli random variable is: it's

$$\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$$

where success is denoted by $X = 1$ and failure is denoted by $X = 0$. But this isn't the interesting thing. What we want to know is: where will the drunk man be after some time? *That* is a much more interesting question to ask. We could do this out by hand. For example, if the drunk man has taken 3 steps, what is the probability distribution for where he will

end?

The first thing that we would have to do is count all the possible ways that he could move. Let us denote L as left and R as right.

$$\begin{array}{cccc} \text{LLL} & \text{LLR} & \text{LRL} & \text{LRR} \\ \text{RRR} & \text{RRL} & \text{RLR} & \text{RLL} \end{array}$$

If we denote the probability of moving right as p and the probability of moving left as $1-p$, then we get that the distribution is

$$\mathbb{P}(\#\text{ of R} = x) = \binom{3}{x} p^x (1-p)^{3-x} \tag{1.2}$$

So we see that the binomial distribution emerges! Now, in general for N steps, we would expect that

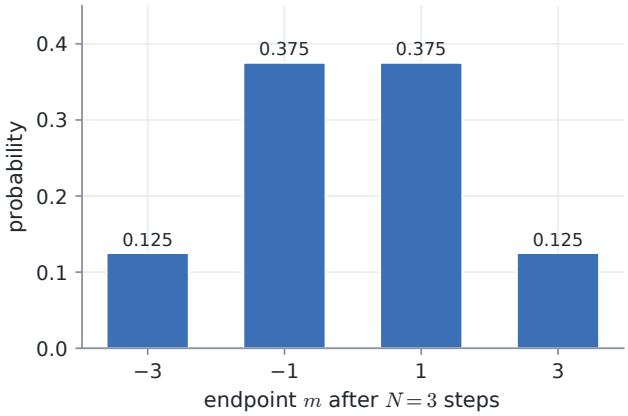


Figure 1.1: Distribution of the endpoint of a fair ($p = 1/2$) random walk after $N = 3$ steps.

$$\mathbb{P}(\#\text{ of R} = x) = \binom{N}{x} p^x (1-p)^{N-x} \tag{1.3}$$

What can we deduce from this? Well, if we wanted to know what this the mean and variance, then we would know that for the binomial distribution

$$\mathbb{E}[X] = Np \quad \text{Var}[X] = Np(1-p) \tag{1.4}$$

This makes sense. But we were interested in the end points of random walks; that is, where do they end up? Well, we already implicitly answered that! We just found out that the distribution of steps is binomially distributed, so if wanted to find the distribution of the *end points*, we would have to do a slight change of variables. If each step is of size 1, and we

say that moving rightward is positive, then that means that moving leftwards is -1. This means that when we quantify the steps, we have the corresponding table of endpoints

$$\begin{array}{cccc} -3 & -1 & -1 & 1 \\ 3 & 1 & 1 & -1 \end{array}$$

The histogram (for a fair walk, $p = 1/2$) can be seen in Figure 1.1.

Problem 1.

If in general the drunk man takes N steps, what is the distribution of the end points? If he takes 7 steps, what is the probability of ending back at the origin? If he takes 8 steps, what is the probability of ending back at the origin?

It's not good enough that we have the distribution, what we really care about is the mean and variance. If we define m to be the displacement; that is,

$$m = n_R - n_L \tag{1.5}$$

where n_R is the number of right steps and n_L is the number of left steps, then we should expect that (using the linearity of expectation)

$$\mathbb{E}[m] = \mathbb{E}[n_R] - \mathbb{E}[n_L] = N(p - q) \tag{1.6}$$

where $q = 1 - p$ for notational simplicity. And the variance?

Problem 2.

Writing the variance in terms of expectations

$$\mathrm{Var}[X] = \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2 \tag{1.7}$$

Find the variance of $m = n_R - n_L$.

We've stumbled upon a deep idea: that simple microscopic models (flipping a coin going left and right) gives rise to nontrivial distributions. But so far, we have only really seen what happens at number of steps. In the world of statistics, we do statistics not because we have small sample sizes (sorta kinda), but because we have *very very large* samples. So does this

idea still hold for when we take the limit that $N \to \infty$? That is, will the distribution still be binomial?

1.2.1 Stirling's Approximation

Before we can answer such a question, we will need a bit of a trick. When we talk about limits, we enter the regime of calculus. We already know how to take limits of the usual functions like sines, cosines, exponentials, blah blah blah. But the distribution that we just saw emerge is the binomial distribution which admits factors involving the factorial. More specifically,

$$\binom{N}{x} = \frac{N!}{x!(N-x)!} \tag{1.8}$$

and it's not immediately obvious how we can do calculus with the factorial. What we can do is use an approximation. We can start with the fact that the factorial is written as

$$n! = n(n-1)(n-2)...(1)$$

which we can compact into

$$n! = \prod_{k=1}^{n} k \tag{1.9}$$

Products are a bit messy to work with, so we can just take the log on both sides. The reason that we do this is because logs of products are sums of logs

$$\ln n! = \sum_{k=1}^{n} \ln k$$

Here is the claim; and this might make some of y'all a bit uneasy. If n is sufficiently large, then the sum can be approximated as an integral

$$\ln n! \approx \int_1^n \ln x \, dx \tag{1.10}$$

Problem 3.

Evaluate the integral

$$\int_1^n \ln x \, dx$$

hint: Use integration by parts.

Ultimately, what we find is that

$$\ln n! \approx n \ln n - n \tag{1.11}$$

You might be asking yourself: isn't this kind of crude? The answer is; sort of. Despite it being an approximation, it works surprisingly well. In fact, you don't even need that large of n for the approximation to work well.

Problem 4.

How good of an approximation is Stirling's formula? For $n = 5$, how accurate is Stirling's approximation? What about $n = 6$?

Problem 4 tells us that even though our methods are approximate, in the limit when we have very large n (this corresponds to the dimension of our data), the errors are vanishingly small relative to the size of the numbers we are dealing with. For example, in a deck of 52 cards, the number of possible combinations you can have is greater than the number of particles in the observable universe! And that's only with 52 cards! So if you are ever queasy about the fact that we are doing approximations, keep this in the back of your mind!

1.2.2 Taking The Limit

We now begin the business of taking the limit of our random walk. To make our lives a bit simpler, we are going to center our variable around that mean. Mathematically, we are always free to do this. Centering a function is just the same thing as doing a variable shift such that the function becomes its parent function. For example, suppose we have a function $f(x) = 5x^2 - 10x + 5$. We could simplify this to $f(x) = 5(x-1)^2$ and say that this is a stretched parabola with the vertex being at $(1, 0)$. If we do the variable substitution $\eta = \sqrt{5}(x-1)$, then we see that $f = \eta^2$, which is the standard parabola. This is purely to make the notation less cluttered. The other reason that we would like to do this is because we know that the binomial distribution has a max at its mean value. Actually, if we look at the ratio of the standard deviation to mean, we see that

$$\frac{\sqrt{\text{Var}[m]}}{\mathbb{E}[m]} = \frac{\sqrt{4Npq}}{N(p-q)} \propto \frac{1}{\sqrt{N}}$$

which tells us that in the limit that N becomes very very large, the displacement is actually centered around the mean with a vanishing relative variance. This localization around the mean will allow us to do a Taylor series expansion at the mean. To keep the algebra clean, we will work with x, the number of right steps, whose mean is Np (the displacement is just

$m = 2x - N$, a shift and a stretch of x). So we expand around the mean as $x = Np + \eta$. This means that the distribution, in the limit that $N \to \infty$ can be well approximated as a distribution centered around the mean

$$f(x) = f(Np) + \left.\frac{df}{dx}\right|_{Np} \eta + \frac{1}{2}\left.\frac{d^2 f}{dx^2}\right|_{Np} \eta^2 + ...$$

Since we just said that the maximum occurs at the mean, the first derivative vanishes and so all that we're really left with is

$$f(x) = f(Np) + \frac{1}{2}\left.\frac{d^2 f}{dx^2}\right|_{Np} \eta^2 + ...$$

Here's the issue though: the distribution that we want to take the limit of is the binomial distribution, which has factorials in it. No matter! We can deploy Stirling's approximation! But to do so, we need to take the log

$$\ln f(x) = \ln f(Np) + \frac{1}{2}\left.\frac{d^2 \ln f}{dx^2}\right|_{Np} \eta^2 + ... \tag{1.12}$$

Let us expand the binomial distribution

$$\ln \mathbb{P} = \ln N! - \ln x! - \ln(N-x)! + x \ln p + (N-x) \ln q$$

$$= N \ln N - N - (x \ln x - x) - ((N-x) \ln(N-x) - (N-x)) + x \ln p + (N-x) \ln q$$

All that we really care about is the second derivative

$$\frac{d^2 \ln \mathbb{P}}{dx^2} = -\frac{N}{x(N-x)}$$

When we evaluate this at the mean $x = Np$ (so $N - x = Nq$)

$$\left.\frac{d^2 \ln \mathbb{P}}{dx^2}\right|_{Np} = -\frac{N}{Np \cdot Nq} = -\frac{1}{Npq}$$

And thus our expansion is

$$\ln \mathbb{P} \approx \ln \mathbb{P}(Np) - \frac{1}{2Npq}(x - Np)^2$$

and our limiting distribution is

$$f(x) = f_0 \exp\left[-\frac{(x-Np)^2}{2Npq}\right] \tag{1.13}$$

Which is just the Gaussian! A couple things to note: In the limit that $N \to \infty$, we no longer have a PMF but rather a PDF and we must change the notation to reflect this. The second thing to note is that we have a normalization constant f_0, which we find by integrating the distribution across all space, and it comes out to $f_0 = 1/\sqrt{2\pi Npq}$. So the number of right steps is approximately $\mathcal{N}(Np, Npq)$, which matches the binomial mean and variance exactly, as it should.

This is an amazing result! What we saw was that if we take random walks, the emergent distribution is a normal distribution!

1.3 The Central Limit Theorem

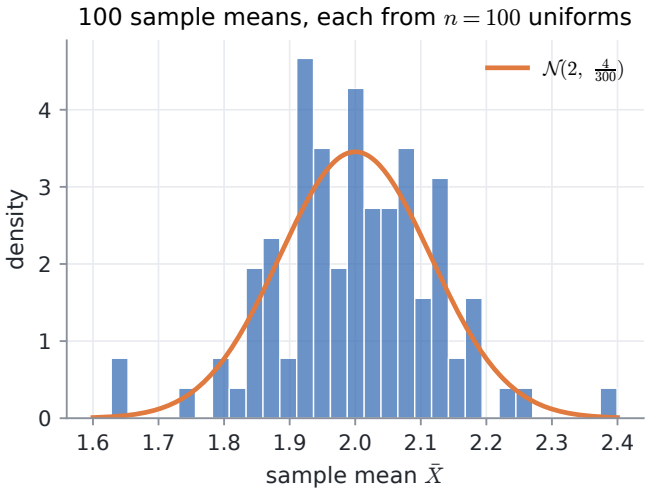


Figure 1.2: Histogram of 100 sample averages, each computed from $n = 100$ draws of a Uniform$(0, 4)$ random variable. The orange curve is the normal distribution the CLT predicts.

The fact that we had a normal distribution emerge from the Bernoulli distribution is a surprising fact about the nature of...well, nature. But how can this help us? So what if one distribution has the normal distribution emerge? Let us do a computational exercise. Suppose we have 100 samples of data points sampled from a, say, uniform distribution on (0,4)

$$X_1, X_2, ..., X_n \sim \text{Uniform}(0, 4)$$

Then, we take the sample average

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{1.14}$$

And we repeat this process, say, 100 times. What is the distribution of the sample means? The python code is

```
xbars = np.array([np.sum(np.random.uniform(0,4,100))/100 for i in range(100)])
plt.hist(xbars,bins=30,density=True)
```

And the result is given in Figure 1.2. But maybe this isn't clear enough, so let's repeat this 1,000 times. Aha! We see something that looks like a normal distribution centered around

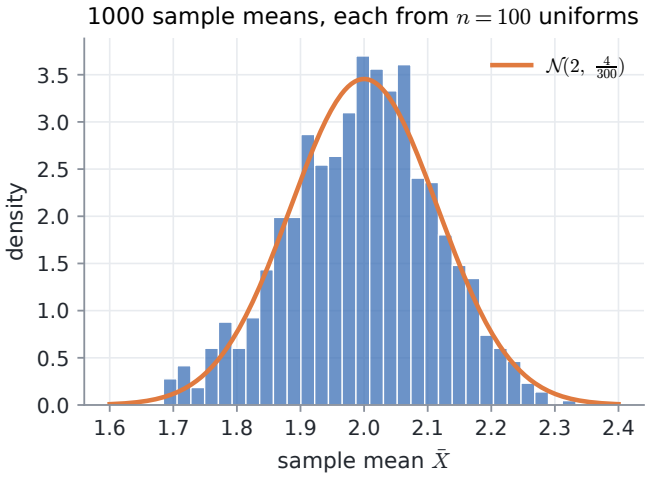


Figure 1.3: The same experiment as Figure 1.2, but with 1,000 sample means.

the mean. Wait a second...The underlying distribution was uniform, how was it that we got the normal distribution? That is the magic of the central limit theorem! To state it blankly

Theorem (Central Limit): Let $\{X_1, X_2, ..., X_n\}$ be a sequence of i.i.d. random variables with mean μ and finite variance σ^2. Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

be the sample mean. The distribution of the sample mean $\bar{X}$ will be a normal random

variable with mean $\mathbb{E}[X]$ and variance $\text{Var}[X]/n$ in the limit that $n \to \infty$. That is

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad n \to \infty \tag{1.15}$$

where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}[X]$.

Notice that the central limit theorem *makes no assumption* on the underlying distribution of the data. In fact, it doesn't even matter! Regardless of what the distribution is, the sample means will *always* be normally distributed in the limit that $n \to \infty$

Problem 5.

We say that the central limit theorem always holds...but is that always true? Suppose we take 100 samples from a Cauchy distribution and find the sample mean. Plot the histogram for 10,000 sample means. Is the distribution normal? If the answer is no, can you deduce why it is not normal? **Hint:** Plot Cauchy distribution.

1.3.1 Revisiting The Random Walk

There are some myths that we need to dispel immediately. The first is that the central limit theorem tells us that if we have a large enough sample, then the distribution becomes normal. That is **NOT** true. The central limit theorem tells us that the **sample means** are normally distributed. Wait what? Then why did the random walk have end points that are normally distributed? Well, look at what the endpoint actually is. Each step is its own little random variable $S_i = \pm 1$, and the endpoint after N steps is the sum $m = S_1 + S_2 + \cdots + S_N$. But a sum is just N times a sample mean, $m = N\bar{S}$, and rescaling a normal random variable gives you another normal random variable. So the endpoint is normal for exactly the same reason that $\bar{S}$ is normal, which is the central limit theorem applied to the steps. The steps themselves are still coin flips no matter how many we take. It's the average (or equivalently the sum) that becomes normal.

Problem 6.

The convergence of the random walk was very very subtle. We implicitly made an assumption about the probability of moving left and moving right. What is this assumption? Test your assumption computationally. That is, run the random walk but

without the assumption that we made. Are the end points still normally distributed?

2

Single Parameter Estimation and Inference

"All the single [parameters], all the single [parameters]!"

– Beyonce

In the previous section, we encountered the biggest problem in statistics: that we don't know what the underlying distributions for our models are. We are able to solve this with the central limit theorem, which tells us that we *don't need to know* what distribution the random variables follow; the distribution of the sample means will always be normal if the sample size is large enough. The backbone of all things statistics will rely on the central limit theorem. But now that we have the CLT, we would like to do something useful with it; namely, we would like to to use it to estimate a quantity relating to a population. For the first part of this chapter, we talk about parameters, how to extract them from the data, and then how to estimate them from a population. The first parameter of interest will be the population mean. Then we will discuss the population proportion, and then move onto estimating the population variance. To quantify how "good" our estimators are, we will generate confidence intervals for each respective parameter.

2.1 Estimators

Before we begin talking about estimators, we should set up the problem that we are trying to solve.

Case 1: Suppose your friend tells you that you are pretty short for your age, but you don't believe them. You say that you're actually around average height. But what is the average height? How would you measure the average height of the people in your town? The first thing that would come to mind is that you go around and measure every single person in the town and then calculate the average. If you lived in small-ish town like me with a population of, say, 14,000 people, then you'd have to spend weeks if not months just to prove your friend wrong. As we can see, this method is pretty impractical. Could we instead look at a smaller sample? Is it possible to deduce properties of the population from a small sample? One would hope so, right?

Case 2: Your friend is from Indonesia, so naturally English is their second language. Whenever you watch a show or read a story or consume any other kind of media together, there will be words that they will not know (for example, the word "zenith"). But if you use a synonym to explain the word, e.g. "peak," then they understand. So they understand what "peak" means but not what "zenith" means. And if they understand what peak means, you can assume that they would know how to say the word "peak" in Indonesian. That implies that you know more words in English than they do in Indonesian (which is typically true because English has the largest lexicon of any language). But how many more do you know? We could open a dictionary and go word for word to figure out how many words you know, but there are almost 1 million words that exist in English and if you read each word for 1 second, it would take almost 2 weeks nonstop. Is it possible to estimate the proportion of words that you know from a smaller sample?

Both these cases illustrate the need to estimate a population parameter (mean or proportion) from a smaller sample. And this is the point of statistics! The great thing about statistics is that it is extremely formulaic. It follows 3 steps:

1). Find an estimator for a parameter of interest.

2). Create a statistic for that estimator.

3). Find the asymptotic distribution for that parameter.

That is a lot of new words to digest, but we will go bit by bit to see what they mean.

2.1.1 Parameters Of A Distribution

Parameters are variables that uniquely define a distribution. Oftentimes, they are intrinsic to the data; that is, we don't get to decide what value the parameter takes on, we have to figure it out. Other times, we get freedom in choosing the value of the parameter, and when this happens, we call it a **hyper-parameter.** Let's go through some basic examples.

For the most simple distribution that we could have, the Bernoulli, the distribution is given as

$$\mathbb{P}(X = x) = p^x(1-p)^{1-x} \tag{2.1}$$

What is the parameter in this case? It is, of course, p. With the way that the distribution is written above, it is actually a *family* of distributions because the value of p is unspecified. If we know that the value of p is $1/2$, then we have a unique distribution

$$\mathbb{P}(X = x) = \left(\frac{1}{2}\right)^x \left(1 - \frac{1}{2}\right)^{1-x}$$

And this could describe the probability of getting heads ($X = 1$) for a fair coin. To give an example of how we might deal with parameters in a real life problem, suppose you have a coin and we want to know if it is fair. What do we mean by fair? Fair means that the probability of heads is equal to the probability of tails (which numerically would be $1/2$). But obviously, we don't know what the probability of getting heads is ahead of time because if we did, then we wouldn't need to set up this experiment. Since p is the parameter that governs the probability of success, that means that we need to *estimate* p by doing experiments (e.g. flipping the coin).

So parameters uniquely define the distribution. But that's not all that they can do! Oftentimes, parameters are directly linked to the mean and variance of the distribution, so if we knew the parameter, then we can read off the mean and variance.

> Problem 1.
>
> Suppose you have a random variable X that is Poisson distributed. Write out the Poisson distribution, identify how many parameters there are as well as what they are, and then show how these parameters relate to the mean and variance of the Poisson distribution.

Distributions can come in a whole myriad of parameters. The one that we are going to familiarize ourselves with are the parameters relating to the normal distribution. This makes sense because we saw that the central limit theorem will provide an asymptotic distribution for the sample means of our data. Let us write out the normal distribution here

$$\rho(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \tag{2.2}$$

Currently there are 2 unknowns: σ^2 and μ. If we knew the value of these symbols, then we

would have a unique normal distribution. Thus, these are the parameters for the normal distribution!

Problem 2.

We made the claim before that the mean and variance can be written in terms of these parameters. Let us test this claim against the normal distribution. Using the definition of the mean

$$\mathbb{E}[X] = \int_D x\rho(x)\, dx \tag{2.3}$$

where D is the domain of the distribution and ρ is the probability density,
a). calculate the average value of the normal distribution in terms of the parameters μ and σ.
b). Calculate the variance of the normal distribution in terms of the parameters μ and σ.

2.1.2 The Maximum Likelihood Estimator

Now that we have an understanding of *what* a parameter is, we would like to know *how* to obtain an estimate of it from the data. The set up is quite simple. Suppose we have a bunch of data points $\{X_1, X_2, ..., X_n\}$ as shown in figure 2.1. Now, we *have* to make an assumption about the distribution of the data. There is no way around this. If we knew exactly what the distribution was, then this class wouldn't exist (that doesn't mean that all assumptions are good though; some assumptions are better than others). Let us make the assumption that the data is normally distributed. The normal distribution has 2 parameters μ and σ^2; what value of μ and σ^2 best fit to the data? Qualitatively, this is asking: which of the green curves best represents the red data points. Quantitatively, we want to know: what is the probability of getting X_1 AND X_2 AND X_3 so and so forth for *some* value of μ and σ^2 (assuming i.i.d.)? We could write this as a **joint probability distribution**

$$\rho(X_1, X_2, ..., X_n; \mu, \sigma^2) = \prod_{i=1}^{n} \rho(X_i; \mu, \sigma^2) \tag{2.4}$$

in the case of a Gaussian, we have

$$\rho(X_1, X_2, ..., X_n; \mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_{i=1}^{n} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right] \tag{2.5}$$

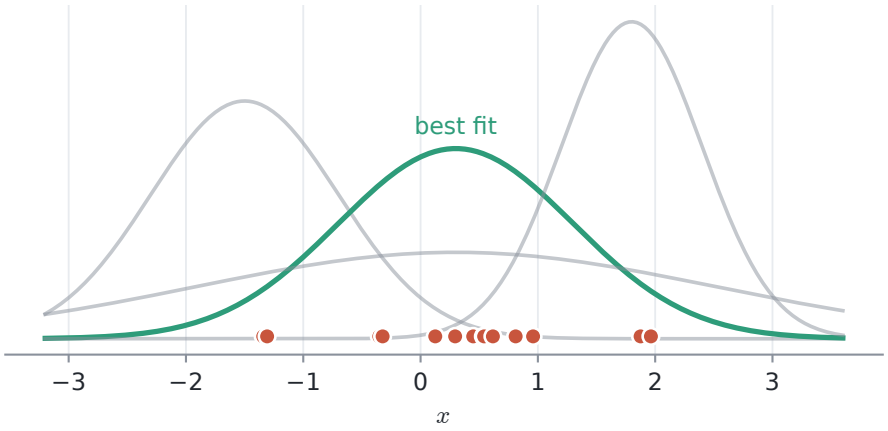


Figure 2.1: A bunch of data points (red) with several candidate Gaussians overlaid. The green one is the one that makes the observed data most probable.

If we are looking to find the probability of getting $X_1, ..., X_n$ for some μ and σ^2, then we call the function in (2.5) **the Likelihood function**

$$L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \prod_{i=1}^{n} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right] \tag{2.6}$$

Notice that the likelihood function is a function of μ and σ^2, not $X_1, ..., X_n$. The difference between the likelihood function and the joint probability distribution is that a joint distribution already has the parameters defined; the likelihood function is what we are going to use to *find* an estimate for the parameters. If we look back at figure 2.1, we see that there does exist some value of μ and σ^2 that best fits to the data, and to find them, we need to find the value of these parameters that **maximizes the likelihood function.** All that this entails is to take a derivative and set it equal to 0. The normal distribution has 2 parameters, so we would need to take a partial derivative with respect to one of the parameters. For now, let us focus on μ. How can we go about taking this derivative? The function looks so ugly! Well, no matter! We can take the log on both sides. The reason for this is because the log of the products is the sum of the logs. The log function is also monotonically increasing, so it preserves the location of the maximum. Let us denote ℓ as the log likelihood

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln\left(2\pi\sigma^2\right) - \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{2\sigma^2}$$

Now we take the derivative and set it equal to 0

$$\frac{\partial \ell}{\partial \mu} = \sum_{i=1}^{n} \frac{(x_i - \mu)}{\sigma^2} = 0$$

Since the σ^2 is a multiplicative factor, we can just factor it out and cancel it when we divide 0 by it. What we are left with is

$$\sum_{i=1}^{n} x_i - \mu = 0$$

Notice that μ has no index, so it's just a constant

$$= \left(\sum_{i=1}^{n} x_i \right) - n\mu = 0 \implies \mu_{\max} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{2.7}$$

So the maximum likelihood estimator for μ is just the sample mean! For now on, we will denote the MLE with a hat $\hat{\mu}$.

Problem 3.

Find the maximum likelihood estimator for σ^2

2.1.3 Properties Of The MLE

The MLE must be pretty important if we are to go through all this length to talk about it. And as it turns out; it is! The MLE is the best estimator that we can come up with (what we mean by best is a topic that is beyond the scope of this course). Let us examine some properties of the MLE that makes it so good.

1). Consistency

Let us denote $\theta_\star$ to be the true value of some parameter θ. Consistency of the MLE $\hat{\theta}$ means that the probability that $\hat{\theta}$ differs from $\theta_\star$ approaches 0. In words, this tells us that the MLE always around the same value as $\theta_\star$, making it "consistent."

2). Asymptotic Normality

Since the MLE is a function of the data which is sampled randomly, then it too must be a random variable. The great thing is that when the sample size is large, we have that

$$\frac{(\hat{\theta} - \theta_\star)}{\text{s. e.}} \sim \mathcal{N}(0, 1) \tag{2.8}$$

where s. e. is called the **standard error**, which is just the standard deviation of the estimator itself

$$\text{s. e.} = \sqrt{\text{Var}[\hat{\theta}]} \tag{2.9}$$

For the sample mean, $\text{Var}[\bar{X}] = \sigma^2/n$, so the standard error is the familiar $\sigma/\sqrt{n}$. The $\sqrt{n}$ lives inside the variance of the estimator, so we don't divide by it a second time. **3. Equivariance**

If $\hat{\theta}$ is the MLE of some parameter, and $\tau = g(\theta)$ is a parameter that is a function of the parameter θ, then $\hat{\tau} = g(\hat{\theta})$ is also the MLE for the parameter τ.

The last property we won't be using immediately, but we will see how it shows up later.

The last property that we need to discuss relates to something called the **bias**. We define the bias as

$$\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta_\star \tag{2.10}$$

We say that an estimator is **unbiased** if the bias is 0. Physically, all that bias tells us is on average, how far off the estimator is from the true value $\theta_\star$. If an estimator is unbiased, then on average it equals the true value. Let us do an example with the MLE for μ in the previous section. Is $\hat{\mu}$ biased or unbiased? We know that

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Now we have to take the average. We can use the linearity property that of the expectation take the expectation inside the sum

$$\mathbb{E}[\hat{\mu}] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i]$$

We know that each X_i are i.i.d. Gaussians with mean μ, so the sum just becomes

$$\mathbb{E}[\hat{\mu}] = \frac{1}{n}\sum_{i=1}^{n} \mu \implies \mathbb{E}[\hat{\mu}] = \mu$$

This makes the bias

$$\text{bias}(\hat{\mu}) = \mu - \mu = 0$$

And thus, the MLE for μ is unbiased. At this point you might be thinking to yourself: are all MLE unbiased? **NO.** Unbiasedness of the MLE would be nice to have, but it is not guaranteed.

Problem 4.

Is the MLE for σ^2 biased or unbiased?

2.2 Inference Of The Mean

Now that we have a way to estimate a certain parameter, we have to move onto step 2 of our 3-step-plan; which is to create a statistic for our estimator. But what is a statistic?

Statistic: A function of a random variable.

... That seems very anticlimatic. The entirety of this field boils down to that??? Yeah sorta. But at the very least, it makes sense. Suppose we want to estimate the value of some population parameter. What would be the statistic? In the case of $\hat{\mu}$, it is the sample mean. **why** do we choose the sample mean? Why not choose, say, the sum? Well, in our case, our statistic will be the sample mean because the sample mean is the MLE and we know some very good properties of the MLE (namely consistency and asymptotic normality). But that's great! If we know that our statistic is going to be the MLE, then we get the asymptotic distribution for free.

2.2.1 Confidence Intervals

It's not good enough to just have an estimate for the population mean. Since the MLE is a random variable, it will have fluctuations and we need a way to encapsulate this fluctuation. The best way to think about this is as an error bar. Suppose you run a lab experiment and you find that you find that .3% of your DNA is neanderthal. Obviously, this isn't exact because we can never know anything exactly. There will be some error in our measurement just due to fluctuations, so we slap on error bars and say that the true percentage of our DNA being neanderthal is $.3\% \pm .01\%$. We would like to do the same for our estimate; we know that it is not exactly $\theta_\star$, but it will fluctuate somewhere around there. If the data is normally distributed, then we can construct an interval such that with 95% probability that the true value will be within this interval (we choose 95% for historical reasons but you can theoretically have any percentage that you want). If our sample size is large enough, then we know that the sample mean will be normally distributed

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \tag{2.11}$$

This is guaranteed to us by the central limit theorem. We have that the distribution follows

$$\rho(x) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{x^2}{2}\right]$$

The question of "what interval encloses 95% percent of the distribution?" is the same question as "for what value of x will this integral be equal to .95?"

$$\int_{-x}^{x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = .95 \tag{2.12}$$

(the standard normal is symmetric about 0, so the lower bound is just the negative of the upper bound). Unfortunately, this integral is not solvable analytically, so we have to resort to numerical integration. But no matter what, we find that the value of x is $x = 1.96$. So the interval is

$$[-x, x] \implies [-1.96, 1.96] \tag{2.13}$$

And this means that to create a confidence interval around our estimate, it must be true that

$$\begin{aligned} \text{C. I.} &= \left[-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right] \\ &= \left[-1.96\frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq 1.96\frac{\sigma}{\sqrt{n}}\right] \\ \text{C. I.} &= \left[\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right] \end{aligned} \tag{2.14}$$

So that we get the confidence 95% confidence interval for the true value of μ to be

$$\mu \in \left[\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right] \tag{2.15}$$

In general, the interval will not be 95%. In that case, (2.12) becomes

$$\int_{-x}^{x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = 1 - \alpha \tag{2.16}$$

Where α is some value that we choose (for historical purposes. Sometimes you will see other texts use just $\alpha/2$ instead of $1 - \alpha/2$). Since the Gaussian is symmetric, we can find the interval numerically as

$$[-x, x] \implies [-z_{\alpha/2}, z_{\alpha/2}] \tag{2.17}$$

where z_α is the **Z-score**. All that it tells us is *where* is the probability of getting *at least* $z_{\alpha/2}$ is $\alpha/2$. So in general, our confidence intervals will be

$$\text{C. I.} = \left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] \tag{2.18}$$

Let's now attempt to do case 1. Suppose you go around and measure the height of 50 people in your town. You find that the average height of those 50 people is 68 inches and you know that the variance of people's height in the town is about 2 square inches. What is What is the 95% confidence interval? What is the 90% confidence interval? The first thing that we need to get straight of the bat is what is our estimate? In this case, we are estimating the population mean using the sample mean. Since our sample is significantly large (the usual rule of thumb is $n \geq 30$), we can use the central limit theorem and say that the sample mean is normally distributed such that

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

This is our statistic and we have the asymptotic distribution! Now we can construct our 95% confidence interval. A 95% interval corresponds to an $\alpha = .05$, thus $1 - \alpha/2 = .975$, which corresponds to a z score of $z_{.025} = 1.96$. So our 95% confidence interval is

$$\left[68 - 1.96 \cdot \frac{\sqrt{2}}{\sqrt{50}}, 68 + 1.96 \cdot \frac{\sqrt{2}}{\sqrt{50}}\right]$$

$$\approx [67.608, 68.392]$$

The way to read this is that if we repeated the whole experiment many times, about 95% of the intervals we build this way would contain the true average (the true average is a fixed number, so it's the interval that is random, not μ). Either way, if your height is 67 inches, then you could confidently (or I guess, inconfidently) say that your height is below average. If you wantes to find the 90% C.I., then that corresponds to $\alpha = .1$, $1 - \alpha/2 = .95$, and $z_{.05} = 1.645$. The result is

$$\left[68 - 1.645 \cdot \frac{\sqrt{2}}{\sqrt{50}}, 68 + 1.645 \cdot \frac{\sqrt{2}}{\sqrt{50}}\right]$$

$$\approx [67.671, 68.329]$$

Yeah it's not looking good for you.

Problem 5.

The production of a nationally marketed detergent results in certain workers receiving prolonged exposures to a Bacillus subtilis enzyme. Nineteen workers were tested to determine the effects of those exposures, if any, on various respiratory functions.

One such function, air-flow rate, is measured by computing the ratio of a person's forced expiratory volume (FEV_1) to his or her vital capacity (VC). (Vital capacity is the maximum volume of air a person can exhale after taking as deep a breath as possible; FEV_1 is the maximum volume of air a person can exhale in one second.) In persons with no lung dysfunction, the "norm" for FEV_1/VC ratios is 0.80 . Based on the following data (164), is it believable that exposure to the Bacillus subtilis enzyme has no effect on the FEV_1/VC ratio? Answer the question by constructing a 95% confidence interval. Assume that FEV_1/VC ratios are normally distributed with $\sigma = 0.09$.

Subject	FEV_1/VC	Subject	FEV_1/VC
RH	0.61	WS	0.78
RB	0.70	RV	0.84
MB	0.63	EN	0.83
DM	0.76	WD	0.82
WB	0.67	FR	0.74
RB	0.72	PD	0.85
BF	0.64	EB	0.73
JT	0.82	PC	0.85
PS	0.88	RW	0.87
RB	0.82		

Problem 6.

What "confidence" would be associated with each of the following intervals? Assume that the random variable Y is normally distributed and that σ is known. (a) $\left(\bar{y} - 1.64 \cdot \frac{\sigma}{\sqrt{n}}, \bar{y} + 2.33 \cdot \frac{\sigma}{\sqrt{n}}\right)$
(b) $\left(-\infty, \bar{y} + 2.58 \cdot \frac{\sigma}{\sqrt{n}}\right)$
(c) $\left(\bar{y} - 1.64 \cdot \frac{\sigma}{\sqrt{n}}, \bar{y}\right)$

2.2.2 Inference With Unknown Variance

In the last section, when we were constructing our confidence intervals, we made the assumption that we knew what the population variance was. This is a very optimistic assumption because in reality, we will never really know what the true population variance is. So do we we call it quits from here? Not yet! Even if we don't know what the true

population variance is, we will have to approximate it using the sample variance. Wait, before we found the MLE for σ^2 for the normal distribution, couldn't we use that? Well, the issue is that it is biased. In fact, the average of the MLE for σ^2 is

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2 < \sigma^2$$

so the MLE for σ^2 actually *underestimates* the true value of σ^2. Could we do better? How can we create an unbiased estimator for the sample variance? Well, if we know that

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \implies \mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2$$

Then to get an unbiased estimator, we should multiply both sides by $\frac{n}{n-1}$ and get

$$S^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \tag{2.19}$$

And to double check that this really is unbiased

$$\mathbb{E}\left[S^2\right] = \frac{n}{n-1}\,\mathbb{E}\left[\hat{\sigma}^2\right] = \frac{n}{n-1}\left(\frac{n-1}{n}\sigma^2\right) = \sigma^2$$

And we conclude that indeed is unbiased! Now that we have our unbiased estimator for the sample variance, will replacing the variance with our estimator have any significant effect? That is, is there any difference between

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \quad \& \quad \frac{\bar{X}-\mu}{S/\sqrt{n}}$$

For the longest time, the answer was thought to be no. And this makes sense because if the sample size is sufficiently large, then any fluctuations from (2.19) would get washed away and you would end up with a constant. What this would entail for our case 1 is that if we did not know the true population variance, we would have to estimate it from our 50 data points using (2.19). But once we have that, then every other calculation is the same.

Problem 7.

Creativity, as any number of studies have shown, is very much a province of the young. Whether the focus is music, literature, science, or mathematics, an individual's best work seldom occurs late in life. Einstein, for example, made his most profound

discoveries at the age of twentysix; Newton, at the age of twenty-three. The following are twelve scientific breakthroughs dating from the middle of the sixteenth century to the early years of the twentieth century (205). All represented high-water marks in the careers of the scientists involved.

Discovery	Discoverer	Year	Age (y)
Earth goes around sun	Copernicus	1543	40
Telescope, basic laws of astronomy	Galileo	1600	34
Principles of motion, gravitation, calculus	Newton	1665	23
Nature of electricity	Franklin	1746	40
Burning is uniting with oxygen	Lavoisier	1774	31
Earth evolved by gradual processes	Lyell	1830	33
Evidence for natural selection controlling evolution	Darwin	1858	49
Field equations for light	Maxwell	1864	33
Radioactivity	Curie	1896	34
Quantum theory	Planck	1901	43
Special theory of relativity, $E = mc^2$	Einstein	1905	26
Mathematical foundations for quantum theory	Schrödinger	1926	39

(a) What can be inferred from these data about the true average age at which scientists do their best work? Answer the question by constructing a 95% confidence interval.
(b) Before constructing a confidence interval for a set of observations extending over a long period of time, we should be convinced that the y_i 's exhibit no biases or trends. If, for example, the age at which scientists made major discoveries decreased from century to century, then the parameter μ would no longer be a constant, and the confidence interval would be meaningless. Plot "date" versus "age" for these twelve discoveries. Put "date" on the abscissa. Does the variability in the y_i 's appear to be random with respect to time?

Problem 8.

Two samples, each of size n, are taken from a normal distribution with unknown mean μ and unknown standard deviation σ. A 90% confidence interval for μ is constructed

with the first sample, and a 95% confidence interval for μ is constructed with the second. Will the 95% confidence interval necessarily be longer than the 90% confidence interval? Explain.

2.2.3 Small Sample Inference

The discussion in the last section brings to light an interesting issue. We said that if the sample size is large enough, then it doesn't matter if we replace σ^2 with S^2. There is a great bit of history here, which Larsen and Marx tell in more detail [4]. The person who noticed that the two ratios

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \quad \text{and} \quad \frac{\bar{X}-\mu}{S/\sqrt{n}}$$

do *not* have the same distribution was William Sealy Gosset, a chemist and mathematician who went to work for the Guinness brewery around 1900. His job was to make brewing more scientific, and he ran into two problems right away. His sample sizes were tiny (often four or five batches), and he never knew the true variance. When he computed the second ratio over and over, it looked bell shaped like the standard normal, but the tails were too fat. Big positive and big negative values showed up way more often than the normal curve said they should.

You can see exactly what he saw in Figure 2.2. On the left we draw 500 samples of size $n = 4$ from a normal distribution and use the true σ, and the standard normal fits beautifully. On the right we use S instead, and the histogram spills out past the normal curve on both sides. The extra spread comes from the fact that S is itself random, and with only four points it is often a lot smaller than σ, which blows the ratio up.

Gosset worked out the density of the second ratio and published it in 1908 under the pen name "Student" [5], because Guinness didn't let its employees publish. Almost nobody cared at first. It was R. A. Fisher who saw how important it was, gave it a rigorous derivation, and named the statistic t, which is why we now call its distribution the Student t distribution.

Despite the derivation for the student t distribution being mathematically beautiful, it's a bit advanced, and so we will not go through all of it. What we will do, though, is state the student t-distribution's density and deduce certain properties.

Student t Distribution:

$$f_t(x;n) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\,\Gamma\left(\frac{n}{2}\right)} \cdot \frac{1}{\left[1+\left(\frac{x^2}{n}\right)\right]^{(n+1)/2}} \tag{2.20}$$

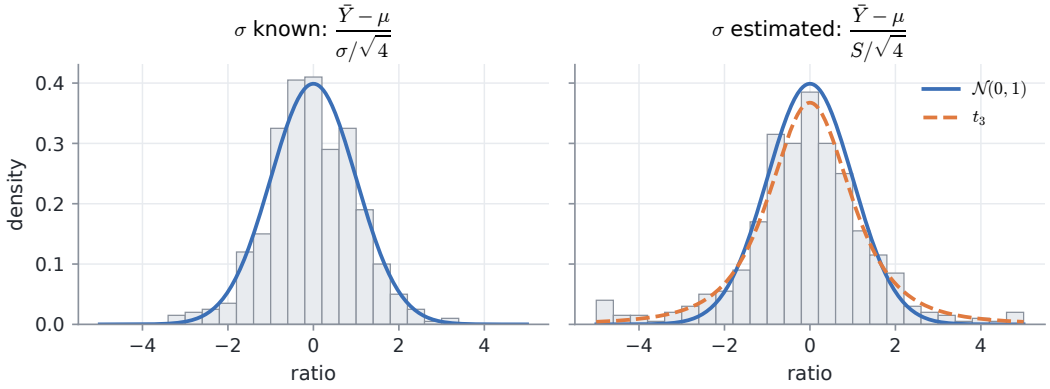


Figure 2.2: Five hundred samples of size $n = 4$ from a normal distribution. Left, the ratio uses the true σ and the standard normal fits perfectly. Right, the ratio uses the sample standard deviation S, and the histogram has much heavier tails than the normal curve. The t distribution with 3 degrees of freedom (dashed) fits.

where $\Gamma(n)$ is the generalized factorial function. That is, if n is a whole number, then $\Gamma(n+1) = n!$. Else, the Gamma function is

$$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1}\, dt$$

More often than not, you don't actually need to know how to compute this integral by hand. If you need to call the Gamma function, we will use a computer to do so. But aside from that, what parameters do we need to know for the t-distribution? Well, it looks like n is our hyperparameter. That is, we have to choose n and the interpretation of n is that it is related to the **degree of freedom.** The degree of freedom just tells you how many unknown parameters you have. Since the t distribution is related to the unbiased sample variance, and the unbiased sample variance has $n-1$ degrees of freedom, we should expect that the t distribution should have $n-1$ degrees of freedom.

Problem 9.

Find the mean of the student t-distribution

We also previously stated that when the sample size is large enough, then we can just go back to using the standard normal distribution. Actually, as it turns out, the student t distribution has a limiting case. In the limit that the degree of freedom goes to infinity, the t-distribution exactly becomes the standard normal distribution.

Problem 10.

Take the limit of the t density as $n \to \infty$ and show that you retrieve the standard normal distribution. You may use Stirling's approximation for the factorial if you need to.

Now that we have the mathematical "know-how" regarding small sample tests, let us go back to creating confidence intervals. The premise is the same, for what interval will I get $1-\alpha\%$ of the distribution

$$\mathbb{P}\left(-t_{\alpha/2,n-1} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t_{\alpha/2,n-1}\right) = 1-\alpha$$

Because the t distribution is symmetric, the critical t value on the lower bound is the same as negative the critical t value on the upper bound. Thus, a $100(1-\alpha)\%$ confidence interval will be

$$\text{C. I.} = \left[\bar{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right] \tag{2.21}$$

To apply this to our case 1: suppose now we don't know the population variance *and* the sample size is small ($n = 6$). What is a 95% confidence interval? For example, suppose we measure the heights (in inches) $\{65, 65, 63, 70, 72, 69\}$. We need to find the sample mean and sample variance

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \approx 67.33$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - 67.33)^2 \approx 12.27$$

The degree of freedom is $n-1 = 5$, so we need to look up $t_{.025,5}$, which is $t_{.025,5} = 2.571$. This makes our confidence interval

$$\begin{aligned}\text{C. I.} &= \left[67.33 - 2.571\frac{\sqrt{12.27}}{\sqrt{6}}, 67.33 + 2.571\frac{\sqrt{12.27}}{\sqrt{6}}\right] \\ &\approx [63.66, 71.01]\end{aligned} \tag{2.22}$$

So if you were 67 inches as stated before, then maybe there is some hope that you are not below average!

Problem 11.

We just did the case of when the sample size is small and the variance is unknown. What if the sample size is small but the variance *is* known? What should our test statistic be and what is the asymptotic distribution? You can answer this question computationally. Sample 6 samples of numbers drawn from some distribution (that is not normally distributed). Standardize it by subtracting the mean and dividing by the standard error. Repeat this 10,000 times and plot the histogram. Overlay a standard normal pdf on top. How well does the standard normal pdf fit to the data? Now suppose the data is sampled from a normal distribution but the sample size is still 6. Is there a difference?

2.3 Inference Of Population Proportions

The central limit theorem goes a long way in allowing us to talk creating statistics and their limiting distributions. However, perhaps the population mean isn't the thing that we are interested in finding; perhaps the parameter we would like to estimate is a population proportion. This takes us back to case 2 at the beginning of the chapter. If we wanted to know the percentage of a dictionary that we know, calculating the sample mean would not be appropriate (because what are we find the mean of?). This is because our model now has binary outputs: each of our data points are either yes or no. So if we skim through a dictionary, we won't see a how continuum of possible values; we only have "yes I do know that word" or "no I do not know that word." This is a fundamentally different question but the name of the game is still the same. The first thing that we need to do is find the parameter we want to estimate. Then, we just need to create a statistic and find its asymptotic distribution.

2.3.1 Finding The MLE

So the output of our model is given by a Bernoulli distribution

$$\mathbb{P}(X = x) = p^x(1-p)^{1-x}$$

The parameter we would like to estimate is p, because that will give us the proportion. Well, we could just use the MLE! We can get a 2-for-1 deal because not only will we get the statistic, but also the asymptotic distribution from the asymptotic normality property. So

let us begin by finding the likelihood

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$

Taking the log to get the log likelihood

$$\ell(p) = \sum_{i=1}^{n} \{x_i \ln p + (1-x_i)\ln(1-p)\} \tag{2.23}$$

Taking derivatives we get

$$\frac{\partial \ell}{\partial p} = \sum_{i=1}^{n} \frac{x_i}{p} - \frac{1-x_i}{1-p} = 0$$

And simplifying gives us

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{2.24}$$

Which, indeed, is the sample mean. We should be clear here: x_i can only take on values of 0 or 1 (corresponding to 0 or yes), so the sample mean really is the **sample proportions.**

2.3.2 The Test Statistic

Well, now that we have our estimator, we can use the asymptotic normality property of the MLE to get the test statistic

$$T = \frac{\hat{p} - p_*}{\text{s. e.}}$$

where p_* is the true population proportion and s.e. is the standard error. To get the standard error

$$\text{s. e.} = \frac{\sqrt{\text{Var}[\hat{p}]}}{\sqrt{n}}$$

So we need to know the variance (the standard deviation, really) of the MLE. No problem! We know that the variance for a Bernoulli random variable is $\text{Var}[X] = p(1-p)$, so using the equivariance property of the MLE, the MLE for the variance is $\hat{p}(1-\hat{p})$. This tells us that the standard error is

$$\text{s. e.} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \tag{2.25}$$

and this makes our test statistic for the population proportions

$$T = \frac{\hat{p} - p_*}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \tag{2.26}$$

And because the MLE has the asymptotic normality property, we know that in the limit that $n \to \infty$,

$$T = \frac{\hat{p} - p_*}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim \mathcal{N}(0,1) \tag{2.27}$$

And because we know that the statistic follows a normal distribution, calculating a confidence interval is actually pretty straightforward

$$\text{C. I.} = \left[\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right] \tag{2.28}$$

2.3.3 Revisiting Case 2

Now we have all the tools to solve case 2. I will create a python script for you to test this for yourself, but for now, I will use made up numbers. Suppose that you randomly sampled 100 words from the dictionary, and you were able to identify approximately 30 words. What is the 95% confidence interval for the true proportion of the dictionary that you know?

As always, we know that an $\alpha = .05$ has a critical z score of $z_{.025} = 1.96$. All that we need to do is find the sample proportions as well as the standard error. Well, if there are 100 words randomly sampled, and you only know 30, then that means

$$\hat{p} = \frac{3}{10}$$

this means that the standard error is

$$\text{s. e.} = \sqrt{\frac{.3(1-.3)}{100}} \approx 0.045$$

And this makes our confidence interval

$$\text{C. I.} = [.3 \pm .09] \approx [.21, .39]$$

Problem 12.

Using the csv file of the words in the Oxford dictionary, randomly sample n words (you may choose n). Count the number of words that you know. Then, create a confidence interval for the true proportion of words that you know.

2.3.4 Small Sample Proportions

We actually have 2 edge cases to deal with so we will talk about them 1 at a time. If we have a small sample, it's very tempting to say that the test statistic will have the student t distribution as an asymptotic distribution but that's actually not the case. We saw from last chapter that if we have a sum of Bernoulli random variables, then then emergent distribution is binomial. In the limit that $N \to \infty$, we have that the binomial approximates a normal distribution. This does not mean that in the small N limit that the binomial is a t distribution; in the small N limit, the binomial is still a binomial distribution (obviously).

> Problem 13.
>
> Find the MLE for p with $\{X_1, X_2, ..., X_n\}$ being i.i.d. binomial random variables.

The test statistic is

$$T = \sum_{i=1}^{n} X_i \sim \text{Binomial}(n, p) \tag{2.29}$$

If we know that the limiting distribution is binomial, then it's actually very easy to read off what the confidence interval is. Recall that the confidence interval is defined as

$$\mathbb{P}\left(c_1 \leq T \leq c_2\right) \approx 1 - \alpha$$

where c_1 is the largest largest number such that $\mathbb{P}(X \leq c_1) \approx .025$ and c_2 is the smallest number such that $\mathbb{P}(X \geq c_2) \approx .025$. Notice that the distribution is no longer symmetric about the origin because the distribution is not centered.

Let us do an example using case 2. Suppose that n is not very large (meaning, you only tested a small number of words). Let's suppose that this number is 15. So you only looked at 15 words, and you recognized around 7 words. If you tried using the test statistic from (2.27), you will get a horrible result. This is a signal that you have to use the exact binomial test. Since we have that $\hat{p} = 7/15 \approx .467$, we can write out all values of the CMF (using python)

```python
import scipy.stats as stats

def binomial_cmf_values(n, p):
    """
    Calculates and prints the binomial cumulative mass function (CMF) values.

    Args:
```

```
    n (int): The number of trials.
    p (float): The probability of success in a single trial.
    """
    for k in range(n + 1):
      cmf_value = stats.binom.cdf(k, n, p)
      print(f"CMF(X <= {k}) = {cmf_value:.4f}")

# Example usage:
n = 15  # Number of trials
p = 7/15 # Probability of success

binomial_cmf_values(n, p)
```

And the output is

```
CMF(X <= 0) = 0.0001
CMF(X <= 1) = 0.0011
CMF(X <= 2) = 0.0076
CMF(X <= 3) = 0.0321
CMF(X <= 4) = 0.0964
CMF(X <= 5) = 0.2201
CMF(X <= 6) = 0.4006
CMF(X <= 7) = 0.6037
CMF(X <= 8) = 0.7813
CMF(X <= 9) = 0.9022
CMF(X <= 10) = 0.9657
CMF(X <= 11) = 0.9910
CMF(X <= 12) = 0.9983
CMF(X <= 13) = 0.9998
CMF(X <= 14) = 1.0000
CMF(X <= 15) = 1.0000
```

So we see that the largest value such that $\mathbb{P}(X \leq c_1) \approx .025$ is $c_1 = 2$, and the smallest value with $\mathbb{P}(X \geq c_2) \approx .025$ is $c_2 = 11$. So if the true proportion really were 7/15, we would expect to know somewhere between 2 and 11 of the 15 words most of the time. That's a massive range, which is just what happens when the sample size is small.

We have to be a little careful though, because $[2, 11]$ is a range for the *count* computed by pretending that $\hat{p}$ is the truth, and that's not quite the same thing as a confidence interval for p. The honest way to do it (called the **Clopper–Pearson** interval) flips the logic around. We collect every value of p for which observing 7 out of 15 would *not* be surprising at the 2.5% level in either tail. You can do this with `scipy.stats.beta.ppf(0.025, 7, 9)` and

`scipy.stats.beta.ppf(0.975, 8, 8)`, and for our example it gives

$$p \in [0.213, 0.734],$$

which is the proportion version of the same story, a very wide interval because 15 words is not much data.

Now we tackle the other edge case; when the sample size is not large *enough*. This typically happens when p is very close to 1 or p is very close to 0. This is typically going to be true for case 2. How do we determine if n is large enough? The general rule of thumb is that if both $np \geq 15$ and $n(1-p) \geq 15$ then you are good. If either np or nq is less than 15, then your estimate will be horrible. The derivation for this test statistic is bit long, and it's not very enlightening but I suggest you look it up if you are curious! There are 2 estimators. The first one is called the **plus-four** estimator (it's a simplified version of the Wilson interval), where we pretend we saw two extra successes and two extra failures

$$\frac{\tilde{p} - p_*}{\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}} \sim \mathcal{N}(0,1) \tag{2.30}$$

which makes the corresponding confidence interval

$$\widetilde{p} \pm z_{\alpha/2}\sqrt{\frac{\widetilde{p}(1-\widetilde{p})}{n+4}} \tag{2.31}$$

where

$$\tilde{p} = \frac{X+2}{n+4}$$

with X being the number of successes. The second is the **Agresti-Coull test statistic**

$$T_{\text{AC}} = \frac{\tilde{p} - p_0}{\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}} \approx N(0,1) \tag{2.32}$$

where

$$\tilde{p} = \frac{X + \frac{a}{2}}{n+a} \tag{2.33}$$

with a being the squared critical z score $a = z^2_{\alpha/2}$, $\tilde{n} = n + a$, and X the number of successes. For a 95% interval $a = 1.96^2 \approx 4$, which is where the plus-four rule comes from. Let us apply the plus-four statistic to case 2 for a 95% confidence interval. Realistically, the proportion of words that you will know is closer to like, less than 10%. In that case, for 100 samples, $np = 10$ which is less than 15 and therefore we should resort to the small sample statistic.

$X = 10$ implies

$$\tilde{p} = \frac{10+2}{100+4} \approx 0.115$$

at 95% confidence, $z_{.025} = 1.96$, and so our confidence interval is

$$\text{C. I.} = [.115 \pm .061] \approx [0.053, .176] \tag{2.34}$$

Problem 14.

Create a 95% confidence interval using the same data as the example but using the Agresti-Coull test statistic. Do you get similar results?

2.4 Inference Of Sample Variances

We've spent a lot of time talking about the estimating sample means (including sample proportions, which are in some sense a mean) but we haven't at all talked about the variance. The only time that we talked about the variance was when we were comparing the standard error for a known population variance and unknown population variance. We came up with the unbiased estimator for the sample variance in the form of

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

But perhaps now we are interested in estimating the population variance and creating confidence intervals for it. Let us set up the problem.

Case 3: My sister and I really like eating at Dave's Hot Chicken and Five Guy's. I generally like Five Guy's more, but the issue is that the food is so inconsistent. On some days, the food is really really good and it's the best meal I've had in a long while. On other days, the fries are really soggy and not very pleasant. Dave's on the other hand, is always very consistent. But I want to quantify these results; I want to know if Five Guy's really has a much more volatile food quality than Dave's.

Suppose my sister and I go to Dave's and Five Guy's 6 times and we both combine our ratings for the meal (out of 10). Can we construct a confidence interval for the variance?

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
Dave's	8	8	7.5	9	8.5	7.5
Five Guy's	6	9	9.5	6	10	8.5

Does Five Guy's really have a higher variance than Dave's (this will actually be answered in a later chapter)?

2.4.1 Transformations Of Random Variables

So we are interested in finding the asymptotic distribution for the estimated population variance. It's very tempting to say that the variance is normally distributed because of the central limit theorem but that's actually not the case. It is true that $\bar{X}$ is asymptotically normally distributed, but $\bar{X}$ lives inside the squared argument; so really the estimated variance is the sum of a bunch of squared normal random variables. It's also very tempting to say that if X is normally distributed, then X^2 has a distribution that is the square of the Gaussian

$$X \sim \mathcal{N}(0,1) \implies f_{X^2}(x) = \frac{1}{2\pi}\left(e^{-x^2/2}\right)^2$$

but that's simply NOT TRUE. Here is a very simple way to disprove this. If you square the above distribution, you get something like

$$f(x) = f_0 e^{-x^2}$$

which is another Gaussian with a different normalization constant f_0. The domain of the Gaussian can admit negative numbers. However, if X is normally distributed, then X^2 can only have positive numbers, and that immediately disproves the fact that X^2 is normally distributed. In fact, whatever distribution X^2 takes on, its domain can *only* be on the positive real numbers! So how do we come up with the distribution? Well, we will have to rely a bit on probability theory. Recall that $\mathbb{P}(X \leq x)$ is called **the cumulative distribution function** (CDF). It can be found by

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)\, dt$$

If we wanted to find the distribution for a transformed random variable, say, X^2, then we know

$$\mathbb{P}(X^2 \leq x) = \mathbb{P}\left(-\sqrt{x} \leq X \leq \sqrt{x}\right) = F(\sqrt{x}) - F(-\sqrt{x})$$

(we need both tails, since a negative X squares to a positive number too) Then to get back the distribution, we just have to take a derivative

$$f_{X^2}(x) = \frac{d}{dx}\left[F(\sqrt{x}) - F(-\sqrt{x})\right] = \frac{f(\sqrt{x}) + f(-\sqrt{x})}{2\sqrt{x}} \tag{2.35}$$

Let us view a concrete example. Suppose we know that X is a standard normal random variable $X \sim \mathcal{N}(0,1)$ and we want to find the distribution of X^2. To do this, we just need to follow the above procedure

$$f_{X^2}(x) = \frac{2}{2\sqrt{x}} \cdot \frac{1}{\sqrt{2\pi}} e^{-x/2}$$

since the standard normal density is symmetric, and simplifying gives

$$f_{X^2}(x) = \frac{1}{\sqrt{2\pi x}} e^{-x/2}, \quad x > 0 \tag{2.36}$$

This distribution has a name and it is called **the χ^2 distribution.**

Problem 15.

Numerically, you can verify that this must be the case. In whatever language you prefer, create a list of 10,000 numbers sampled from a normal distribution. Now square each element and plot the histogram. Overlay the χ^2_1 density on top and see how well it fits.

So the square of a standard normal gives us a χ^2 random variable. What if we had a sum of a bunch of squared normal random variables? Well, as it turns out, the result would still be a χ^2 random variable but now with n degrees of freedom, where n is the number of terms in the sum. To spell it out, if $\{X_1, ..., X_n\}$ are i.i.d. standard normal random variables, then

$$\sum_{i=1}^{n} X_i^2 \sim \chi^2_n \tag{2.37}$$

is a χ^2 random variable with n degrees of freedom. We won't derive the general form of the χ^2 because, again, it is beyond the scope of the course, but I will give it to you here

$$f(x;k) = \begin{cases} \frac{x^{k/2-1}e^{-x/2}}{2^{k/2}\Gamma\left(\frac{k}{2}\right)}, & x > 0 \\ 0, & \text{otherwise} \end{cases} \tag{2.38}$$

Problem 16.

Double-check the χ^2_n density. In whatever language you prefer, write n (you choose n,

but make it larger than 1) lists of 10,000 numbers sampled from the standard normal distribution. Square the elements of every list, add up every list, and then plot the histogram. Overlay equation 2.38 and check that it aligns with the histogram.

Problem 17.

Is it believable that the numbers 65,30 , and 55 are a random sample of size 3 from a normal distribution with $\mu = 50$ and $\sigma = 10$? Answer the question by using a chi square distribution. [**Hint:** Let $Z_i = (Y_i - 50)/10$ and use a χ^2 table for look up]

2.4.2 The Test Statistic

Let us now derive the test statistic. Let us start with the definition of the sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

We can introduce a 0 term by adding and subtracting by μ, which signifies the true population average

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \mu + \mu - \bar{X}\right)^2$$

And then expand

$$\implies S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \mu\right)^2 - 2\left(X_i - \mu\right)\left(\bar{X} - \mu\right) + \left(\bar{X} - \mu\right)^2$$

We can divide both sides by σ^2 and then distribute the sum

$$\frac{S^2}{\sigma^2} = \frac{1}{n-1}\left[\sum_{i=1}^{n}\left(\frac{X_i - \mu}{\sigma}\right)^2 - 2\left(\frac{\bar{X} - \mu}{\sigma}\right)\sum_{i=1}^{n}\frac{X_i - \mu}{\sigma} + \sum_{i=1}^{n}\left(\frac{\bar{X} - \mu}{\sigma}\right)^2\right]$$

A couple things to note. The first is that

$$\left(\frac{X_i - \mu}{\sigma}\right) = Z$$

is the standard normal by definition of standardization. The second thing is that the middle sum evaluates to $(n\bar{X} - n\mu)/\sigma$ so it just combines with the factored out term. This means that the middle sum and the last sum can be combined (remember, the last term is just a

constant). So we get

$$\frac{S^2}{\sigma^2} = \frac{1}{n-1}\left[\sum_{i=1}^{n} Z_i^2 - n\left(\frac{\bar{X}-\mu}{\sigma}\right)^2\right]$$
$$= \frac{1}{n-1}\left[\sum_{i=1}^{n} Z_i^2 - \left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2\right]$$

And we know that the right hand term inside the parenthesis is a standard normal (exactly, since the data is normal, and approximately otherwise by the central limit theorem). Squaring it means that it is a χ_1^2 with 1 degree of freedom. The left hand sum is a χ_n^2 with n degrees of freedom as we saw previously, so the difference between the 2 terms must be a χ^2 with $n-1$ degrees of freedom! (Strictly speaking, this step also needs $\bar{X}$ and S^2 to be independent, which is a special property of normal data that we prove in Part II.) Multiplying both sides by $n-1$, we see that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \tag{2.39}$$

In fact, this will form our test statistic for estimating the population variance

$$T = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \tag{2.40}$$

Problem 18.

Use the fact that $(n-1)S^2/\sigma^2$ is a χ^2 random variable with $n-1$df to prove that

$$\text{Var}\left[S^2\right] = \frac{2\sigma^4}{n-1}$$

(**Hint:** Use the fact that the variance of a χ^2 random variable with $k$df is $2k$.)

2.4.3 The Confidence Interval

The χ^2 distribution is not symmetric about the origin, so we can't say that the upper bound and lower bounds are negatives of each other. However, that doesn't change what the definition of a confidence interval is! This just means that we have to do something similar to the exact binomial test. Let us start with the definition of the confidence interval

$$\mathbb{P}(c_1 \leq T \leq c_2) = 1 - \alpha$$

Our test statistic is the one above, so we need to find the interval such that

$$\mathbb{P}\left(c_1 \leq \frac{(n-1)S^2}{\sigma^2} \leq c_2\right) = 1-\alpha$$

We know that the distribution of the test statistic is χ^2_{n-1} so we can actually just find the critical values c_1 and c_2. With our upper-tail convention, the lower one is $\chi^2_{1-\alpha/2,n-1}$ and the upper one is $\chi^2_{\alpha/2,n-1}$

$$\implies \mathbb{P}\left(\chi^2_{1-\alpha/2,n-1} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right) = 1-\alpha$$

So to create the confidence interval for σ^2, we just need to rearrange

$$\implies \mathbb{P}\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right) = 1-\alpha$$

which allows us to read out the $1-\alpha$ confidence interval as

$$\text{C. I.} = \left[\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right] \tag{2.41}$$

If we revisit the original problem of Dave's and Five Guy's, we can use a computer to find the sample variance for Dave's and Five Guy's as

$$S^2_{\text{Dave's}} \approx 0.342 \quad S^2_{\text{FG's}} \approx 3.067$$

So in terms of estimates, Five Guy's does seem to have a higher variance (we will test whether or not this is a fluke in the next chapter). What about their confidence intervals? Well, we just need the critical χ^2 values for $n-1=5$ degrees of freedom and $\alpha = .05$ which are 12.83 and .831 respectively for the upper bound and lower bound. Plugging this into (2.41) gives

$$\text{C. I.}_{\text{Dave's}} = \left[\frac{5 \cdot 0.342}{12.83}, \frac{5 \cdot 0.342}{.831}\right] \approx [.133, 2.05]$$

And doing the same for Five Guy's

$$\text{C. I.}_{\text{FG's}} = \left[\frac{5 \cdot 3.067}{12.83}, \frac{5 \cdot 3.067}{.831}\right] \approx [1.20, 18.45]$$

Problem 19.

How much interest certificates of deposit (CDs) pay varies by financial institution and also by length of the investment. A large sample of national one-year CD offerings in 2009 showed an average interest rate of 1.84 and a standard deviation $\sigma = 0.262$. A five-year CD ties up an investor's money, so it usually pays a higher rate of interest. However, higher rates might cause more variability. The table lists the five-year CD rate offerings from $n = 10$ banks in the northeast United States. Find a 95% confidence interval for the standard deviation of 5-year CD rates. Do these data suggest that interest rates for five-year CDs are more variable than those for one-year certificates?

Bank	Interest Rate (%)
Domestic Bank	2.21
Stonebridge Bank	2.47
Waterfield Bank	2.81
NOVA Bank	2.81
American Bank	2.96
Metropolitan National Bank	3.00
AIG Bank	3.35
iGobanking.com	3.44
Discover Bank	3.44
Intervest National Bank	3.49

Problem 20.

The derivation for the confidence interval for population variances relied on the fact that the data was normally distributed. In fact, that is how we were able to extract the asymptotic distribution being χ^2_{n-1}. Let us relax the assumption that the data is normally distributed. Generate n random numbers sampled from a uniform(0,4) distribution and then find the sample variance. Repeat this process 10,000 times and plot the histogram for $(n-1)S^2/\sigma^2$. Does it follow a χ^2_{n-1} distribution? If it doesn't, for what value of n does it begin to resemble a χ^2_{n-1} (if ever)?

3

Single Parameter Hypothesis Testing

"We didn't infer you were stupid. We implied it. You then inferred it."

– Sheldon Cooper

Chapter 2 laid the groundwork for a lot of what we are going to be doing for the rest of the course, which means that chapter 3 will actually go by relatively quickly. In the last chapter, we talked about what a statistic is and how we can use statistics to estimate population parameters. We will continue a similar line of thought but instead of asking "what is the population parameter?" we are going to ask "Is the population parameter ...?" This is a fundamentally different question, which is why we need to develop the tools to do **hypothesis testing.** What we will realize, though, is that the solutions to this problem has the same fundamental solution. Let us set up the cases that we are trying to solve.

Case 1: In 1964, Peter Higgs theorized the existence of a scalar field particle that would later be named after him called the Higgs boson. The significance of the Higgs boson is that it is the fundamental particle that gives all other particles "mass." Essentially, everything would be massless if not for this particle. Popular science dubbed it "the God particle." In wouldn't be until decades later that the LHC would discover the Higgs boson. The question is: how did they do it? How do physicists determine detection of a new particle from "noise" in the data?

Case 2: Before Covid-19, the antivax community was a pretty fringe group of conspiracy theorists. After Covid-19, many more people start becoming antivaxers for a myriad of reasons but unified in their distrust of the government. A frequently cited reason is that vaccines (specifically the covid vaccine) is very dangerous. If someone comes up to you and says "10% of this age group who took the vaccine died (compared to a smaller death rate when not taking the vaccine)." How do you go about testing this claim?

For case 2, in order for us to look at the data in more depth, we need to develop more tools. In this chapter, we will begin the work to answer these questions, but the problem of the covid vaccine will be a case that we will revisit over and over again in the following chapters.

3.1 The Decision Rule

The first thing that we are going to do is set up the systematic procedure for doing hypothesis testing. The purpose of hypothesis testing is to make conclusions about an estimate of a population parameter. Since the data is random, it should follow a well behaved distribution. Even if we don't know what the underlying distribution is, the central limit theorem allows us to say that the sample means will be normally distributed. The format goes as follows:

$$H_0 : \theta = \# \quad H_a : \theta \neq \# \tag{3.1}$$

or

$$H_0 : \theta \leq \# \quad H_a : \theta > \# \tag{3.2}$$

or

$$H_0 : \theta \geq \# \quad H_a : \theta < \# \tag{3.3}$$

where H_0 is called **the null hypothesis** and H_a is **the alternative hypothesis**, and θ is some parameter that we want to make conclusions of. If we want to make claims about the population parameter, we will need to estimate it. But we know from the last chapter that estimators of the population parameters are random variables and therefore follow some distribution! If we get a number that seems like it deviates from the null hypothesis, how do we determine whether it is a statistical fluke or if the null hypothesis is actually wrong? This is why we need to establish the idea of **statistical significance.** Obviously, our estimates will fluctuate around the true parameter value, but if it strays very far away, we can quantify the probability of getting at least that value. If the probability getting our estimate is below some cut off, then we can safely reject the null the hypothesis. What is the cut off? That is up to you! For historical reasons, people choose a probability of 5%. This **significance level** is denoted by α.

So it seems like the process is pretty straightforward enough. We can even put it in a 3 step plan!

1). Write down the null hypothesis and the alternative hypothesis.

2). Find a test statistic for the parameter we are testing as well as its asymptotic distribution. Use the data to estimate this statistic.

3). Find the probability of getting that statistic assuming the null is true. If it is below the significance level, we reject the null. Otherwise, we fail to reject the null.

3.1.1 1-Sided Tests

Hypothesis testing admits 2 possible tests: 1-sided or 2-sided tests. We will focus on the 1-sided tests first. 1-sided tests are typically associated with inequalities in the hypotheses. The easiest way to illustrate this is the following example.

Suppose you want to measure the height of people in your town (again), but this time, you make the claim that people in your town are actually not that tall. You believe that the average height is less than 68 inches. What would the hypothesis test be?

$$H_0 : \mu \leq 68 \quad H_a : \mu > 68$$

Voila! We are done with step 1. But what do we mean by 1 sided tests? It means that we only reject the null in 1 direction. For example, say that you measure the average height of people to be 65 inches. Well, that's well within the null hypothesis so you don't really care (because you didn't learn anything new from what you claimed). However, if you measured the average height to be 70 inches, now that might be grounds for statistical significance! In this example, we would only reject the null hypothesis when the estimate is greater than 68, because that is where the alternative hypothesis is. We don't reject when the estimate is less.

So now we move onto the second step: Finding a test statistic for the population parameter and estimating it. In this example, our claim is that the population average is less than 68 inches, so that means we want to estimate the population mean. We know a good estimate for the population mean: it's the sample average! And from the last chapter, we know that the test statistic will be

$$T = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

Of course, we would replace σ with S if the population variance is unknown and of course if n is not large enough then we have to resort to using the t distribution. Nothing new!

The only thing left to do is to estimate the mean from the data. You go and you collect averages from 50 people in your town and you find that the sample average is 69.6 inches. The population variance is 2. Now we can calculate our test statistic *under the null*

$$T = \frac{69.6 - 68}{\sqrt{2}/\sqrt{50}} = 8.0$$

The last step is determining the significance. Again, we are going to use a $\alpha = .05$ significance level but the choice is up to you. We know that our test statistic follows a normal standard normal distribution, so the cut off 5% or less happens when $x = 1.645$ or more. That is, the critical z score is $z_{.05} = 1.645$. Comparing that to our test statistic 8.0, we find that our test statistic is much much greater than the critical cut off, and therefore we would reject the null hypothesis. (in fact, a z score of 8 corresponds to a probability that is numerically indistinguishable from 0 on most calculators).

Problem 1.

We found that the probability of getting at least 69.6 was astronomically small. But what average height must we measure such that we are on the border? That is, value of $\bar{X}$ must we measure such that the probability of getting at least $\bar{X}$ is 5%?

3.1.2 2-Sided Tests

We've now done 1-sided tests; what about 2-sided tests? The only difference between a 2-sided test and a 1-sided test is that now there are 2 possible directions to reject. This type of hypothesis testing usually corresponds to equalities within the hypotheses. For example, going back to the average height measurement, maybe you say: "so the average height is not less than 68 inches. I hypothesize that the average is actually 69 inches!" Now there are 2 possible ways to reject the null: either we get an average that is significantly less than the null or we get an average that is significantly larger than the null. If we want to total probability of rejecting the null to be α, then that corresponds a significance level of $\alpha/2$ above or $\alpha/2$ below. But wait..isn't that how we constructed the confidence interval? Yes! In fact, the complement of the confidence interval forms the **rejection region** for a 2-sided hypothesis test!

Let's examine the height case more numerically. You just made the claim that the true average height is 69 inches. This makes the hypothesis test

$$H_0 : \mu = 69 \quad H_a : \mu \neq 69$$

Our test statistic is the same. To estimate it, we get

$$T = \frac{69.6 - 69}{\sqrt{2}/\sqrt{50}} \approx 3$$

The only thing different is that since we are dealing with 2 tails, the upper bound and the lower bound cut offs are higher. In fact, for $\alpha = .05$, the critical z scores are 1.96 or -1.96. In our case, we got a test statistic that is still greater than the critical 1.96, so therefore we *reject the null hypothesis.*

Problem 2.

An herbalist is experimenting with juices extracted from berries and roots that may have the ability to affect the Stanford-Binet IQ scores of students afflicted with mild cases of attention deficit disorder (ADD). A random sample of twenty-two children diagnosed with the condition have been drinking Brain-Blaster daily for two months. Past experience suggests that children with ADD score an average of 95 on the IQ test with a standard deviation of 15 . If the data are to be analyzed using the $\alpha = 0.06$ level of significance, what values of $\bar{y}$ would cause H_0 to be rejected? Assume that H_1 is two-sided.

Problem 3.

If $H_0 : \mu = \mu_o$ is rejected in favor of $H_1 : \mu > \mu_o$, will it necessarily be rejected in favor of $H_1 : \mu \neq \mu_o$? Assume that α remains the same.

We've also mentioned that the complement of the confidence interval is the rejection region. That is, if our test statistic is not within the 95% confidence interval, then we must reject the null hypothesis. Let's see this clearly. Since we have our test statistic, we can find create a 95% C.I. by

$$\text{C. I.} = [\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$$

Plugging in the numbers, we see that

$$\text{C. I.} = \left[69.6 - 1.96\frac{\sqrt{2}}{\sqrt{50}}, 69.6 + 1.96\frac{\sqrt{2}}{\sqrt{50}}\right] \approx [69.208, 69.992]$$

And since our null hypothesis is that $\mu = 69$, it is not in the 95% C.I. and we therefore

reject the null hypothesis.

Problem 4.

Which of the following differences is larger? Explain.

$$t_{.05,n} - t_{.10,n} \quad \text{or} \quad t_{.10,n} - t_{.15,n}$$

3.1.3 p Values

p values are deeply connected to the idea of a test statistic. In fact, the definition of p value is that it is the probability of observing *at least* the absolute value of the test statistic or more (that's for a 2 sided test. For a 1 sided test you drop the absolute value and only look in the direction of the alternative, which is why the 2 sided p value is *twice* the 1 sided one). Before, when we calculated the test statistic, we only compared it to the critical z or t scores. If we wanted to convert the test statistic to being a probability with respect to the null, then we would need to compare the p value to the critical α value. For example, in our 2 sided test, we found that the sample mean's test statistic was 3. If we take that the test statistic follows a normal distribution, what is the probability of observing at least 3? That is, what is $\mathbb{P}(|X| \geq 3)$? This probability is 0.0027. If we compare this to the critical $\alpha = .05$, then we see that the p value is much smaller than the α level and therefore we reject the null hypothesis.

Problem 5.

A random sample of 25 automobile owners is asked to keep a record of the kilometers they travel. The random sample showed an average of 23,500 kilometers and a standard deviation of 3,900 kilometers. It is planned to test the above with parametric-based hypothesis testing. Assume a 5% confidence level.

a). What should be the null and alternate hypothesis?

b). Which test statistic: z, t, or χ^2 test is applicable in this case?

c). According to your test statistic, calculate the critical value.

d). From the sample data, calculate the test value.

e). Decide if the null hypothesis is to be accepted or rejected.

Problem 6.

Why are p-values misleading? Suppose you are a statistician, and you want to calculate the average height of men in a town. Let's say that the true average height is 70 inches. You think that that is a bit too short, so you form a hypothesis that the average height of men is actually greater than 70. Generate a list of 10 heights sampled from a normal distribution with mean $\mu = 70$ and standard deviation $\sigma = 1$. Calculate the sample average, and then calculate the p-value (do this for each sample size). How many times do you have to do this until you get a p-value less than .05? (it should actually be really low).

Now let's go one step more. A friend says to you: this is not very suspicious, because under the null hypothesis, every p-value is equally likely (true statement), so a p-value of .05 is no more suspicious than a p-value of .99 (not true). You want to prove them wrong. Run the experiment 10,000 times and tally the number of times you get a p-value between .04-.05 and separately tally then number of times you get a p-value between 0-.01 and plot your tallies as a 2 bin histogram. What do you see? Now, change the true population height to 71 while still holding that null should be 70 inches. Replot the tally histograms. What do you see?

The point of this problem is to show that the critical value $p = .05$ is arbitrary, and often times not very helpful. In physics, we use 5 standard deviations σ away from the mean to determine statistical significance. So if you see a paper that uses p values, especially p values with relatively high values, you should be suspicious!

3.2 Testing Population Proportions

At this point, you should start noticing the structure. In fact, you can probably even deduce how to do hypothesis testing for population proportions. The only difference between testing population proportions and testing μ is that the form of the standard error is different. Let us attempt to do case 2 with our 3 step plan.

The claim is that 10% of people within a certain age bracket died from the Covid vaccine. You, as a health statistician, go out and conduct a survey and found that of the 200 people you surveyed, 15 died from the Covid vaccine (family member informed you). What is the hypothesis test?

$$H_0 : p = .1 \quad H_a : p \neq .1$$

The test statistic is almost the one from chapter 2, with one small change. When we test, we compute everything *assuming the null is true*, so we use the null value p_0 in the standard

error instead of $\hat{p}$,

$$T = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim \mathcal{N}(0,1)$$

(For confidence intervals there is no null value to plug in, which is why we used $\hat{p}$ there.) Now we just have to calculate the test statistic. From there, we can either compare the test statistic to the critical z score or we can compute the p value. I will go with comparing the critical z scores. Let us take the significance level to be $\alpha = .05$ and since this is a 2-sided test, $\pm z_{\alpha/2} = \pm 1.96$. We should double check that the large n condition is satisfied, and it is, since $np_0 = 20$ and $n(1-p_0) = 180$ are both bigger than 15. Computing our test statistic gives us

$$\hat{p} = 0.075 \implies T = \frac{.075 - .1}{\sqrt{\frac{.1(1-.1)}{200}}} \approx -1.18$$

And since this is not in the rejection region, we *do not reject the null.*

Problem 7.

Efforts to find a genetic explanation for why certain people are right-handed and others left-handed have been largely unsuccessful. Reliable data are difficult to find because of environmental factors that also influence a child's "handedness." To avoid that complication, researchers often study the analogous problem of "pawedness" in animals, where both genotypes and the environment can be partially controlled. In one such experiment (27), mice were put into a cage having a feeding tube that was equally accessible from the right or the left. Each mouse was then carefully watched over a number of feedings. If it used its right paw more than half the time to activate the tube, it was defined to be "right-pawed." Observations of this sort showed that 67% of mice belonging to strain A/J are right-pawed. A similar protocol was followed on a sample of thirty-five mice belonging to strain A/HeJ. Of those thirty-five, a total of eighteen were eventually classified as right-pawed. Test whether the proportion of right-pawed mice found in the A/HeJ sample was significantly different from what was known about the A/J strain. Use a two-sided alternative and let 0.05 be the probability associated with the critical region.

3.2.1 Small Sample Proportions

Doing hypothesis testing for small sample proportions is no different either. The only thing that we have to do is to remember the test statistic that we developed for small sample proportions all the way back in chapter 2. Depending on the type of data that we have, we

could either use the exact distribution if we assume that the data is binomially distributed. The other option is to use Wilson's adjusted or Agresti-Coull test statistic

$$T_W = \frac{\tilde{p} - p_*}{\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}} \sim \mathcal{N}(0,1) \tag{3.4}$$

$$T_{\text{AC}} = \frac{\tilde{p} - p_0}{\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}} \sim N(0,1) \tag{3.5}$$

In the latter case, both test statistics are approximately normal and therefore we can do hypothesis testing around that.

Problem 8.

Using both the Agresti-Coull and Wilson's test statistic, redo the covid example but with 50 participants in the survey instead of 200. Do you still have enough evidence to reject the null?

Problem 9.

Suppose $H_0 : p = 0.75$ is to be tested against $H_1 : p < 0.75$ using a random sample of size $n = 7$ and the decision rule "Reject H_0 if $k \leq 3$."
(a) What is the test's level of significance?
(b) Graph the probability that H_0 will be rejected as a function of p.

3.3 Type 1 and Type 2 Errors

Most of the results from hypothesis testing is not very exciting because it's almost a restatement of the things that we have done before in chapter 2. The only novel thing that arises with hypothesis testing that I believe is worth mentioning are the errors that plague hypothesis testing. In class, a student asked why does physics require such a strong condition to determine statistical significance but in statistics, the standard is an α level of .05. The answer was two-fold; the first is historical. Traditionally, statisticians have always used .95 as the standard for confidence intervals, and that has stuck for the past hundred or so years. The second reason is a bit more methodical, and that is because some fields can't risk pushing the α value to being that low. A very clear example of this is in law. If we push the critical α to being very small, then that means that only a very very small

portion of defendants will actually be found guilty, and we would be a lot more susceptible to letting guilty defendants go. On the other hand, if we were a bit more liberal with where we decide the cutoff is, then we run the risk of putting innocent people into prison. These are examples of what we call **type 1** and **type 2 errors.** To define it a bit more rigorously,

Type 1 Error: The probability of rejecting the null hypothesis given that the null is true

$$\mathbb{P}(\text{Rejecting the null}|\text{Null is true}) \tag{3.6}$$

To examine this a bit more quantitatively, let us first consider the null hypothesis. If the null hypothesis is true, when do we reject the null? Well, obviously, the rejection region! And the rejection region is the region where the p value for the statistic is less than the critical α level. But wait...that's nothing more than just α! So by construction, the probability of committing a type 1 error is α!

Let us examine an example with covid. As a health researcher, you are interested in the average duration of sickness for patients infected with covid. You believe that the mean duration is 3.5 days. You go and you collect 50 patients with covid and you measure the duration of illness and find a sample average of 4.1 days with a variance of 1 day. As a medical professional, you would like to be a bit more strict. How should you change the critical α level to reflect this? What is the hypothesis test?

First, to set up the hypothesis test, we have

$$H_0 : \mu = 3.5 \quad H_a : \mu > 3.5$$

Now, we address the problem: if we want to be strict, what does that mean? In this example, being strict means that you want to reduce the probability of making a false positive (that is, you don't want to say that the duration is longer when it really is 3.5 days). A consequence of making this error is that we treat the patient longer than we need to and that could harm the patient both financially and physically. Looking at the effect that α has on this, if we increase α, then you've increased the probability of saying that the duration is longer than in reality it actually is 3.5. *You've increased the probability of a type 1 error.* But that's not what we want, so we must *decrease* the critical α value for us to avoid making a type 1 error.

That's all and well, if we could just decrease the α value to our pleasing, why wouldn't we just make the α value arbitrarily small so that we are *very sure* of statistical significance when we see it?

Type 2 Error: The probability of **NOT** rejecting the null hypothesis given that the true

average is some other quantity

$$\mathbb{P}(\text{NOT rejecting null}|\mu \text{ is something else}) \tag{3.7}$$

A type 2 error is typically the harder of the 2 for people to comprehend. It is NOT the complement of a type 1 error; meaning, it does not simply equal $1-\alpha$. Let's see an example how.

We revisit the covid duration example that we set up previously. We saw that the probability of making a type 1 error is exactly equal to α, so if we wanted to decrease the probability of a type 1 error, we would decrease α. But now, let us examine the probability of us making a type 2 error. We have to ask ourselves: when do we *not* reject? That is, based on our choice α, where is the rejection region and its complement? Since the hypothesis test involves an inequality, the test must be 1 tailed (more specifically, upper). We create our test statistic as

$$T_0 = \frac{\bar{X}_{\text{crit}} - \mu_0}{S/\sqrt{n}}$$

The rejection boundary happens when this statistic equals the critical z score

$$\implies \frac{\bar{X}_{\text{crit}} - \mu_0}{S/\sqrt{n}} = z_\alpha$$

If we solve for $\bar{X}$, we find

$$\bar{X}_{\text{crit}} = \mu_0 + z_\alpha \frac{S}{\sqrt{n}} \tag{3.8}$$

Now, if we are committing a type 2 error, then we are saying that relative to the *alternative hypothesis* (that the true mean is actually some other number), what is the probability of not rejecting values less than the critical sample average that we calculated in equation (3.8). To find this probability, we need to find the cumulative probability associated with the alternative statistic

$$T_a = \frac{\bar{X}_{\text{crit}} - \mu_a}{S/\sqrt{n}} \tag{3.9}$$

To add numbers for visualization, let us use the numbers that we stated before as well as choosing the critical $\alpha = .05$ making $z_\alpha = 1.645$. To find the critical sample average

$$\bar{X}_{\text{crit}} = 3.5 + 1.645\frac{1}{\sqrt{50}} \approx 3.73$$

Thus, for any sample average less than 3.73, we do not reject the null. Now suppose the true average is really 3.8 days. To calculate the probability of a type 2 error, find the alternative

statistic

$$T_a = \frac{3.73 - 3.8}{1/\sqrt{50}} \approx -0.48$$

which has a corresponding cumulative probability of approximately 0.32. So even though the truth is different from the null, about a third of the time we would fail to notice! Immediately, we see that this is not the same thing as $1 - \alpha = 0.95$. (If the true average were all the way out at 4.1 days, $T_a \approx -2.6$ and β drops to about 0.004, so β depends a lot on *which* alternative is true.) In fact, the probability of a type 2 error (the cumulative probability

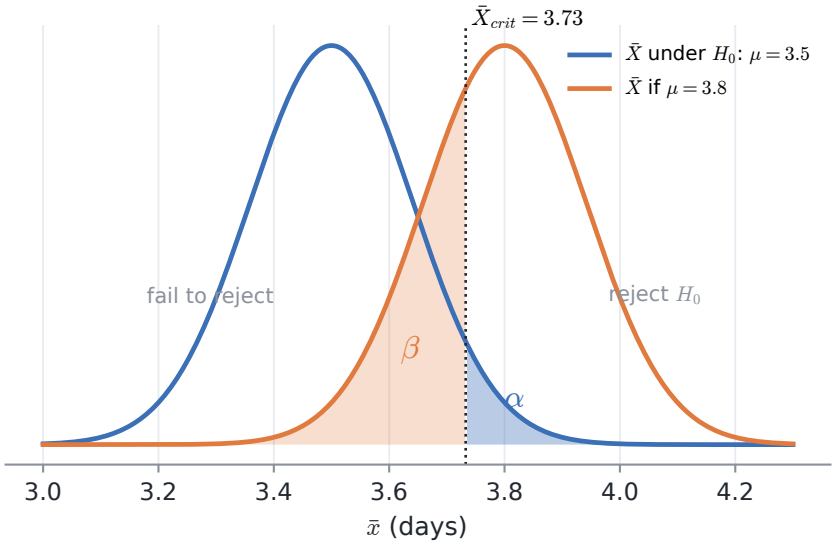


Figure 3.1: The sampling distribution of $\bar{X}$ under the null ($\mu = 3.5$, blue) and under the alternative $\mu = 3.8$ (orange), with $\sigma = 1$ and $n = 50$. The blue tail past the cutoff is α, and the orange area below the cutoff is β.

associated with the alternative statistic) is called β. To get a relationship between α and β, we plug the critical sample average into the alternative statistic

$$T_a = \frac{\mu_0 + z_\alpha \frac{S}{\sqrt{n}} - \mu_a}{S/\sqrt{n}}$$

$$T_a = \frac{\mu_0 - \mu_a}{S/\sqrt{n}} + z_\alpha \tag{3.10}$$

To get β, find the cumulative probability

$$\beta = \mathbb{P}\left(X \leq \frac{\mu_0 - \mu_a}{S/\sqrt{n}} + z_\alpha\right) \tag{3.11}$$

where X can be a normal random variable or a t random variable. So there is an inverse relation between β and α. If you decrease α, you increase β. If you increase α, you decrease β.

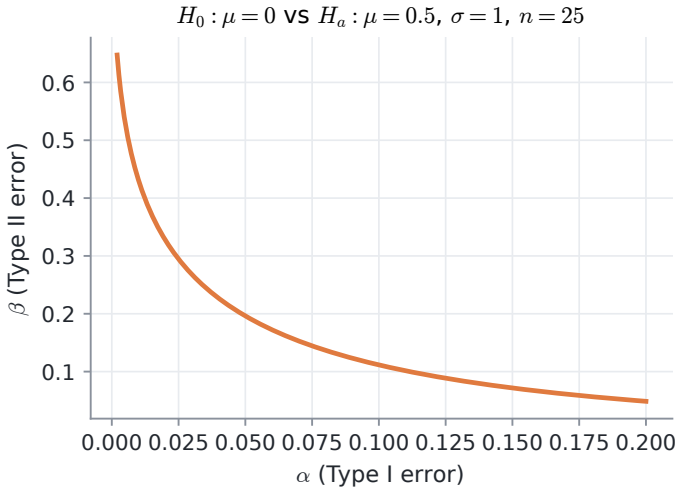


Figure 3.2: Plot of β as a function of α for $\mu_0 = 0$, $\mu_a = .5$, $S = 1$, and $n = 25$ for visualization purposes.

Problem 10.

The analysis that we have done so far is for a 1 sided test ($>$). Show how the alternative statistic and, consequently, β would change if instead the hypothesis test involved $<$ or a 2 sided test with $\neq$.

Problem 11.

Suppose a sample of size 1 is taken from the pdf $f_Y(y) = (1/\lambda)e^{-y/\lambda}, y > 0$, for the purpose of testing

$$H_0 : \lambda = 1$$
versus
$$H_1 : \lambda > 1$$

The null hypothesis will be rejected if $y \geq 3.20$.
a). Calculate the probability of committing a Type I error.

b). Calculate the probability of committing a Type II error when $\lambda = \frac{4}{3}$.
c). Draw a diagram that shows the α and β calculated in parts (a) and (b) as areas.

Problem 12.

Polygraphs used in criminal investigations typically measure five bodily functions: (1) thoracic respiration, (2) abdominal respiration, (3) blood pressure and pulse rate, (4) muscular movement and pressure, and (5) galvanic skin response. In principle, the magnitude of these responses when the subject is asked a relevant question ("Did you murder your wife?") indicate whether he is lying or telling the truth. The procedure, of course, is not infallible, as a recent study bore out (82). Seven experienced polygraph examiners were given a set of forty records-twenty were from innocent suspects and twenty from guilty suspects. The subjects had been asked eleven questions, on the basis of which each examiner was to make an overall judgment: "Innocent" or "Guilty." The results are as follows:

	Suspect's True Status	
Examiner's Decision	**Innocent**	**Guilty**
"Innocent"	131	15
"Guilty"	9	125

What would be the numerical values of α and β in this context? In a judicial setting, should Type I and Type II errors carry equal weight? Explain.

3.3.1 Power Of A Statistic

We discussed in the previous section that β is not the complement of α despite our intuition. If so, then what is the complement of β? Does it even have an interpretation? If β is the probability of not rejecting the null given that the alternative is true, then the complement would be the probability of *successfully rejecting the null given that the alternative is true.* When we do statistics, we aren't really interested in saying "we accept the null." We can only say that we failed to reject the null or we reject the null. Thus, the probability of successfully rejecting the null given that the alternative is true is called the **power** of the statistic. Mathematically, we can denote the power as

$$\text{Power} = 1 - \beta \tag{3.12}$$

We saw before that β is a function of both α and some alternative parameter μ_a. If the power is defined this way, then that means that the power of the statistic must also be a function of the alternative parameter μ_a. In fact, we can plot the power of a statistic as a function of μ_a just by using the formula for β. These plots are called **power curves.** Power

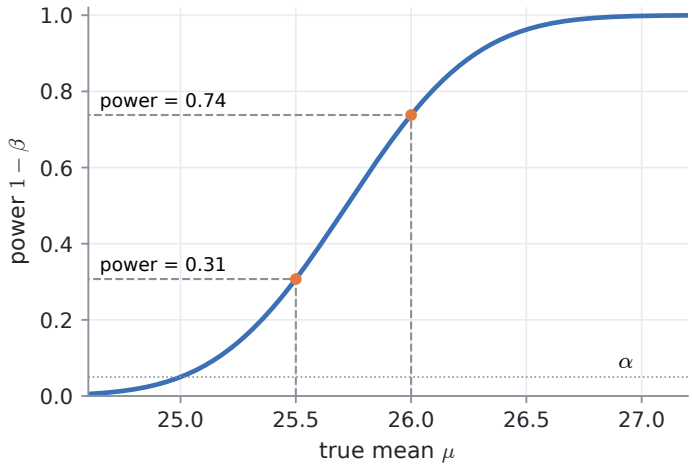


Figure 3.3: The power curve of the one-sided test of $H_0 : \mu = 25$ with $\sigma = 2.4$, $n = 30$ and $\alpha = 0.05$. At $\mu = 25.5$ the test only catches the difference 31% of the time, and at $\mu = 26$ it catches it 74% of the time.

curves can sometimes be useful because they can help us distinguish 2 different procedures. When we are doing hypothesis testing, there are a myriad of different procedures (some more computationally costly than others) that we can perform for a myriad of different μ_0 and μ_a. Each procedure has its own associated power curve. Thus, if the goal is to separate the null from the alternative, then we should choose the procedure that produces the steepest power curve. In Figure 3.4, we see 2 procedures A and B produce 2 different power curves for the same critical α. Why might we want to choose procedure B over procedure A? If we are interested in separating the null from the alternative, then what method B tells is that for every value of μ that is different from the null parameter μ_0, the procedure is able to successfully reject the null with a much higher probability.

Problem 13.

We've discussed the effects of α on β (which also means the effects of α on the power), let us look at the effects of n and σ on the power. Suppose an experimenter wishes to

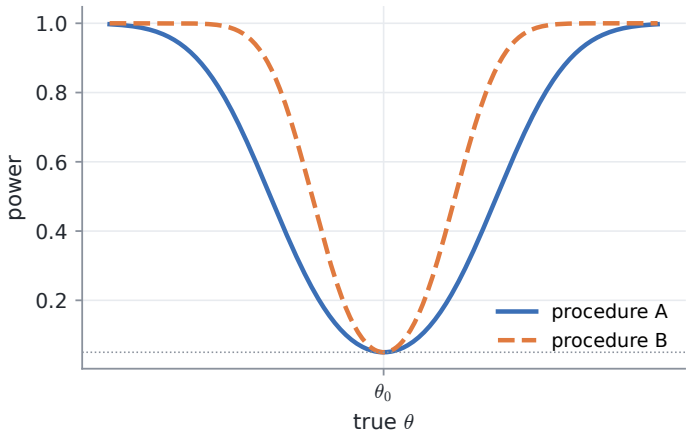


Figure 3.4: Power curves for 2 different two-sided procedures A and B with the same α. Procedure B rejects with higher probability at every $\theta \neq \theta_0$.

test

$$H_0 : \mu = 100$$

versus

$$H_1 : \mu > 100$$

at the $\alpha = 0.05$ level of significance and wants $1 - \beta$ to equal 0.60 when $\mu = 103$.
a). What is the smallest (i.e., cheapest) sample size that will achieve that objective? Assume that the variable being measured is normally distributed with $\sigma = 14$.
b). Now we fix n to be a constant. Does there exist a smallest/largest σ value that achieves this objective?

Problem 14.

Will $n = 45$ be a sufficiently large sample to test $H_0 : \mu = 10$ versus $H_1 : \mu \neq 10$ at the $\alpha = 0.05$ level of significance if the experimenter wants the Type II error probability to be no greater than 0.20 when $\mu = 12$? Assume that $\sigma = 4$.

3.4 Testing Population Variance

We always save the best for last. Except this is not really the best. Nor is it really that interesting, to be honest. Doing hypothesis testing for sample variances is no different than doing hypothesis testing for the mean. The only difference is that you will be using a different

test statistic, but we've already discussed what that was back in chapter 2. Remember that the test statistic for the sample variance was

$$T_{\text{var}} = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1} \tag{3.13}$$

The χ^2 distribution is not symmetric, so the upper and lower bounds for the confidence intervals were not just signed versions of each other. But that's no matter! It doesn't change the fact that it still is the same procedure as doing hypothesis testing for the mean. As mentioned before, the case of a 2 sided hypothesis test for the population variance has already been done; the rejection region is just the complement of the confidence interval for some α level. Thus, let us do an example of a 1 sided test.

Consider the hypothesis test

$$H_0 : \sigma^2 = 25 \quad H_a : \sigma^2 < 25 \tag{3.14}$$

When do we reject the null hypothesis? We reject the null hypothesis when our sample variance S^2 is significantly smaller than 25. If we choose α to be the cutoff, then we would need to compare the test statistic with the lower critical value $\chi^2_{1-\alpha,n-1}$. To put some numbers to this, let $n = 25$ and the sample variance be $S^2 = 21$. Do we reject the null? The only missing ingredient is the value for the critical χ^2 value, which, more precisely, would be $\chi^2_{.95,24}$ if we choose the standard $\alpha = .05$. Looking up this critical χ^2 value gives approximately 13.85. Thus, if our test statistic is less than 13.85, we will reject the null. Computing the test statistic gives

$$T_{\text{var}} = \frac{24 \cdot 21}{25} \approx 20.2$$

which is greater than the critical χ^2 value, so we do NOT reject the null.

Problem 15.

For the example that we just did, what is the probability of a type 1 error and what is the probability of a type 2 error?

Problem 16.

Motivation of drug dealers. Refer to the Applied Psychology in Criminal Justice (Sept.

2009) study of the personality characteristics of convicted drug dealers. A random sample of 100 drug dealers had a mean Wanting Recognition (WR) score of 39 points, with a standard deviation of 6 points. Recall that the WR score is a quantitative measure of a person's level of need for approval and sensitivity to social situations. (Higher scores indicate a greater need for approval.) A criminal psychologist claims that the range of WR scores for the population of convicted drug dealers is 42 points. Do you believe the psychologist's claim? (Hint: Assume the population of WR scores is normally distributed.)

Problem 17.

A stock analyst claims to have devised a mathematical technique for selecting high-quality mutual funds and promises that a client's portfolio will have higher average ten-year annualized returns and lower volatility; that is, a smaller standard deviation. After ten years, one of the analyst's twenty-four-stock portfolios showed an average ten-year annualized return of 11.50% and a standard deviation of 10.17%. The benchmarks for the type of funds considered are a mean of 10.10% and a standard deviation of 15.67%. (a) Let μ be the mean for a twenty-four-stock portfolio selected by the analyst's method. Test at the 0.05 level that the portfolio beat the benchmark; that is, test $H_0 : \mu = 10.1$ versus $H_1 : \mu > 10.1$. (b) Let σ be the standard deviation for a twenty-four stock portfolio selected by the analyst's method. Test at the 0.05 level that the portfolio beat the benchmark; that is, test $H_0 : \sigma = 15.67$ versus $H_1 : \sigma < 15.67$.

Problem 18.

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n from a normal distribution having mean μ and variance σ^2. What is the smallest value of n for which the following is true?

$$P\left(\frac{S^2}{\sigma^2} < 2\right) \geq 0.95$$

(Hint: Use a trial-and-error method.)

4

2 Sample Inference And Hypothesis Testing

"And then there were 2"

– Various

For the past 2 chapters, we built the foundation to discuss statistical inference and hypothesis testing, but we only considered the 1 sample case. In practice, this is very boring. The types of questions that we are interested in asking are not as simple as "what is the population mean" or "what is the population variance?" For a great many people, especially those in the medical and health sciences, they are interested in comparing the effects on 2 samples. Or in general, we might be interested in finding out how 2 samples might be related to each other. Before we delve into the mechanics, let us set up the problems.

Case 1: In the health sciences, researchers might be interested in the effect of a drug on 2 different groups. Suppose there is a new drug that is to treat ADHD by blocking the uptake of neurotransmitters such as dopamine and serotonin. You randomly select 2 groups of people, and they might be of different sizes, but you give 1 group the test drug and the other a placebo (a sugar cube or something). Once you've given them the drug, you measure their productivity by asking them to perform a certain task. You expect that those that take the drug will have an average productivity time of μ and the placebo to have a productivity time of some number less than μ. How can you determine whether or not there was a statistically significant effect on the average productivity time between the 2 groups?

Case 2: I would like to believe that if you are taking this class, then you are interested in learning (even if it means that learning is difficult). My goal for the class is for everyone to have learned something about statistics. But you're skeptical about the efficacy of my teaching, so you want to conduct a test to determine whether or not people actually learned. At the beginning of the semester, I give you guys an exam. At the end of the semester, I give you guys the exact same exam. If we could show that the average is greater after me teaching, then that means I have done my job. How can we determine the statistical significance of the before and after effects?

Case 3: Let us revisit the Covid example. We've talked about vaccine skeptics, but now let's actually be able to examine the numbers. Suppose you have 2 groups of people: those that took the vaccine and those that didn't. For both cases, you would like to measure the morbidity rate (i.e. how many of those that took the vaccine experienced severe effects from covid versus those that didn't). How can we test whether or not the covid vaccine was effective in reducing the morbidity rate.

In a lot of these cases, the formalism that we will need have already been developed in previous chapters. Thus, there will be a bit of repetition. But obviously, similar to the case of calculus, going from 1 dimension to 2 dimensions can be a bit challenging but after that, the generalization to arbitrary numbers is very straightforward.

4.1 2 Sample Means

If we think back to inferences of single parameters, in the case of the mean, we typically had that the test statistic was

$$T = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \tag{4.1}$$

Using the central limit theorem, we know that the above statistic is normally distributed, the MLE for μ is the sample average $\bar{X}$, and the variance (if known) is σ^2/n. If we wanted to generalize this to comparing 2 means μ_1 and μ_2, we could naively write that the test statistic should be

$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sigma/\sqrt{n}}$$

And you would be correct in spirit but the subtle details are missing. The most obvious one is the fact if you have 2 samples with different sizes n and m, should there still be a $\sqrt{n}$ dividing the variance? And speaking of the variance, could it be possible that the variance for each sample average be different? These are the details that we are missing.

4.1.1 Adding Variances

If we can answer the question of how standard deviations add, then we get both ingredients in the standard error for free. It might be very tempting to say that the standard deviation adds linearly because the sample means are subtracted linearly. But that's actually not the case, and to see that, consider the 2 random variables $\bar{X}$ and $\bar{Y}$. We know that the variance for the sample average is σ_X^2/n_X and σ_Y^2/n_Y respectively. Thus, if we sum the 2 random variables and take the variance

$$\begin{aligned}\sigma_{X-Y}^2 &= \text{Var}\left[\bar{X} - \bar{Y}\right] \\ &= \frac{1}{n_X}\sigma_X^2 + \frac{1}{n_Y}\sigma_Y^2 \end{aligned} \tag{4.2}$$

and thus we find that the standard error is

$$\text{s. e.} = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \tag{4.3}$$

We say that the standard deviation **adds in quadrature.** From here, we see that there are 3 special cases to look at. The first is when the variance is the same for both samples. In which case, (4.3) reduces down to

$$\text{s. e.} = \sigma\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}} \tag{4.4}$$

The second case is if the sample size for both is the same

$$\text{s. e.} = \sqrt{\frac{\sigma_X^2 + \sigma_Y^2}{n}} \tag{4.5}$$

which is not that exciting. The last case is when both the sample size and the variances are the same. This reduces the standard error completely to just

$$\text{s. e.} = \frac{\sigma}{\sqrt{2n}} \tag{4.6}$$

Problem 1.

We mentioned before that in physics, a discovery is a 5-σ event. However, it oftentimes is very difficult to measure a single sample with a 5-σ deviation. If you have 2 detectors,

each measuring their own respective sample averages, what do the individual standard errors have to be to get a 5-σ event?

4.1.2 Large Independent Samples

Now that we have solved the issue of the standard error, we can write the test statistic for 2 independent samples completely

$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} \tag{4.7}$$

All that's left to do is find the asymptotic distribution. The theory of additions of random variables is beyond the scope of this course, so we won't develop it (if you take MA582, we will talk more about it in depth). This is when we can use computers to make our lives a bit easier!

Problem 2.

Create 2 lists of 4000 or so numbers generated from a standard normal distribution with mean μ (of your choosing). For simplicity, also choose the same variance for both of them, so let's say $\sigma^2 = 1$.
a). Using the expectation $\mathbb{E}$, what is the expectation $\mathbb{E}[X_1 + X_2]$?
b). What is $\mathrm{Var}[X_1 + X_2]$?
c). Now add both lists that you've generated and plot it as a histogram (with density=True). Using the equation for the pdf of the Gaussian with mean and variance calculated in parts a). and b)., overlay the histogram with the plot of said Gaussian. How well does it line up?

As problem 2 has demonstrated, linear combinations of i.i.d. normal random variables are themselves normal random variables. Thus, we see that the asymptotic distribution for the statistic (4.7) is a standard normal!

We should now be able to answer case 1 using a hypothesis test. Under the null, we should expect that the drug has the same effect as the placebo. To determine if there is a significance, the alternative must be that the average productivity time for the drug is longer (if it were shorter, that would be problematic)

$$H_0 : \mu_{\text{drug}} = \mu_{\text{placebo}} \quad H_a : \mu_{\text{drug}} > \mu_{\text{placebo}}$$

Now we actually conduct the experiment and find that $\bar{X}_{\text{drug}} = 6$ hours and $\bar{Y}_{\text{placebo}} = 2$ hours. The sample sizes for both groups are the same (50), and for now, let us assume that the variances for the drug group is 2 and the variance for the placebo group is 1. Under the null, we should expect that the statistic (4.7) becomes

$$\frac{\bar{X}_{\text{drug}} - \bar{Y}_{\text{placebo}} - 0}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_X^2}{n_X}}} \sim \mathcal{N}(0,1) \tag{4.8}$$

$$\implies T = \frac{6-2}{\sqrt{\frac{4}{50} + \frac{1}{50}}} \approx 12.65$$

The p-value for this is astronomically tiny and therefore we should reject the null hypothesis and say that the drug actually does increase the productivity of individuals.

Problem 3.

An executive has two routes that she can take to and from work each day. The first is by interstate; the second requires driving through town. On the average it takes her 33 minutes to get to work by the interstate and 35 minutes by going through town. The standard deviations for the two routes are 6 and 5 minutes, respectively. Assume the distributions of the times for the two routes are approximately normally distributed.
a). What is the probability that on a given day, driving through town would be the quicker of her choices?
b). What is the probability that driving through town for an entire week (ten trips) would yield a lower average time than taking the interstate for the entire week?

Problem 4.

Does rudeness really matter in the workplace? A study in the Academy of Management Journal (Oct. 2007) investigated how rude behaviors influence a victim's task performance. College students enrolled in a management course were randomly assigned to one of two experimental conditions: rudeness condition (45 students) and control group (53 students). Each student was asked to write down as many uses for a brick as possible in five minutes; this value (total number of uses) was used as a performance measure for each student. For those students in the rudeness condi-

tion, the facilitator displayed rudeness by berating the students in general for being irresponsible and unprofessional (due to a late-arriving confederate). No comments were made about the late-arriving confederate for students in the control group. The number of different uses of a brick for each of the 98 students is shown in the table. Conduct a statistical analysis (at $\alpha = .01$) to determine if the true mean performance level for students in the rudeness condition is lower than the true mean performance level for students in the control group. Use the following data (the complete data set is found on blackboard) to perform your analysis.

Control Group:

1 24 5 16 21 7 20 1 9 20 19 10 23 16 0 4 9 13
17 13 0 2 12 11 7 1 19 9 12 18 5 21 30 15 4 2
12 11 10 13 11 3 6 10 13 16 12 28 19 12 20 3 11

Rudeness Condition:

4 11 18 11 9 6 5 11 9 12 7 5 7 3 11 1 9 11
10 7 8 9 10 7 11 4 13 5 4 7 8 3 8 15 9 16
10 0 7 15 13 9 2 13 10

4.1.3 Confidence Intervals

Now that we have the test statistic and the asymptotic distribution, creating the confidence is as straightforward as it gets. Remember that the definition of the $1-\alpha$ confidence interval for a normal distribution is

$$\text{C. I.} = \bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \tag{4.9}$$

If we also have the standard error for 2 independent samples, then we can construct the confidence interval *for the difference in means* $\mu_1 - \mu_2$ as

$$\text{C. I.} = \left(\bar{X} - \bar{Y}\right) \pm z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \tag{4.10}$$

For case 1, we can construct a confidence interval for the differences in productivity time by plugging in the numbers to (4.10). Using a standard $1 - \alpha = .95$ confidence level, we find that the confidence interval is

$$\begin{aligned}\text{C. I.} &= (6-2) \pm 1.96\sqrt{\frac{4}{50} + \frac{1}{50}} \\ &\approx [3.38, 4.62]\end{aligned} \tag{4.11}$$

What you will quickly realize is that the construction of the confidence intervals and the hypothesis testing will follow the exact same procedure for all the statistics that we will develop.

4.1.4 Small Independent Samples

Of course things will not always be squeaky clean. Similar to when we did the small sample inference in chapter 2 with the t test, we should expect that for inferences on 2 small samples, the asymptotic distribution will be t-distributed. Looking back to the 1 sample t statistic

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \tag{4.12}$$

as well as equation (4.7), perhaps we can conclude that the small sample statistic must be

$$t = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n_X} + \frac{S_X^2}{n_Y}}} \tag{4.13}$$

where S^2 is the unbiased estimator for the sample variance. You would be *almost* correct. As it turns out, the above statistic (which is known as the **Welch t-Test**) is not exactly t-distributed; it is only approximately t-distributed. The actual distribution is some crazy complicated mess. Is all hopes for an exact t-distributed statistic gone? Not all! First, let us examine this special case and then discuss the general case in the following subsection. For now, we will make the assumption that $X_1, .., X_n$ are i.i.d. $\mathcal{N}(\mu_X, \sigma^2)$ and $Y_1, .., Y_n$ are i.i.d. $\mathcal{N}(\mu_Y, \sigma^2)$. That is, both sample data are normally distributed *with the same (unknown) variance*. If that is the case, then we can pool the sample variances of each dataset together and factor it out of the square root in the denominator to say that the test statistic is

$$t = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}} \tag{4.14}$$

where S_p is known as the **pooled sample variance.** To derive the estimator for the pooled sample variance, we just have to remember back to when we derived the unbiased estimator for the sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

We argued that the average of the sum of squared error was $(n-1)\sigma^2$, so to make the estimator unbiased, we had to divide by $n-1$. So if we combined the 2 sample variances

into 1 pooled variance by adding them (to get the *total* sample variance)

$$\underbrace{\sum_{i=1}^{n_X}\left(X_i-\bar{X}\right)^2}_{=(n_X-1)s_X^2}+\underbrace{\sum_{j=1}^{n_Y}\left(Y_j-\bar{Y}\right)^2}_{=(n_Y-1)s_Y^2}$$

we would get

$$(n_X-1)S_X^2+(n_Y-1)S_Y^2=\sum_{i=1}^{n_X}\left(X_i-\bar{X}\right)^2+\sum_{i=1}^{n_Y}\left(Y_i-\bar{Y}\right)^2$$

Using the linearity of the expectation (and using the i.i.d. property)

$$\mathbb{E}\left[S_X^2+S_Y^2\right]\propto(n_X-1)\sigma^2+(n_Y-1)\sigma^2=(n_X+n_Y-2)\,\sigma^2$$

and we see that in order to make the pooled variance an unbiased estimator, we have to divide by (n_X+n_Y-2)

$$S_p^2=\frac{(n_X-1)S_X^2+(n_Y-1)S_Y^2}{n_X+n_Y-2} \tag{4.15}$$

So with this unbiased estimator for the pooled sample variance, the statistic (4.14) is exactly t-distributed with n_X+n_Y-2 degrees of freedom.

Revisiting case 1, if the sample size that we chose was not 50 for each, but say 10 for the drug group and 17 for the non drug group with sample variances 4 and 2 respectively, using all the relevant assumptions about the data, we can redo the hypothesis testing but this time use the pooled t-test

$$S_p^2=\frac{(10-1)4+(17-1)2}{25}=2.72$$

$$\implies t=\frac{6-2}{2.72\sqrt{\frac{1}{10}+\frac{1}{17}}}\approx 3.69$$

For 25 degrees of freedom, the critical .95 α level t value is 1.708, so we still reject the null hypothesis even though the test statistic gave us a more reasonable probability.

Problem 5.

Using the likelihood function, find the MLE for σ^2 for the 2 sample Gaussian model.

Is the estimator biased or unbiased?

Problem 6.

Poverty Point is the name given to a number of widely scattered archaeological sites throughout Louisiana, Mississippi, and Arkansas. These are the remains of a society thought to have flourished during the period from 1700 to 500 b.c. Among their characteristic artifacts are ornaments that were fashioned out of clay and then baked. The following table shows the dates (in years b.c.) associated with four of these baked clay ornaments found in two different Poverty Point sites, Terral Lewis and Jaketown (86). The averages for the two samples are 1133.0 and 1013.5, respectively. Is it believable that these two settlements developed the technology to manufacture baked clay ornaments at the same time? Set up and test an appropriate H_0 against a two-sided H_1 at the $\alpha = 0.05$ level of significance. For these data $s_x = 266.9$ and $s_y = 224.3$.

Problem 7.

a). Suppose $H_0 : \mu_X = \mu_Y$ is to be tested against $H_1 : \mu_X \neq \mu_Y$. The two sample sizes are 6 and 11. If $s_p = 15.3$, what is the smallest value for $|\bar{x} - \bar{y}|$ that will result in H_0 being rejected at the $\alpha = 0.01$ level of significance?
(b) What is the smallest value for $\bar{x} - \bar{y}$ that will lead to the rejection of $H_0 : \mu_X = \mu_Y$ in favor of $H_1 : \mu_X > \mu_Y$ if $\alpha = 0.05, s_P = 214.9, n = 13$, and $m = 8$?

4.1.5 The Welch Statistic

In the last section, we worked with the assumption that the 2 sample variances came from the same population variance. This was because if both samples had the same variance, we could pull it out into a pooled variance and the test statistic would be exactly t-distributed. In the case where the variances might be different, the distribution for the Welch statistic is no longer exactly t but could very well be approximated as t. Finding the exact distribution is an unsolved problem called the **Behren-Fisher Problem**.

To understand the Welch statistic, we need to start with a few reminders. Remember that the unbiased estimators for the sample variance is given by sum of squared deviations from

the mean divided by $n-1$. We also know from the end of chapter 2 that

$$\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Looking at the Welch statistic given in (4.13), if we divide the top and bottom by the following

$$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{S_X^2}{n}+\frac{S_Y^2}{m}}} = \frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \div \frac{\sqrt{\frac{S_X^2}{n}+\frac{S_Y^2}{m}}}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \tag{4.16}$$

The reason we do this is because in the numerator, we have a standard normal random variable. Since $\bar{X}$ and $\bar{Y}$ are independent from each other and their respective sample variances, the quantity in the denominator would look something similar to

$$\frac{\frac{S_X^2}{n}+\frac{S_Y^2}{m}}{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}$$

which is χ^2 distributed if σ was known. This in turn would make (4.16) exactly t-distributed if σ was known. But the issue is is that σ is oftentimes unknown, so we have to resort to using approximations. We can do a bit of simplification on (4.16) to get the statistic

$$T = \frac{(\bar{X}-\bar{Y})/\sqrt{\sigma_X^2/n_X+\sigma_Y^2/n_Y}}{\sqrt{\left(s_X^2/n_X\right)/\left(\sigma_X^2/n_X\right)+\left(s_Y^2/n_Y\right)/\left(\sigma_Y^2/n_Y\right)}} \tag{4.17}$$

then we can approximate it as a t-distribution with degrees of freedom

$$df = \frac{\left(\frac{s_X^2}{n}+\frac{s_Y^2}{m}\right)^2}{\frac{\left(s_X^2/n\right)^2}{n-1}+\frac{\left(s_Y^2/m\right)^2}{m-1}} \tag{4.18}$$

Problem 8.

Define the quantity

$$V = \frac{s_X^2}{n_X}+\frac{s_Y^2}{n_Y}$$

a). Find the expectation $\mathbb{E}[S]$ and variance $\mathrm{Var}[S]$ to subsequently find the expectation and variance of S

b). The idea of Satterthwaite's approximation relies on $V \sim \kappa\chi^2_\nu$ (χ^2 distributed with ν degrees of freedom) and κ being some constant. Using the fact that the mean and variance of a χ^2_ν random variable with degrees of freedom ν is ν and 2ν respectively,

write a system of 2 equations relating ν to σ_X and σ_Y.
c). We have 2 unknowns: κ and ν. Using the system of equations in part b). to solve for κ, and then use the expectation of V to subsequently find ν in terms of σ_X, σ_Y, and the sample sizes n and m
d). In the case where σ_X and σ_Y are unknown, we replace the σ^2 with their respective sample variances S^2. Show that when you do this, you retrieve the degrees of freedom equation found in equation (4.18).

4.1.6 Paired Differences

We first started out with a boat load of assumptions on the data and slowly relaxed each of them to reveal how this relaxation affects the statistic and its distribution. We do the same here and relax the independence condition on the means. But this could mean a lot of different things, so what do we mean by independence? Let us examine case 2; if we tried to use the previous tests to solve case 2, we would fail. The test could very well conclude that there is no statistical significance in the difference in exam scores despite *every student* doing better than the previously did. Why? Because second sample (the exam scores at the end of the semester) are *not independent of the other sample.* In fact, for each student, their before and after scores come in pairs, so the appropriate statistic to use would be called the **paired t-test.** The most intuitive solution would be to find not the difference between sample averages, but the average of the differences. That is, suppose we define X_i to be the exam scores before the start of the semester and Y_i are the scores after. Define a new variable X_d as the pairwise difference in scores

$$Z_i \equiv X_i - Y_i \tag{4.19}$$

then we can construct the pair test statistic

$$t_d = \frac{\bar{Z} - \mu_d}{S_d/\sqrt{n_d}} \tag{4.20}$$

where μ_d is the average of the pairwise difference, S_d is the standard deviation of the pairwise differences, and n_d is the number of pairs. At first glance, it's not immediately clear what the asymptotic distribution is; but if we look at how we defined (4.19), it's just a linear combination of normal random variables; which makes $\bar{Z}$ also a normal random variable because it is also a linear combination of normal random variables. For similar reasons, the sample variance is χ^2_{n-1} distributed, so knowing what we know before, (4.20) must be t_{n-1}

distributed

$$\frac{\bar{Z} - \mu_d}{S_d/\sqrt{n_d}} \sim t_{n-1} \tag{4.21}$$

Now we can begin solving case 2. The hypothesis test is

$$H_0 : \mu_d = 0 \quad H_a : \mu_d \neq 0$$

Suppose that the scores for 8 students before are $\{60, 75, 75, 50, 100, 90, 70, 80\}$. The after scores are $\{75, 80, 80, 60, 100, 90, 70, 85\}$. Is there enough evidence to conclude that I made a difference in their understanding? We need to find the pairwise differences, which is $\{-15, -5, -5, -10, 0, 0, 0, -5\}$, the average of the pairwise difference is then 5 and the unbiased standard deviation is 5.34. The test statistic is

$$T = \frac{5}{5/\sqrt{8}} \approx 2.8$$

For a 95% significance level, the critical t value is about 2.3, and since our test statistic is greater, we reject the null. So I in fact did have an impact!

Problem 9.

In the example that we just did, I used a 2 tailed test. Do you believe that this was appropriate? If not, why?

Problem 10.

To combat red-light-running crashes, many states are adopting photo-red enforcement programs. In these programs, red light cameras installed at dangerous intersections photograph the license plates of vehicles that run the red light. How effective are photo-red enforcement programs in reducing red-light-running crash incidents at intersections? The Virginia Department of Transportation (VDOT) conducted a comprehensive study of its newly adopted photo-red enforcement program and published the results in a June 2007 report. In one portion of the study, the VDOT provided crash data both before and after installation of red light cameras at several intersections. The data (measured as the number of crashes caused by red light running per intersection per year) for 13 intersections in Fairfax County, Virginia, are given in the table. Analyze the data for the VDOT. What do you conclude?

Intersection	Before Camera	After Camera
1	3.60	1.36
2	0.27	0
3	0.29	0
4	4.55	1.79
5	2.60	2.04
6	2.29	3.14
7	2.40	2.72
8	0.73	0.24
9	3.15	1.57
10	3.21	0.43
11	0.88	0.28
12	1.35	1.09
13	7.35	4.92

Table 4.1: Average delay (in minutes) at each intersection before and after camera installation.

4.2 2 Sample Proportions

At this point, it should be straight to the point on how we can find the estimator and its asymptotic distribution. The 2 sample proportion is no different; it is just a simple generalization of the 1 sample case.

4.2.1 Large Independent Samples

Recall that the 1 sample proportion test statistic was

$$T = \frac{\hat{p} - p_0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim \mathcal{N}(0,1) \tag{4.22}$$

If there's anything that the previous section has taught us, it's that the only thing that we need to do to generalize this to 2 samples is to reconstruct the standard error. This one is pretty easy because the model is a Bernoulli model which only admits 1 parameter to estimate. This means that we can estimate the variance for free if we have an estimate for p. Remembering our discussion about how standard deviations add in quadrature, we should

expect that the standard error would become

$$\text{s. e.} = \sqrt{\frac{\hat{p}_X(1-\hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1-\hat{p}_Y)}{n_Y}} \tag{4.23}$$

This time there is no trickery. This truly is the standard error. This means that we should expect the test statistic to be

$$T = \frac{\hat{p}_X - \hat{p_Y} - (p_{0,X} - p_{0,Y})}{\sqrt{\frac{\hat{p}_X(1-\hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1-\hat{p}_Y)}{n_Y}}} \sim \mathcal{N}(0,1) \tag{4.24}$$

and is standard normally distributed for all the reasons that we mentioned before. Sometimes you will see in textbooks that if you are doing hypothesis testing for the special case where $p_X = p_Y$, then the best estimator the population proportion can just be aggregated into the average of the 2 estimators

$$\hat{p} = \frac{n_X\hat{p}_X + n_Y\hat{p}_Y}{n_X + n_Y} \tag{4.25}$$

which leads to a test statistic

$$\implies T = \frac{\hat{p}_X - \hat{p}_Y}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_X} + \frac{1}{n_Y}\right)}} \sim \mathcal{N}(0,1) \tag{4.26}$$

Problem 11.

Using the likelihood function and maximizing it, show that the maximum likelihood estimator for when $p_X = p_Y$ is the average

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}, \quad \text{or} \quad \hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$$

Problem 12.

Teeth defects and stress in prehistoric Japan. Linear enamel hypoplasia (LEH) defects are pits or grooves on the tooth surface that are typically caused by malnutrition, chronic infection, stress, and trauma. A study of LEH defects in prehistoric Japanese cultures was published in the American Journal of Physical Anthropology (May 2010). Three groups of Japanese people were studied: Yayoi farmers (early agriculturists),

eastern Jomon foragers (broad-based economy), and western Jomon foragers (wet rice economy). LEH defect prevalence was determined from skulls of individuals obtained from each of the three cultures. Of the 182 Yayoi farmers in the study, 63.1% had at least one LEH defect; of the 164 Eastern Jomon foragers, 48.2% had at least one LEH defect; and, of the 122 Western Jomon foragers, 64.8% had at least one LEH defect. Two theories were tested. Theory 1 states that foragers with a broad-based economy will have a lower LEH defect prevalence than early agriculturists. Theory 2 states that foragers with a wet rice economy will not differ in LEH defect prevalence from early agriculturists. Use the results to test both theories, each at $\alpha = .01$.

Problem 13.

A utility infielder for a National League club batted .260 last season in three hundred trips to the plate. This year he hit .250 in two hundred at-bats. The owners are trying to cut his pay for next year on the grounds that his output has deteriorated. The player argues, though, that his performances the last two seasons have not been significantly different, so his salary should not be reduced. Who is right?

The next logical step would be to ask what happens in the small n limit. We will actually defer this conversation to the chapter on categorical data analysis. The statistic is called **Fisher's exact test**. I am deferring it because the context around it is needed to fully appreciate the test.

4.3 2 Sample Variances

Estimating and hypothesizing on 2 sample variances is a bit more interesting, and it's due to the fact that variances are strictly nonnegative. This implies that we can't just find the difference in the 2 variances and hope that there exists some asymptotic distribution. Aside from the fact that variances are non-negative, they scale quite drastically. Recall the Dave's vs. Five Guys example. Seemingly small differences in fluctuation can cause the variance to blow up due to the fact that estimator for variance is involves a sum of squares. We can, however, get rid of the scaling issue by finding a ratio.

$$H_0 : \frac{\sigma_X^2}{\sigma_Y^2} = 1 \quad H_a : \frac{\sigma_X^2}{\sigma_Y^2} \neq 1 \tag{4.27}$$

I should clarify that while we haven't gotten rid of the scaling property entirely, we've incorporated it into the problem instead of it being the problem. The only difficulty that this brings up is that now we need to find the random variable associated with the ratio of sample variances.

4.3.1 The F Distribution

Deriving this distribution analytically is a nightmare. I am going to save myself (and you) the pain and dance around this a little bit. Why do we even need this new distribution? Remember the first step 1 in the 3 step program: we need to find the estimator. If the quantity that we are trying to test is σ_X^2/σ_Y^2. The estimator for each of the variances is the unbiased sample variance. Thus, the estimator must be

$$\hat{F} = \frac{S_X^2}{S_Y^2} \tag{4.28}$$

We know previously that the sample variances follow a χ^2 distribution with $n-1$ degrees of freedom. Thus, the new random variable must be

$$F = \frac{\chi^2_{n_X-1}/(n_X-1)}{\chi^2_{n_Y-1}/(n_Y-1)} \tag{4.29}$$

Problem 14.

We discovered the form of the χ^2 distribution. We are going to build another random variable from it. Create 3 lists of 10,000 elements of χ^2 with degrees 1,5,8. Then take the ratio of 2 χ^2 distributions with their respective degrees of freedom. That is, let U_{d_1} and U_{d_2} be independent χ^2 distributions with degrees of freedom d_1 and d_2 respectively. Create a new list such that the elements are

$$F = \frac{U_{d_1}/d_1}{U_{d_2}/d_2} \tag{4.30}$$

Plot a histogram of the different χ^2 distributions. What do you see? These new random variables follow what we call **F-distributions.**

There is a bit of a caveat, and that is that when we are taking the ratio of χ^2 distributions in the F test, the order cannot be in any order that we want. The χ^2 random variable that appears in the denominator must be the smaller of the 2. That is, the denominator is $\min(\chi^2_{n_X-1}, \chi^2_{n_Y-1})$. This is because if it happens that the varaince of one of them is much

larger than the other, the F statistic will be very very small and it will be hard to determine statistical significance. If you're curious about what the exact pdf is, it's

$$f_F(x; d_1, d_2) = \frac{1}{B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{d_1/2} x^{\frac{d_1}{2}-1} \left(1 + \frac{d_1}{d_2}x\right)^{-\frac{d_1+d_2}{2}}, \quad x > 0 \tag{4.31}$$

with

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

4.3.2 The F Test

Now that we have the estimator, the statistic, and the distribution, it really is just smooth sailing from here. To do an example, let us revisit the Dave's vs. Five Guy's case. The data is given as Our hypothesis test will be

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6
Dave's	8	8	7.5	9	8.5	7.5
Five Guy's	6	9	9.5	6	10	8.5

$$H_0 : \frac{\sigma^2_{\text{FG}}}{\sigma^2_{\text{Dave's}}} = 1 \quad H_a : \frac{\sigma^2_{\text{FG}}}{\sigma^2_{\text{Dave's}}} > 1$$

Which is a 1 sided test. The F statistic is

$$F = \frac{S^2_{\text{FG}}}{S^2_{\text{Dave's}}}$$

Plugging in the numbers that we had before, we find that the F statistic is

$$F = \frac{3.037}{.342} \approx 8.88$$

The only thing missing is the critical $F_{d_X, d_Y, 1-\alpha}$ score, where d_X is the degree of freedom for the X sample variance and d_Y is the other degree of freedom. Looking up what this is, we find that the critical F score is 5.05. Comparing our F statistic to this critical F value, we conclude that we should reject the null and say that Five Guy's does indeed have a greater inconsistency than Dave's.

Problem 15.

Refer to the Journal of Food Engineering (Sept. 2013) study of the characteristics of fried sweet potato chips. Recall that a sample of 6 sweet potato slices fried at 130° using a vacuum fryer yielded the following statistics on internal oil content (measured in gigagrams): $\bar{x}_1 = .178$ g/g and $s_1 = .011$ g/g. A second sample of 6 sweet potato slices was obtained, only these were subjected to a two-stage frying process (again, at 130°) in an attempt to improve texture and appearance. Summary statistics on internal oil content for this second sample follows: $\bar{x}_2 = .140$ g/g and $s_2 = .002$ g/g. The researchers want to compare the mean internal oil contents of sweet potato chips fried with the two methods using a t-test. Do you recommend the researchers carry out this analysis? Explain. (Recall your answer to Exercise 8.122.)

Problem 16.

Two popular forms of mortgage are the thirty-year fixed-rate mortgage, where the borrower has thirty years to repay the loan at a constant rate, and the adjustablerate mortgage (ARM), one version of which is for five years with the possibility of yearly changes in the interest rate. Since the ARM offers less certainty, its rates are usually lower than those of fixed-rate mortgages. However, such vehicles should show more variability in rates. Test this hypothesis at the 0.10 level of significance using the following samples of mortgage offerings for a loan of $\$160,000$ (the borrower needs $\$200,000$, but must pay $\$40,000$ up front).

$160,000 Mortgage Rates	
30-Year Fixed	ARM
5.500	3.875
5.500	5.125
5.250	5.000
5.125	4.750
5.875	4.375
5.625	
5.250	
4.875	

5

ANOVA

"When stars die, they go super(A)NOVA"

–

The last topic in our discussion of frequentist statistics is ANOVA, which is an acronym for the **analysis of variance.** All that it really is is a generalization of the hypothesis testing that we have done. More specifically, instead of testing whether or not 2 samples have the same mean, we will test whether n samples all have the same mean and the alternative will be that at least one of the samples has a different mean. If we have already developed the mechanics to talk about testing 2 sample means, why couldn't we just compare the means of every sample and determine whether or not they are different from each other? The answer to this is almost obvious; or at least, the first answer is obvious. Suppose you have n samples that you need to test. If you went ahead and tested pairs of them individually, how many tests would you need to do? Well, it would be n choose 2, which is of order $O(n^2)$. If there were 20 samples you had to compare, you would need to conduct 400 different hypothesis tests. This is unwieldy. The second reason why doing individual testing might not be practical is because there is an increased risk of committing a type 1 error. Remember that when we do hypothesis testing of the mean, we are comparing differences in the sample mean. If each experiment is independent from each other and we repeat this for $O(n^2)$ possible tests, then we can see how this would increase the chances of committing a type 1 error. This problem also tells us that comparing the means for all these samples could be, at best, difficult and at worst incorrect. Hence, we have to resort to the analysis of variance. Now let us set up the cases for this problem.

Case 1: A doctor prescribes 3 diets to you; diet A, diet B, and diet C. You are a skeptic so you question whether or not these diets are effective, or if there really is a difference between the diets. So what you do is you ask 30 different people to try this diet for a month; 10 will be on diet A, 10 will be on diet B, and 10 will be on diet C. You measure for each participant. Now how do we show whether or not the diets are different from each other?

Case 2: The doctor (from case 1) then not only prescribes the 3 diets, but also whether or not you exercised. This is included in order to maximize the effects of the diets. Now, on top of the 3 diets, you also ask the participants exercise and not exercise. You measure the weight loss of the participants at the end of the month. On top of the diets, how does exercising affect the weight loss of the participants?

5.1 One-Way ANOVA

To be honest, I think ANOVA is kinda boring, so I don't really want to spend more time than I need to on doing these problems. But before we can get started in discussing ANOVA, we need to talk about the model.

5.1.1 The Model

Before, the data was simple enough to where we could just say that the data is i.i.d. (some random variable), and we can use the central limit theorem to do blah blah blah. The problems relating to ANOVA are slightly different. We typically model the data as

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij} \tag{5.1}$$

Y_{ij} is the j-th observation of the i-th group. In the context of case 1, for example, i would index 1 of the 3 possible diets and j would index the participants on that i-th diet. When we say observation, we mean the weight loss. μ is some global average that all the observations share (under the null, they must have all the same global means), τ_i is called the **effect** of the i-th group; i.e. a random variable that causes the i-th group to deviate from the true mean, and ϵ_{ij} is the noise from each observation. To elaborate on ϵ_{ij}, it is the random variable that causes fluctuations in weight loss for each participant. The only assumption that we make on the ϵ_{ij} is that it is a normal random variable with 0 mean and variance σ^2. Any background noise that has this property is called **white noise** (or sometimes Gaussian noise). It's clear, then, that the effect of ϵ_{ij} on the participants is to cause fluctuations. Note here that we assume that *the variance for every observation is the same.* Based on

this model, it's clear that what we are really testing is

$$H_0 : \tau = \tau_2 = ... = \tau_k = 0 \tag{5.2}$$

and the alternative hypothesis would be that *at least one* of the τ_i would be different. Obviously this makes sense because it is the τ_i in this model that causes each group average to be different from each other.

5.1.2 Some Notation

We would like to be consistent with the notation used from the statistics community. Firstly, if an index is replaced by a dot, that means that that index has been summed over. For example, if T_{ij} is a matrix with indices i and j, then

$$T_{.j} = \sum_{i=1}^{k} T_{ij} \tag{5.3}$$

You can think of it as a column trace

$$T_{i.} = \sum_{j}^{n_i} T_{ij} \tag{5.4}$$

and lastly, if we have a double dot, then that means we have added up all all the values of the observations

$$T_{..} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} T_{ij} \tag{5.5}$$

Problem 1.

Let A be a 3 by 3 matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ -5 & 10 & 0 \\ 3 & -1 & -1 \end{bmatrix}$$

a). Find $A_{.j}$
b). Find $A_{i.}$
c). Find $A_{..}$

Problem 2.

Suppose now we have a 3 index object T_{ijk}. Sometimes you will hear people call this a **tensor** but it is NOT a tensor! It's just a multidimensional array. The result of $T_{.jk}$ is what kind of object?

5.1.3 Why Variance?

To see why we need to analyze the variance to determine differences in population mean, let us start with the total variance

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n}\left(Y_{ij} - \bar{Y}_{..}\right)^2 \tag{5.6}$$

Let us split up this sum by introducing a 0 term. For any observation,

$$Y_{ij} - \bar{Y}_{..} = \left(Y_{ij} - \bar{Y}_{i.}\right) + \left(\bar{Y}_{i.} - \bar{Y}_{..}\right)$$

The first term in the parentheses on the right has the interpretation of the deviation of every observation from its group's average. We call this deviation the **within group** deviation. The second term on the right in parentheses has the interpretation of the deviation of every group's average to the total average. This is called the **between group** deviation

$$Y_{ij} - \bar{Y}_{..} = \underbrace{\left(Y_{ij} - \bar{Y}_{i.}\right)}_{\text{within-group deviation}} + \underbrace{\left(\bar{Y}_{i.} - \bar{Y}_{..}\right)}_{\text{between-group deviation}} \tag{5.7}$$

Let us square both sides and then sum over every index

$$\sum_{i=1}^{k}\sum_{j=1}^{n}\left(Y_{ij} - \bar{Y}_{..}\right)^2 = \sum_{i=1}^{k}\sum_{j=1}^{n}\underbrace{\left(Y_{ij} - \bar{Y}_{i.}\right)^2}_{\text{(A)}} + \underbrace{\left(\bar{Y}_{i.} - \bar{Y}_{..}\right)^2}_{\text{(B)}} + 2\left(Y_{ij} - \bar{Y}_{i.}\right)\left(\bar{Y}_{i.} - \bar{Y}_{..}\right)$$

The left hand side of the above hasn't changed; it's still the total sum of squared deviations. On the right hand side, we can make a couple of simplifications term by term. Starting with the A term, we can call this term the **sum of squared deviations within groups** (SSW). As for the B term, notice that the quantity inside the parentheses don't depend on the index j, therefore it is a constant with respect to that index

$$\sum_{i=1}^{k}\sum_{j=1}^{n}\left(\bar{Y}_{i.} - \bar{Y}_{..}\right) = \sum_{i=1}^{k} n\left(\bar{Y}_{i.} - \bar{Y}_{..}\right)^2 \tag{5.8}$$

We call this term the **sum of squared deviations between groups** (SSB). As a simple exercise, one can show that the sum on the remaining term evaluates to 0

Problem 3.

Show that

$$\sum_{i=1}^{k}\sum_{j=1}^{n} 2\left(Y_{ij}-\bar{Y}_{i.}\right)\left(\bar{Y}_{i.}-\bar{Y}_{..}\right)=0$$

So we can reduce 4.6 down to

$$SST = SSW + SSB \tag{5.9}$$

Which is great and all, but why did we do this? Remember what it is that we are trying to test whether or not $\tau_1 = \tau_2 = ... = \tau_k = 0$. If the τ_i in our model represents the effect of each group away from the population mean, then we can write $\tau_i = \mu - \mu_i$ where μ_i is the average of each group. Under the null, we should expect that the distance of each group's average away from the population average should be very small/comparable to the SSW. The reason we took the square is because if we didn't, then we would always get an average 0 displacement and that does not carry much information for us. However, with the alternative is true, then the distance between every group's average to the total average should be significantly large/larger than the SSW. Immediately, we see what the statistic should be: it should be the ratio of the SSW to the SSB! (or at least, related to the SSB and SSW)

5.1.4 The Test Statistic

In the previous section, we laid the foundation for what the test statistic should be: if the means for each group truly were different, then we would expect that the sum of squared errors for the between groups to be larger than the sum of squared errors within group. We can measure this difference not as a difference, but as a ratio

$$T \propto \frac{SSB}{SSW} = \frac{\sum_{i=1}^{k} n\left(\bar{Y}_{i.}-\bar{Y}_{..}\right)^2}{\sum_{i=1}^{k}\sum_{j=1}^{n}\left(Y_{ij}-\bar{Y}_{i.}\right)^2} \tag{5.10}$$

The only thing that we have yet to do is find the asymptotic distribution. Remember that we modeled the observations Y_{ij} as (4.1). We see that the randomness is only encapsulated by the ϵ_{ij} term; but we know that this is a normal random variable. The other 2 terms have the effect of shifting the mean, and so we expect that $Y_{ij} \sim \mathcal{N}(\mu, \sigma^2)$. That means that the SSB and SSW are sum of squared normal random variables, making them both χ^2

random variables! The ratio in (4.10) then is a ratio of χ^2 random variables (up to a factor of their degrees of freedom), making it an F random variable! The only (2) parameters that are unknown to us at the current moment are the degrees of freedom for both χ^2 variables. Let's go over the easy one first. If we look at the SSW, we see

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n} \left(Y_{ij} - \bar{Y}_{i.}\right)^2$$

each Y_{ij} is a normal random variable, and the double sum over the 2 indices gives us nk degrees of freedom, which we can just denote as N. That is, N is total number of observations that we have. Similarly for $\bar{Y}_{i.}$ term, there are k of these normal random variables, and so we should expect that the total degree of freedom is $N - k$.

Problem 4.

Using the same argument as above, find what the degrees of freedom for the SST and subsequently find the degrees of freedom for the SSB

Solving problem 4, we see that the degrees of freedom for the SSB is $k - 1$. This completes all the missing information, and we can say that the test statistic is found in (4.10) and has an asymptotic distribution

$$T = \frac{SSB/(k-1)}{SSW/(N-k)} \sim F_{N-k,k-1} \tag{5.11}$$

We now have all that we need to solve case 1. Sometimes in textbooks you will see something called "ANOVA tables" and all that it really is is just to help you keep track of all numbers. We won't really emphasize that here. Consider the following table of weight loss All units

	1	2	3
Diet A	2.3	2	1.1
Diet B	3.3	1	2.5
Diet C	1	2	2

are in kilograms. First, we need to find the total average $\bar{Y}_{..}$. This can be done by summing up every term and dividing by the total number of 9 observations. The total average is $\bar{Y}_{..} = 1.911$. The group averages for diets A, B, and C respectively are 1.8, 2.267, and 1.667 kilograms. Using (4.10), we find the SSB to be $3(1.911-1.8)^2+3(1.911-2.267)^2+3(1.911-1.677)^2 = 0.596$. The degrees of freedom for this quantity is $3 - 1 = 2$. The second quantity

that we need to measure is the SSW. All that we have to do is add up all the deviations for each group. For example, for diet A, we have $(2.3-1.8)^2+(2-1.8)^2+(1.1-1.8)^2=0.78$. Doing this for the remaining groups gives a total of 4.173. The corresponding degree of freedom is $9-3=6$. We can calculate our test statistic numerically as

$$T=\frac{0.596/2}{4.173/6}=0.428$$

Choosing an $\alpha=0.05$, we find that the critical F score for degrees of freedom 2 and 6 is 5.143. Based on this, we do not have enough evidence to claim that there is a significant difference in the diets.

Problem 5.

The following are the gas mileages recorded during a series of road tests with four new models of Japanese luxury sedans. Test the null hypothesis that all four models, on the average, give the same mileage. Let $\alpha=0.05$. Will the conclusion change if $\alpha=0.10$?

Model			
A	B	C	D
22	28	29	23
26	24	32	24
	29	28	

Problem 6.

Restoring self-control when intoxicated. Does coffee or some other form of stimulation really allow a person suffering from alcohol intoxication to "sober up"? Psychologists investigated the matter in Experimental and Clinical Psychopharmacology (Feb. 2005). A sample of 44 healthy male college students participated in the experiment. Each student was asked to memorize a list of 40 words (20 words on a green list and 20 words on a red list). The students were then randomly assigned to one of four different treatment groups (11 students in each group). Students in three of the groups were each given two alcoholic beverages to drink prior to performing a word completion task. Students in Group A received only the alcoholic drinks. Participants in Group AC had caffeine powder dissolved in their drinks. Group AR participants received

a monetary award for correct responses on the word completion task. Students in Group P (the placebo group) were told that they would receive alcohol, but instead received two drinks containing a carbonated beverage (with a few drops of alcohol on the surface to provide an alcoholic scent). After consuming their drinks and resting for 25 minutes, the students performed the word completion task. Their scores (simulated on the basis of summary information from the article) are reported in the table. (Note: A task score represents the difference between the proportion of correct responses on the green list of words and the proportion of incorrect responses on the red list of words.)

Table 5.1: Sample statistics by group

AR	AC	A	P
0.51	0.50	0.16	0.58
0.58	0.30	0.10	0.12
0.52	0.47	0.20	0.62
0.47	0.36	0.29	0.43
0.61	0.39	−0.14	0.26
0.00	0.22	0.18	0.50
0.32	0.20	−0.35	0.44
0.53	0.21	0.31	0.20
0.50	0.15	0.16	0.42
0.46	0.10	0.04	0.43
0.34	0.02	−0.25	0.40

a). What type of experimental design is employed in this study?

b). Analyze the data for the researchers, using $\alpha = .05$. Are there differences among the mean task scores for the four groups?

c). What assumptions must be met in order to ensure the validity of the inference you made in part b). ?

Problem 7.

Do the following data appear to violate the assumptions underlying the analysis of variance? Explain.

Treatment			
A	B	C	D
16	4	26	8
17	12	22	9
16	2	23	11
17	26	24	8

5.2 Two-Way ANOVA

Now that we have the special case of the one-way ANOVA understood, the matter of understanding two-way ANOVA follows pretty seamlessly. The only real change is that we have to modify the model to include interaction terms. Other than that, the process is the same.

5.2.1 The Generalized Model

In the one-way ANOVA model, we had that the observations Y_{ij} be normal random variables modeled by equation (4.1). When we add another factor, we also need to include another index. What exactly does the second factor mean? In case 2, we not only look at whether or not the diets A, B, and C have an effect on the weight loss of the N participants, but also whether or not exercising would also affect weight loss. The second factor in this example would be exercising or not exercising. The fully generalized model becomes

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{5.12}$$

where Y_{ijk} is the k-th observation of the i-th level of the first factor and j-th level of the second factor. μ is the population average, α_i is the effect of the first factor's averages, β_j is the effect of the second factor's averages, and $(\alpha\beta)_{ij}$ is the effect of the interactions between the first and second factor's averages. Lastly, we have ϵ_{ijk}, which is the same noise term as before. So really, what is different? Aside from the introduction of more indices, nothing really changes. We still have to compute the SSW, but now we have 2 additional SSB to compute: the SSB for the second factor and the SSB for the interactions between the 2 factors. Perhaps the only challening thing would be finding out what the degrees of freedoms must be for each of the terms. First, for illustrative purposes, we should write out

explicitly what we mean by the additional SSW's

$$\text{SST} = \text{SS}_A + \text{SS}_B + \text{SS}_{AB} + \text{SS}_E \tag{5.13}$$

where we haved changed the name SSB to SSE as to not confuse ourselves with the sum of squares for the second factor B. Going term by term, we can find the degrees of freedom starting with factor A by writing the sum

$$\text{SS}_A = rb \sum_{i=1}^{a} \left(\bar{Y}_{i..} - \bar{Y}_{...}\right)^2 \tag{5.14}$$

For the same reasons as before, when we look at the SS_A, the sum does not depend on the j and k indices and so therefore the triple sum reduces down to a single sum over the i index. Thus immediately we see that the degrees of freedom corresponding to SS_A is $a-1$, where a is the number of groups for the A factor. We do a similar thing for the B factor and get

$$\text{SS}_B = ra \sum_{j=1}^{b} \left(\bar{Y}_{.j.} - \bar{Y}_{...}\right)^2 \tag{5.15}$$

and we get a degree of freedom $b-1$. The interaction term is a bit difficult. We write it as

$$\text{SS}_{AB} = r \sum_{i=1}^{a} \sum_{j=1}^{b} \left[\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}\right]^2 \tag{5.16}$$

Here is the first thing that we can note. For each factor, it only has $n-1$ degrees of freedom; and this is just due to the fact every index when summed over has the total average $\bar{Y}_{...}$ subtracted from it. Thus for the first sum, we have a factor of $(b-1)$ and then the second sum picks up a second factor of $(a-1)$. In total, the degrees of freedom for the interaction term is $(a-1)(b-1)$.

> Problem 8.
>
> Using the fact that the SST still has degrees of freedom $N-1$ where N is the total number of observations, find the degree of freedom for the SS_E term.

Once we have all of the sum of squares, we can create F statistics for each of the factors. Namely, we can create an F statistic for F_A, F_B, and F_{AB}. Remember that this statistic is

$$T_\nu = \frac{SS_\nu}{SS_E} \sim F \tag{5.17}$$

where the index ν indexes the factor that we want to test. For example, if $\nu = A$, then that means we are testing whether or not the means from factor A are statistically significant. To help keep notational consistency with the literature, let us define the MS_ν as the mean of the sum of squares for the ν index

$$MS_\nu = \frac{1}{df_\nu} SS_\nu$$

where df_ν is the degree of freedom for the ν factor. For example,

$$MSB = \frac{ra}{b-1} \sum_{j=1}^{b} \left(\bar{Y}_{\cdot j \cdot} - \bar{Y}_{\cdots} \right)^2$$

The reason we might be interested in doing this is because the it is ultimately the ratio of the MS_ν that will give us the F statistic. Going back to our case 2, our data might look something like this

	No Exercise	**Yes Exercise**
Diet A	2, 3, 4	5, 6, 5
Diet B	3, 4, 3	7, 8, 7
Diet C	4, 5, 5	9, 10, 9

If that is the case, then the ANOVA table would look like

Source	SS	df	MS	F
Diet (A)	24.29	2	12.15	27.61
Exercise (B)	60.28	1	60.28	137.00
A × B	4.33	2	2.17	4.93
Error	5.32	12	0.44	
Total	94.22	17		

On an individual level, we can test whether the diets alone made a difference, exercising alone made a difference, and if combining them together makes a difference. Based on the numbers, dieting and exercising both have a big impact. The interaction is much smaller, but it isn't nothing. Its F statistic of 4.93 on $(2, 12)$ degrees of freedom is bigger than the critical value $F_{.05,2,12} = 3.89$ (the p value is about 0.03), so at the 5% level the effect of exercise does depend a little on which diet you're on.

Problem 9.

Large companies have the option of limiting their growth, but does doing so lead to higher profitability? The table below gives the profitability for a sample twenty-one top-ranked companies, where profitability is expressed in terms of annual profit as a percentage of total company assets. The firms are divided into three groups by size of assets- $50 billion or less, between $51 and $100 billion, and over $100 billion. Test the hypothesis that small- and medium-size companies are as profitable as large companies. Let $\alpha = 0.10$

Table 5.2: Size of Assets (billions of $)

Size of Assets (billions of $)		
$50 or Less	Between $51 and $100	Greater than $100
7.2	11.3	14.8
6.5	5.6	11.3
5.7	5.3	9.2
4.4	5.3	4.8
3.4	10.4	3.9
3.4	6.2	10.2
7.8	5.3	7.3

Problem 10.

Fill in the entries missing from the following ANOVA table.

Table 5.3: ANOVA Table

Source	df	SS	MS	F
Treatment	4			6.40
Error			10.60	
Total		377.36		

6

Linear Regression

"All models are wrong, but some are useful."

– George Box

So far every question we have asked was about a single variable. What's the average height in town? Is the proportion of words I know bigger than 10%? Is Five Guys more inconsistent than Dave's? Even ANOVA, which compared several groups, was really asking about one variable measured in different places. But a lot of the interesting questions in science are about *relationships*. Does studying more actually raise your exam score, and by how much? If a state spends more per student, do more of its students graduate? This chapter is about the simplest model of a relationship between two variables, the straight line, and about how to do honest inference with it. The good news is that the recipe hasn't changed. We find an estimator, we find its sampling distribution, and then we use that distribution to build intervals and run tests.

6.1 Setting Up The Problem

Suppose we collect pairs of observations $(x_1, Y_1), (x_2, Y_2), \ldots, (x_n, Y_n)$. To make it concrete, let x_i be the number of hours student i studied for an exam and Y_i be their score. When we plot them (Figure 6.1), the points seem to drift upward, but they certainly don't sit on a line. Two students who studied for the same amount of time can get very different scores, because of sleep and luck and a hundred other things we didn't measure.

So we write down a model that has both pieces in it, a deterministic trend and some random noise around it. The **simple linear regression model** says that

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2). \tag{6.1}$$

There's a lot packed into this line, so let's unpack it. The x_i are treated as fixed numbers (we chose them, or we just condition on them), and all the randomness lives in the errors ε_i. The parameters are the intercept β_0, the slope β_1 and the noise variance σ^2, and none of them are known. Written another way, the model says that for each fixed x,

$$Y \mid x \sim \mathcal{N}(\beta_0 + \beta_1 x,\ \sigma^2), \tag{6.2}$$

so the *mean* of Y moves along a straight line as x changes, while the spread around that line stays the same everywhere. That model packs in four assumptions, and they will matter later when we check whether the model is any good. The relationship is linear, the errors are independent, they have constant variance, and they are normally distributed. The slope β_1 is the thing we usually care about, since it's the expected change in Y when x goes up by one unit.

6.2 Least Squares Estimators

How do we estimate β_0 and β_1 from the data? We want the line that is, in some sense, closest to all the points at once. For a candidate line $b_0 + b_1 x$, the vertical distance from point i to the line is the **residual** $Y_i - b_0 - b_1 x_i$. We can't just add up the residuals, since positive and negative ones would cancel, so we square them and add them up. This gives the sum of squared errors,

$$\text{SSE}(b_0, b_1) = \sum_{i=1}^{n} \left(Y_i - b_0 - b_1 x_i\right)^2, \tag{6.3}$$

and the **least squares estimators** $\hat{\beta}_0$ and $\hat{\beta}_1$ are the values of b_0 and b_1 that make this as small as possible. To find them we take partial derivatives and set them equal to zero,

$$\frac{\partial\ \text{SSE}}{\partial b_0} = -2 \sum_{i=1}^{n} (Y_i - b_0 - b_1 x_i) = 0, \tag{6.4}$$

$$\frac{\partial\ \text{SSE}}{\partial b_1} = -2 \sum_{i=1}^{n} x_i (Y_i - b_0 - b_1 x_i) = 0. \tag{6.5}$$

These two equations are called the **normal equations**. The first one says $\sum Y_i = nb_0 + b_1 \sum x_i$, and dividing by n gives $b_0 = \bar{Y} - b_1 \bar{x}$. Plugging that into the second one and doing a bit of algebra (which you'll do in Problem 1), we get

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}, \tag{6.6}$$

where we use the shorthand $S_{xx} = \sum (x_i - \bar{x})^2$, $S_{yy} = \sum (Y_i - \bar{Y})^2$ and $S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$. The second formula says that the fitted line always passes through the point of averages $(\bar{x}, \bar{Y})$, which is a nice sanity check. Once we have the estimates, the **fitted values** are $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the residuals are $e_i = Y_i - \hat{Y}_i$.

Problem 1.

Starting from the two normal equations, derive the formulas for $\hat{\beta}_1$ and $\hat{\beta}_0$ in (6.6). Then show that the residuals always satisfy $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.

6.2.1 Least Squares Is Maximum Likelihood

Squaring the residuals might feel a bit arbitrary. Why not absolute values? Here's the reason it isn't arbitrary at all. Under the model (6.1), each Y_i is normal with mean $\beta_0 + \beta_1 x_i$ and variance σ^2, so the likelihood of the data is

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(Y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right], \tag{6.7}$$

and the log-likelihood is

$$\ell(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2. \tag{6.8}$$

The first term doesn't involve the β's at all, and the second term is $-\,\mathrm{SSE}\,/2\sigma^2$. So maximizing the likelihood over β_0 and β_1 is *exactly* the same as minimizing the sum of squares. Least squares is the maximum likelihood estimator when the errors are normal, which means everything we know about MLEs (from Chapter 2, and in much more detail in Part II) applies here too.

Problem 2.

Find the maximum likelihood estimator of σ^2 under the model (6.1). Is it unbiased? (Compare your answer with the estimator S^2 we define below.)

6.2.2 An Example

Let's make up a dataset of 25 students. We generate it from a true line $Y = 52 + 3.5x$ with noise of standard deviation $\sigma = 6$, so we can see how close the estimates get to the truth. Here is the code.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(7)
n = 25
x = np.round(rng.uniform(1, 10, n), 1)            # hours studied
y = 52 + 3.5 * x + rng.normal(0, 6, n)            # exam scores

xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar) ** 2)
Sxy = np.sum((x - xbar) * (y - ybar))

b1 = Sxy / Sxx
b0 = ybar - b1 * xbar
print(f"b0 = {b0:.3f}, b1 = {b1:.3f}")
```

```
b0 = 54.234, b1 = 2.862
```

The fitted line is drawn in Figure 6.1. We didn't get exactly 52 and 3.5, and we shouldn't expect to, because the estimates are computed from random data. A different class of 25 students would give a different line. That wiggle from sample to sample is exactly what we need to quantify next.

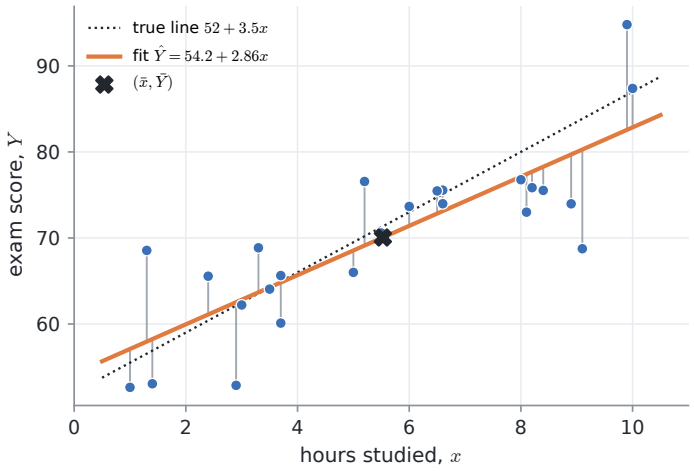


Figure 6.1: Exam score against hours studied for 25 made-up students, with the least squares line. The grey segments are the residuals, whose squares the line makes as small as possible. The dotted line is the true line $52 + 3.5x$ that generated the data.

6.3 Inference On The Slope

The key observation is that $\hat{\beta}_1$ is a *linear combination* of the Y_i. Since $\sum(x_i - \bar{x}) = 0$, the $\bar{Y}$ in the numerator of S_{xy} drops out, and we can write

$$\hat{\beta}_1 = \sum_{i=1}^{n} c_i Y_i, \qquad c_i = \frac{x_i - \bar{x}}{S_{xx}}. \tag{6.9}$$

The c_i are fixed numbers, and the Y_i are independent normals, so $\hat{\beta}_1$ is normal too (a linear combination of independent normals is normal). All that's left is its mean and variance. For the mean, using $\sum c_i = 0$ and $\sum c_i x_i = 1$,

$$\begin{aligned} \mathbb{E}[\hat{\beta}_1] &= \sum_{i=1}^{n} c_i \, \mathbb{E}[Y_i] = \sum_{i=1}^{n} c_i(\beta_0 + \beta_1 x_i) \\ &= \beta_0 \sum_{i=1}^{n} c_i + \beta_1 \sum_{i=1}^{n} c_i x_i \\ &= \beta_1, \end{aligned} \tag{6.10}$$

so $\hat{\beta}_1$ is unbiased. For the variance, independence lets us add the variances,

$$\operatorname{Var}[\hat{\beta}_1] = \sum_{i=1}^{n} c_i^2 \operatorname{Var}[Y_i] = \sigma^2 \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}. \tag{6.11}$$

So all together,

$$\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right). \tag{6.12}$$

This formula is worth staring at for a second, because it tells you how to design a good experiment. The variance gets smaller when the noise σ^2 is small, which makes sense, and also when S_{xx} is big, meaning the x values are spread out. If every student studied for about 5 hours, it would be really hard to tell how the score changes with hours. Spreading out the x's gives the line more leverage.

6.3.1 Estimating σ^2

Of course we don't know σ^2, so just like in Chapter 2 we have to estimate it. The natural thing to do is look at how spread out the residuals are. We use

$$S^2 = \frac{1}{n-2}\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2 = \frac{\text{SSE}}{n-2}, \tag{6.13}$$

which is the regression version of the sample variance. Why $n-2$ instead of $n-1$? In Chapter 2 we lost one degree of freedom because we had to estimate one parameter (μ) before we could measure spread around it. Here we had to estimate two (β_0 and β_1), and the residuals satisfy two constraints (Problem 1), so we lose two. Dividing by $n-2$ makes S^2 unbiased for σ^2, and (this is the part we won't prove here, see Larsen and Marx [4] or Part II) it turns out that

$$\frac{(n-2)S^2}{\sigma^2} \sim \chi^2_{n-2}, \tag{6.14}$$

independently of $\hat{\beta}_0$ and $\hat{\beta}_1$. The square root S is often called the **residual standard error**. And now the recipe is exactly the one from Chapter 2. Standardize, and replace σ by S, which turns the normal into a t,

$$T = \frac{\hat{\beta}_1 - \beta_1}{S/\sqrt{S_{xx}}} \sim t_{n-2}. \tag{6.15}$$

The denominator $S/\sqrt{S_{xx}}$ is the **standard error of the slope**, $\text{s.e.}(\hat{\beta}_1)$.

6.3.2 Confidence Intervals And Tests

From (6.15), a $100(1-\alpha)\%$ confidence interval for the slope is

$$\hat{\beta}_1 \pm t_{\alpha/2,\,n-2}\,\frac{S}{\sqrt{S_{xx}}}. \tag{6.16}$$

And to test $H_0 : \beta_1 = \beta_{1,0}$ against a two-sided alternative, we compute $T = (\hat{\beta}_1 - \beta_{1,0})/\,\mathrm{s.e.}(\hat{\beta}_1)$ and reject if $|T| > t_{\alpha/2,n-2}$. By far the most common version of this test is

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_a : \beta_1 \neq 0. \tag{6.17}$$

What does this test actually ask? Under the null, the mean of Y doesn't change with x at all, so x is useless for predicting Y (at least linearly). Rejecting the null says there's evidence of a linear association. Notice what it does *not* say. It doesn't say x causes Y. If the students who study more also sleep more, the slope could be picking up the effect of sleep. Regression on observational data measures association, and causation needs either a randomized experiment or a lot more care.

```
y_hat = b0 + b1 * x
resid = y - y_hat
S = np.sqrt(np.sum(resid ** 2) / (n - 2))      # residual standard error
se_b1 = S / np.sqrt(Sxx)

t_stat = b1 / se_b1                            # test H0: beta1 = 0
p_value = 2 * stats.t.sf(abs(t_stat), df=n - 2)
t_crit = stats.t.isf(0.025, df=n - 2)
print(f"S = {S:.3f}, se(b1) = {se_b1:.3f}")
print(f"t = {t_stat:.2f}, p-value = {p_value:.2e}")
print(f"95% CI for beta1: [{b1 - t_crit*se_b1:.3f}, {b1 + t_crit*se_b1:.3f}]")
```

```
S = 5.787, se(b1) = 0.420
t = 6.81, p-value = 6.08e-07
95% CI for beta1: [1.992, 3.732]
```

The interval contains the true slope 3.5, and the p value is tiny, so there is overwhelming evidence that studying is associated with higher scores (in our made-up world, anyway). Each extra hour of studying goes with somewhere between about 2 and 3.7 more points, on average.

Problem 3.

Show that $\hat{\beta}_0 = \sum_i d_i Y_i$ with $d_i = \frac{1}{n} - \bar{x}c_i$, and use it to prove that $\hat{\beta}_0$ is unbiased with

$$\mathrm{Var}[\hat{\beta}_0] = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) = \frac{\sigma^2 \sum_i x_i^2}{n\, S_{xx}}.$$

Write down the t statistic for testing $H_0 : \beta_0 = \beta_{0,0}$. When is a test on the intercept actually meaningful? (Think about whether $x = 0$ is a sensible value in the exam example.)

Problem 4.

Insect flight ability can be measured in a laboratory by attaching the insect to a nearly frictionless rotating arm with a thin wire. The "tethered" insect then flies in circles until exhausted, and the nonstop distance flown can be calculated from the number of revolutions made by the arm. The following are measurements of this sort made on *Culex tarsalis* mosquitos of four different ages. The response variable is the average distance flown until exhaustion for forty females of the species. Fit a straight line to these data and test that the slope is zero. Use a two-sided alternative and the 0.05 level of significance.

Age, x (weeks)	Distance flown, y (thousand meters)
1	12.6
2	11.6
3	6.8
4	9.2

Problem 5.

The best straight line through the Massachusetts funding and graduation rate data from Larsen and Marx [4] has the equation $y = 81.088 + 0.412x$, with $s = 11.78848$. (You'll need n and S_{xx} from the data table in the book.)

(1) Construct a 95% confidence interval for β_1.

(2) What does your answer to part (a) imply about the outcome of testing $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 \neq 0$ at the $\alpha = 0.05$ level of significance?

(3) How would you summarize these data, and their implications, to a meeting of the state School Board?

6.4 Estimating The Mean And Predicting New Values

Once we have a fitted line, there are two very different questions we might ask about a particular value x_0. The first is "what is the *average* score of all students who study x_0 hours?" That's a question about the parameter $\mu_0 = \beta_0 + \beta_1 x_0$. The second is "what score will *this one new* student who studies x_0 hours get?" That's a question about a future random variable $Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$. Both have the same point estimate, $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$, but very different uncertainties.

For the mean, $\hat{Y}_0$ is again a linear combination of the Y_i, and a short calculation (Problem 6) gives

$$\mathrm{Var}[\hat{Y}_0] = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right), \tag{6.18}$$

so a **confidence interval for the mean response** at x_0 is

$$\hat{Y}_0 \pm t_{\alpha/2,\, n-2}\, S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}. \tag{6.19}$$

For a new observation, we have to account for two sources of uncertainty. We don't know exactly where the line is, and even if we did, the new student has their own noise ε_0, which is independent of everything we've seen. The variances add, $\mathrm{Var}[Y_0 - \hat{Y}_0] = \sigma^2 + \mathrm{Var}[\hat{Y}_0]$, so the **prediction interval** is

$$\hat{Y}_0 \pm t_{\alpha/2,\, n-2}\, S \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}. \tag{6.20}$$

That extra 1 under the square root makes a huge difference. As $n \to \infty$ the confidence interval for the mean shrinks to nothing, because we learn the line perfectly, but the prediction interval never gets narrower than about $\pm 1.96\sigma$, because individual students are still noisy. Both intervals are narrowest at $x_0 = \bar{x}$ and fan out as we move away from the center of the data (Figure 6.2). That fanning is also a warning about **extrapolation**. Predicting the score of someone who studied 30 hours using data that stops at 10 is asking the line to hold up in a region where we have no evidence it does.

Problem 6.

Use $\hat{Y}_0 = \bar{Y} + \hat{\beta}_1 (x_0 - \bar{x})$, together with the fact that $\bar{Y}$ and $\hat{\beta}_1$ are uncorrelated, to derive $\mathrm{Var}[\hat{Y}_0]$. Then compute a 95% confidence interval for the mean score and a

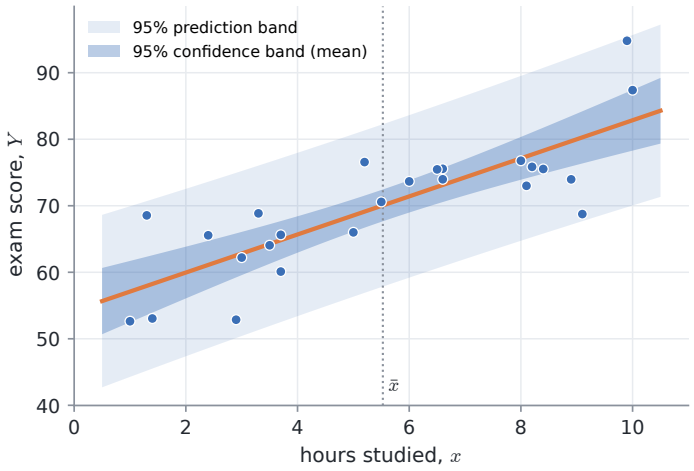


Figure 6.2: The fitted line with a 95% confidence band for the mean response (dark) and a 95% prediction band for a new observation (light). Both are narrowest at $\bar{x}$. The prediction band has to be wide enough to contain individual students, so it's much wider.

95% prediction interval for a single student at $x_0 = 6$ hours, using the example data.

6.5 How Good Is The Fit?

6.5.1 Partitioning The Variability

Here is an idea that connects regression right back to ANOVA. The total variability in the Y's around their mean can be split into a piece that the line explains and a piece it doesn't,

$$\underbrace{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}_{\text{SST}} = \underbrace{\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}_{\text{SSR}} + \underbrace{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}_{\text{SSE}}. \tag{6.21}$$

To see this, write $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i)$, square, and sum. The cross term is $2\sum(\hat{Y}_i - \bar{Y})e_i = 2\hat{\beta}_1 \sum(x_i - \bar{x})e_i$, which is zero by Problem 1. The ratio of explained to total variability is the **coefficient of determination**,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \tag{6.22}$$

the fraction of the variation in Y that is accounted for by its linear relationship with x. It lives between 0 (the line explains nothing) and 1 (every point is on the line). For simple

linear regression it is also exactly the square of the sample correlation coefficient,

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}, \qquad R^2 = r^2. \tag{6.23}$$

Be careful about reading too much into R^2. A small R^2 doesn't mean the slope is zero (with enough data you can detect a tiny but real slope), and a big R^2 doesn't mean the model is right (a curved relationship can have a big R^2 for a straight line that is still the wrong shape).

6.5.2 The ANOVA Table And The F Test

Just like in Chapter 5, we can organize the sums of squares into an ANOVA table. The regression sum of squares has 1 degree of freedom (one slope), the error sum of squares has $n - 2$, and the total has $n - 1$. Dividing each sum of squares by its degrees of freedom gives a mean square, and the ratio

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \sim F_{1,\,n-2} \quad \text{under } H_0 : \beta_1 = 0. \tag{6.24}$$

In simple regression this F test is exactly the same test as the two-sided t test for the slope, since $F = T^2$ (Problem 7). The F version earns its keep in multiple regression, where it tests whether *any* of several slopes is nonzero.

Source	SS	df	MS	F
Regression	SSR	1	$\text{MSR} = \text{SSR}$	MSR / MSE
Error	SSE	$n - 2$	$\text{MSE} = S^2$	
Total	SST	$n - 1$		

```python
SST = np.sum((y - ybar) ** 2)
SSE = np.sum(resid ** 2)
SSR = SST - SSE
F = (SSR / 1) / (SSE / (n - 2))
print(f"R^2 = {SSR/SST:.3f}, r = {np.corrcoef(x, y)[0, 1]:.3f}")
print(f"F = {F:.2f} (t^2 = {t_stat**2:.2f}), p-value = {stats.f.sf(F, 1, n - 2):.2e}")
```

```
R^2 = 0.668, r = 0.818
F = 46.34 (t^2 = 46.34), p-value = 6.08e-07
```

Problem 7.

Show that $\text{SSR} = \hat{\beta}_1^2 S_{xx}$, and use it to prove that $F = T^2$, where T is the t statistic for $H_0 : \beta_1 = 0$. Then show that $R^2 = r^2$, and verify the computing formula $\text{SSE} = \sum_i Y_i^2 - \hat{\beta}_0 \sum_i Y_i - \hat{\beta}_1 \sum_i x_i Y_i$.

6.5.3 Checking The Assumptions

Everything we did leaned on the four assumptions in the model. If they fail, the t intervals and p values can be way off, even though the least squares line can still be computed. The best tool for checking them is the **residual plot**, residuals against fitted values (or against x). If the model is right, the residuals should look like a structureless cloud centered at zero with roughly constant spread, as in the left panel of Figure 6.3. A curve in the residuals means the relationship isn't linear. A funnel shape, where the spread grows with the fitted value, means the variance isn't constant. For normality we can make a normal quantile plot of the residuals (right panel), which should look like a straight line. Independence is the one we usually can't see in a plot. It's mostly about how the data was collected, for example whether the same student was measured twice, or whether observations were taken over time and drift.

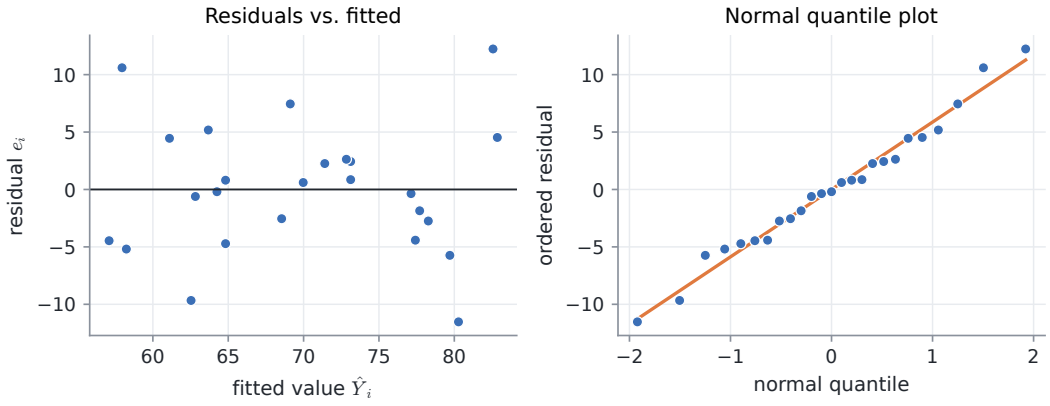
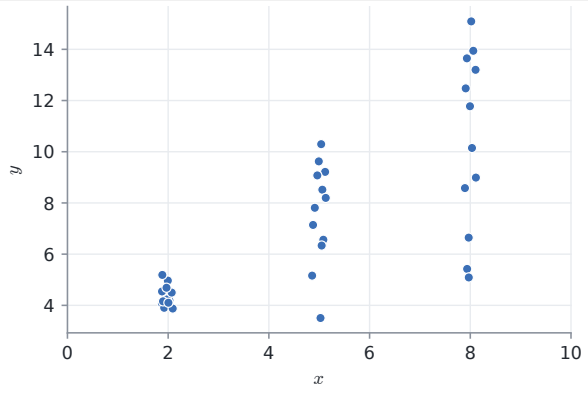


Figure 6.3: Diagnostics for the exam score regression. Left, residuals against fitted values show no pattern and no funnel. Right, the normal quantile plot of the residuals is close to a straight line, so the normality assumption looks reasonable.

Problem 8.

Which, if any, of the assumptions of the simple linear model appear to be violated in the following scatterplot? Which, if any, appear to be satisfied? Which, if any, cannot be assessed by looking at the scatterplot?



6.6 Multiple Linear Regression

A single predictor is rarely the whole story. A student's score probably depends on hours studied *and* hours slept *and* whether they went to lecture. The **multiple linear regression model** with p predictors is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2). \tag{6.25}$$

Now each slope has a more careful interpretation. The coefficient β_j is the expected change in Y when x_j goes up by one unit *while all the other predictors are held fixed.* That "holding fixed" is the whole point. It's how regression tries to separate the effect of studying from the effect of sleeping, even when the two are correlated in the data.

6.6.1 The Normal Equations In Matrix Form

Writing out $p+1$ partial derivatives by hand gets old fast, so we switch to matrices. Stack the responses into a vector $\mathbf{Y}$, the coefficients into a vector $\boldsymbol{\beta}$, and the predictors into the

$n \times (p+1)$ **design matrix X**,

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}. \tag{6.26}$$

The column of ones is there to multiply the intercept. Then the whole model is the single line $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, and the sum of squared errors is $\text{SSE}(\mathbf{b}) = (\mathbf{Y} - \mathbf{X}\mathbf{b})^T(\mathbf{Y} - \mathbf{X}\mathbf{b})$. Expanding and taking the gradient with respect to $\mathbf{b}$,

$$\begin{aligned} \nabla_{\mathbf{b}} \,\text{SSE} &= \nabla_{\mathbf{b}} \left(\mathbf{Y}^T\mathbf{Y} - 2\mathbf{b}^T\mathbf{X}^T\mathbf{Y} + \mathbf{b}^T\mathbf{X}^T\mathbf{X}\mathbf{b} \right) \\ &= -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\mathbf{b} = \mathbf{0}, \end{aligned} \tag{6.27}$$

which gives the normal equations $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{Y}$, and as long as the columns of $\mathbf{X}$ are linearly independent,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}. \tag{6.28}$$

With $p = 1$ this is exactly the pair of formulas (6.6), just written without indices.

6.6.2 Inference On The Coefficients

The estimator is a matrix times $\mathbf{Y}$, so it's a linear combination of independent normals, so it's (multivariate) normal. Its mean is $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$, so it's unbiased, and its covariance matrix is

$$\text{Cov}[\hat{\boldsymbol{\beta}}] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}. \tag{6.29}$$

We estimate σ^2 with $S^2 = \text{SSE}\,/(n-p-1)$, since we used up $p+1$ degrees of freedom on the coefficients. Then each coefficient gets its own t test, exactly as before,

$$T_j = \frac{\hat{\beta}_j - \beta_j}{S\sqrt{[(\mathbf{X}^T\mathbf{X})^{-1}]_{jj}}} \sim t_{n-p-1}, \tag{6.30}$$

and the overall F test of $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ uses $F = \frac{\text{SSR}\,/p}{\text{SSE}\,/(n-p-1)} \sim F_{p,\,n-p-1}$.

Let's add a second predictor to our example, hours of sleep the night before. In our made-up world, sleep helps a lot, and the students who study more also tend to sleep a bit less (crunch time). Watch what happens to the slope on studying.

```
rng = np.random.default_rng(11)
n = 60
```

```
study = rng.uniform(1, 10, n)
sleep = np.clip(8 - 0.3 * study + rng.normal(0, 1, n), 3, 10)
score = 30 + 3.0 * study + 4.0 * sleep + rng.normal(0, 5, n)

def ols(X, y):
    XtX_inv = np.linalg.inv(X.T @ X)
    beta = XtX_inv @ X.T @ y
    resid = y - X @ beta
    df = len(y) - X.shape[1]
    S2 = resid @ resid / df
    se = np.sqrt(S2 * np.diag(XtX_inv))
    p = 2 * stats.t.sf(np.abs(beta / se), df)
    return beta, se, p

X1 = np.c_[np.ones(n), study]                 # study only
X2 = np.c_[np.ones(n), study, sleep]          # study and sleep
for name, X in [("study only", X1), ("study + sleep", X2)]:
    beta, se, p = ols(X, score)
    print(name, " beta:", np.round(beta, 2), " se:", np.round(se, 2), " p:", np.round(p,
     ↪ 4))
```

```
study only  beta: [65.03  1.36]  se: [1.47 0.26]  p: [0. 0.]
study + sleep  beta: [37.06  2.66  3.24]  se: [5.09 0.31 0.57]  p: [0. 0. 0.]
```

With study hours alone, the slope comes out at about 1.4, way smaller than the true 3.0, because studying more goes with sleeping less, and the missing sleep drags the scores down. The study-only model blames studying for some of the damage done by lost sleep. Once sleep is in the model, the slope on studying is estimated holding sleep fixed, and it lands at about 2.7, much closer to the truth. This is called **omitted variable bias**, and it's the reason a regression slope is only as trustworthy as the list of variables you thought to include.

Problem 9.

(1) Show that with $p = 1$, the matrix $\mathbf{X}^T\mathbf{X}$ is $\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$, and that the bottom-right entry of $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ is σ^2/S_{xx}, which matches the variance of $\hat{\beta}_1$ we found by hand.

(2) In the study and sleep example, what happens to $\mathbf{X}^T\mathbf{X}$ and to the standard errors if the two predictors are almost perfectly correlated? This problem is called

multicollinearity. Try it by making `sleep` an almost exact linear function of `study`.

6.6.3 Adjusted R^2

In multiple regression, $R^2 = 1 - \text{SSE} / \text{SST}$ has an annoying property. Adding a predictor can never make it go down, even if the predictor is pure noise, because least squares can always set the new coefficient to zero and do no worse. So R^2 always rewards bigger models. The **adjusted** R^2 fixes this by comparing mean squares instead of sums of squares,

$$R^2_{\text{adj}} = 1 - \frac{\text{SSE}/(n-p-1)}{\text{SST}/(n-1)}, \tag{6.31}$$

which only goes up when a new predictor reduces the error variance by more than you'd expect from chance. It's a rough tool, and in practice people compare models with cross-validation or information criteria too, but it is a lot more honest than plain R^2 when the number of predictors changes.

Problem 10.

Add 10 columns of pure noise (`rng.normal(size=(n, 10))`) to the study and sleep design matrix. Compute R^2 and R^2_{adj} before and after. Which one went up, and by how much? How many of the noise coefficients come out "significant" at the 5% level, and how many would you expect to?

7

Categorical Data Analysis

"We're in the end game now."

– Tony Stark

In this chapter, we will extend the notion of "levels" but slightly different (maybe even orthogonal). We might be interested in, say, the number of students in a school, and we might even compare the students between schools, but the difference here is that for categorical data, you will have to put, say, your students, in 1 box or the other. So instead of the model itself having a nested or level property like with hierarchal modeling, it is the outcome of our model now that will have this nested or levels property. This also implies that all of the probability distribution and structure will have to be discrete.

7.1 Problem Set Up

Before we start yapping about the mathematics, I wanted to give some context as to what problems we are trying to solve. I think this is good because it will make it less like I'm pulling these equations out of thin air, and it will give some historical context on the development of modern statistics.

Case 1: In 1988, a study was conducted by the Physicians' Health Study research group of Harvard Medical School to understand the relationship between the usage of Aspirin and Heart attack. Below is the data based on a 5-year observation study: Based on the data,

the researchers were interested in understanding whether regular aspirin intake reduces mortality from cardiovascular disease.

	Fatal attack	**Non-fatal attack**	**No attack**
Placebo	18	171	10,845
Aspirin	5	99	10,933

Here, two categorical variables are X: whether a patient takes aspirin or not (placebo) and Y:the status of heart attack (fatal, non-fatal, or no attack). How do we show correlation? At what point can we say that taking aspirin *does* decreases the mortality rate? At what point can we say that there is not just a correlation, but also a causation of X on Y? The causation part is very difficult to do, and has an entire field of study around it called **causal inferece.** For interest of time, we won't be able to go too much into it.

Case 2: One of the first studies to understand the association between smoking and lung cancer was conducted by Doll and Hill (1950, British Med. Journal). They collected observations from 709 patients with lung cancer from 20 hospitals in London and 709 patients without lung cancer from the same hospitals.

	Lung cancer	**No lung cancer**
Smoker	688	650
Non-smoker	21	59
Total	709	709

Here, gender and age were controlled, i.e., you can imagine that the gender distribution and histogram of age among patients with lung cancer are very similar to those without lung cancer. This is to prevent some confounding factors, i.e., if all the patients with lung cancer were male, and all the patients without lung cancer were female, then we cannot be sure whether it is gender or smoking habit that is associated with the lung cancer.

In both of these case studies, we are interested in the effects of X on Y, or more accurately, what we can say about the correlation structure between X and Y. In both cases, we are examining 2 groups: placebo and aspirin, smoker and non-smoker. Within each group, there are different categories that we can put the subjects in: fatal, nonfatal, no attack, and lung cancer, no lung cancer. These categories that we put members of the different groups in are universal; as in, across different groups, we have to categorize them with same...well, categories. The tables that we formed for both cases are called **contingency tables**, and these are going to be the main objects of interest in our studies (sorta like how the likelihood function was the main player in regression analysis).

7.2 Estimation Of Categorical Data

Now that we know what we are looking for, let us start the journey of developing the tools to answer such questions. What we will learn is that a lot of the same concepts from previous chapters will arise, but they will take on different forms. For starters, the idea that any 2 variables are correlated to each other typically means independence.

7.2.1 Probability Structure

The Bayesian intuition (and honestly, the most logical intuition) for independence is the statement that

$$\mathbb{P}(Y|X) = \mathbb{P}(Y) \tag{7.1}$$

Which is a very simple statement that gets the point across: the probability of Y given some event X is just the probability of Y. X has no impact on the probability of Y occurring, so we can say that X and Y are **statistically independent**. Some people will call this a corollary of the definition of independence, but I think that is a stupid take. I will just take (4.1) to be the definition of independence and see what we can milk out of it. Using the definition of a conditional probability, we can say

$$\mathbb{P}(Y|X) = \frac{\mathbb{P}(X \cap Y)}{\mathbb{P}(X)} \tag{7.2}$$

Which is just a statement of: if you are in X, how much of Y do you see (hence the observation). To get probabilities out, we have to include that our sample space is no longer the entire space but restricted to X, so we have to divide by the probability of X. Nothing crazy, nothing new. If we multiply both sides by $\mathbb{P}(X)$, we see that we get a corollary condition

$$\mathbb{P}(X \cap Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$$

and if X and Y are statistically independent, then (4.1) holds and we get

$$\mathbb{P}(X \cap Y) = \mathbb{P}(Y|X)\mathbb{P}(X) \tag{7.3}$$

How can we apply this to our problem set ups? Along the way, we will introduce new notation, so bear with us. If we look at the contingency tables in case 1 and case 2, what is the natural way to construct probabilities? And probabilities of *what*? Let us look at the case of lung cancer, because it's a simple 2x2. If we were interested in, say, the probability of getting lung cancer, how might we construct that? The most natural thing to do is just to count the number of items in a bin and find how often that item occurs in the bigger "sample

space." So in the case of asking: "what is the probability of getting lung cancer?" We would have to **marginalize** the **joint probability mass** function over the other variable Y = a smoker or not a smoker

$$\mathbb{P}(X) = \sum_{Y=y} \mathbb{P}(X, Y)$$

(Technically it's usually an integral but since all the distributions are discrete for categorical data, we will just stick with summation notation). In the case of the lung cancer patients, if we marginalize over Y, we find that the probability of X = lung cancer or no lung cancer is

$$\mathbb{P}(X = \text{Lung Cancer}) = \frac{709}{1418} = \frac{1}{2} \quad \mathbb{P}(X = \text{No Lung Cancer}) = \frac{709}{1418} = \frac{1}{2}$$

So the probability of getting lung cancer and not getting lung cancer in general is just a 50/50 coin toss (kinda wack; imagine just going about your life and God decides to flip a coin to decide whether or not you have lung cancer). Let us now introduce new. Let π_{ij} denote the joint probability mass for an arbitrary contingency table for I columns and J rows

$$\pi_{i,j} = \mathbb{P}(X = i, Y = j) \quad 1 \leq i \leq I, 1 \leq j \leq J \tag{7.4}$$

Let us denote the marginal probabilities over X and Y be π_{i+} and π_{+j} respectively

$$\begin{aligned} \pi_{i+} &= \sum_{j=1}^{J} \pi_{i,j} = \mathbb{P}(X = i) \\ \pi_{+j} &= \sum_{i=1}^{I} \pi_{i,j} = \mathbb{P}(Y = j) \end{aligned} \tag{7.5}$$

This is just to be consistent with the statistics literature. Then, this will allow us to write the conditional probabilities as

$$\begin{aligned} \pi_{j|i} &= \mathbb{P}(Y = j \mid X = i) = \frac{\pi_{ij}}{\pi_{i+}} \\ \pi_{i|j} &= \mathbb{P}(X = i \mid Y = j) = \frac{\pi_{i,j}}{\pi_{+j}} \end{aligned} \tag{7.6}$$

And these arise naturally from our definitions of conditional probabilities as in equation (4.2). Then, for some contingency table, we can say that independence occurs if either

$$\mathbb{P}(X = i, Y = j) = \pi_{ij} = \mathbb{P}(X = i)\mathbb{P}(Y = j) = \pi_{i+}\pi_{+j} \quad \forall i, j \tag{7.7}$$

or

$$\mathbb{P}(Y = j \mid X = i) = \pi_{j|i} = \pi_{+j} \quad \forall i, j \tag{7.8}$$

Remember that (4.7) and (4.8) are equivalent statements because they arise from each other.

7.2.2 Distribution Sampling

The story of statistics is 1 part sad but also 1 part beautiful. If we knew what the true underlying distributions and probabilities were, then we would know everything there is to know about the data and just do probability theory. The sad part is, like with estimation and regression, we will (likely) never know what the true probabilities for these contingency tables are. The beautiful part is that we can use the estimation techniques that we learned previous and apply to here. So even if we don't know the true probabilities π_{ij}, we could estimate them from the data. Once we have an estimation, then we can form a statistic and an asymptotic distribution and do all the inferences that we want. The only thing that we don't have yet at the moment are the test statistic and the asymptotic distribution. For that, we will need to build models.

Model 1: Poisson

The simplest model that we can construct is the Poisson model, which just tells us that the number of observations for each entry of the contingency table is $n_{ij} \sim \text{Poi}(\lambda_{ij})$. The Poisson random variable characterizes probabilities of "rare" events, so this could work. However, there is one caveat that is because if n_{ij} are random variables, then that means that the total n is also a random variable. We would need to develop an analysis that is capable of handling fluctuating totals.

Problem 1.

Suppose X_1, ..., $X_n \sim Poi(\lambda)$. Define $S_n = X_1 + X_2 + ... + X_n$ to be the sum of these n i.i.d. random variables. Show that conditioned on $S_n = s$, the random variables $\{X_1, ..., X_n\}$ follows a multinomial distribution. Determine the parameters of the multinomial distribution. This is the key motivation why we generally use multinomial distribution when we assume the total number of samples is fixed.

Solution:

We did something similar at the end of chapter 3. This is just probability theory. What we want to know is $\mathbb{P}(X_1, ..., X_n | S_n)$ and we can do this by using the definition of conditional probability

$$\mathbb{P}(X_1, ..., X_n | S_n) = \frac{\mathbb{P}(\vec{X} \cap S_n)}{\mathbb{P}(S_n)}$$

where for lack of notational clutter, $\vec{X} = \{X_1, ..., X_n\}$. Notice that $\vec{X} = \{X_1 = x_1, X_2 = x_2, ..., X_n = x_n\}$ implies that $S_n = \sum_{i=1}^n x_i$. That is, if $\sum_{i=1}^n x_i = s$, then $\vec{X}$ implies $S_n = s$ (because otherwise if $\sum_{i=1}^n x_i \neq s$ then $\mathbb{P}(\vec{X} \cap \{S_n = s\}) = 0$. So this means

$$\mathbb{P}(X_1, ..., X_n | S_n) = \frac{\mathbb{P}(\vec{X})}{\mathbb{P}(S_n)}$$

Let's focus on the numerator. The joint probability for i.i.d. Poisson random variables is just the product of the individual Poisson distributions

$$\mathbb{P}(X_1 = x_1, ..., X_n = x_n) = e^{-n\lambda} \frac{\lambda^{x_1} \lambda^{x_2} ... \lambda^{x_n}}{x_1! x_2! ... x_n!}$$

Now what about the denominator $\mathbb{P}(S_n)$? As it turns out, the sum of i.i.d. Poisson random variables is itself a Poisson random variable with $\lambda_{S_n} = n\lambda$ (left as an exercise)

$$\mathbb{P}(S_n = s) = \frac{e^{-n\lambda}(n\lambda)^s}{s!}$$

Now we can divide these 2 and get

$$\begin{aligned} \mathbb{P}(X_1, ..., X_n | S_n) &= \frac{s!}{e^{-n\lambda}(n\lambda)^s} e^{-n\lambda} \frac{\lambda^{x_1} \lambda^{x_2} ... \lambda^{x_n}}{x_1! x_2! ... x_n!} \\ &= \frac{s!}{x_1! x_2! ... x_n!} p^s \end{aligned}$$

where $p = \frac{\lambda}{n\lambda} = \frac{1}{n}$. As we see, we retrieve the multinomial distribution. Because the Poissons are i.i.d., each p will just be $1/n$ naturally. In general, we would have that

$$p_i = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j}$$

Model 2: Multinomial

We saw in problem 1 that constraining the total number of subjects (for whatever reason) and assuming a Poisson model will lead to a multinomial distribution. Now we have a density for our model

$$\mathbb{P}\left(\{n_{ij}\}_{1 \leq i \leq I, 1 \leq j \leq J} \mid \{\pi_{ij}\}_{1 \leq i \leq I, 1 \leq j \leq J}\right) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J \pi_{ij}^{n_{ij}} \tag{7.9}$$

And once we have the density, we can read off the log likelihood function as

$$\ell_n\left(\{\pi_{ij}\}_{1\leq i\leq I, 1\leq j\leq J}\right) = \sum_i \sum_j n_{ij} \ln \pi_{ij} + \ln\left(\frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!}\right) \tag{7.10}$$

if we're interested in finding the $\hat{\pi}_{MLE}$ then we don't really care about the second term

$$\hat{\pi}_{ij}^{MLE} = \underset{\{\pi_{ij}\}_{1\leq i\leq I, 1\leq j\leq J}}{\operatorname{argmax}} \left[\sum_i \sum_j n_{ij} \ln \pi_{ij}\right] \tag{7.11}$$

But there is a subtle issue. Suppose we did take the derivative of the likelihood function and set it to 0

$$\frac{\partial \ell}{\partial \pi_{ij}} = \frac{n_{ij}}{\pi_{ij}} = 0$$

But this would imply that $\hat{\pi}_{ij} = \infty$, or at the very least, $\hat{\pi}_{ij} = \frac{n_{ij}}{0}$. What happened?? Did we make a mistake somewhere?? Well, yes, but not completely. The issue with the log likelihood function as seen in equation (4.10) is that it puts *no constraints* on what π_{ij} can be. For all we know, π_{ij} could be 10000000. But this simply cannot be true because the π_{ij} are supposed to represent probabilities. If that's the case, then we have our first constraint

$$\sum_{i=1}^I \sum_{j=1}^J \pi_{ij} = 1$$

The other condition is that probabilities cannot be negative $\pi_{ij} > 0 \ \ \forall\, i, j$. This condition won't really get much attention because if we are taking the maximum of the log likelihood, then $\ln(\pi_{ij})$ will necessarily force $\pi_{ij} > 0$ because you can't take the log of a negative number. So now what? Do we just take a Lagrange multiplier and retry? Maybe in a second. But there is still one more trick that we can squeeze out of this. Since we know that all the probabilities must sum to 1, that actually reduces the number of free parameters that we have. More specifically, if we sum all the probabilities from 1 to $I-1$ and 1 to $J-1$, we know from the unity condition that

$$\pi_{IJ} = 1 - \sum_{i,j\neq I,J} \pi_{ij}$$

so we have 1 less free parameter. This means that the maximization of our log likelihood function becomes

$$\left\{\hat{\pi}_{ij}^{\text{MLE}}\right\}_{ij\neq IJ} = \underset{\{\pi_{ij}\}_{1\leq i\leq I, 1\leq j\leq J}}{\operatorname{argmax}} \left[\sum_{ij\neq IJ} n_{ij} \log \pi_{ij} + n_{IJ} \log\left(1 - \sum_{ij\neq IJ} \pi_{ij}\right)\right]$$

And now, if we take the derivative and set it to 0, we find

$$\frac{\partial \ell}{\partial \pi_{ij}} = \frac{n_{ij}}{\pi_{ij}} - \frac{n_{IJ}}{\left(1 - \sum_{ij \neq IJ} \pi_{ij}\right)} = 0$$

Doing a bit of algebra

$$\frac{n_{ij}}{n_{IJ}} = \frac{\pi_{ij}}{1 - \sum_{ij \neq IJ} \pi_{ij}}$$

If you remember from machine learning, this is similar to how the softmax function is generated. With a little bit of help (there is a trick, but the details of the trick aren't so important), we get

$$\hat{\pi}_{ij} = \frac{n_{ij}}{n} \tag{7.12}$$

where n is the total number of subjects. This makes sense because we know that the MLE for a binomial gives us proportions so the multinomial should also give us something is a proportion.

Model 3: Row-wise Multinomial

We can continue to put more constraints on our samples. In the last section, we forced the total number of samples n to be fixed. We can also force the row totals n_{i+} to be fixed. Well, we saw that if we conditioned a sum of Poisson random variables to add up to a fixed number, we got a multinomial distribution. Now if we fix the row totals, each row will be its own multinomial. When we do this, the MLE will no longer be a simple proportion.

Problem 2.

Suppose we first fixed the row sums to be $(n_{1+}, \ldots, n_{I+})$ (whereas, if you recall, in standard multinomial sampling, we only fix the total number of samples) and then sample observations in i^{th} row of the observed contingency table as:

$$(n_{i1}, \ldots, n_{iJ}) \sim \text{Mult}\left(n_{i+}, \pi_{1|i}, \ldots, \pi_{J|i}\right)$$

where $\pi(j \mid i) = \mathbb{P}(Y = j \mid X = i)$. Show that $\hat{\pi}_{j|i}^{\text{MLE}} = n_{ij}/n_{i+}$ for all $1 \leq i \leq I$ and $1 \leq j \leq J$.

Solution:

We know that each sample for a given row i follows a multinomial distribution

$$\mathbb{P}(n_{i1}, n_{i2}, ..., n_{iJ}) = \frac{n_{i+}!}{n_{i1}! n_{i2}! ... n_{iJ}!} \pi_{1|i}^{n_{i1}} \pi_{2|i}^{n_{i2}} ... \pi_{J|i}^{n_{iJ}}$$

And this makes the log likelihood function

$$\ell_i(\pi_j) = \sum_{j=1}^{J} n_{ij} \ln \pi_{j|i}$$

We've ignored the factorial part because that is not important to us when we are maximizing the log likelihood. We could either do the Lagrange multiplier trick, or we could do the summation trick from before. Let us do the Lagrange multiplier trick for completeness. We know that the sum over columns must sum to a fixed n_{i+}. We can write the condition as

$$\sum_{i=1}^{I} n_{ij} = n_{i+} \implies n_{i+} - \sum_{i=1}^{I} n_{ij} = n_{i+} \left(1 - \sum_{i=1}^{I} \pi_{j|i}\right) = 0$$

where we used the fact that $\pi_{j|i} = \frac{n_{ij}}{n_{i+}}$. We can add this Lagrange multiplier into our log likelihood

$$\ell_i(\pi_j) = \sum_{j=1}^{J} n_{ij} \ln \pi_{j|i} + \lambda \left(1 - \sum_{i=1}^{I} \pi_{j|i}\right)$$

Now we can take derivatives of the log likelihood

$$\frac{\partial \ell}{\partial \pi_{j|i}} = \frac{n_{ij}}{\pi_{j|i}} - \lambda = 0$$

And if we do this, then we find that

$$\hat{\pi}_{j|i} = \frac{n_{ij}}{\lambda}$$

But remember that $\lambda = n_{i+}$ and we get

$$\boxed{\hat{\pi}_{j|i} = \frac{n_{ij}}{n_{i+}}}$$

7.2.3 Odds, Risk, Relative Risks, and Odds Ratio

We've built a lot of probability mechanics to be able to describe independence. But perhaps independence isn't the only thing that we are interested in looking for. Let us go back to the lung cancer versus smoker contingency table. Later on when we do inference, sure we might want to test using the data whether or not getting lung cancer is independent from smoking, but perhaps we would also like to quantify that relationship. What are some ways

we can do that? Let us take $\mathbb{P}(Y = 1|X = 1)$ to be the probability that you have lung cancer given that you are a smoker. That would make $\mathbb{P}(X = 1|Y = 2)$ be the probability of you *not* having lung cancer given that you are a smoker. If we take the ration

$$\text{RR} = \frac{\mathbb{P}(Y = 1 \mid X = 1)}{\mathbb{P}(Y = 1 \mid X = 2)} \tag{7.13}$$

What information does this quantity tell us? It tells us how much more likely you are to have lung cancer given that you are a smoker than to have lung cancer given that you are a non-smoker. We coin this term the **relative risk.** If we were to estimate the probabilities using the "common sense" estimator (MLE), we can find the estimated risk ratio as

$$\text{RR} = \frac{\mathbb{P}(\text{Lung cancer}|\text{smoker})}{\mathbb{P}(\text{Lung cancer}|\text{non-smoker})} = \frac{\hat{\pi}_{1|1}}{\hat{\pi}_{1|2}} = \frac{n_{11}n_{2+}}{n_{1+}n_{21}} = \frac{n_{11}(n_{21} + n_{22})}{n_{21}(n_{11} + n_{12})} \tag{7.14}$$

If we use the numbers from the contingency table, we find that the risk ratio is $\frac{688}{21}\frac{80}{1338} \approx 1.96$. So the relative risk of getting lung cancer given that you are a smoker is 2 times greater than if you were a non smoker. This begs the question: are we able to tell independence from looking at the relative risk? It seems almost obvious; if a smoker is 2 times more likely to get lung cancer than a non-smoker, then surely they can't be independent events? It turns out that the answer is *yes* but only in the 2x2 case. To see this, let us recall the definition of independence to be $\mathbb{P}(Y = 1 \mid X = 1) = \mathbb{P}(Y = 1 \mid X = 2) = \mathbb{P}(Y = 1)$. This implies that

$$\text{RR} = \frac{\mathbb{P}(Y = 1 \mid X = 1)}{\mathbb{P}(Y = 1 \mid X = 2)} = 1 \tag{7.15}$$

in the case of independence. So to reiterate: if the probabilities are independent, then the risk ratio is 1. But is the backwards case true? If the risk ratio is 1, does that necessarily imply that the events are independent? Let us examine the last equality of numerator and denominator $\mathbb{P}(Y = 1 \mid X = 1) = \mathbb{P}(Y = 1 \mid X = 2)$. To get $\mathbb{P}(Y = 1)$ we will use the law of total probability

$$\begin{aligned}
\mathbb{P}(Y = 1) &= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) + \mathbb{P}(Y = 1 \mid X = 2)\mathbb{P}(X = 2) \\
&= \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) + \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 2) \\
&= \mathbb{P}(Y = 1 \mid X = 1)\{\mathbb{P}(X = 1) + \mathbb{P}(X = 2)\} \\
&= \mathbb{P}(Y = 1 \mid X = 1)
\end{aligned}$$

If we multiply both sides by $\mathbb{P}(X = 1)$, then we get

$$\mathbb{P}(Y = 1)\mathbb{P}(X = 1) = \mathbb{P}(Y = 1 \mid X = 1)\mathbb{P}(X = 1) = \mathbb{P}(Y = 1, X = 1)$$

We can do this for the other group; but the answer is yes! In the case of a 2x2 contingency table, independence happens if and only if the relative risk is 1.

Problem 3.

A study (E. G. Krug et al., Internat. J. Epidemiol., 27: 214-221, 1998) reported that the number of gun-related deaths per 100,000 people in 1994 was 14.24 in the United States, 4.31 in Canada, 2.65 in Australia, 1.24 in Germany, and 0.41 in England and Wales. Use the relative risk to compare the United States with the other countries. Interpret.

Solution:

We write the definition of relative risk here once again

$$\text{RR} = \frac{\mathbb{P}(\text{dying}|\text{country})}{\mathbb{P}(\text{dying}|\text{non-country})}$$

We saw in (4.14) that we only really need to compare the number of deaths in each group as well as their marginals. However, because the numbers are standardized to per 100,000, the marginals for all groups are the same. So the relative risk of dying from guns in United States compared to each country is given as

Canada	$\frac{14.24}{4.31} \approx 4.31$
Australia	$\frac{14.24}{2.65} \approx 5.37$
Germany	$\frac{14.24}{1.24} \approx 11.48$
England	$\frac{14.24}{0.41} \approx 34.73$

The relative risk compared the probability of the same effect across different groups. But of course, we can compare the probability of different effects within the same group

$$\Omega(1) = \frac{\mathbb{P}(Y = 1 \mid X = 1)}{\mathbb{P}(Y = 2 \mid X = 1)} = \frac{\mathbb{P}(Y = 1 \mid X = 1)}{1 - \mathbb{P}(Y = 1 \mid X = 1)} \tag{7.16}$$

If we insist on using the lung cancer table, then we would say that we are comparing the ratio of the probability of getting lung cancer given that you are a smoker to the probability of not getting lung cancer given that you are a smoker. This ratio is called the **odds.** Is there a way to get an estimate of the odds from the data? Using the definition of the conditional probability

$$\frac{\mathbb{P}(Y = 1 \mid X = 1)}{1 - \mathbb{P}(Y = 1 \mid X = 1)} = \frac{\pi_{1|1}}{1 - \pi_{1|1}} = \frac{n_{11}}{n_{1+}(1 - \frac{n_{11}}{n_{1+}})} = \frac{n_{11}}{n_{1+} - n_{11}} = \frac{n_{11}}{n_{12}} \tag{7.17}$$

The odds for group 2 $\Omega(2)$ is defined similarly. Using the lung cancer data, we see the odds of getting lung cancer given that you're a smoker is $\frac{688}{650} \approx 1.058$. Surprisingly, the odds of getting lung cancer given that you are a smoker isn't supper high. But perhaps that's the point? Maybe the odds are not what we're supposed to look at. Maybe, we should be looking at the ratio of the odds

$$\theta \equiv \frac{\Omega(1)}{\Omega(2)} = \frac{\mathbb{P}(Y=1 \mid X=1)/\mathbb{P}(Y=2 \mid X=1)}{\mathbb{P}(Y=1 \mid X=2)/\mathbb{P}(Y=2 \mid X=2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}} \tag{7.18}$$

Where the last equality comes from Baye's theorem. Because all the probabilities are normalized by the same constant n (the total number of subjects), we can actually just estimate the odds ratio from the number of subjects in each entry of the contingency table

$$\theta = \frac{n_{11}n_{22}}{n_{12}n_{21}} \tag{7.19}$$

Almost like a determinant. But physically, what does this quantity mean? Well, the odds tell you how much more likely 1 group is to get lung cancer. If we take the ratio, then that is telling us how much larger the odds of getting lung cancer is for smokers compared to nonsmokers. Using the data, we find that $\theta = \frac{688*59}{650*21} \approx 2.97$. So the interpretation here is that the odds of smoker getting lung cancer is almost 3 times that of a non smoker getting lung cancer.

Problem 4.

For adults who sailed on the Titanic on its fateful voyage, the odds ratio between gender (female, male) and survival (yes, no) was 11.4. (For data, see R. J. M. Dawson, J. Statist. Ed. 3, 1995.)
a). What is wrong with the interpretation, "The probability of survival for females was 11.4 times that for males"? Give the correct interpretation. When would the quoted interpretation be approximately correct?
b). The odds of survival for females equaled 2.9. For each gender, find the proportion who survived.
Solution:
a). The odds ratio is not the same thing as the ratio of the *conditional* probabilities groups. The odds ratio is simply the ratio of the the probability of success between 2 groups. Therefore, the correct interpretation of the of the odds ratio is "there were 11.4 times more alive females to dead females than alive males to dead males." The interpretation that they gave was the relative risk. But the relative risk and the odds

ratio only coincide when the ratio is 1.

b). We are given

$$\frac{\mathbb{P}(Y = \text{Yes}|X = \text{F})}{\mathbb{P}(Y = \text{No}|X = \text{M})} = 2.9$$

We can divide the odds of survival for females by the odds ratio from part a to get the odds of survival for men ≈ 0.254. To get the proportion of people that survived for each gender, the total number of people for each gender is the number of survival plus the number of dead. The number of survival is directly proportional to the odds of survival, more specifically it is $n_s = n_m(1 + \text{odds})$. To get the proportion, we just divide the odds by the total number of people in each gender, but that's just odds/$(1 + \text{odds})$. So for males

$$p_{s,m} = \frac{.254}{1 + .254} \approx .202$$

and for females

$$p_{s,f} = \frac{2.9}{(1 + 2.9)} \approx .74$$

Problem 4 gets at the heart a source of a lot of confusion: what is the difference between the odds ratio and the relative risk (sometimes called the risk ratio)? The fundamental difference is that the risk ratio only considers one of the cases $Y = 1$ while the odds ratio considers both outputs $Y = 1$ and $Y = 2$. In fact, we can see when the odds ratio and risk ratio coincide by looking at equation (4.19) and (4.15)

$$\text{OR} = \text{RR} \times \frac{\mathbb{P}(Y = 2 \mid X = 2)}{\mathbb{P}(Y = 2 \mid X = 1)} = \text{RR} \times \frac{1 - \mathbb{P}(Y = 1 \mid X = 2)}{1 - \mathbb{P}(Y = 1 \mid X = 1)} \tag{7.20}$$

7.2.4 Simpson's Paradox

In 1973, U.C. Berkeley had a lawsuit filed against them for sex-based discrimination in college admissions. It was reported that the university admitted 44% of male applicants while only admitting 35% of the female applicants. Seems a bit strange looking back on it that almost half of all males were accepted. Anyways, in this case, the input variable is X = sex and the response was Y = acceptance rate. When the university was being investigated by a group of statisticians, what they found was that across almost all majors, the percentage of accepted females was significantly higher than the percentage of males. So at the university level, more males were accepted but on an individual major basis, more females were accepted? What is going on here?

The problem arose because we neglected a third variable: the major. In all of our analysis thus far, we've only considered the 2 variables X and Y. If we lived in a world where those were the only variables that existed, then all is good. However in reality, we need to consider the possibility of a third confounding variable. As we can see, if we neglect this third variable $Z =$ major in the case of U.C. Berkeley, then we would see that might have been a basis for sex based discrimination in the admissions process. However, if we included the third variable Z, then all of a sudden the discrimination is reversed. This phenomenon is called **Simpson's paradox**. It's actually not a logical paradox, just a bit of a sleight of hand in probability theory.

In terms of probability, the question that we are trying to answer is: is it possible that

$$\begin{gathered}\mathbb{P}(Y = 1 \mid X = 1, Z = 1) > \mathbb{P}(Y = 1 \mid X = 2, Z = 1) \\ \text{and} \\ \mathbb{P}(Y = 1 \mid X = 1, Z = 2) > \mathbb{P}(Y = 1 \mid X = 2, Z = 2)\end{gathered} \tag{7.21}$$

While having that

$$\mathbb{P}(Y = 1 \mid X = 1) < \mathbb{P}(Y = 1 \mid X = 2) \tag{7.22}$$

(or the other way around). The answer is yes. But it's probably much easier to see this if we did a problem.

		Death penalty		
Victim's race	Defendant's race	Yes	No	Percent yes
White	White	19	132	12.6
	Black	11	52	17.5
Black	White	0	9	0.0
	Black	6	97	5.8
Total	White	19	141	11.9
	Black	17	149	10.2

Table 7.1: Death penalty verdicts in Florida by defendant's and victim's race, $n = 326$. Source: M. L. Radelet, *Amer. Sociol. Rev.* **46**: 918–927 (1981), as reported in Agresti [1].

Problem 5.

The table below is from an early study on the death penalty in Florida. Analyze these

data and show that Simpson's paradox occurs.

Solution:

Ultimately, we have to show that both of the inequalities above are satisfied (but with the directions reversed relative to the marginal one), using Table 7.1. Let Y = Sentence, X = Defendant's race, and Z = Victim's race. First, let us find $\mathbb{P}(Y = 1 \mid X = 1, Z = 1)$. We can extend our definition of conditional probability to multivariables as

$$\mathbb{P}(Y|X, Z) = \frac{\mathbb{P}(X, Y, Z)}{\mathbb{P}(X, Z)} \tag{7.23}$$

We can use our common sense estimators with the total number of subjects $n = 326$. So

$$\mathbb{P}(Y = 1 \mid X = 1, Z = 1) = \frac{19}{326}\frac{326}{151} \approx 0.126$$

$$\mathbb{P}(Y = 1 \mid X = 2, Z = 1) = \frac{11}{63} \approx 0.174$$

So $\mathbb{P}(Y = 1 \mid X = 1, Z = 1) < \mathbb{P}(Y = 1 \mid X = 2, Z = 1)$. Similarly for $Z = 2$ (black victims)

$$\mathbb{P}(Y = 1 \mid X = 1, Z = 2) = 0$$

$$\mathbb{P}(Y = 1 \mid X = 2, Z = 2) = \frac{6}{103} \approx 0.018$$

so $\mathbb{P}(Y = 1 \mid X = 1, Z = 2) < \mathbb{P}(Y = 1 \mid X = 2, Z = 2)$. We shouldn't be mindlessly plugging numbers; we have to ask: what does this mean? What we've shown s far is that the probability of getting the death penalty if you're a white is less than if you're black *regardless* of who the victim was. But if we marginalize over the victims

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{19}{326}\frac{326}{160} \approx 0.118$$

$$\mathbb{P}(Y = 1 \mid X = 2) = \frac{17}{326}\frac{326}{166} \approx 0.102$$

so $\mathbb{P}(Y = 1 \mid X = 1) > \mathbb{P}(Y = 1 \mid X = 2)$. What did we just show? What we just showed was that black defendants are more likely to get the death penalty regardless of who their victims were, and yet when we neglect the victims, white defendants are more likely to get the death penalty.

So which picture should we look at? Why do we see the trends that we see? In the case of U.C. Berkeley, it could be possible that there are majors with massive massive student body count that are male dominated (for example, engineering). But in every other major, it is female dominated but the number of students is smaller. If we marginalized over the

majors, we would make the incorrect assumption that there is discrimination based on sex because engineering takes up a significant chunk of the university, and therefore it isn't very representative of all the departments in the university. Similarly, in problem 5, we saw that the as a whole, white people are more likely to get the death penalty. But that's just white on white crimes are just much more than any other combo and there are less white people (160). So if we neglect the victim's race, then it would seem like white people are more likely to get the death penalty. However, if we add in the victim's race, we see that black people are more likely to get the death penalty. The analysis that we have done is called **partial association analysis**, and the crux of it relies on a third confounding variable Z.

If we want to introduce a new variable Z to the analysis that we have done so far, we should generalize the probability structures from before to be able to handle it, but the generalization is nothing new, really. For example, we say that *2 random variables are independent given* Z if

$$\mathbb{P}(X = i, Y = j \mid Z = k) = \mathbb{P}(X = i \mid Z = k)\mathbb{P}(Y = j \mid Z = k) \quad \forall\, i, j, k \tag{7.24}$$

We can similarly define the conditional relative risk by saying when X and Y has 2 levels and Z has K levels, then the conditional risk ratio of (X, Y) given Z = k (for any $1 \leq k \leq K$) is defined as:

$$\mathrm{RR}_{XY}(k) = \frac{\mathbb{P}(Y = 1 \mid X = 1, Z = k)}{\mathbb{P}(Y = 1 \mid X = 2, Z = k)} \tag{7.25}$$

and the odds ratio becomes

$$\theta_{XY}(k) = \frac{\mathbb{P}(Y = 1 \mid X = 1, Z = k)/\mathbb{P}(Y = 0 \mid X = 1, Z = k)}{\mathbb{P}(Y = 1 \mid X = 2, Z = k)/\mathbb{P}(Y = 0 \mid X = 2, Z = k)} \tag{7.26}$$

There is a new measure of association when we introduce a third variable and it is called the **homogeneous association.** We say that 2 categorical variables are homogeneous with respect to a third variable Z if the conditional odds ratio are all the same across levels of Z

$$\theta_{XY}(1) = \theta_{XY}(2) = \cdots = \theta_{XY}(K) \tag{7.27}$$

Problem 6.

For a diagnostic test of a certain disease, π_1 denotes the probability that the diagnosis is positive given that a subject has the disease, and π_2 denotes the probability that the diagnosis is positive given that a subject does not have it. Let ρ denote the probability that a subject does have the disease.

a). Given that the diagnosis is positive, show that the probability that a subject does have the disease is

$$\pi_1\rho/\left[\pi_1\rho + \pi_2(1-\rho)\right]$$

b). Suppose that a diagnostic test for HIV + status has both sensitivity and specificity equal to 0.95 , and $\rho = 0.005$. Find the probability that a subject is truly HIV+, given that the diagnostic test is positive. To better understand this answer, find the joint probabilities relating diagnosis to actual disease status, and discuss their relative sizes.

Solution:

a). This is nothing more than Bayes' theorem. We state Bayes' theorem as

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} \tag{7.28}$$

In our case, B = positive diagnostic and A = does have disease. First, we need to find $\mathbb{P}(B|A) = \mathbb{P}$(positive result|actually having disease). The problem states that π_1 is the probability the diagnostic is positive given the subject has the disease, so $\mathbb{P}(B|A) = \pi_1$. The problem also gives us that ρ is the probability of having the disease, so $\mathbb{P}(A) = \rho$. The only thing that we need is $\mathbb{P}(B) = \mathbb{P}$(getting positive diagnostic). We can use the law of total probability

$$\mathbb{P}(B) = \sum_{a\in A} \mathbb{P}(B|A=a)\mathbb{P}(A=a)$$

but we have $\mathbb{P}(B|A)$, it's π_1, and that makes $\mathbb{P}(B|A^c) = \pi_2$. We've also seen before that $\mathbb{P}(A) = \rho$, and therefore $P(A^c) = 1-\rho$. Thus, all together, we have

$$\boxed{\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{\pi_1\rho}{\pi_1\rho + \pi_2(1-\rho)}}$$

b). It's saying that $\pi = .95$ and $\rho = .005$. Plugging this into our formula, we see

$$\boxed{\frac{.95 * .005}{.95 * .005 + .05 * .995} \approx .08}$$

A seemingly very very small number for such a high accuracy. To understand this better, we should find the joint probabilities

$$\mathbb{P}(A,B) = \mathbb{P}(B|A)\mathbb{P}(A) = \pi_1\rho \approx .00475$$

so the probability of getting both HIV and a correct diagnostic is very very small, thus the conditional probability must necessarily be very very small.

7.2.5 Generalization To IxJ Contingency Tables

The idea of a marginal/relative risk so far has relied on the fact that the contingency tables be 2x2. Is it possible for us to generalize these ideas to tables larger than 2x2? The difficulty in saying yes immediately is due to the fact that we wouldn't be able to say that a single number captures all the information. You would need to compute these numbers from the 2x2 sub-tables. For example, given any two levels of X (say $\{i_1, i_2\}$) and any two levels of Y (say $\{j_1, j_2\}$) you could write

$$\theta_{i_1,i_2,j_1,j_2} = \frac{\mathbb{P}\left(Y = j_1 \mid X = i_1\right)/\mathbb{P}\left(Y = j_2 \mid X = i_1\right)}{\mathbb{P}\left(Y = j_1 \mid X = i_2\right)/\mathbb{P}\left(Y = j_2 \mid X = i_2\right)} = \frac{\pi_{i_1 j_1}\pi_{i_2 j_2}}{\pi_{i_1 j_2}\pi_{i_2 j_1}} \tag{7.29}$$

which means that the number of possible odds ratios or risk ratios that you can compute is $\binom{J}{2}\binom{I}{2}$ which is of order J^2I^2. This can be very very large. It turns out, however, that you don't need to compute all of them. You only need to compute $(I-1)(J-1)$ pairs (which still is a lot, don't get me wrong) but reduces the number of computations greatly. Using these pairs, one could derive all the remaining odds ratio. The local odds ratio can be computed as

$$\theta_{ij} \triangleq \theta_{i,i+1,j,j+1} = \frac{\mathbb{P}(Y = j \mid X = i)/\mathbb{P}(Y = j+1 \mid X = i)}{\mathbb{P}(Y = j \mid X = i+1)/\mathbb{P}(Y = j+1 \mid X = i+1)} = \frac{\pi_{ij}\pi_{(i+1)(j+1)}}{\pi_{i(j+1)}\pi_{(i+1)j}} \tag{7.30}$$

7.3 Inference On Categorical Data

We've done a great deal of estimation, but estimation is only really half the story. Once we have our estimates for our parameters of interest, we want to be able to do inference on them. For example, we might want to conduct hypothesis tests to see whether or not X and Y are independent

$$H_0 : X \perp Y, \quad H_1 : X \not\perp Y$$

In the last section, we know how to extract independence from quantities like the risk ratio and odds ratio. The only things that we are missing right now are the test statistics and the asymptotic distributions.

7.3.1 Fisher, Pearson, And The MLE

If independence is really what we are seeking out, then we should recall that the independence from a contingency table states that

$$\mathbb{P}(X = i, Y = j) = \mathbb{P}(X = i)\mathbb{P}(Y = j) \implies \pi_{ij} = \pi_{i+}\pi{+j}$$

We can use the common sense estimates for the probabilities and this would imply that if X and Y were truly independent, then

$$\frac{n_{ij}}{n} \approx \frac{n_{i+}}{n}\frac{n_{+j}}{n} \tag{7.31}$$

Which we could further simplify to say that

$$n_{ij} \approx \frac{n_{i+}n_{+j}}{b} \equiv \hat{\mu}_{ij} \tag{7.32}$$

So under the null, n_{ij} is equal to $\hat{\mu}_{ij}$. What kind of test statistic can we construct out of this, and what distribution would it follow? Well, what we know is that we assume that the n_{ij} are Poisson distributed. If that is the case, then we can read off the mean and variance immediately because they are one and the same. If we denote $\hat{\mu}_{ij}$ to be the estimated mean, then that means that $\hat{\mu}_{ij}$ is also the estimated variance (remember! when we are doing hypothesis testing, the distribution is always the distribution of the null). So let us rely on the central limit theorem and construct something like a standardized Z statistic

$$T_n = \sum_{i=1}^{I}\sum_{j=1}^{J}\frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}} \tag{7.33}$$

We squared the numerator because we don't want to be working with square roots of the estimate. But regardless, this is called the **Pearson's χ^2 test**. Disappointingly, the name kinda gives away what the asymptotic distribution is but it's pretty obvious what it should be. If we invoke the CLT, then the sum of standard normals will be χ^2 distributed. What about the degree of freedom? It's a simple calculation. If you put no constraints on the model, then we would expect the degree of freedom to be $IJ - 1$, since there are IJ possible probabilities for a IxJ contingency table and there's a $-\hat{\mu}_{ij}$ term, so all together $IJ - 1$ free parameters. But there in fact *is* a constraint, and that constraint is the fact that independence demands the I's are treated separately from the J's *as well as* the constraint that the probabilities have to sum up to 1. So really, we only have $(I - 1)$ free parameters for I and $(J - 1)$ free parameters for J. If we the product is all that we care about (for independence), then we should expect that the degree of freedom is $(I - 1)(J - 1)$, and indeed, it is. Thus, the Pearson's test statistic is $\chi^2_{(I-1)(J-1)}$ distributed.

Problem 7.

For a 2×2 table, consider $H_0 : \pi_{11} = \theta^2,\ \pi_{12} = \pi_{21} = \theta(1-\theta),\ \pi_{22} = (1-\theta)^2$
a). Show that the marginal distributions are identical and that independence holds.
b). For a multinomial sample, under H_0 show that $\hat{\theta} = (p_{1+} + p_{+1})/2$.
c). Explain how to test H_0. Show that df $= 2$ for the test statistic.
Solution:
a). First, we need to find the marginals

$$\boxed{\pi_{1+} = \theta^2 + \theta(1-\theta)} \quad \boxed{\pi_{2+} = \theta(1-\theta) + (1-\theta)^2}$$

$$\boxed{\pi_{+1} = \theta^2 + \theta(1-\theta)} \quad \boxed{\pi_{+2} = \theta(1-\theta) + (1-\theta)^2}$$

so indeed, the marginal distributions are identical. But what about independence? Independence means $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i and j (equation (4.31)). So let's start with the fact that $\theta^2 + \theta(1-\theta) = \theta$. Then

$$\pi_{1+}\pi_{+1} = \theta\theta = \theta = \pi_{11}$$

Now for the other. We need to use the fact that $\theta(1-\theta) + (1-\theta)^2 = 1 - \theta$

$$\pi_{1+}\pi_{+2} = \theta(1-\theta) = \pi_{12} = \pi_{21}$$

and lastly,

$$\pi_{2+}\pi_{+2} = (1-\theta)(1-\theta) = (1-\theta)^2 = \pi_{22}$$

thus, fully independent.
b). Assuming multinomial sampling, we would have

$$f(x;\theta) = \frac{n!}{n_{11}!n_{12}!n_{21}!n_{22}!}\pi_{11}^{n_{11}}\pi_{12}^{n_{12}}\pi_{21}^{n_{21}}\pi_{22}^{n_{22}}$$

Let us find the log likelihood (dropping extraneous constants)

$$\ell(\theta) = 2n_{11}\ln(\theta) + n_{21}\ln(\theta(1-\theta)) + n_{12}\ln(\theta(1-\theta)) + 2n_{22}\ln(1-\theta)$$

taking the derivative and setting it equal to 0

$$\frac{\partial \ell}{\partial \theta} = \frac{2n_{11}}{\theta} - \frac{(2\theta-1)(n_{12}+n_{21})}{\theta - \theta^2} - \frac{2n_{22}}{1-\theta} = 0$$

$$\implies 2n_{11}(1-\theta) - (2\theta - 1)(n_{12} + n_{21}) - 2\theta n_{22} = 0$$

$$\implies 2n_{11}(1-\theta) - (2\theta n_{12} + 2\theta n_{21} - n_{12} - n_{21}) - 2\theta n_{22} = 0$$

$$\implies \hat{\theta} = \frac{2n_{11} + n_{12} + n_{21}}{2(n_{11} + n_{12} + n_{21} + n_{22})} = \frac{p_{+1} + p_{1+}}{2}$$

c). As with what we know now, we should use the Pearson's statistic which follows a χ^2. Since the off diagonals are identical, you have 1 less parameter and therefore you go from 3 in the full model down to 2. Hence the degree of freedom is 2.

The derivation of the Pearson's test statistic seems a bit ad-hoc. Is it possible for us to construct another test statistic, say using the MLE? Well, if we know that our data is multinomially distributed $(n_{11}, \ldots, n_{IJ}) \sim \text{mult}\,(n, \pi_{11}, \ldots, \pi_{IJ})$, then the test statistic is just going to be the likelihood ratio test statistic. This is immediately evident because from before, if we take the maximum likelihood of the full model (using the fact that the MLE is n_{ij}/n), then we get that the full model's maximum likelihood is

$$\ell_{\max} = \log\left(\frac{n!}{\Pi_{ij} n_{ij}!}\right) + \sum_{i,j} n_{ij} \log\left(n_{ij}/n\right) \tag{7.34}$$

What about the null model? Well, the null model is the model that X and Y are independent, so the MLE is $\hat{\pi}_{i+}\hat{\pi}_{+j}$ and thus the maximum likelihood of the null model is

$$\ell_{\text{null}} = \log\left(\frac{n!}{\Pi_{ij} n_{ij}!}\right) + \sum_{i,j} n_{ij} \log\left(n_{i+} n_{+j}/n^2\right) \tag{7.35}$$

The ratio of the likelihoods is the difference of the log likelihoods, and so we can create a test statistic

$$\begin{aligned} T_n &= 2(\text{ maximum log-likelihood of full model } - \text{ maximum log-likelihood of null model }) \\ &= 2\sum_{i,j} n_{ij} \log\left(n_{ij}/\hat{\mu}_{ij}\right) \end{aligned} \tag{7.36}$$

And this shouldn't be anything new because we have already encountered this in regression analysis for GLMs. We know, then, that this test statistic must be χ^2 distributed with the same degrees of freedom $(I-1)(J-1)$ because under the null, X and Y are independent. Remember! This only works in the asymptotic limit when $n \to \infty$ when we can approximate it as a normal distribution and we get the emergence of the χ^2 statistic. This statistic is known as the **maximum likelihood test statistic.**

Like we mentioned before; the distributions asymptotically approach χ^2 in the limit of $n \to \infty$. But obviously, we can never have an infinite number of samples, and what that means is that there will always be some sort of approximation error. How much error is tolerable is dependent on the user. In regression analysis for GLMs, we have sharp upper bounds for these errors, but back in the good ol' days, the did not have these upper bounds for the errors yet. But that's no matter! R.A. Fisher was able to come up with an exact distribution for the test statistic, but *only in the case of 2x2.* We name this statistic after him, and call it **Fisher's exact test statistic.** Let us visualize with a 2x2 contingency table. Suppose we know n_{1+}, n_{+1}, and n. If we know what n_{11} is, then we know everything there is to know about the contingency table. The exercise is really straight forward because $n_{12} = n_{1+} - n_{11}, n_{21} = n_{+1} - n_{11}$ and finally $n_{22} = n - (n_{11} + n_{12} + n_{21}) = n - n_{1+} - n_{+1} + n_{11}$. But if we didn't know what n_{11} was, we could come up with an exact distribution for n_{11}, and once we know that, we know everything. Under the null (X and Y are independent), the distribution is

$$\mathbb{P}(n_{11} = k) = \frac{\binom{n_{1+}}{k}\binom{n-n_{1+}}{n_{+1}-k}}{\binom{n}{n_{+1}}} \triangleq f(k; n, n_{1+}, n_{+1}) \tag{7.37}$$

which is a **hypergeometric** distribution. So how can we come up with a test statistic? We know from previous problems that the odds ratio determines independence (when $\theta = 1$). We also know that the estimate for the odds ratio has the form

$$\hat{\theta} = \frac{n_{11}n_{22}}{n_{21}n_{12}}$$

So what? Well, we see that if n_{11} is sufficiently large, $\hat{\theta}$ will deviate from 1. Now that we have a distribution for n_{11}, we can construct a test statistic around θ and n_{11} determine for what value of n_{11} would we reject the null.

Problem 8.

For testing independence, show that $\chi^2 \leq n \min(I-1, J-1)$. Hence $V^2 = \chi^2/[n \min(I-1, J-1)]$ falls between 0 and 1 (Cramér 1946). For 2×2 tables, X^2/n is often called phi-squared; it equals Goodman and Kruskal's tau. Other measures based on X^2 include the contingency coefficient $\left[\chi^2/\left(\chi^2 + n\right)\right]^{1/2}$ (Pearson 1904).

Solution:

We know that Pearson's test statistic is

$$T_n = \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}}$$

Now, if we factor out an n, we get

$$T_n = n\sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(\pi_{ij} - \hat{\pi}_{+j}\hat{\pi}_{i+})^2}{\hat{\pi}_{i+}\hat{\pi}_{+j}}$$

We know that the test statistic above is χ^2 distributed with $(I-1)(J-1)$ degrees of freedom (assuming independence). If we take the square root of the summand, we know that it follows a standard normal distribution

$$\frac{\pi_{ij} - \hat{\pi}_{+j}\hat{\pi}_{i+}}{\sqrt{\hat{\pi}_{+j}\hat{\pi}_{i+}}} \sim \mathcal{N}(0,1)$$

The reason that we do this is because the sum squared sum of the above is a χ^2, but we can always decompose this sum using singular value decomposition. That is,

$$X^2 = n\|Z\|^2 = n\sum_{k=1}^{r}\lambda_k^2$$

where λ are the singular values of the covariance matrix. Now, the covariance matrix is not of full rank, and the rank of the covariance matrix is dependent on the column the column space, which just means that it is upper bounded by whichever one is smaller: $I-1$ or $J-1$. The singular values of the covariance matrix are all less than 1, and thus we can force an upper bound in the sum as

$$\boxed{\chi^2 \leq n \sum_{k=1}^{\min I-1, J-1} 1 = n\min\{I-1, J-1\}}$$

we can divide through by both sides $n\min\{I-1, J-1\}$

$$\boxed{V = \frac{\chi^2}{n\min\{I-1, J-1\}} \in [0,1]}$$

7.3.2 The Delta Method

In the last section, we said that if we knew n_{11}, then we would know everything there is to know about the 2x2 contingency table. Even if we didn't, we have the distribution of n_{11} and we have asymptotic distributions of the n_{ij}, and we used that to make inferences about quantities like the odds ratio. But this seems a bit implicit and it would be nicer if we could construct a test statistic for the odds ratio directly. If we have the distribution for n_{ij}, does that mean that the asymptotic distributions for the odds ratio and the risk ratio follow the same distribution? The answer is not as straightforward because transformations of random variables are not a simple transformation of the distribution. But perhaps we could tackle this problem perturbatively, and that's what we are going to do. This method has a name and is called the **the Delta method.**

Suppose $\hat{\theta}$ is a consistent and asymptotically normal estimator of θ_0. If $g(.)$ is a smooth (and by smooth we typically mean like, to second order) with $g(\theta_0) \neq 0$, then

$$\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{D} \mathcal{N}(0, \sigma^2(g'(\theta_0))) \tag{7.38}$$

The proof of this is not so important; all that it is is just a Taylor series expansion and then using Slutsky's theorem. But this tells us something really neat: that smooth transformations of the MLE also preserve the asymptotically normal structure of the distribution. To see an example, let's use the binomial data, so $X_1, X_2, ..., X_n \sim \text{Bernoulli}(p)$. We know that the maximum likelihood estimator for the Bernoulli data is $\hat{p}$ which is just the sample mean. The estimated sample variance is

$$\hat{\sigma}^2 = \hat{p}(1 - \hat{p})$$

If we let our transformation function

$$g(x) = \ln\left(\frac{x}{1-x}\right)$$

which means that

$$g'(p) = \frac{1}{p(1-p)}$$

then the Delta method tells us that

$$\sqrt{n}\left(\ln\left(\frac{\hat{p}}{1-\hat{p}}\right) - \ln\left(\frac{p_0}{1-p_0}\right)\right) \sim \mathcal{N}\left(0, \frac{1}{p_0(1-p_0)}\right) \tag{7.39}$$

Obviously this generalizes to higher dimensions as well. Suppose, instead, that $\hat{\theta}$ is a vector

quantity $\hat{\theta} \in \mathbb{R}^p$. If $g(.) : \mathbb{R}^p \to \mathbb{R}$, then

$$\sqrt{n}\left(g(\hat{\theta}) - g\left(\theta_0\right)\right) \xrightarrow{D} \mathcal{N}\left(0, \nabla g\left(\theta_0\right)^T \Sigma \nabla g\left(\theta_0\right)\right) \tag{7.40}$$

where Σ is the covariance matrix. Let's see what this holds in store for us. We will use the Delta method to derive the asymptotic distribution of the log odds ratio. If $\theta_0 \equiv \{\pi_{11}, \pi_{12}, \pi_{21}, \pi_{22}\}$ are the unknown parameters we want to estimate, then the log odds ratio is defined as

$$g\left(\theta_0\right) = \log \theta_{0,1} - \log \theta_{0,2} - \log \theta_{0,3} + \log \theta_{0,4} = \log \pi_{11} + \log \pi_{22} - \log \pi_{12} - \log \pi_{21} \tag{7.41}$$

The MLEs are $\hat{\theta} = (\hat{\pi}_{11}, \hat{\pi}_{12}, \hat{\pi}_{21}, \hat{\pi}_{22})$ and we know that they are multinomially distributed. We need to know the covariance matrix before we can use the Delta method, but the covariance matrix is pretty simple because it's just a 4x4

$$\begin{aligned} \Sigma_{ii} &= \theta_{0,i}\left(1 - \theta_{0,i}\right) \quad 1 \leq i \leq 4, \\ \Sigma_{ij} &= -\theta_{0,i}\theta_{0,j} \quad 1 \leq i \neq j \leq 4 \end{aligned}$$

The gradient is

$$\nabla g\left(\theta_0\right) = \left(\frac{1}{\theta_{0,1}}, -\frac{1}{\theta_{0,2}}, -\frac{1}{\theta_{0,3}}, \frac{1}{\theta_{0,4}}\right)$$

Which just means that the sample variance is

$$\nabla g\left(\theta_0\right)^\top \Sigma \nabla g\left(\theta_0\right) = \frac{1}{\theta_{0,1}} + \frac{1}{\theta_{0,2}} + \frac{1}{\theta_{0,3}} + \frac{1}{\theta_{0,4}} = \frac{1}{\pi_{11}} + \frac{1}{\pi_{12}} + \frac{1}{\pi_{21}} + \frac{1}{\pi_{22}}$$

The Delta method tells us

$$\sqrt{n}\left(g(\hat{\theta}) - g\left(\theta_0\right)\right) \xrightarrow{D} \mathcal{N}\left(0, \frac{1}{\pi_{11}} + \frac{1}{\pi_{12}} + \frac{1}{\pi_{21}} + \frac{1}{\pi_{22}}\right) \tag{7.42}$$

We can replace all the π_{ij} with their maximum likelihood estimates

$$\hat{\sigma} = \frac{1}{n_{11}/n} + \frac{1}{n_{12}/n} + \frac{1}{n_{21}/n} + \frac{1}{n_{22}/n}$$

and thus we get

$$\frac{\sqrt{n}\left(g(\hat{\theta}) - g\left(\theta_0\right)\right)}{\sqrt{\frac{1}{n_{11}/n} + \frac{1}{n_{12}/n} + \frac{1}{n_{21}/n} + \frac{1}{n_{22}/n}}} = \frac{\left(g(\hat{\theta}) - g\left(\theta_0\right)\right)}{\sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}} \xrightarrow{D} \mathcal{N}(0, 1) \tag{7.43}$$

So if you wanted to find the asymptotic distribution of a transformation of the MLE, then

the Delta method is the way to go!

Problem 9.

Using the delta method, show that the Wald confidence interval for the logit of a binomial parameter π is

$$\log[\hat{\pi}/(1-\hat{\pi})] \pm z_{\alpha/2}/\sqrt{n\hat{\pi}(1-\hat{\pi})}.$$

Explain how to use this interval to obtain one for π itself. [Newcombe (2001) noted that the sample logit is also the midpoint of the score interval for π, on the logit scale. He showed that this logit interval contains the score interval.]

Solution:

We saw from the first example that the logit transformation of the Bernoulli MLE $\hat{p}$ gives rise to a centered normal with variance

$$\sigma^2 = \frac{1}{p_0(1-p_0)}$$

We can standardize the transformation and using the estimated variance

$$\frac{\left(\ln\left(\frac{\hat{p}}{1-\hat{p}}\right) - \ln\left(\frac{p_0}{1-p_0}\right)\right)}{\sqrt{\frac{1}{n\hat{p}(1-\hat{p})}}} \sim \mathcal{N}(0,1)$$

And now we can create confidence intervals with recognizing that the standard error is $1/(\sqrt{n}\hat{p}(1-\hat{p})$

$$\boxed{\text{CI} : \ln\left(\frac{\hat{p}}{1-\hat{p}}\right) \pm \frac{z_{\alpha/2}}{\sqrt{n\hat{p}(1-\hat{p})}}}$$

7.3.3 Small Sample Tests

In all the previous sections before, asymptotic normality was assumed, and we worked with the asymptotic distributions. Obviously, in the case where we have small sample sizes, assuming normality might actually be a very bad assumption, and that's why we came up with Fisher's exact test. In the last section, we wanted to find the asymptotic distribution for the transformation of the MLE so that we could construct the asymptotic distribution for the odds ratio. Now we ask: can we combine the 2 ideas? Can we have devise a test statistic to determine not just when the odds ratio is 1, but when the odds ratio is *any*

number? Can we do a transformation on the hypergeometric random variable? Of course we can! Such a transformation results in the **non-centered hypergeometric** distribution

$$f\left(t; n, n_{1+}, n_{+1}; \tau\right) = \mathbb{P}\left(n_{11} = t \mid \text{OR} = \tau\right) = \frac{\binom{n_{1+}}{t}\binom{n-n_{1+}}{n_{+1}-t}\tau^t}{\sum_u \binom{n_{1+}}{u}\binom{n-n_{1+}}{n_{+1}-u}\tau^u} \tag{7.44}$$

and this will allow us to test *any* odds ratio value. To create a rejection interval, we just need

$$\sum_{t=t_1}^{t_2} f\left(t; n, n_{1+}, n_{+1}; \tau\right) \approx 1 - \alpha \tag{7.45}$$

and the rejection region is when n_{11} falls outside of $[t_1, t_2]$

7.4 Inference On Ordinal Data

8

Logistic Regression

"For a successful technology, reality must take precedence over public relations, for nature cannot be fooled."

– Richard Feynman, *Appendix F of the Rogers Commission Report*

In the last chapter, the response Y was a number that could be anything, like an exam score. But a lot of the time the thing we want to explain is a yes or no. Did the patient survive? Did the student pass? Did the part fail? In the categorical data chapter we handled questions like this when the explanatory variable was also categorical, by putting everything in a contingency table and looking at odds ratios. That works great for smokers versus nonsmokers, but what if the explanatory variable is a temperature, or a dosage, or an age? We can't make a contingency table with a column for every possible temperature. What we want is a regression model for a binary response, and the most common one by far is **logistic regression**. It's going to borrow the idea of the odds from the categorical data chapter and the idea of a linear predictor from the regression chapter and glue them together.

8.1 Why Not Just Use Linear Regression?

Let $Y_i \in \{0, 1\}$, and let $p_i = \mathbb{P}(Y_i = 1 \mid x_i)$. Since Y_i is a Bernoulli random variable, its mean is $\mathbb{E}[Y_i \mid x_i] = p_i$. So the first thing we might try is the linear model from the last chapter, $p_i = \beta_0 + \beta_1 x_i$, and fit it with least squares. People do this, and it even has a name (the linear probability model), but it has two pretty serious problems. The first is that a straight line is unbounded, so for large or small enough x the model will happily predict

probabilities bigger than 1 or less than 0, which doesn't make any sense. The second is that the variance of a Bernoulli is $\text{Var}[Y_i] = p_i(1-p_i)$, which changes with x_i whenever p_i does. That breaks the constant variance assumption, so all the t intervals and tests we built in the last chapter are off. We need a model that keeps the probability inside $[0, 1]$ and that uses the right variance.

8.2 The Logistic Regression Model

8.2.1 Odds And Log-Odds

Here's the trick. Instead of modeling the probability directly, we model something that is allowed to be any real number. Recall from the categorical data chapter that the **odds** of an event with probability p are $p/(1-p)$. When p goes from 0 to 1, the odds go from 0 to ∞, so the odds can't be negative, but they are unbounded above. If we then take the log, the **log-odds** (also called the **logit**)

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \tag{8.1}$$

goes from $-\infty$ to ∞ (Figure 8.1). That's exactly the range of a linear function, so the logistic regression model says

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i, \qquad Y_i \mid x_i \sim \text{Bernoulli}(p_i), \tag{8.2}$$

with the Y_i independent. If we solve for p_i, we get

$$p_i = \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}} = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}, \tag{8.3}$$

and the function $\sigma(z) = 1/(1+e^{-z})$ is called the **logistic** (or **sigmoid**) function. It's an S-shaped curve that is always strictly between 0 and 1, so the first problem is gone for free. And because the Y_i are modeled as Bernoulli, the variance $p_i(1-p_i)$ is built in, which takes care of the second problem too.

8.2.2 Interpreting The Coefficients

In linear regression, β_1 was the change in the mean of Y per unit change in x. Here, (8.2) says that β_1 is the change in the *log-odds* per unit change in x, which is not the most intuitive thing in the world. It gets a lot better once we exponentiate. Compare the odds at

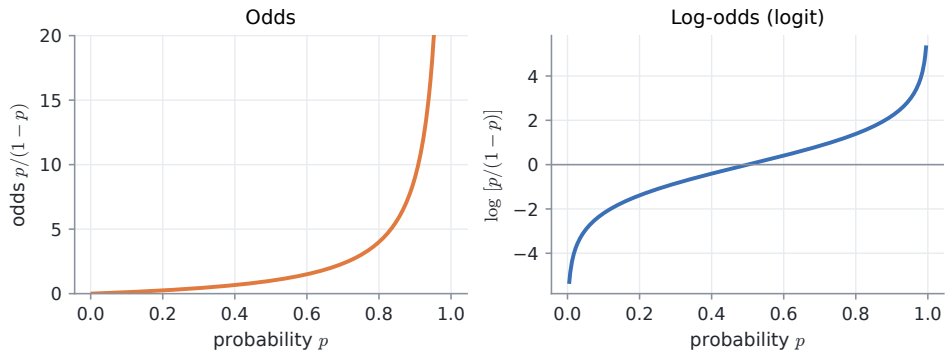


Figure 8.1: The odds stretch probabilities in $(0, 1)$ out to $(0, \infty)$, and the log-odds stretch them out to the whole real line. Logistic regression puts the linear predictor on the log-odds scale.

$x + 1$ with the odds at x,

$$\frac{\text{odds}(x+1)}{\text{odds}(x)} = \frac{e^{\beta_0+\beta_1(x+1)}}{e^{\beta_0+\beta_1 x}} = e^{\beta_1}. \tag{8.4}$$

So e^{β_1} is an **odds ratio**, the same quantity θ from the categorical data chapter. Each one-unit increase in x multiplies the odds of $Y = 1$ by e^{β_1}, no matter where you start. If $\beta_1 = 0$, the odds ratio is 1 and x has no effect, which is the logistic version of independence. Notice that the effect on the *probability* isn't constant though. Since the curve is an S, a one-unit change in x moves p the most near $p = 1/2$ and barely at all out in the tails. In fact the slope of the curve is $\frac{dp}{dx} = \beta_1 p(1-p)$, which is biggest at $p = 1/2$.

Problem 1.

Suppose the explanatory variable is itself binary, $x \in \{0, 1\}$ (for example $x = 1$ for smokers), so that the data can be summarized by a 2×2 table with counts n_{ij}, where row i is x and column j is Y. Show that the maximum likelihood estimates satisfy

$$e^{\hat{\beta}_1} = \frac{n_{11} n_{00}}{n_{10} n_{01}},$$

which is exactly the sample odds ratio from the categorical data chapter. (Hint: with only two values of x, the model can fit each row's proportion exactly.)

8.2.3 An Example: The Challenger O-Rings

On January 28, 1986, the space shuttle *Challenger* broke apart 73 seconds after launch, killing all seven crew members. The cause was a failure of the O-ring seals in one of the solid rocket boosters, and the launch happened on an unusually cold morning, with an air temperature of about 31°F. The night before, engineers had argued about whether cold temperatures made O-ring damage more likely, and the data they had looked at was from the 23 previous flights where the boosters were recovered. For each one we know the launch temperature and whether any O-ring showed damage (Dalal, Fowlkes and Hoadley [3]),

Temp (°F)	53	57	58	63	66	67	67	67	68	69	70	70
Damage	1	1	1	1	0	0	0	0	0	0	1	1
Temp (°F)	70	70	72	73	75	75	76	76	78	79	81	
Damage	0	0	0	0	1	0	0	0	0	0	0	

Just eyeballing it, all four of the coldest launches had damage, and only three of the nineteen warmer ones did. We'll fit model (8.2) with x the temperature and see whether that pattern holds up as statistical evidence. But first we need to know how to fit it.

8.3 The Generalized Linear Model

Before we fit anything, it's worth stepping back, because linear regression and logistic regression turn out to be two members of one big family of models. Both of them have three ingredients. There is a random component, the distribution of Y_i (normal in one case, Bernoulli in the other). There is a linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$. And there is a **link function** g that connects the mean $\mu_i = \mathbb{E}[Y_i]$ to the linear predictor through $g(\mu_i) = \eta_i$. For linear regression the link is the identity, $g(\mu) = \mu$, and for logistic regression it's the logit. A model with these three pieces is called a **generalized linear model** (GLM), and the distributions that make the theory work nicely form the exponential family.

8.3.1 The Exponential Family

A distribution belongs to the **exponential family** if its density (or mass function) can be written in the form

$$f(y;\theta,\phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right], \tag{8.5}$$

where θ is called the **natural parameter**, ϕ is a **dispersion parameter**, and a, b and c are known functions. The one rule that really matters is that the support of the distribution (the set of y where $f > 0$) is not allowed to depend on θ. Let's see how the Bernoulli fits in.

With $\mathbb{P}(Y = y) = p^y(1-p)^{1-y}$ for $y \in \{0, 1\}$,

$$\begin{aligned} p^y(1-p)^{1-y} &= \exp\left[y \log p + (1-y)\log(1-p)\right] \\ &= \exp\left[y \log\left(\frac{p}{1-p}\right) + \log(1-p)\right], \end{aligned} \tag{8.6}$$

so the natural parameter is $\theta = \log(p/(1-p))$, the log-odds! Solving for p gives $p = e^\theta/(1+e^\theta)$, so $\log(1-p) = -\log(1+e^\theta)$, and we can read off $b(\theta) = \log(1+e^\theta)$, $a(\phi) = 1$ and $c(y, \phi) = 0$.

Problem 2.

Is the normal distribution with mean μ and variance σ^2 part of the exponential family? If so, what are θ, $a(\phi)$, $b(\theta)$ and $c(y, \phi)$? If not, explain why not.

Problem 3.

Consider the density

$$f(y; \theta) = \frac{1}{\theta} e^{1-y/\theta}, \qquad y \geq \theta.$$

Does this distribution belong to the exponential family? If yes, find the natural parameter and the functions $a(\phi)$, $b(\theta)$ and $c(y, \phi)$. If not, explain why not.

8.3.2 Mean And Variance From $b(\theta)$

The function $b(\theta)$ is more useful than it looks, because it secretly knows the mean and variance of Y. Since f is a density, it integrates to one for every θ, so its derivative with respect to θ integrates to zero,

$$0 = \int \frac{\partial f}{\partial \theta}\, dy = \int \frac{y - b'(\theta)}{a(\phi)} f(y; \theta, \phi)\, dy = \frac{\mathbb{E}[Y] - b'(\theta)}{a(\phi)}, \tag{8.7}$$

and so $\mathbb{E}[Y] = b'(\theta)$. Differentiating one more time (Problem 4) gives $\mathrm{Var}[Y] = a(\phi)\, b''(\theta)$. For the Bernoulli, $b'(\theta) = e^\theta/(1+e^\theta) = p$ and $b''(\theta) = p(1-p)$, which are exactly the mean and variance we know and love. These two identities are the same ones we'll meet again in Part II, where they show up as the facts that the score has mean zero and variance equal to the Fisher information.

Problem 4.

(1) Differentiate $\int f(y;\theta,\phi)\,dy = 1$ twice with respect to θ to show that $\text{Var}[Y] = a(\phi)\,b''(\theta)$.
(2) Write the Poisson distribution in exponential family form and use $b(\theta)$ to find its mean and variance.

8.3.3 The Canonical Link

Since $\mu = b'(\theta)$, the natural parameter is a function of the mean, $\theta = (b')^{-1}(\mu)$. The **canonical link** is the choice $g = (b')^{-1}$, which makes the natural parameter equal to the linear predictor, $\theta_i = \mathbf{x}_i^T\boldsymbol{\beta}$. For the normal distribution the canonical link is the identity, which gives linear regression back. For the Bernoulli it's the logit, which is the real reason logistic regression uses the log-odds and not some other S-shaped curve. For the Poisson it's the log, which gives Poisson regression for count data. You don't have to use the canonical link (probit regression uses the normal CDF instead of the logistic), but it makes the likelihood especially nice, as we're about to see.

Problem 5.

Find the canonical link for the Poisson distribution, and write down the resulting model for $\mathbb{E}[Y_i]$ in terms of $\mathbf{x}_i^T\boldsymbol{\beta}$. How would you interpret e^{β_1} in this model?

8.4 Maximum Likelihood Estimation

In linear regression we found the estimates with least squares and got a closed form. For logistic regression we go straight to maximum likelihood. With independent $Y_i \sim \text{Bernoulli}(p_i)$ and $\text{logit}(p_i) = \eta_i = \mathbf{x}_i^T\boldsymbol{\beta}$ (where $\mathbf{x}_i$ includes a leading 1 for the intercept), the likelihood is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} p_i^{Y_i}(1-p_i)^{1-Y_i}, \tag{8.8}$$

and using the exponential family form, the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n}\left[Y_i\,\mathbf{x}_i^T\boldsymbol{\beta} - \log\left(1+e^{\mathbf{x}_i^T\boldsymbol{\beta}}\right)\right]. \tag{8.9}$$

Taking the gradient, and using the fact that the derivative of $\log(1+e^{\eta})$ is $\sigma(\eta) = p$, the **score** is

$$\nabla \ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} (Y_i - p_i)\, \mathbf{x}_i = \mathbf{X}^T(\mathbf{Y} - \mathbf{p}), \tag{8.10}$$

where $\mathbf{X}$ is the same design matrix as in multiple regression and $\mathbf{p}$ is the vector of p_i's. This is a really pretty formula. Compare it with the normal equations from the last chapter, which said $\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$. It's the same thing, "the residuals are orthogonal to the columns of $\mathbf{X}$," except that the fitted values are now $\sigma(\mathbf{X}\boldsymbol{\beta})$ instead of $\mathbf{X}\boldsymbol{\beta}$. And that's exactly the problem. Because the p_i depend on $\boldsymbol{\beta}$ through the sigmoid, the equation $\mathbf{X}^T(\mathbf{Y} - \mathbf{p}) = \mathbf{0}$ can't be solved in closed form, so we have to solve it numerically.

8.4.1 Newton's Method And Fisher Scoring

The standard way to do this is Newton's method. To find a root of the score, we expand it to first order around the current guess $\boldsymbol{\beta}^{(t)}$ and solve for where the linear approximation is zero, which gives the update

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \left[\nabla^2 \ell(\boldsymbol{\beta}^{(t)})\right]^{-1} \nabla \ell(\boldsymbol{\beta}^{(t)}). \tag{8.11}$$

For the Hessian, the derivative of p_i with respect to $\boldsymbol{\beta}$ is $p_i(1-p_i)\mathbf{x}_i$, so

$$\nabla^2 \ell(\boldsymbol{\beta}) = -\sum_{i=1}^{n} p_i(1-p_i)\, \mathbf{x}_i \mathbf{x}_i^T = -\mathbf{X}^T \mathbf{W} \mathbf{X}, \qquad \mathbf{W} = \operatorname{diag}\big(p_1(1-p_1), \ldots, p_n(1-p_n)\big). \tag{8.12}$$

Two nice things fall out of this. First, $\mathbf{X}^T\mathbf{W}\mathbf{X}$ is positive definite (as long as $\mathbf{X}$ has full rank), so the Hessian is negative definite and the log-likelihood is strictly concave. That means there's at most one maximum and Newton's method won't get stuck at the wrong one. Second, the Hessian doesn't involve the Y_i at all, so it equals its own expectation. The negative expected Hessian is the **Fisher information matrix**, $\mathbf{I}(\boldsymbol{\beta}) = \mathbf{X}^T\mathbf{W}\mathbf{X}$, and Newton's method with the Fisher information in place of the Hessian is called **Fisher scoring**. For a canonical link the two are the same algorithm. Putting it together, the update is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \left(\mathbf{X}^T\mathbf{W}\mathbf{X}\right)^{-1} \mathbf{X}^T(\mathbf{Y} - \mathbf{p}), \tag{8.13}$$

where $\mathbf{W}$ and $\mathbf{p}$ are evaluated at $\boldsymbol{\beta}^{(t)}$. With a little bit of algebra (Problem 6) this can be rewritten as a weighted least squares problem that gets re-solved at every step, which is why this algorithm is also called **iteratively reweighted least squares** (IRLS). It's what R's `glm` and Python's `statsmodels` do under the hood.

Problem 6.

Define the "working response" $\mathbf{z} = \mathbf{X}\boldsymbol{\beta}^{(t)} + \mathbf{W}^{-1}(\mathbf{Y} - \mathbf{p})$. Show that the Fisher scoring update can be written as

$$\boldsymbol{\beta}^{(t+1)} = \left(\mathbf{X}^T\mathbf{W}\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{W}\mathbf{z},$$

which is the weighted least squares solution for regressing $\mathbf{z}$ on $\mathbf{X}$ with weights $\mathbf{W}$. Why does it make sense that observations with p_i near 0 or 1 get small weight?

Let's fit the Challenger data.

```
import numpy as np
from scipy import stats

temp = np.array([53, 57, 58, 63, 66, 67, 67, 67, 68, 69, 70, 70,
                 70, 70, 72, 73, 75, 75, 76, 76, 78, 79, 81])
damage = np.array([1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 1,
                   0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0])
X = np.c_[np.ones(len(temp)), temp]
sigmoid = lambda z: 1 / (1 + np.exp(-z))

beta = np.zeros(2)
for t in range(8):
    p = sigmoid(X @ beta)
    W = p * (1 - p)
    info = X.T @ (W[:, None] * X)                # Fisher information X^T W X
    beta = beta + np.linalg.solve(info, X.T @ (damage - p))
    print(t, np.round(beta, 6))
```

```
0 [ 9.619048 -0.149524]
1 [13.655738 -0.211247]
2 [14.938289 -0.2306  ]
3 [15.042291 -0.232154]
4 [15.042902 -0.232163]
5 [15.042902 -0.232163]
6 [15.042902 -0.232163]
7 [15.042902 -0.232163]
```

Newton's method converges really fast. After about five steps the estimates stop changing in the sixth decimal place, and we get $\hat{\beta}_0 \approx 15.04$ and $\hat{\beta}_1 \approx -0.232$. The slope is negative, so warmer launches have lower odds of damage. Each extra degree multiplies the odds of

damage by $e^{-0.232} \approx 0.79$, or turned around, each degree *colder* multiplies the odds by about 1.26. The fitted curve is in Figure 8.2.

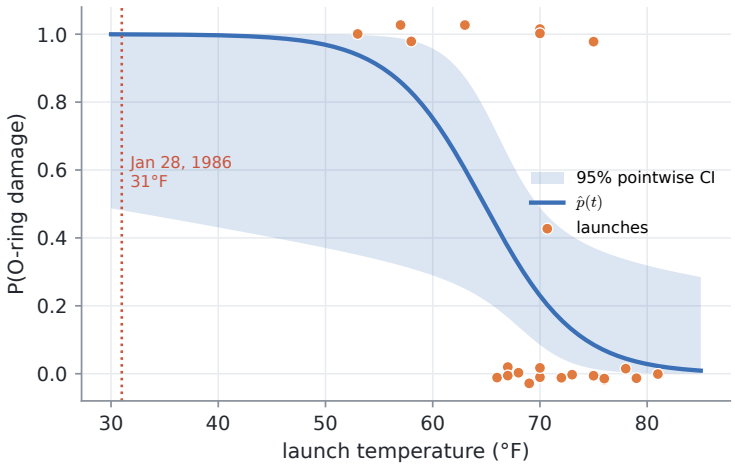


Figure 8.2: The fitted logistic curve for the Challenger data, with a 95% pointwise confidence band built on the log-odds scale. The points are jittered vertically so you can see the ties. The launch on January 28, 1986 was at 31°F, far colder than any previous flight.

8.5 Inference On The Coefficients

8.5.1 Wald Tests And Confidence Intervals

In linear regression we had exact t distributions for the coefficients. For logistic regression nothing is exact, but the MLE is asymptotically normal (we'll prove this carefully in Part II), with covariance matrix given by the inverse Fisher information,

$$\hat{\boldsymbol{\beta}} \overset{\text{approx}}{\sim} \mathcal{N}\left(\boldsymbol{\beta},\ \left(\mathbf{X}^T \mathbf{W} \mathbf{X}\right)^{-1}\right). \tag{8.14}$$

This is the logistic version of $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ from the last chapter. We estimate it by plugging $\hat{\boldsymbol{\beta}}$ into $\mathbf{W}$, and the standard error of $\hat{\beta}_j$ is the square root of the jth diagonal entry. Then, just like in Chapter 2, we have a **Wald test** of $H_0 : \beta_j = 0$,

$$Z = \frac{\hat{\beta}_j}{\text{s.e.}(\hat{\beta}_j)} \overset{\text{approx}}{\sim} \mathcal{N}(0,1), \tag{8.15}$$

and a $100(1-\alpha)\%$ Wald interval $\hat{\beta}_j \pm z_{\alpha/2}\,\text{s.e.}(\hat{\beta}_j)$. Notice it's a z and not a t. There is no separate σ^2 to estimate here, because for a Bernoulli the variance is completely determined by the mean. To get an interval for the odds ratio e^{β_j}, we just exponentiate the endpoints.

That's legitimate because e^x is increasing, so the event "β_j is in $[L, U]$" is the same event as "e^{β_j} is in $[e^L, e^U]$". The same trick works for the probability at a new x_0. We build the interval for the log-odds $\mathbf{x}_0^T\boldsymbol{\beta}$, whose estimate has variance $\mathbf{x}_0^T(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{x}_0$, and then push both endpoints through the sigmoid. That's how the band in Figure 8.2 was made, and it's why the band never leaves $[0, 1]$.

```
p = sigmoid(X @ beta)
cov = np.linalg.inv(X.T @ ((p * (1 - p))[:, None] * X))
se = np.sqrt(np.diag(cov))
z = beta / se
print("se:", np.round(se, 4), " z:", np.round(z, 3), " p-value:", np.round(2 * stats.norm.
    ↪ sf(np.abs(z)), 4))

lo, hi = beta[1] - 1.96 * se[1], beta[1] + 1.96 * se[1]
print(f"odds ratio per degree: {np.exp(beta[1]):.3f}, 95% CI [{np.exp(lo):.3f}, {np.exp(
    ↪ hi):.3f}]")

x0 = np.array([1, 31])                          # the morning of the launch
eta0, se0 = x0 @ beta, np.sqrt(x0 @ cov @ x0)
print(f"P(damage at 31F) = {sigmoid(eta0):.4f}, "
      f"95% CI [{sigmoid(eta0 - 1.96*se0):.4f}, {sigmoid(eta0 + 1.96*se0):.4f}]")
```

```
se: [7.3786 0.1082]  z: [ 2.039 -2.145]  p-value: [0.0415 0.032 ]
odds ratio per degree: 0.793, 95% CI [0.641, 0.980]
P(damage at 31F) = 0.9996, 95% CI [0.4816, 1.0000]
```

The Wald test for temperature gives $|Z| \approx 2.1$ and a p value around 0.03, so at the 5% level there is evidence that colder launches are more likely to have O-ring damage. The interval for the odds ratio per degree lies entirely below 1. And the estimated probability of damage at 31°F is about 0.9996. Before you take that number too seriously, look at how wide its interval is and remember the warning about extrapolation from the last chapter. The coldest launch in the data was 53°F, so the model is being asked about a temperature more than 20 degrees outside anything it has seen. But every reasonable model fit to these data says the same qualitative thing, which is that 31°F was way out in the region where damage was more likely than not. The 23 data points were enough to see that, if someone had plotted all of them (and not just the flights with damage, which is what was looked at the night before).

8.5.2 Deviance And The Likelihood Ratio Test

The Wald test is easy, but it's known to behave badly in small samples and when the estimates are large. The better tool is the likelihood ratio test, and to set it up we need the idea of deviance. The best a model could possibly do is fit every observation exactly, with a separate parameter for each data point. This is called the **saturated** model, and it sets $\hat{\mu}_i = Y_i$. The **deviance** of a fitted model measures how far its log-likelihood falls short of the saturated one,

$$D = 2\left[\ell_{\text{sat}} - \ell(\hat{\boldsymbol{\beta}})\right]. \tag{8.16}$$

For a normal linear model with known σ^2, the deviance is just SSE/σ^2, so the deviance is the GLM version of the residual sum of squares. For binary data, the saturated model puts $\hat{p}_i = Y_i \in \{0, 1\}$, so every term in its log-likelihood is $\log 1 = 0$, and the deviance is simply $D = -2\ell(\hat{\boldsymbol{\beta}})$. The **null deviance** D_0 is the deviance of the intercept-only model, which predicts $\hat{p}_i = \bar{Y}$ for everybody (Problem 7).

Now suppose we want to test whether a set of q coefficients is all zero. We fit the model with them (the "full" model) and without them (the "reduced" model). The difference in deviances is

$$D_{\text{reduced}} - D_{\text{full}} = 2\left[\ell(\hat{\boldsymbol{\beta}}_{\text{full}}) - \ell(\hat{\boldsymbol{\beta}}_{\text{reduced}})\right] = -2\log\Lambda, \tag{8.17}$$

where Λ is the ratio of the maximized likelihoods, and a famous theorem of Wilks (which we'll prove in Part II) says that under the null,

$$D_{\text{reduced}} - D_{\text{full}} \overset{\text{approx}}{\sim} \chi^2_q. \tag{8.18}$$

So we reject when the drop in deviance is bigger than $\chi^2_{\alpha,q}$. This plays the same role as the F test in linear regression. Comparing the null model with the model that has temperature in it is a test with $q = 1$,

```python
def loglik(beta, X, y):
    eta = X @ beta
    return np.sum(y * eta - np.log1p(np.exp(eta)))

D = -2 * loglik(beta, X, damage)                          # residual deviance
ybar = damage.mean()
D0 = -2 * np.sum(damage * np.log(ybar) + (1 - damage) * np.log(1 - ybar))
print(f"null deviance = {D0:.3f}, residual deviance = {D:.3f}")
print(f"LR statistic = {D0 - D:.3f}, p-value = {stats.chi2.sf(D0 - D, 1):.4f}")
```

```
null deviance = 28.267, residual deviance = 20.315
LR statistic = 7.952, p-value = 0.0048
```

The likelihood ratio test gives a smaller p value than the Wald test, which is typical. Both lead to the same conclusion at the 5% level here.

One thing that trips people up is that for binary data, the residual deviance by itself is *not* a good goodness-of-fit statistic. You might hope to compare D with a χ^2_{n-p-1}, the way you'd compare a sum of squares with its degrees of freedom, but with 0/1 responses the asymptotics that justify that don't kick in (the saturated model has as many parameters as data points, so it never settles down). Differences of deviances between nested models are fine, which is all the LRT uses. If the data are grouped, with several trials at each value of x, then D does have an approximate χ^2 distribution and can be used to check the fit, which is the same idea as Pearson's χ^2 test from the categorical data chapter.

Problem 7.

For a general exponential family, write the log-likelihood in terms of the means μ_i. Show that the MLE of the means in the saturated model is $\hat{\mu}_i = Y_i$, and that the MLE in the null model (one common mean for everyone) is $\hat{\mu} = \bar{Y}$.

Problem 8.

Suppose your responses are Poisson counts and you use the canonical link. Show that the residual deviance is

$$D = 2\sum_{i=1}^{n}\left[Y_i \log\left(\frac{Y_i}{\hat{\mu}_i}\right) - (Y_i - \hat{\mu}_i)\right],$$

with the convention $0 \log 0 = 0$, and find the null deviance. If the model includes an intercept, the second term sums to zero. Why?

Problem 9.

Remove the four coldest launches from the Challenger data and refit the model. What happens to $\hat{\beta}_1$, its standard error, and the Wald and likelihood ratio p values? What does this tell you about where the evidence in this dataset comes from?

8.6 More Than Two Categories

Everything above extends to more predictors without any changes, since we already wrote it in matrix form. Each e^{β_j} is then the odds ratio for a one-unit change in x_j holding the other predictors fixed, just like in multiple linear regression.

When the response has $K > 2$ unordered categories, the usual approach is to pick one category as the baseline (say category K) and fit a separate logistic regression for the log-odds of each other category against it,

$$\log\left(\frac{\mathbb{P}(Y = k \mid \mathbf{x})}{\mathbb{P}(Y = K \mid \mathbf{x})}\right) = \mathbf{x}^T\boldsymbol{\beta}_k, \qquad k = 1, \ldots, K-1. \tag{8.19}$$

Solving for the probabilities gives

$$\mathbb{P}(Y = k \mid \mathbf{x}) = \frac{e^{\mathbf{x}^T\boldsymbol{\beta}_k}}{1 + \sum_{j=1}^{K-1} e^{\mathbf{x}^T\boldsymbol{\beta}_j}}, \tag{8.20}$$

with $\boldsymbol{\beta}_K = \mathbf{0}$, which is called the **baseline-category logit** model (in machine learning the same model goes by *softmax regression*). It's fit by maximum likelihood with Newton's method, exactly like before, just with a bigger parameter vector. The $K = 2$ case is the logistic regression we've been doing the whole chapter.

Problem 10.

Show that in the baseline-category logit model, the log-odds of category j versus category k (neither of them the baseline) is $\mathbf{x}^T(\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)$, so the choice of baseline doesn't change the fitted probabilities. Then check that $K = 2$ reduces to (8.2).

Part II

Mathematical Statistics

9

Moments And The MGF

"Gotta live in the moment"

– Hieu

In the first half of the book, we used a lot of results without really proving them. We said that the sample mean is approximately normal, that $(n-1)S^2/\sigma^2$ is χ^2_{n-1}, that maximum likelihood estimators are good, and that the likelihood ratio statistic is approximately χ^2. The second half of the book is about why all of those things are true. Before we can get started talking about anything like that though, we need some machinery, and the first and most useful piece of machinery is the moment generating function. It's going to let us identify distributions, find the distribution of sums, and (in Chapter 11) prove the central limit theorem.

Throughout Part II, you are expected to know probability theory at the level of a first course. What this means is that you should know multivariable calculus, and you should be comfortable with us saying things like "X is a random variable with density $f(x)$." For a discrete random variable $f(x) = \mathbb{P}(X = x)$ is the probability mass function, and for a continuous one $f(x) = \frac{d}{dx}F(x)$ is the probability density function, where $F(x) = \mathbb{P}(X \leq x)$ is the **cumulative distribution function**. (It's not called the cumulative *density* function, because discrete random variables have a CDF too.) Two series will come up over and over

again, so it's worth having them in front of us,

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \quad |x| < 1 \quad \text{(geometric series)}, \qquad \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x \quad \text{(Taylor series of } e^x\text{)}. \tag{9.1}$$

We'll also write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

9.1 The Moment List

We can start with the definition of the kth (power) **moment** of a random variable,

$$m_k = \mathbb{E}\left[X^k\right] = \begin{cases} \sum_x x^k f(x) & X \text{ discrete} \\ \int_{\mathbb{R}} x^k f(x)\, dx & X \text{ continuous} \end{cases} \tag{9.2}$$

The most trivial moment is the zeroth one, $m_0 = \mathbb{E}[X^0] = \mathbb{E}[1] = 1$, which is true for every random variable. The first moment is the mean, $m_1 = \mathbb{E}[X] = \mu$. And as we remember, the variance can be written as

$$\mathrm{Var}[X] = \underbrace{\mathbb{E}\left[X^2\right]}_{m_2} - \underbrace{\mathbb{E}[X]^2}_{m_1^2}, \tag{9.3}$$

so the variance is related to the second moment. Higher moments exist too (as long as the sums or integrals converge), and they do matter to an extent. The third moment is related to the skewness of a distribution and the fourth one to its peakedness, or *kurtosis*. We can go one step further and collect all of the moments into a single object, the **moment list**

$$\mathcal{M}_X = \{m_0, m_1, m_2, \dots\}, \tag{9.4}$$

which can often be written as one formula in k. The reason we bring this up is because of the really cool fact that (under mild conditions, which we'll make precise in a second) the moment list uniquely determines the distribution. That is, if X and Y are random variables with $\mathcal{M}_X = \mathcal{M}_Y$, then X and Y have the same distribution. But this does NOT mean that X and Y are the same random variable. Take our good friends Alice and Bob. Suppose you give Alice a fair coin that she can toss 10 times, and likewise for Bob. The number of heads for both of them is Binomial$(10, 1/2)$, so they have the same distribution. But this does NOT mean that Alice will get the same outcome as Bob every single time.

Let's do an actual example. Suppose that $X \sim \text{Uniform}(a, b)$, so that $f(x) = \frac{1}{b-a}$ on $[a, b]$.

To find the moment list, we just do the integral for an arbitrary k,

$$\begin{aligned} m_k = \mathbb{E}\left[X^k\right] &= \int_a^b x^k \frac{1}{b-a}\,dx \\ &= \frac{x^{k+1}}{(k+1)(b-a)}\Bigg|_a^b \\ &= \frac{b^{k+1}-a^{k+1}}{(k+1)(b-a)}. \end{aligned} \tag{9.5}$$

We can go even further with this result. Because of the uniqueness, if you were handed the moment list $m_k = \frac{3^{k+1}-1}{2(k+1)}$, it should be immediately obvious that X is a Uniform$(1,3)$ random variable.

Problem 1.

Show that the moment list of the uniform distribution in (9.5) can be written as

$$m_k = \frac{1}{k+1}\sum_{j=0}^{k} a^j b^{k-j}.$$

Let's try another one. Suppose $X \sim \text{Exp}(\lambda)$, with density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, where λ has the interpretation of a **failure rate**. (You'll also see it parametrized by the mean $\beta = 1/\lambda$.) What is the 88th moment? Doing integration by parts 88 times would be cumbersome, to say the least, so it would be nice to have a formula for the whole moment list. The integral we need is

$$m_k = \lambda \int_0^\infty x^k e^{-\lambda x}\,dx, \tag{9.6}$$

and doing it directly requires pulling the Gamma function $\Gamma(x)$ out of thin air. Let's see if there's a lazier way.

9.2 The Moment Generating Function

Here's the cheat. Let's take the expectation of e^{tX} for a real number t. Using the Taylor series of e^x in (9.1) and the linearity of expectation (and assuming we can swap the sum and the expectation, which is fine whenever the result is finite near $t = 0$),

$$\mathbb{E}\left[e^{tX}\right] = \mathbb{E}\left[\sum_{k=0}^{\infty}\frac{(tX)^k}{k!}\right] = \sum_{k=0}^{\infty}\mathbb{E}\left[X^k\right]\frac{t^k}{k!} = \sum_{k=0}^{\infty} m_k \frac{t^k}{k!}. \tag{9.7}$$

So the kth moment is encoded as the coefficient of the kth term in a power series in t. This is worth giving a name.

Definition. The **moment generating function** (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}\left[e^{tX}\right], \tag{9.8}$$

defined for all t where the expectation is finite. We say X *has* an MGF if $M_X(t)$ is finite on some open interval $(-h, h)$ around $t = 0$.

Let's compute the MGF of the exponential distribution to see how this works,

$$\begin{aligned} M_X(t) = \mathbb{E}\left[e^{tX}\right] &= \lambda \int_0^\infty e^{tx} e^{-\lambda x}\, dx \\ &= \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx \\ &= \frac{\lambda}{\lambda - t}, \qquad t < \lambda. \end{aligned} \tag{9.9}$$

A quick note. We have to assume $t < \lambda$, because for $t \geq \lambda$ the integrand doesn't decay and the integral blows up to infinity. But since $\lambda > 0$, the MGF exists on an interval around $t = 0$, which is the only part we'll need.

Problem 2.

Let $X \sim \text{Uniform}(a, b)$. Find its MGF (be careful at $t = 0$), check that it is continuous at $t = 0$, and verify its zeroth, first and second moments.

But why do we care? What is so special about the MGF? The first thing to notice is that every MGF satisfies $M_X(0) = \mathbb{E}[e^0] = 1$. The second, and most important, thing is the **moment generating property**. Equation (9.7) says that m_k is the kth Taylor coefficient of M_X at $t = 0$ (times $k!$), and we know how to get Taylor coefficients, by taking derivatives. So

$$m_k = \frac{d^k}{dt^k} M_X(t)\bigg|_{t=0} = M_X^{(k)}(0), \qquad k \in \mathbb{N}_0. \tag{9.10}$$

Let's do a quick calculation. Suppose $M_X(t) = e^{t^2}$. If we wanted the variance, we would need the first two moments, but with the MGF all we need is two derivatives,

$$M_X'(t) = 2te^{t^2} \;\Rightarrow\; M_X'(0) = 0, \qquad M_X''(t) = \left(2 + 4t^2\right) e^{t^2} \;\Rightarrow\; M_X''(0) = 2, \tag{9.11}$$

so by (9.3),

$$\mathrm{Var}[X] = M_X''(0) - M_X'(0)^2 = 2. \tag{9.12}$$

(We'll see shortly that this is the MGF of a $\mathcal{N}(0, 2)$, so that's the right answer.)

9.2.1 Properties Of The MGF

Let's go back to the exponential distribution and finally answer the question of the 88th moment. Taking 88 derivatives of $\lambda/(\lambda - t)$ is better than integrating by parts 88 times, but not by much. Instead, divide the top and bottom by λ and use the geometric series backwards (for $|t| < \lambda$),

$$M_X(t) = \frac{1}{1 - t/\lambda} = \sum_{k=0}^{\infty} \left(\frac{t}{\lambda}\right)^k. \tag{9.13}$$

A power series is unique, so we can match this term by term with (9.7),

$$\frac{m_k}{k!} = \frac{1}{\lambda^k} \quad \Longrightarrow \quad m_k = \frac{k!}{\lambda^k}, \qquad k \in \mathbb{N}_0. \tag{9.14}$$

So the 88th moment is $88!/\lambda^{88}$, and we got the entire moment list without doing a single integral by parts. This trick of expanding the MGF and matching coefficients is often the fastest way to get moments.

We said before that the moment list (usually) determines the distribution, and since the MGF packages up the whole moment list, the MGF should determine the distribution too. This is true, and it's the property we'll use the most. We'll call it MGF uniqueness.

Theorem (MGF uniqueness). Suppose X and Y have MGFs that are finite on an interval $(-h, h)$ with $h > 0$. If $M_X(t) = M_Y(t)$ for all $t \in (-h, h)$, then X and Y have the same distribution.

The condition that the MGF exists near zero is not just a technicality. There are distributions (the lognormal is the famous one) that have every moment finite but no MGF, and for those the moment list does *not* pin down the distribution. That's the "mild condition" we promised. The proof of the theorem uses the theory of Laplace transforms, and we won't need it. The other property we'll use constantly is how MGFs handle sums.

Property. If X and Y are independent, then

$$M_{X+Y}(t) = \mathbb{E}\left[e^{tX}e^{tY}\right] = \mathbb{E}\left[e^{tX}\right]\mathbb{E}\left[e^{tY}\right] = M_X(t)M_Y(t), \tag{9.15}$$

where the middle equality is the fact that the expectation of a product of independent random variables is the product of the expectations. The MGF of a sum of independent

random variables is the product of their MGFs. Compare this with trying to find the density of $X + Y$ directly, which requires a convolution integral. For example, if $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$ are independent, then

$$M_{X+Y}(t) = \frac{\lambda}{\lambda - t} \cdot \frac{\mu}{\mu - t}, \qquad t < \min\{\lambda, \mu\}, \tag{9.16}$$

where we need t to be less than both rates for both MGFs to be finite. One last property is how MGFs handle shifting and scaling. If $Y = a + bX$, then

$$M_{a+bX}(t) = \mathbb{E}\left[e^{ta}e^{(bt)X}\right] = e^{at}M_X(bt). \tag{9.17}$$

9.3 MGFs Of The Famous Distributions

9.3.1 Discrete Distributions

Let's start with the binomial, $\mathbb{P}(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $k = 0, \ldots, n$. Plugging into the definition,

$$M_X(t) = \sum_{k=0}^{n} e^{tk}\binom{n}{k}p^k(1-p)^{n-k} = \sum_{k=0}^{n}\binom{n}{k}\left(pe^t\right)^k(1-p)^{n-k}, \tag{9.18}$$

and the trick is to use the binomial theorem for the binomial distribution (crazy, right?). This gives

$$M_X(t) = \left(pe^t + 1 - p\right)^n = \left(1 + p\left(e^t - 1\right)\right)^n, \qquad t \in \mathbb{R}. \tag{9.19}$$

We could have also gotten this for free. A binomial is the sum of n i.i.d. Bernoulli(p) random variables, each with MGF $\mathbb{E}[e^{tX_i}] = (1-p) + pe^t$, so by (9.15) the MGF is that to the nth power.

Next is the geometric distribution, which counts the number of trials up to and including the first success, $\mathbb{P}(X = k) = p(1-p)^{k-1}$ for $k = 1, 2, \ldots$ where p is the success probability. The calculation is

$$M_X(t) = \sum_{k=1}^{\infty} e^{tk}p(1-p)^{k-1} = pe^t\sum_{k=1}^{\infty}\left[(1-p)e^t\right]^{k-1} = \frac{pe^t}{1-(1-p)e^t}, \tag{9.20}$$

where we used the geometric series, which only converges if $(1-p)e^t < 1$, so we need $t < -\log(1-p)$. The negative binomial counts the number of trials up to and including the rth success. Remembering its pmf might be cumbersome, but if we remember that it's the

sum of r i.i.d. geometric random variables, then (9.15) gives the MGF for free,

$$M_X(t) = \left(\frac{pe^t}{1-(1-p)e^t}\right)^r, \qquad t < -\log(1-p). \tag{9.21}$$

What about the last and most important discrete random variable? Obviously it's the Poisson. It counts the number of rare events in a fixed time interval and arises as a limit of the binomial (we'll prove this in Chapter 11). With $\mathbb{P}(X=k) = e^{-\lambda}\lambda^k/k!$,

$$\begin{aligned} M_X(t) &= \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} \\ &= e^{-\lambda}e^{\lambda e^t} = e^{\lambda\left(e^t-1\right)}, \qquad t \in \mathbb{R}. \end{aligned} \tag{9.22}$$

And here's a side note that shows off the uniqueness theorem. If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent, then

$$M_{X+Y}(t) = e^{\lambda(e^t-1)}e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)}, \tag{9.23}$$

which is the MGF of a Poisson$(\lambda+\mu)$. So by MGF uniqueness, the sum of independent Poissons is Poisson, and we didn't have to do a single convolution.

9.3.2 Continuous Distributions

We've already done the exponential, and the gamma comes for free in the same way the negative binomial did. A Gamma(α,λ) with integer shape α is the sum of α i.i.d. Exp(λ) random variables, so

$$M_X(t) = \left(\frac{\lambda}{\lambda-t}\right)^{\alpha}, \qquad t < \lambda, \tag{9.24}$$

and doing the integral directly shows this formula holds for any real $\alpha > 0$. That leaves the most important distribution of quite possibly all time, the normal. Let $Z \sim \mathcal{N}(0,1)$. Its MGF is

$$M_Z(t) = \int_{-\infty}^{\infty} e^{tz}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}\,dz, \tag{9.25}$$

and we can *complete* this calculation by *completing* the square (ba dum tss),

$$\begin{aligned} M_Z(t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\exp\left[-\frac{1}{2}\left(z^2-2tz+t^2\right)+\frac{t^2}{2}\right]dz \\ &= e^{t^2/2}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-t)^2}\,dz \\ &= e^{t^2/2}, \qquad t \in \mathbb{R}, \end{aligned} \tag{9.26}$$

since the last integral is the integral of a $\mathcal{N}(t,1)$ density, which is 1. For a general normal, we write it in terms of its standardization. If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $X = \mu + \sigma Z$, and by (9.17),

$$M_X(t) = e^{\mu t} M_Z(\sigma t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right), \qquad t \in \mathbb{R}. \tag{9.27}$$

A consequence of how well behaved the normal is (and of MGF uniqueness) is that if $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then

$$M_{X_1+X_2}(t) = \exp\left((\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2\right), \tag{9.28}$$

so $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. Combined with (9.27), this means that any linear combination of independent normals is normal, which is a fact we used over and over in Part I.

Problem 3.

Use the MGFs of this section to find the third moment $\mathbb{E}[X^3]$ of the binomial, Poisson, exponential and normal distributions. Then use (9.27) to find the fourth moment of $X \sim \mathcal{N}(\mu, \sigma^2)$. (For the normal, expanding the exponential in a series is faster than taking derivatives.)

9.3.3 The χ^2 Distribution

MGFs also make finding the distribution of a transformation easier. Let $Z \sim \mathcal{N}(0,1)$ and define $X = Z^2$. Then

$$M_{Z^2}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz^2} e^{-z^2/2}\, dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(1-2t)z^2}\, dz. \tag{9.29}$$

For $t < 1/2$ we can do the u-substitution $u = z\sqrt{1-2t}$, $du = \sqrt{1-2t}\, dz$, which turns this into

$$M_{Z^2}(t) = \frac{1}{\sqrt{1-2t}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du = \frac{1}{\sqrt{1-2t}}, \qquad t < \frac{1}{2}. \tag{9.30}$$

Back in Chapter 2 we defined the χ^2 distribution with ν degrees of freedom as the distribution of $\sum_{i=1}^{\nu} Z_i^2$ for i.i.d. standard normals, so by (9.15),

$$M_{\chi^2_\nu}(t) = (1-2t)^{-\nu/2}, \qquad t < \frac{1}{2}. \tag{9.31}$$

Compare this with the gamma MGF. With $\alpha = \nu/2$ and $\lambda = 1/2$, $\left(\frac{1/2}{1/2-t}\right)^{\nu/2} = (1-2t)^{-\nu/2}$, so a χ^2_ν is exactly a Gamma$(\nu/2, 1/2)$. We work with the MGF here because the χ^2 density is rather ugly, and if what we want are moments, there's no reason to integrate an ugly density when we could differentiate a nice function.

Problem 4.

(1) Use the MGF to show that $\mathbb{E}[\chi^2_\nu] = \nu$ and $\text{Var}[\chi^2_\nu] = 2\nu$.
(2) Show that if $X \sim \chi^2_\nu$ and $Y \sim \chi^2_\mu$ are independent, then $X + Y \sim \chi^2_{\nu+\mu}$.
(3) Show that for $n \in \mathbb{N}$, the χ^2_{2n} distribution is the same as the distribution of the sum of n i.i.d. Exp$(1/2)$ random variables.

9.4 Standardization And The Sample Mean

As we know, not all random variables have the same mean and variance. But golly wouldn't it be nice if they did? If they all did, then proving a result for one random variable would prove it for all of them. The way we make this happen is **standardization**, shifting and scaling a random variable so that its mean is 0 and its variance is 1. If X has mean μ and finite nonzero variance σ^2, its standardization is

$$Z = \frac{X - \mu}{\sigma}, \tag{9.32}$$

and it's easy to see that $\mathbb{E}[Z] = 0$ and $\text{Var}[Z] = 1$. Since this is just a shift and a scale, it doesn't change the shape of the distribution. A standardized Bernoulli still only takes two values, and a standardized exponential is still skewed to the right. Going backwards, we can always write $X = \mu + \sigma Z$, which is how we got the general normal MGF.

To see why we care about this, suppose $X_1, X_2, \ldots, X_n$ are i.i.d. with mean μ and variance σ^2, and define the **sample mean**

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{9.33}$$

We'll write $\bar{X}_n$ when we want to remember that it depends on n, and just $\bar{X}$ when it doesn't matter. Using the linearity of expectation and the i.i.d. property,

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mu, \tag{9.34}$$

and since the variance of a sum of independent random variables is the sum of the variances,

$$\mathrm{Var}[\bar{X}_n] = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}[X_i] = \frac{\sigma^2}{n}. \tag{9.35}$$

Notice that we made no statement about what distribution the X_i have, and yet the sample mean always has expectation μ and variance σ^2/n. As $n \to \infty$ the variance goes to zero, which suggests that the sample mean settles down onto the true mean. That's the **law of large numbers**, which we'll prove in the next chapter. Its standardization is

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{X_i - \mu}{\sigma}, \tag{9.36}$$

which has mean 0 and variance 1 for every n. The central limit theorem says that Z_n is approximately $\mathcal{N}(0,1)$ for large n, no matter what the distribution of the X_i is. In Part I we took this on faith. To prove it, we first need to say what "approximately" means for random variables, which is what the next two chapters are about. But we can already do the case where the answer is exact.

Problem 5.

Suppose $X_1, \ldots, X_n$ are i.i.d. $\mathcal{N}(\mu, \sigma^2)$. Use MGFs to prove that the standardized sample mean Z_n in (9.36) is *exactly* $\mathcal{N}(0,1)$ for every n. Then show that if the X_i are i.i.d. Exp(λ) instead, Z_n is never exactly normal. (Hint: what is the distribution of $\sum X_i$?)

Problem 6.

The **cumulant generating function** is $K_X(t) = \log M_X(t)$. Show that $K_X'(0) = \mu$ and $K_X''(0) = \sigma^2$. Then compute K_X for the Poisson and the normal, and explain why the cumulant generating function of a sum of independent random variables is the sum of their cumulant generating functions.

10

Inequalities And The Law Of Large Numbers

"Life is unfair"

– Probably your parents at some point

The study of probability inequalities is an important one, and the reason it's important is because more often than not, it's not possible to compute a probability exactly. This is especially true when we talk about (potentially infinite) sequences of random variables, like the sample mean $\bar{X}_n$ as n grows. What *is* usually possible is to bound the probability. In the same way that in real analysis it's easier to talk about convergence of a sequence of numbers using inequalities like the triangle inequality, talking about convergence of random variables is easier using probability inequalities. This is where we will begin, and by the end of the chapter we'll have used them to prove the law of large numbers.

10.1 Markov And Chebyshev

10.1.1 Markov's Inequality

We want to relate the probability of an event to an average. Let $X \geq 0$ be a nonnegative random variable, and let g be a nonnegative function that is increasing on $[0, \infty)$. Then for

every $t > 0$ with $g(t) > 0$, **Markov's inequality** says

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[g(X)]}{g(t)}. \tag{10.1}$$

The left hand side has the interpretation of a survival probability (the probability that a lifetime X makes it past time t). To prove it, we use the definition of the expectation and the fact that the integrand is nonnegative, so throwing away part of the domain can only make the integral smaller,

$$\begin{aligned} \mathbb{E}[g(X)] = \int_0^\infty g(x)f(x)\,dx &\geq \int_t^\infty g(x)f(x)\,dx \\ &\geq \int_t^\infty g(t)f(x)\,dx \\ &= g(t)\,\mathbb{P}(X \geq t), \end{aligned} \tag{10.2}$$

where the second inequality is where we need g to be increasing, since then $g(x) \geq g(t)$ for every $x \geq t$. Dividing through by $g(t)$ gives (10.1). The most common version just takes $g(x) = x$, which gives $\mathbb{P}(X \geq t) \leq \mathbb{E}[X]/t$.

> Problem 1.
>
> We just did the continuous case. Now prove Markov's inequality for a discrete random variable. Then show that it can be written without any sums or integrals as $g(t)\mathbf{1}\{X \geq t\} \leq g(X)$, and take expectations of both sides.

Let's do a cute example. Suppose the MGF of $X \geq 0$ is finite at some point $s > 0$. Then with $g(x) = e^{sx}$, for every $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}\left[e^{sX}\right]}{e^{st}} = M_X(s)\,e^{-st}. \tag{10.3}$$

Since $M_X(s)$ is just a constant, this says that the survival probability dies off *exponentially fast* in t. So having an MGF is a strong statement about the tails of a distribution. It forces them to be at least as thin as an exponential, which is another way of seeing why heavy-tailed things like the lognormal don't have one. This trick of applying Markov to e^{sX} and then picking the best s is called a **Chernoff bound**.

Problem 2.

Let $Z \sim \mathcal{N}(0,1)$.

(1) Apply Markov's inequality to e^{sZ} and minimize over $s > 0$ to show that $\mathbb{P}(Z \geq t) \leq e^{-t^2/2}$ for $t > 0$.

(2) Prove that $\mathbb{P}(|Z| \geq t) \leq e^{-t^2/3}$ for all $t > 0$. (Hint: the bound from part 1 handles large t. For small t, compare with Chebyshev or just with 1.)

10.1.2 Chebyshev's Inequality

Another cute application of Markov's inequality is one that most of you have seen before if you took undergraduate statistics. Suppose X has mean μ and finite variance σ^2. The random variable $|X - \mu|$ is nonnegative, so we can apply Markov to it with $g(x) = x^2$, which is increasing on $[0, \infty)$,

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\mathbb{E}\left[|X - \mu|^2\right]}{t^2} = \frac{\sigma^2}{t^2}. \tag{10.4}$$

We squared because it lets us drop the absolute value, and $\mathbb{E}[(X - \mu)^2]$ is by definition the variance. This is **Chebyshev's inequality**. If we take t to be a multiple of the standard deviation, $t \to t\sigma$, it becomes the form most people know,

$$\mathbb{P}\left(\left|\frac{X - \mu}{\sigma}\right| \geq t\right) \leq \frac{1}{t^2}, \qquad \text{or equivalently} \qquad \mathbb{P}\left(\left|\frac{X - \mu}{\sigma}\right| < t\right) \geq 1 - \frac{1}{t^2}. \tag{10.5}$$

In words, the probability of being more than t standard deviations away from the mean is *at most* $1/t^2$, so the probability of being within t standard deviations is *at least* $1 - 1/t^2$. This lets us make statements about *all* random variables with a finite variance. For example, suppose we have a midterm. At least 75% of the class will get a score within 2 standard deviations of the mean ($t = 2$). That's an absolute fact regardless of the distribution of the scores. It has nothing to do with normal distributions, the empirical rule, hoopla hoopla. The price we pay for that generality is that the bound is usually loose. For a normal, the actual probability of being within 2 standard deviations is about 95%, not 75%.

10.2 Jensen's Inequality

Suppose g is a convex function (concave up, like x^2). If we draw the tangent line to g at any point, the tangent line always stays below the function. And if we draw a chord between two points a and b on the graph, the chord always stays above the function (Figure 10.1). The

chord picture already says something interesting. The midpoint of the chord is at height $\frac{g(a)+g(b)}{2}$, and it sits above the function at the midpoint, so

$$g\left(\frac{a+b}{2}\right) \leq \frac{g(a)+g(b)}{2}. \tag{10.6}$$

If we think of a and b as the two equally likely values of a random variable X, the left side is $g(\mathbb{E}[X])$ and the right side is $\mathbb{E}[g(X)]$. So we conjecture that $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$ for any convex g, and it turns out that's true.

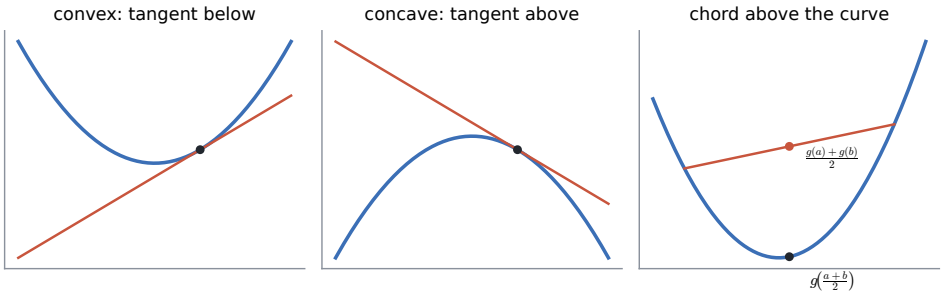


Figure 10.1: Left, a convex function always lies above its tangent lines. Right, the chord between $(a, g(a))$ and $(b, g(b))$ lies above the function, so g at the average is below the average of g.

Theorem (Jensen's inequality). If g is convex and X has a finite mean, then

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]. \tag{10.7}$$

If g is strictly convex, the inequality is strict unless X is a constant. For concave g (like $\log$ or $\sqrt{\cdot}$), the inequality flips.

To prove it, let $\mu = \mathbb{E}[X]$ and use the tangent line at μ. Convexity means the function is above the tangent line everywhere, so (for differentiable g)

$$g(X) \geq g(\mu) + g'(\mu)(X - \mu). \tag{10.8}$$

This is an inequality between two random variables, and expectation preserves inequalities, since if $U \leq V$ then $V - U \geq 0$, so $\mathbb{E}[V] - \mathbb{E}[U] = \mathbb{E}[V - U] \geq 0$. So we can slap expectations on both sides,

$$\begin{aligned} \mathbb{E}[g(X)] &\geq g(\mu) + g'(\mu)\,\mathbb{E}[X - \mu] \\ &= g(\mu) + g'(\mu) \cdot 0 \\ &= g(\mathbb{E}[X]), \end{aligned} \tag{10.9}$$

and we have our proof. For the strict version, if g is strictly convex then $g(X) > g(\mu) + g'(\mu)(X - \mu)$ whenever $X \neq \mu$, and if $\mathbb{P}(U < V) > 0$ for $U \leq V$, then $\mathbb{E}[U] < \mathbb{E}[V]$ (Problem 3).

Jensen's inequality is going to show up a lot. Here are two quick consequences. With $g(x) = x^2$, it says $\mathbb{E}[X]^2 \leq \mathbb{E}[X^2]$, which is just the statement that the variance is nonnegative. And with $g(x) = 1/x$, which is convex on $(0, \infty)$, it says $\mathbb{E}[1/X] \geq 1/\mathbb{E}[X]$ for positive X. That second one is going to come back to bite us in Chapter 12, where it tells us that the natural estimator of an exponential rate is biased.

Problem 3.

Let $U \leq V$ be random variables with finite means. Prove that if $\mathbb{P}(U < V) > 0$ then $\mathbb{E}[U] < \mathbb{E}[V]$. Do the discrete case first, then the continuous case (which is a bit more difficult).

Problem 4.

Let $x_1, \ldots, x_n$ be positive numbers. The arithmetic, geometric and harmonic means are

$$\text{AM} = \frac{1}{n}\sum_i x_i, \qquad \text{GM} = \left(\prod_i x_i\right)^{1/n}, \qquad \text{HM} = \left(\frac{1}{n}\sum_i \frac{1}{x_i}\right)^{-1}.$$

Let X be uniform on $\{x_1, \ldots, x_n\}$. Apply Jensen's inequality with $g = \log$ to show GM $\leq$ AM, and then apply that result to the numbers $1/x_i$ to show HM $\leq$ GM. When do the inequalities become equalities?

10.3 Cauchy–Schwarz

One more inequality, which we'll need for the Cramér–Rao bound in Chapter 13. For any two random variables with finite second moments,

$$(\mathbb{E}[UV])^2 \leq \mathbb{E}\left[U^2\right]\mathbb{E}\left[V^2\right]. \tag{10.10}$$

The proof is a classic. For any real number s, the random variable $(U - sV)^2$ is nonnegative, so its expectation is too,

$$0 \le \mathbb{E}\left[(U - sV)^2\right] = \mathbb{E}\left[U^2\right] - 2s\,\mathbb{E}[UV] + s^2\,\mathbb{E}\left[V^2\right]. \tag{10.11}$$

The right side is a quadratic in s that is never negative, so it can have at most one real root, which means its discriminant is ≤ 0. That is $4\mathbb{E}[UV]^2 - 4\mathbb{E}[U^2]\mathbb{E}[V^2] \le 0$, which is the inequality. Applying it to $U = X - \mathbb{E}[X]$ and $V = Y - \mathbb{E}[Y]$ gives

$$\mathrm{Cov}[X, Y]^2 \le \mathrm{Var}[X]\,\mathrm{Var}[Y], \tag{10.12}$$

which says that the correlation $\rho = \mathrm{Cov}[X, Y]/\sqrt{\mathrm{Var}[X]\,\mathrm{Var}[Y]}$ always satisfies $|\rho| \le 1$. Equality happens exactly when $U = sV$ for some s, meaning Y is a linear function of X.

10.4 Convergence In Probability

Now we can talk about what it means for a sequence of random variables to settle down. Suppose you want to find the average height of BU students. Obviously, we can't go out and ask 30,000 or so students their height. What we can do is take a sample of n students and compute $\bar{X}_n$. We'd like to say that $\bar{X}_n$ "converges" to the true mean μ as n gets large. But $\bar{X}_n$ is random, so it's not a sequence of numbers, and the usual definition from calculus doesn't apply. There's always some tiny chance that we happen to sample only the basketball team. What we can hope for is that the chance of being far from μ goes to zero.

Definition. A sequence of random variables $Y_1, Y_2, \ldots$ **converges in probability** to a random variable Y, written $Y_n \xrightarrow{p} Y$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|Y_n - Y| \ge \varepsilon\right) = 0. \tag{10.13}$$

In words, the probability that Y_n and Y differ by more than any fixed amount goes to zero. Most of the time the limit Y will just be a constant c.

10.4.1 The Weak Law Of Large Numbers

Now we can prove one of the biggest results in probability.

Theorem (Weak law of large numbers). Let $X_1, X_2, \ldots, X_n$ be i.i.d. (sometimes called a **random sample of size** n) with mean μ and finite variance σ^2. Then $\bar{X}_n \xrightarrow{p} \mu$. That is, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|\bar{X}_n - \mu| \ge \varepsilon\right) = 0. \tag{10.14}$$

The proof is almost free now. We showed in the last chapter that $\mathbb{E}[\bar{X}_n] = \mu$ and $\text{Var}[\bar{X}_n] = \sigma^2/n$. So Chebyshev's inequality (10.4), applied to $\bar{X}_n$, gives

$$\mathbb{P}\left(|\bar{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\text{Var}[\bar{X}_n]}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}. \tag{10.15}$$

No matter how small we choose $\varepsilon > 0$, the right side goes to zero as $n \to \infty$, and we're done. (The finite variance assumption isn't actually needed, but the proof without it takes more work.) Notice that we also got a rate. The bound shrinks like $1/n$, so to cut the bound on the probability of a miss in half, you need twice as many samples. This is the theorem that justifies the whole idea of estimating a population mean with a sample mean.

10.4.2 Properties Of Convergence In Probability

Convergence in probability plays nicely with algebra, which is going to let us build up complicated convergence results from simple ones. Suppose $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$. Then

(1) $X_n + Y_n \xrightarrow{p} X + Y$,
(2) $aX_n \xrightarrow{p} aX$ for any constant a,
(3) $X_n Y_n \xrightarrow{p} XY$,
(4) $1/Y_n \xrightarrow{p} 1/Y$ if $\mathbb{P}(Y = 0) = 0$,
(5) $X_n/Y_n \xrightarrow{p} X/Y$ if $\mathbb{P}(Y = 0) = 0$.

The first one is a nice use of the triangle inequality. If $|(X_n + Y_n) - (X + Y)| \geq \varepsilon$, then at least one of $|X_n - X|$ or $|Y_n - Y|$ has to be at least $\varepsilon/2$, so

$$\mathbb{P}\left(|(X_n + Y_n) - (X + Y)| \geq \varepsilon\right) \leq \mathbb{P}\left(|X_n - X| \geq \tfrac{\varepsilon}{2}\right) + \mathbb{P}\left(|Y_n - Y| \geq \tfrac{\varepsilon}{2}\right) \to 0. \tag{10.16}$$

The fifth one follows from the third and fourth, since $X_n/Y_n = X_n \cdot (1/Y_n)$. All of these are actually special cases of a single result.

Theorem (Continuous mapping). If $Y_n \xrightarrow{p} Y$ and h is a continuous function, then $h(Y_n) \xrightarrow{p} h(Y)$.

We'll prove it in the case that matters most for us, where $Y = c$ is a constant. Fix $\varepsilon > 0$. Since h is continuous at c, there is a $\delta > 0$ such that $|y - c| < \delta$ implies $|h(y) - h(c)| < \varepsilon$. Turning this around, $|h(Y_n) - h(c)| \geq \varepsilon$ can only happen if $|Y_n - c| \geq \delta$, so

$$\mathbb{P}\left(|h(Y_n) - h(c)| \geq \varepsilon\right) \leq \mathbb{P}\left(|Y_n - c| \geq \delta\right) \to 0. \tag{10.17}$$

So for example, the law of large numbers immediately gives $\bar{X}_n^2 \xrightarrow{p} \mu^2$, $e^{\bar{X}_n} \xrightarrow{p} e^{\mu}$, and $1/\bar{X}_n \xrightarrow{p} 1/\mu$ as long as $\mu \neq 0$.

Problem 5.

Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2, and with a finite fourth moment. Use the law of large numbers (applied to the X_i^2) and the properties above to show that

$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}_n^2 \xrightarrow{p} \sigma^2.$$

Then show that the unbiased sample variance $S_n^2 = \frac{n}{n-1}\hat{\sigma}_n^2$ and the sample standard deviation S_n also converge in probability to σ^2 and σ.

Problem 6.

Let $X_1, X_2, \ldots$ be i.i.d. Bernoulli(p) and let $\hat{p}_n = \bar{X}_n$. Use Chebyshev to find an n that guarantees $\mathbb{P}(|\hat{p}_n - p| \geq 0.05) \leq 0.05$ for *every* p. (Hint: $p(1-p) \leq 1/4$.) Compare your answer with the sample size the normal approximation from Part I would suggest. Why is Chebyshev's answer so much bigger?

11

Convergence In Distribution And The CLT

"The law would have been personified by the Greeks and deified, if they had known of it."

– Francis Galton, *Natural Inheritance*

The law of large numbers told us that $\bar{X}_n$ collapses onto μ. That's great, but it's not enough to build a confidence interval. For that we needed to know the *shape* of the fluctuations of $\bar{X}_n$ around μ, and in Part I we got that shape from the central limit theorem. The trouble is that $\bar{X}_n - \mu$ converges in probability to the constant zero, so to see any shape at all, we have to blow it up by the right factor. That gives a sequence of random variables whose *distributions* settle down to a limit, even though the random variables themselves don't settle down to anything. That's a different kind of convergence, and this chapter is about it.

11.1 Convergence In Distribution

Let X be a random variable with CDF $F_X(t) = \mathbb{P}(X \leq t)$. The CDF contains all of the information about the distribution of X. If X is continuous we can get the density by taking a derivative, and if X is discrete we can read off the pmf as the size of the jump of the CDF at each point. (Every CDF is also right continuous, which is a small note for the analysis people, but it's going to matter in a second.) So it's natural to say that a sequence of distributions converges if the CDFs converge.

The first thing we might try is to say X_n converges in distribution to X if $F_n(t) \to F(t)$ for every $t \in \mathbb{R}$. But this is a little too strict. Here's a counterexample. Let $X_n = 1/n$ with probability one. Obviously X_n is getting closer and closer to the constant $X = 0$, so we'd like to say its distribution converges to the point mass at 0. The CDFs are

$$F_n(t) = \mathbb{P}\left(\tfrac{1}{n} \le t\right) = \begin{cases} 1 & t \ge \frac{1}{n} \\ 0 & t < \frac{1}{n} \end{cases} \qquad F(t) = \mathbb{P}(0 \le t) = \begin{cases} 1 & t \ge 0 \\ 0 & t < 0 \end{cases} \tag{11.1}$$

For any $t > 0$, eventually $1/n \le t$, so $F_n(t) \to 1 = F(t)$. For $t < 0$ everything is zero. But at $t = 0$, $F_n(0) = 0$ for every n, while $F(0) = 1$. So the convergence fails at exactly one point, the point where F jumps. The failure here isn't so much a failure of the sequence, it's more of a failure of the definition. It's like shooting at a target that moves at the last second. If the bullet lands where the target would have been, is it really fair to say you're a bad marksman? So we fix the definition by ignoring the points where F jumps.

Definition. Let X_n have CDF F_n and X have CDF F, and let C_F be the set of points where F is continuous. We say X_n **converges in distribution** to X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_n(t) = F(t) \quad \text{for every } t \in C_F. \tag{11.2}$$

If the limit X is continuous (like a normal), then C_F is all of $\mathbb{R}$ and the exceptions don't come up. Notice that convergence in distribution is only a statement about the distributions. The random variables X_n and X don't even have to live on the same probability space.

11.1.1 An Example: The Uniform Maximum

Suppose $X_1, \ldots, X_n$ are i.i.d. Uniform$(0, \theta)$, and let $M_n = \max_i X_i$ be the largest observation. Intuitively, M_n should creep up toward θ, and the question is how fast. To find the distribution of a maximum, we use a really handy fact. The maximum is at most t exactly when every observation is at most t, so by independence,

$$F_{M_n}(t) = \mathbb{P}(X_1 \le t, \ldots, X_n \le t) = \prod_{i=1}^{n} \mathbb{P}(X_i \le t) = \left(F_X(t)\right)^n. \tag{11.3}$$

For the uniform, $F_X(t) = t/\theta$ on $[0, \theta]$. It turns out the right scaling for the gap $\theta - M_n$ is n (not $\sqrt{n}$), so let's look at $n(\theta - M_n) \ge 0$. For $t \ge 0$ (and n big enough that $t/n < \theta$),

$$\begin{aligned} \mathbb{P}\left(n(\theta - M_n) \le t\right) &= \mathbb{P}\left(M_n \ge \theta - \frac{t}{n}\right) \\ &= 1 - \left(\frac{\theta - t/n}{\theta}\right)^n \end{aligned}$$

$$= 1 - \left(1 - \frac{t/\theta}{n}\right)^n. \tag{11.4}$$

Now we use the limit definition of e, $\lim_{n\to\infty}\left(1+\frac{c}{n}\right)^n = e^c$, which gives

$$\lim_{n\to\infty} \mathbb{P}\left(n(\theta - M_n) \le t\right) = 1 - e^{-t/\theta}. \tag{11.5}$$

That's the CDF of an exponential distribution with mean θ. So

$$n(\theta - M_n) \xrightarrow{d} Y, \qquad Y \sim \text{Exp}(1/\theta), \tag{11.6}$$

where we're writing the exponential in terms of its rate $1/\theta$. Figure 11.1 shows how fast this kicks in. Roughly speaking, the gap between the maximum and θ is about θ/n, which is much faster than the $1/\sqrt{n}$ rate we get for the sample mean. That's going to make the uniform family an oddball in a lot of the theory that follows.

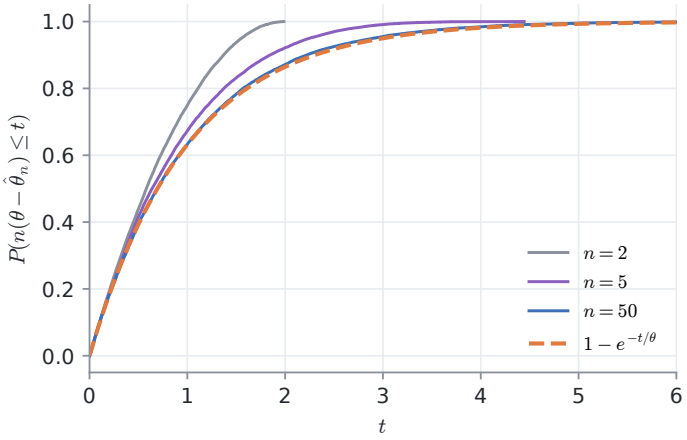


Figure 11.1: Empirical CDFs of $n(\theta - M_n)$ for samples from Uniform$(0, 1)$, against the limiting Exp(1) CDF. Even at $n = 5$ the curves are close.

Problem 1.

Fix $\lambda > 0$. Each day, you perform n independent Bernoulli(λ/n) trials. Let X_n be the number of trials up to and including the first win. Find the distribution of X_n and its expectation. Then let $Y_n = X_n/n$ be the number of days up to the first win (as a fraction of a day). Show that $Y_n \xrightarrow{d} \text{Exp}(\lambda)$. (Hint: work with the survival function $H(t) = 1 - F(t)$, and use $\mathbb{P}(X_n > nt) = (1 - \lambda/n)^{\lfloor nt \rfloor}$.)

11.1.2 Comparing To Convergence In Probability

How does this relate to convergence in probability from the last chapter? The short answer is that convergence in probability is stronger.

Theorem. If $X_n \xrightarrow{p} X$, then $X_n \xrightarrow{d} X$.

To prove it, let $t \in C_F$, fix $\varepsilon > 0$, and let $a_n = \mathbb{P}(|X_n - X| \geq \varepsilon)$, which goes to zero by assumption. If $X_n \leq t$, then either X is within ε of X_n, in which case $X \leq t + \varepsilon$, or it isn't. So, splitting $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$,

$$\begin{aligned} F_n(t) = \mathbb{P}(X_n \leq t) &= \mathbb{P}(X_n \leq t,\ |X_n - X| < \varepsilon) + \mathbb{P}(X_n \leq t,\ |X_n - X| \geq \varepsilon) \\ &\leq \mathbb{P}(X \leq t + \varepsilon) + \mathbb{P}(|X_n - X| \geq \varepsilon) \\ &= F(t + \varepsilon) + a_n. \end{aligned} \tag{11.7}$$

Doing the same thing to $1 - F_n(t) = \mathbb{P}(X_n > t)$, which forces $X > t - \varepsilon$ unless $|X_n - X| \geq \varepsilon$, gives $1 - F_n(t) \leq 1 - F(t - \varepsilon) + a_n$. Putting the two together, we have a squeeze

$$F(t - \varepsilon) - a_n \leq F_n(t) \leq F(t + \varepsilon) + a_n. \tag{11.8}$$

Let $n \to \infty$, so $a_n \to 0$, and then let $\varepsilon \to 0$. Since F is continuous at t, both $F(t - \varepsilon)$ and $F(t + \varepsilon)$ go to $F(t)$, and by the squeeze theorem $F_n(t) \to F(t)$.

The converse is false. Let $Z \sim \mathcal{N}(0, 1)$, and let $X_n = Z$ for every n. Since $-Z$ is also standard normal, $X_n \xrightarrow{d} -Z$ trivially (every X_n has exactly the distribution of $-Z$). But $|X_n - (-Z)| = 2|Z|$, which doesn't shrink at all, so X_n does not converge in probability to $-Z$. Kind of a trivial example, but at least it's a counterexample. The one important case where the converse does hold is when the limit is a constant.

> Problem 2.
>
> Let $c \in \mathbb{R}$ be a constant. Prove that $X_n \xrightarrow{p} c$ if and only if $X_n \xrightarrow{d} c$. (Hint: the CDF of the constant c is continuous everywhere except at c. Write $\mathbb{P}(|X_n - c| \geq \varepsilon)$ in terms of $F_n(c - \varepsilon)$ and $F_n(c + \varepsilon/2)$.)

11.1.3 Slutsky's Theorem

Convergence in distribution does not play as nicely with algebra as convergence in probability did. If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$, it's not true in general that $X_n + Y_n \xrightarrow{d} X + Y$ (think about $Y_n = -X_n$ in the counterexample above). But if one of the sequences converges to a *constant*,

everything works out. This result is going to get used all over the place, so we'll state it here without proof.

Theorem (Slutsky). If $X_n \xrightarrow{d} X$, $A_n \xrightarrow{p} a$ and $B_n \xrightarrow{p} b$, where a and b are constants, then

$$A_n X_n + B_n \xrightarrow{d} aX + b. \tag{11.9}$$

Here's an example of how it gets used. We'll see in the next chapter that the maximum M_n is a biased estimator of θ, and that the fix is $\theta'_n = \frac{n+1}{n} M_n$. Does the fix change the limiting distribution? With a bit of algebra,

$$\begin{aligned} n(\theta - \theta'_n) &= n\left(\theta - M_n\right) + n\left(M_n - \frac{n+1}{n} M_n\right) \\ &= n\left(\theta - M_n\right) - M_n. \end{aligned} \tag{11.10}$$

The first term converges in distribution to $Y \sim \text{Exp}(1/\theta)$, and $M_n \xrightarrow{p} \theta$ (Problem 3), so by Slutsky,

$$n(\theta - \theta'_n) \xrightarrow{d} Y - \theta. \tag{11.11}$$

So the bias correction just shifts the limit so it has mean $\mathbb{E}[Y] - \theta = 0$.

> Problem 3.
>
> Show directly from (11.3) that $M_n \xrightarrow{p} \theta$ for the Uniform$(0, \theta)$ maximum. Then show it a second way, using $n(\theta - M_n) \xrightarrow{d} Y$, Slutsky (with $A_n = 1/n$), and Problem 2.

11.2 Convergence Of MGFs

Remember MGF uniqueness from Chapter 9. If $M_X(t) = M_Y(t)$ near zero, then X and Y have the same distribution. Presumably, then, if M_{X_n} gets close to M_Y near zero, the distribution of X_n should get close to the distribution of Y. That turns out to be true.

Theorem (MGF convergence). Suppose there's a $\delta > 0$ such that for every $t \in (-\delta, \delta)$, $M_{X_n}(t)$ is finite and $M_{X_n}(t) \to M_Y(t) < \infty$. Then $X_n \xrightarrow{d} Y$.

Unfortunately this is not an analysis class, so we won't prove it. But it's the tool that makes the rest of the chapter work. Here's our first use of it. Fix $\lambda > 0$ and let $X_n \sim \text{Bin}(n, \lambda/n)$, so that the number of trials grows while the success probability shrinks, and the mean stays

at λ. Using the binomial MGF,

$$M_{X_n}(t) = \left[1 + \frac{\lambda}{n}\left(e^t - 1\right)\right]^n = \left[1 + \frac{\lambda(e^t-1)}{n}\right]^n \underset{n\to\infty}{\longrightarrow} e^{\lambda(e^t-1)}, \tag{11.12}$$

by the limit definition of e again. That's the MGF of a Poisson(λ), so $X_n \xrightarrow{d}$ Poisson(λ). This is the **Poisson limit theorem**. To connect it with the version you might have seen in a probability course, fix $k \in \mathbb{N}_0$. Since X_n is integer valued and $k \pm \frac{1}{2}$ are continuity points of the Poisson CDF,

$$\mathbb{P}(X_n = k) = F_n\left(k + \tfrac{1}{2}\right) - F_n\left(k - \tfrac{1}{2}\right) \longrightarrow \frac{\lambda^k}{k!}e^{-\lambda}, \tag{11.13}$$

so the binomial probabilities converge to the Poisson ones (Figure 11.2). This is why the Poisson is a good model for counts of rare events, like typos on a page or goals in a soccer game.

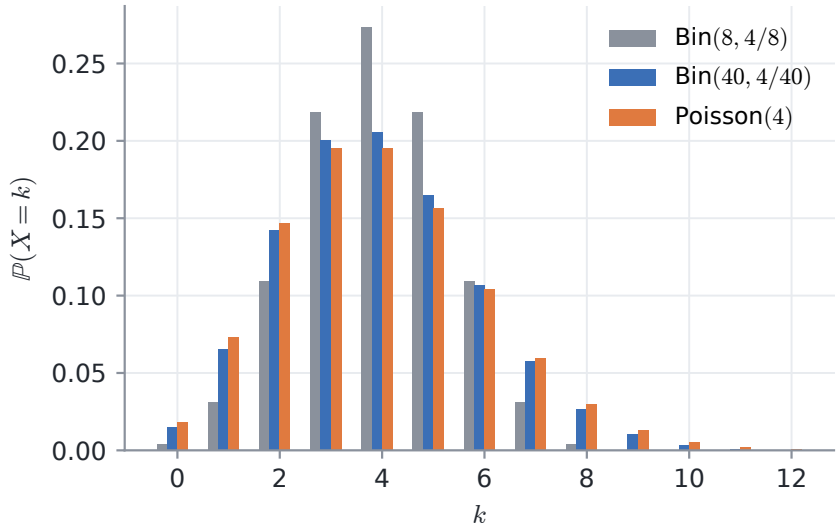


Figure 11.2: The Bin($n, 4/n$) pmf for $n = 8$ and $n = 40$, against the Poisson(4) pmf.

11.3 The Central Limit Theorem

There are many climaxes in this book, and we will now prove one of them, the "adult" central limit theorem.

Theorem (Central Limit Theorem). Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and finite

nonzero variance σ^2. Then

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1). \tag{11.14}$$

We'll prove it under the extra assumption that the X_i have an MGF (the theorem is true without it, but the proof needs characteristic functions instead). As we saw in (9.36), we can write $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i$, where $Y_i = (X_i - \mu)/\sigma$ are i.i.d. standardized random variables with mean 0 and variance 1. Let M be the MGF of the Y_i. Using (9.17) and (9.15),

$$M_{Z_n}(t) = M_{\sum Y_i}\left(\frac{t}{\sqrt{n}}\right) = \left[M\left(\frac{t}{\sqrt{n}}\right)\right]^n. \tag{11.15}$$

For any fixed t, the argument $t/\sqrt{n}$ is getting very small, so we can Taylor expand M around 0,

$$M\left(\frac{t}{\sqrt{n}}\right) = M(0) + M'(0)\frac{t}{\sqrt{n}} + \frac{M''(0)}{2}\frac{t^2}{n} + o\left(\frac{1}{n}\right). \tag{11.16}$$

But what are the derivatives of the MGF at zero? They're the moments! $M(0) = 1$ always. $M'(0) = \mathbb{E}[Y] = 0$ because we standardized. And $M''(0) = \mathbb{E}[Y^2] = \mathrm{Var}[Y] = 1$, again because we standardized. Putting everything back,

$$M_{Z_n}(t) = \left[1 + \frac{t^2/2}{n} + o\left(\frac{1}{n}\right)\right]^n \underset{n\to\infty}{\longrightarrow} e^{t^2/2}, \tag{11.17}$$

where the small leftover term doesn't affect the limit. That's the MGF of a standard normal, so by MGF convergence, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. And we're done!

Look at how little of the distribution of X survived. The proof only used the first two moments, and everything else about the shape of the distribution got washed out into the $o(1/n)$ term. That's why the theorem works for any distribution with a finite variance (Figure 11.3). The skewness does show up in the third-order term, which goes to zero like $1/\sqrt{n}$, and that's why very skewed distributions need a bigger n before the normal approximation is good.

For some, this is not the form of the CLT you saw in a probability course, so let's make contact with the past. Since the normal CDF Φ is continuous, convergence in distribution means that for every $a < b$,

$$\mathbb{P}\left(a < \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \le b\right) \underset{n\to\infty}{\longrightarrow} \Phi(b) - \Phi(a) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz. \tag{11.18}$$

Let's try one example. Roll a fair six-sided die 10,000 times. Why? Because you've got nothing better to do. What is the probability that the average roll is between 3.48 and 3.53?

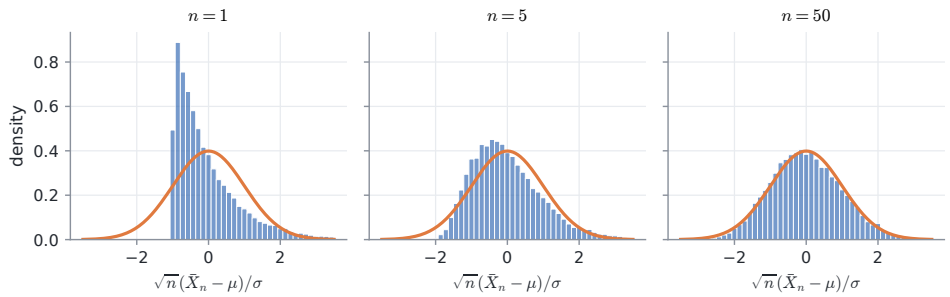


Figure 11.3: Histograms of the standardized mean of n i.i.d. Exp(1) random variables, against the $\mathcal{N}(0,1)$ density. The skew from the exponential fades as n grows.

Each roll has $\mu = 3.5$ and $\sigma^2 = 35/12$ (the variance of a discrete uniform on $\{1,\dots,6\}$), so $\sigma/\sqrt{n} = \sqrt{35/12}/100 \approx 0.01708$, and the CLT tells us

$$\mathbb{P}\left(3.48 \le \bar{X}_n \le 3.53\right) \approx \Phi\left(\frac{3.53-3.50}{0.01708}\right) - \Phi\left(\frac{3.48-3.50}{0.01708}\right) = \Phi(1.76) - \Phi(-1.17) = 0.8398. \tag{11.19}$$

You might be curious why we were cavalier about $<$ versus $\le$. It's because the limit distribution is continuous, so it makes no difference in the limit. Anyway, there's about an 84% chance. Pretty good, yeah?

11.3.1 The Normal Approximation To The Binomial

If $X_n \sim \text{Bin}(n,p)$ with p fixed (not shrinking like in the Poisson limit), then $X_n = \sum_{i=1}^{n} I_i$, where I_i is the indicator that trial i is a success. The I_i are i.i.d. Bernoulli(p) with mean p and variance $p(1-p)$, so X_n/n is a sample mean and the CLT gives

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0,1). \tag{11.20}$$

This is the result we used for proportions back in Chapter 2. There's one subtlety when you use it for actual numbers. The binomial is discrete, so $\mathbb{P}(X \le 220)$ and $\mathbb{P}(X < 221)$ are the same thing, but the normal approximation gives them different answers. The standard fix is the **continuity correction**, which splits the difference and uses the half-integer in between. For example, toss a fair coin 400 times and find the probability of getting between 210 and 220 heads, inclusive. Here $np = 200$ and $\sqrt{np(1-p)} = 10$, so

$$\mathbb{P}(210 \le X \le 220) = \mathbb{P}(209.5 < X < 220.5) \approx \Phi\left(\frac{220.5-200}{10}\right) - \Phi\left(\frac{209.5-200}{10}\right) = 0.151. \tag{11.21}$$

The exact binomial answer is 0.1509, so the approximation is excellent.

Problem 4.

Redo the coin example without the continuity correction, using the bounds 210 and 220 directly, and then using 209 and 221. Compare both with the exact answer. Then explain why the continuity correction matters less and less as n grows.

11.3.2 Unknown Variance

In practice, we almost never know σ, so we replace it with the sample standard deviation S_n. What happens to the limit? Slutsky answers this right away. We know from the last chapter that $S_n \xrightarrow{p} \sigma$, so $\sigma/S_n \xrightarrow{p} 1$, and

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} = \frac{\sigma}{S_n} \cdot \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} 1 \cdot Z \sim \mathcal{N}(0,1). \tag{11.22}$$

So for large samples, the studentized mean is still approximately standard normal. This is the justification for the large-sample intervals in Part I. When the data themselves are normal, Gosset's result from Chapter 2 says that this ratio is *exactly* t_{n-1} for every n, and we just showed that t_{n-1} converges to $\mathcal{N}(0,1)$ as n grows, which is the picture in Figure 2.2.

Problem 5.

Let $X_1, \dots, X_n$ be i.i.d. Poisson(λ). Use the CLT and Slutsky to show that

$$\frac{\bar{X}_n - \lambda}{\sqrt{\bar{X}_n/n}} \xrightarrow{d} \mathcal{N}(0,1),$$

and use it to build an approximate 95% confidence interval for λ. Then simulate: with $\lambda = 2$ and $n = 20$, what fraction of 10,000 such intervals actually contain λ?

12

Parametric Families And Estimators

"The existence of parametric statistics implies the existence of dia-empirical statistics."

– Hieu

For the rest of the book, we're going to work inside **parametric statistics**. A parametric model makes an assumption about the data before we've seen it, namely that it came from a distribution in some family that is described by a handful of numbers, and then it uses the data to figure out those numbers. Typically, the more we assume, the stronger the results we can get, since we've placed more restrictions on what the world can look like. The cost is that if the assumptions are wrong, so are the results. In this chapter we set up the language and figure out what makes an estimator good.

12.1 Parametric Families

Definition. Let $\Theta \neq \varnothing$ be a set called the **parameter space**, and suppose that for every $\theta \in \Theta$ there is a density (or pmf) $f(x;\theta)$. Then $\mathcal{F} = \{f(\cdot\,;\theta) : \theta \in \Theta\}$ is a **parametric family** of distributions.

For a quick example, the exponential family (not to be confused with the exponential family of Chapter 8, sadly) is $\mathcal{F} = \{\lambda e^{-\lambda x} : \lambda > 0\}$. This collects all the possible distributions we would call "the exponential distribution." If we knew that $\lambda = 7$, then we would know

everything there is to know about the distribution and we'd be all gucci. But we will seldom ever know λ, and so we have to estimate it.

In order to estimate anything, we first have to assume which family the data came from. If a crime was committed in Brookline and the spokesperson said that the last name of the criminal is Weiner, then that reduces the space of all possible criminals dramatically, down to just the Weiner family. In general, the setup for the rest of the book is

(1) $X_1, X_2, \ldots, X_n$ are i.i.d. from some $f \in \mathcal{F} = \{f(\cdot\,;\theta) : \theta \in \Theta\}$,
(2) so there is a true $\theta \in \Theta$ with $f = f(\cdot\,;\theta)$, and our objective is to use the data to find it.

12.1.1 The Five Families

We'll keep coming back to five famous parametric families (like in the Godfather movies).

- The **Bernoulli** family $\{\text{Bernoulli}(\theta) : 0 < \theta < 1\}$, which collects all coin flips. (The binomial is a sum of these.) We drop $\theta = 0$ and $\theta = 1$ since always winning or always losing is uninteresting, so $\Theta = (0, 1)$ is an open interval.
- The **Poisson** family $\{\text{Poisson}(\theta) : \theta > 0\}$, with $\Theta = (0, \infty)$.
- The **exponential** family $\{\text{Exp}(\theta) : \theta > 0\}$, with $\Theta = (0, \infty)$.
- The **uniform** family $\{\text{Uniform}(0, \theta) : \theta > 0\}$. This one is the scapegoat. Any time something weird happens in the theory, it's because of the uniform family. It rarely shows up in reality, so it's mostly here for pedagogy (and as a warning).
- The **normal** family $\{\mathcal{N}(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma^2 > 0\}$. This one breaks into cases. If σ^2 is known and only $\theta = \mu$ is unknown, then $\Theta = \mathbb{R}$. If μ is known and only $\theta = \sigma^2$ is unknown, then $\Theta = (0, \infty)$. If both are unknown, then $\theta = (\mu, \sigma^2)$ and the parameter space is the Cartesian product $\Theta = \mathbb{R} \times (0, \infty)$.

12.2 Estimators

We start with some definitions.

Definition. A **parameter** is a population-based number, computable directly from the distribution $f(\cdot\,;\theta)$. A **statistic** is a sample-based number, a function $T(X_1, \ldots, X_n)$ that we can compute after observing the data. An **estimator** of θ is a statistic designed to guess the true value of θ well, and we write it as $\hat{\theta}_n = T(X_1, \ldots, X_n)$.

In some ways, parameters and statistics occupy opposing battlegrounds. Everyone wants to know the parameter, but anyone can get the statistic. The question is how to turn one into a guess for the other, and how to know if the guess is any good. The simplest way to come up with an estimator is what we'll call the **Golden Rule of Estimation**: do to the sample

what you would have done to the whole population. If we wanted the population mean, we'd add up everyone and divide by the population size. We don't have the whole population, so we add up the sample and divide by n. This is also called the **plug-in principle**. Is it guaranteed to give the best estimator? No. But it's a good start, and we can modify it from there.

For example, if $X \sim \text{Poisson}(\lambda)$, then $\lambda = \mathbb{E}[X]$ is the population mean, and the Golden Rule says to use $\hat{\lambda}_n = \bar{X}_n$. How do we know this is good? We know that $\mathbb{E}[\bar{X}_n] = \mathbb{E}[X] = \lambda$, so on average the estimator hits the true value. And by the law of large numbers, $\bar{X}_n \xrightarrow{p} \lambda$, so with enough data the estimator gets arbitrarily close to the truth. These are the two most basic kinds of "goodness."

Definition. An estimator $\hat{\theta}_n$ is **unbiased** for θ if $\mathbb{E}_\theta[\hat{\theta}_n] = \theta$ for every $\theta \in \Theta$ and every n. The **bias** is $\text{bias}(\hat{\theta}_n) = \mathbb{E}_\theta[\hat{\theta}_n] - \theta$.

Definition. An estimator $\hat{\theta}_n$ is **consistent** for θ if $\hat{\theta}_n \xrightarrow{p} \theta$ for every $\theta \in \Theta$.

The subscript in $\mathbb{E}_\theta$ is there to remind us that the expectation is computed assuming θ is the true parameter. Notice that both definitions say "for every θ". An estimator that always guesses $\hat{\theta} = 7$ is perfect when $\theta = 7$, but it's useless, because we don't know that $\theta = 7$. So the sample mean is unbiased and consistent for the Poisson mean. But obviously things are never that simple.

12.2.1 The Uniform Maximum

Suppose $X_1, \ldots, X_n$ are i.i.d. Uniform$(0, \theta)$ for $\theta > 0$. Here θ has the interpretation of the population maximum, since it's the largest value the distribution can produce. So the Golden Rule says to use the sample maximum, $\hat{\theta}_n = M_n = \max\{X_1, \ldots, X_n\}$. Is it consistent? Is it unbiased?

For consistency, pick $\varepsilon \in (0, \theta)$. Since M_n can never be bigger than θ, the event $|M_n - \theta| \geq \varepsilon$ is the same as $M_n \leq \theta - \varepsilon$, and using the CDF of the maximum (11.3),

$$\mathbb{P}\left(|M_n - \theta| \geq \varepsilon\right) = \mathbb{P}(M_n \leq \theta - \varepsilon) = \left(\frac{\theta - \varepsilon}{\theta}\right)^n \underset{n \to \infty}{\longrightarrow} 0, \tag{12.1}$$

since $(\theta - \varepsilon)/\theta < 1$. So M_n is consistent. For the bias, differentiate the CDF $(t/\theta)^n$ to get the density nt^{n-1}/θ^n on $[0, \theta]$, so

$$\mathbb{E}[M_n] = \int_0^\theta t \cdot \frac{nt^{n-1}}{\theta^n}\, dt = \frac{n}{n+1}\,\theta. \tag{12.2}$$

So M_n is biased. It *always* underestimates θ on average, which makes sense, because the sample maximum can never be bigger than the population maximum. The fix is to scale it up,

$$\theta'_n = \frac{n+1}{n} M_n, \tag{12.3}$$

which is unbiased by construction. As $n \to \infty$ the factor $n/(n+1)$ goes to 1, so M_n is at least **asymptotically unbiased**.

12.2.2 The Sample Variance

Let's do another one, and this one we've been using since Chapter 2. In general, the variance is $\sigma^2 = \mathbb{E}[(X-\mu)^2]$. Following the Golden Rule, we replace the expectation with a sample average and the unknown μ with $\bar{X}_n$, which gives

$$\hat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}_n^2. \tag{12.4}$$

Consistency is pretty easy, and you showed it in Problem 5 of Chapter 10. By the law of large numbers, $\frac{1}{n}\sum X_i^2 \xrightarrow{p} \mathbb{E}[X^2]$ and $\bar{X}_n^2 \xrightarrow{p} \mu^2$, so $\hat{\sigma}_n^2 \xrightarrow{p} \mathbb{E}[X^2] - \mu^2 = \sigma^2$. What about the bias? We'll use a trick called **pivoting**, adding and subtracting μ inside the square,

$$\sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2 = \sum_{i=1}^{n} \left[(X_i - \mu) - (\bar{X}_n - \mu)\right]^2 = \sum_{i=1}^{n} (X_i-\mu)^2 - n(\bar{X}_n - \mu)^2, \tag{12.5}$$

where the cross term combined with the last term because $\sum_i (X_i - \mu) = n(\bar{X}_n - \mu)$. Now take expectations. The first term is $n\sigma^2$, and the second is $n\,\mathrm{Var}[\bar{X}_n] = \sigma^2$, so

$$\mathbb{E}\left[\hat{\sigma}_n^2\right] = \frac{1}{n}\left(n\sigma^2 - \sigma^2\right) = \frac{n-1}{n}\,\sigma^2. \tag{12.6}$$

So, maybe unfortunately, the plug-in variance is biased low. Who do we blame? We should blame $\bar{X}_n$. If we knew μ, then $\frac{1}{n}\sum (X_i - \mu)^2$ would be unbiased. But we didn't, so we replaced μ with $\bar{X}_n$, and $\bar{X}_n$ is by construction the number that makes $\sum (X_i - c)^2$ as small as possible. So the spread around $\bar{X}_n$ is always a little smaller than the spread around μ. To undo this, we divide out the factor, which gives the sample variance from Part I,

$$S_n^2 = \frac{n}{n-1}\hat{\sigma}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2, \tag{12.7}$$

which is unbiased. The $n-1$ is the **degrees of freedom**. In crude terms, the n data points carry n pieces of information, and we used one of them up estimating μ.

Problem 1.

The **mean squared error** of an estimator is $\mathrm{MSE}(\hat{\theta}_n) = \mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]$.
(1) Show that $\mathrm{MSE}(\hat{\theta}_n) = \mathrm{Var}[\hat{\theta}_n] + \mathrm{bias}(\hat{\theta}_n)^2$.
(2) For the Uniform$(0, \theta)$ family, compute the MSE of M_n and of $\theta'_n = \frac{n+1}{n} M_n$. Which is smaller? (You'll need $\mathbb{E}[M_n^2]$.)
(3) For normal data, $\mathrm{Var}[S_n^2] = 2\sigma^4/(n-1)$. Compute the MSE of S_n^2 and of $\hat{\sigma}_n^2$. Which one wins? What does this say about insisting on unbiasedness?

Problem 2.

Show that if $\hat{\theta}_n$ is asymptotically unbiased and $\mathrm{Var}[\hat{\theta}_n] \to 0$, then $\hat{\theta}_n$ is consistent. (Hint: Chebyshev, applied with the MSE in place of the variance.) Use this to give a second proof that M_n is consistent for the uniform.

12.3 Asymptotic Normality

Consistency says that $\hat{\theta}_n$ gets close to θ, but it doesn't say how close, or what the errors look like. For a confidence interval we need more. The CLT gives us exactly this for the sample mean. Multiplying the standardized mean by σ, we can write it as

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2). \tag{12.8}$$

In words, the error $\bar{X}_n - \mu$ is roughly normal with standard deviation $\sigma/\sqrt{n}$. This shape is so common that it gets a name.

Definition. An estimator $\hat{\theta}_n$ is **consistent and asymptotically normal** (CAN) for θ if there's a function $V : \Theta \to (0, \infty)$ such that

$$\sqrt{n}\left(\hat{\theta}_n - \theta\right) \xrightarrow{d} \mathcal{N}(0, V(\theta)) \quad \text{for every } \theta \in \Theta. \tag{12.9}$$

$V(\theta)$ is called the **asymptotic variance**.

(A CAN estimator is automatically consistent, by Slutsky. Write $\hat{\theta}_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(\hat{\theta}_n - \theta)$, which converges in distribution to $0 \cdot \mathcal{N}(0, V) = 0$, and convergence in distribution to a constant is convergence in probability.) So the sample mean is CAN for μ with $V = \sigma^2$. For the Poisson, $V(\lambda) = \lambda$, and for the exponential with mean $\beta = 1/\lambda$, $V(\beta) = \beta^2$. The

uniform maximum is *not* CAN, since its error shrinks like $1/n$ and the limit is exponential, not normal.

12.3.1 The Delta Method

Often the parameter we care about is a function of something we already have a CAN estimator for. For the exponential, the CLT hands us $\bar{X}_n$ for the mean β, but the parameter in the family is the rate $\lambda = 1/\beta$. The Golden Rule says to use $\hat{\lambda}_n = 1/\bar{X}_n$, but is *that* CAN, and if so, what's its asymptotic variance? The answer is the delta method.

Theorem (Delta method). Suppose $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, V(\theta))$, and let g be differentiable at θ with $g'(\theta) \neq 0$. Then

$$\sqrt{n}\left(g(\hat{\theta}_n) - g(\theta)\right) \xrightarrow{d} \mathcal{N}\left(0,\ g'(\theta)^2\, V(\theta)\right). \tag{12.10}$$

In a slogan, CAN in, CAN out, asymptotic variance modified. To see why, expand g to first order around the true θ,

$$g(\hat{\theta}_n) \approx g(\theta) + g'(\theta)\left(\hat{\theta}_n - \theta\right), \tag{12.11}$$

which is a good approximation for large n because $\hat{\theta}_n$ is consistent, so it's close to θ. Subtracting $g(\theta)$ and multiplying by $\sqrt{n}$,

$$\sqrt{n}\left(g(\hat{\theta}_n) - g(\theta)\right) \approx g'(\theta) \cdot \sqrt{n}\left(\hat{\theta}_n - \theta\right). \tag{12.12}$$

The right side is a constant times something converging to $\mathcal{N}(0, V(\theta))$, so it converges to $\mathcal{N}(0, g'(\theta)^2 V(\theta))$. (Making the $\approx$ rigorous uses the mean value theorem and Slutsky.) Geometrically, near θ the function g looks like a straight line with slope $g'(\theta)$, and a straight line just stretches the normal by a factor of $|g'(\theta)|$.

Let's do the three classic examples.

1. The exponential rate. With $\beta = \mathbb{E}[X]$ and $V(\beta) = \beta^2$, take $g(t) = 1/t$, so $g'(\beta) = -1/\beta^2$. Then

$$\sqrt{n}\left(\frac{1}{\bar{X}_n} - \frac{1}{\beta}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{\beta^4}\beta^2\right) = \mathcal{N}\left(0, \lambda^2\right), \tag{12.13}$$

so $\hat{\lambda}_n = 1/\bar{X}_n$ is CAN for λ with $V(\lambda) = \lambda^2$. There's a catch here, though. The asymptotic variance depends on the parameter we're trying to estimate. That's not a disaster (we can plug $\hat{\lambda}_n$ in for λ and use Slutsky, like we did with S_n), but it would be nicer if it didn't.

2. Variance stabilization. Let $X \sim \text{Poisson}(\lambda)$, so $\bar{X}_n$ is CAN with $V(\lambda) = \lambda$. Using

$g(t) = \sqrt{t}$ and $g'(t) = 1/(2\sqrt{t})$,

$$\sqrt{n}\left(\sqrt{\bar{X}_n} - \sqrt{\lambda}\right) \xrightarrow{d} \mathcal{N}\left(0, \lambda\left(\frac{1}{2\sqrt{\lambda}}\right)^2\right) = \mathcal{N}\left(0, \frac{1}{4}\right). \tag{12.14}$$

Now the variance is a *constant*! This is called a **variance stabilizing transformation**. It gives us a confidence interval for $\sqrt{\lambda}$ that doesn't require plugging anything in, $\sqrt{\bar{X}_n} \pm z_{\alpha/2}/(2\sqrt{n})$, and squaring the endpoints gives one for λ.

3. A squared mean. For any data with mean μ and variance σ^2, take $g(t) = t^2$, $g'(t) = 2t$. Then

$$\sqrt{n}\left(\bar{X}_n^2 - \mu^2\right) \xrightarrow{d} \mathcal{N}\left(0, 4\mu^2\sigma^2\right). \tag{12.15}$$

Notice the condition $g'(\theta) \neq 0$. If $\mu = 0$ the asymptotic variance is zero, which means $\sqrt{n}$ is the wrong scaling. In that case $n\bar{X}_n^2/\sigma^2 \xrightarrow{d} \chi_1^2$ instead (Problem 4).

Problem 3.

Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(p) and $\hat{p}_n = \bar{X}_n$.

(1) Find the asymptotic distribution of the estimated log-odds $\log \frac{\hat{p}_n}{1-\hat{p}_n}$. Compare with the variance of the log odds ratio from the categorical data chapter.

(2) Find a function g that stabilizes the variance of $\hat{p}_n$. (Hint: you need $g'(p)^2 p(1-p)$ to be constant. Try $g(p) = \arcsin\sqrt{p}$.)

Problem 4.

Suppose $\mu = 0$. Use the CLT and the continuous mapping theorem (which also holds for convergence in distribution) to show that $n\bar{X}_n^2/\sigma^2 \xrightarrow{d} \chi_1^2$.

12.3.2 Jensen Strikes Back

Let's go back to the exponential rate $\hat{\lambda}_n = 1/\bar{X}_n$. It's CAN, but is it unbiased? Since $g(t) = 1/t$ is strictly convex on $(0, \infty)$, Jensen's inequality says

$$\mathbb{E}\left[\hat{\lambda}_n\right] = \mathbb{E}\left[\frac{1}{\bar{X}_n}\right] > \frac{1}{\mathbb{E}[\bar{X}_n]} = \frac{1}{\beta} = \lambda, \tag{12.16}$$

so it's biased high. This is a general lesson. Even if $\hat{\theta}_n$ is unbiased for θ, $g(\hat{\theta}_n)$ is almost never unbiased for $g(\theta)$ unless g is linear. How bad is it here? The sum of n i.i.d. Exp(λ) random variables is $Y = \sum X_i \sim \text{Gamma}(n, \lambda)$, and for $n \geq 2$ a direct integral shows $\mathbb{E}[1/Y] = \lambda/(n-1)$ (Problem 5). So

$$\mathbb{E}\left[\hat{\lambda}_n\right] = \mathbb{E}\left[\frac{n}{Y}\right] = \frac{n}{n-1}\lambda. \tag{12.17}$$

The bias goes away as $n \to \infty$, but we can also just fix it for every n by scaling,

$$\lambda'_n = \frac{n-1}{n}\hat{\lambda}_n = \frac{n-1}{\sum_{i=1}^n X_i}, \tag{12.18}$$

which is unbiased. Did the fix break the asymptotic normality? Using the same trick as with the uniform,

$$\begin{aligned}
\sqrt{n}\,(\lambda'_n - \lambda) &= \sqrt{n}\left(\lambda'_n - \hat{\lambda}_n\right) + \sqrt{n}\left(\hat{\lambda}_n - \lambda\right) \\
&= \sqrt{n}\left(\frac{n-1}{n} - 1\right)\hat{\lambda}_n + \sqrt{n}\left(\hat{\lambda}_n - \lambda\right) \\
&= -\frac{\hat{\lambda}_n}{\sqrt{n}} + \sqrt{n}\left(\hat{\lambda}_n - \lambda\right).
\end{aligned} \tag{12.19}$$

The first term goes to 0 in probability, since $\hat{\lambda}_n \xrightarrow{p} \lambda$ and $1/\sqrt{n} \to 0$, and the second term converges to $\mathcal{N}(0, \lambda^2)$. So by Slutsky, $\sqrt{n}(\lambda'_n - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda^2)$. The bias correction is invisible at the $\sqrt{n}$ scale, which is typical. Biases of order $1/n$ get swamped by random errors of order $1/\sqrt{n}$.

Sometimes there's no fix at all. The sample standard deviation $S_n = \sqrt{S_n^2}$ is the natural estimator of σ, but $\sqrt{\cdot}$ is strictly concave, so Jensen gives

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sqrt{S_n^2}\right] < \sqrt{\mathbb{E}[S_n^2]} = \sigma, \tag{12.20}$$

and S_n underestimates σ on average, even though S_n^2 is unbiased for σ^2. For normal data there's a correction factor that depends on n, but there's no correction that works for every distribution at once.

Problem 5.

Let $Y \sim \text{Gamma}(n, \lambda)$ with density $\frac{\lambda^n}{\Gamma(n)} y^{n-1} e^{-\lambda y}$ for $y > 0$. Show that $\mathbb{E}[1/Y] = \lambda/(n-1)$ for $n \geq 2$. What goes wrong when $n = 1$?

13

Maximum Likelihood

> *"To go after Fisher's early writings is somewhat like going after Jefferson's original draft of the Declaration of Independence, or the Continental Congress' final version because they didn't practice or intend the meaning of the words as written."*
>
> – Some Redditor

In the last chapter we came up with estimators using the Golden Rule, which works, but it's by no means systematic. For the Poisson it was obvious to use $\bar{X}_n$, and for the uniform it was obvious to use the maximum. But what would you do for a family where the parameter isn't the mean or the max of anything? What we want is a recipe that takes any parametric family and spits out an estimator. The most important such recipe is the **maximum likelihood estimator** (MLE). It's not the only one (the method of moments is another), but it has some really nice properties, and in a precise sense it's the best you can do when n is large. For machine learning people (like myself) it shows up everywhere. For classical statisticians, it's deep in their blood. We've already been using it in Part I, in Chapter 2 to justify $\bar{X}$ and $\hat{\sigma}^2$ for normal data, and in Chapters 6 and 8 to fit regressions. Now we'll see why it works.

13.1 The Likelihood

Here's the idea in one sentence. For your estimate of the parameter, use the value of the parameter that assigns the greatest probability to the data you actually observed.

Let's do an example. Suppose we toss a coin 10 times and get 7 heads. The number of heads is Binomial$(10, \theta)$, so the probability of what we saw, as a function of θ, is

$$L(\theta) = \binom{10}{7}\theta^7(1-\theta)^3. \tag{13.1}$$

The MLE principle says to choose the θ that makes this as big as possible. Taking the derivative and setting it equal to zero,

$$\begin{aligned} L'(\theta) &= \binom{10}{7}\left[7\theta^6(1-\theta)^3 - 3\theta^7(1-\theta)^2\right] \\ &= \binom{10}{7}\theta^6(1-\theta)^2\left[7(1-\theta) - 3\theta\right] = 0. \end{aligned} \tag{13.2}$$

For $0 < \theta < 1$ the factors in front aren't zero, so $7(1-\theta) = 3\theta$, and $\hat{\theta} = 7/10$. (You should check that this is a max, not a min, which you can do with the second derivative or just by noticing $L = 0$ at both endpoints.) That's the answer everyone would have guessed, which is reassuring.

In general, let $X_1, \ldots, X_n$ be i.i.d. from $f(x;\theta)$. The **likelihood function** is the joint density (or pmf) of the data, viewed as a function of θ with the data held fixed,

$$L(\theta) = L(\theta; X_1, \ldots, X_n) = \prod_{i=1}^{n} f(X_i;\theta), \tag{13.3}$$

where the product comes from independence. For discrete data, this is literally the probability of the observed data. For continuous data, the probability of any exact point is zero, so we use the density instead, which is the probability per unit volume. Remember, the observations have already been observed, so the only free variable is θ.

Definition. The **maximum likelihood estimator** is

$$\hat{\theta}_n = \arg\max_{\theta\in\Theta} L(\theta). \tag{13.4}$$

Since a product of n terms is a pain to differentiate, we almost always work with the **log-likelihood** $\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(X_i;\theta)$. Since log is strictly increasing, it doesn't move the location of the maximum, and it turns the product into a sum, which lets us use the linearity of the derivative. The derivative $\ell'(\theta)$ is called the **score**, and setting it to zero gives the **likelihood equation** $\ell'(\theta) = 0$.

13.1.1 Some Examples

Poisson. Let $X_1, \ldots, X_n$ be i.i.d. Poisson(λ). Then

$$\ell(\lambda) = \sum_{i=1}^{n} \left[X_i \log \lambda - \lambda - \log\left(X_i!\right) \right], \qquad \ell'(\lambda) = \sum_{i=1}^{n} \left[\frac{X_i}{\lambda} - 1 \right] = 0, \tag{13.5}$$

which gives $\hat{\lambda}_n = \bar{X}_n$. The second derivative is $-\sum X_i/\lambda^2 < 0$, so it's a max. This matches the Golden Rule.

Uniform. But nothing is ever so easy, and this is our cue to turn to our least favorite family. Let $X_1, \ldots, X_n$ be i.i.d. Uniform$(0, \theta)$. The density is $1/\theta$, but only on $[0, \theta]$, and that "only" is where everything goes wrong. If we forget it, we'd write $L(\theta) = \theta^{-n}$, which has no maximum at all. The correct likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} \mathbf{1}\{0 \le X_i \le \theta\} = \begin{cases} \theta^{-n} & \theta \ge \max_i X_i \\ 0 & \theta < \max_i X_i, \end{cases} \tag{13.6}$$

because if any observation is bigger than θ, the data are impossible under that θ. The plot is in Figure 13.1. The likelihood is zero up to $\max X_i$, jumps up, and then decreases forever. There is no point where the derivative is zero. Instead, the maximum is at the jump, so

$$\hat{\theta}_n = \max\{X_1, \ldots, X_n\}, \tag{13.7}$$

which again matches the Golden Rule. The lesson is that "set the derivative to zero" is a shortcut that only works when the likelihood is nice, and the uniform isn't nice.

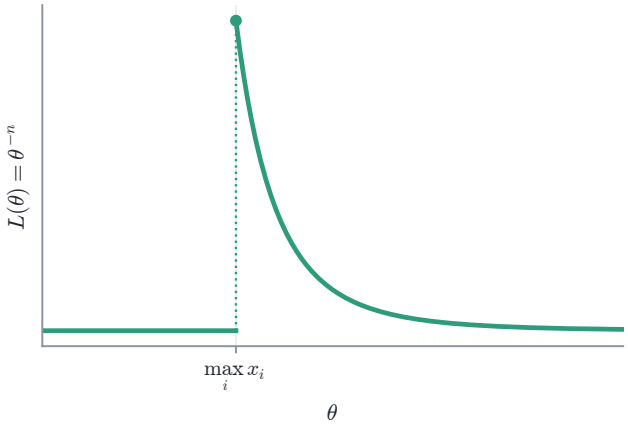


Figure 13.1: The likelihood for Uniform$(0, \theta)$ data. It's zero until θ reaches the largest observation, and decreasing after that, so the maximum is at the jump.

Laplace. Let $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$, which is a symmetric, pointy distribution centered at θ. Then $\ell(\theta) = -n\log 2 - \sum_i |X_i - \theta|$, and except at the data points,

$$\ell'(\theta) = \sum_{i=1}^{n} \operatorname{sgn}(X_i - \theta) = \#\{i : X_i > \theta\} - \#\{i : X_i < \theta\}. \tag{13.8}$$

This is zero (or changes sign) when there are as many observations above θ as below it, which means θ is a median of the data. So the MLE is the **sample median**, $\hat{\theta}_n = \text{Median}\{X_1, \ldots, X_n\}$. This is the first time the MLE has given us something different from the Golden Rule, which would have said "θ is the mean, so use $\bar{X}_n$." We'll see shortly that the MLE wins.

Problem 1.

Find the MLE of the parameter for each of the following families, and check that it's a maximum.

(1) Bernoulli(θ), and Exp(λ) with density $\lambda e^{-\lambda x}$.
(2) $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown. (You should get $\bar{X}_n$ and $\hat{\sigma}_n^2$, the biased one.)
(3) Geometric(p) on $\{1, 2, \ldots\}$.
(4) The shifted exponential $f(x;\theta) = e^{-(x-\theta)}$ for $x \geq \theta$. (Careful, this one is like the uniform.)

13.2 Regularity Conditions

The examples show that sometimes the likelihood is as well behaved as the Poisson, and sometimes it isn't. To guarantee that the MLE is easy to compute and has good properties, we need some assumptions on the family. These are called the **regularity conditions**.

R0. Identifiability. Different parameters give different distributions, so $\theta_1 \neq \theta_2$ implies $f(\cdot\,;\theta_1) \neq f(\cdot\,;\theta_2)$. If two parameters produce the same distribution, no amount of data can tell them apart. The Poisson is identifiable, since $f(0;\lambda) = e^{-\lambda}$ is different for different λ.

R1. Open parameter space. Θ is an open set, so it contains none of its endpoints. Every family in our list is fine. A Poisson with the restriction $\lambda \geq 4$ is not, and its MLE is $\max\{4, \bar{X}_n\}$, which sits on the boundary whenever $\bar{X}_n < 4$.

R2. Parameter-free support. The support $\{x : f(x;\theta) > 0\}$ is the same for every θ. The Poisson has support $\mathbb{N}_0$ for every λ, which is fine. The uniform has support $[0, \theta]$, which

moves with θ, so it fails.

R3. Smoothness. For every x, $f(x;\theta)$ is differentiable in θ, nicely enough that we can swap derivatives and integrals. The Poisson is fine. The Laplace fails at $\theta = x$, where $|x-\theta|$ has a corner.

R4. Twice differentiable. The same as R3, but for the second derivative $\partial_\theta^2 f(x;\theta)$.

A family satisfying R0 through R4 is called a **very regular family**. All of the exponential-family distributions (Bernoulli, Poisson, exponential, normal) are very regular. The uniform fails R2, and that's ultimately the reason it keeps misbehaving.

13.3 Consistency Of The MLE

Why should maximizing the likelihood lead you to the truth? Here's the key fact, which is basically our license to practice MLE.

Theorem (License to practice MLE). Assume R0 and R2, let θ_0 be the true parameter, and let $\theta_1 \neq \theta_0$ be any other parameter. Then

$$\lim_{n\to\infty} \mathbb{P}_{\theta_0}\left(L(\theta_1) < L(\theta_0)\right) = 1. \tag{13.9}$$

In words, with enough data the true parameter beats any specific wrong parameter in the likelihood contest, with probability going to one. To prove it, rewrite the event using logs,

$$L(\theta_1) < L(\theta_0) \iff \frac{1}{n}\sum_{i=1}^{n} \log \frac{f(X_i;\theta_1)}{f(X_i;\theta_0)} < 0. \tag{13.10}$$

Call the left side V_n. It's a sample mean of i.i.d. random variables, so by the law of large numbers,

$$V_n \xrightarrow{p} c = \mathbb{E}_{\theta_0}\left[\log \frac{f(X;\theta_1)}{f(X;\theta_0)}\right]. \tag{13.11}$$

Now we use Jensen. The log is strictly concave, and the ratio $f(X;\theta_1)/f(X;\theta_0)$ isn't constant. (If it were constant, say k, then integrating $f(x;\theta_1) = kf(x;\theta_0)$ over x gives $k = 1$, so the two densities would be the same, which R0 rules out.) So

$$c < \log \mathbb{E}_{\theta_0}\left[\frac{f(X;\theta_1)}{f(X;\theta_0)}\right] = \log \int \frac{f(x;\theta_1)}{f(x;\theta_0)} f(x;\theta_0)\,dx = \log \int f(x;\theta_1)\,dx = \log 1 = 0, \tag{13.12}$$

where R2 is what lets the integral run over the whole support of $f(\cdot\,;\theta_1)$. So V_n converges in probability to a strictly negative number c. Taking $\varepsilon = -c/2 > 0$, the event $|V_n - c| < \varepsilon$

forces $V_n < c/2 < 0$, so

$$1 \underset{n\to\infty}{\longleftarrow} \mathbb{P}\left(|V_n - c| < \varepsilon\right) \leq \mathbb{P}(V_n < 0) \leq 1, \tag{13.13}$$

and by the squeeze theorem $\mathbb{P}(V_n < 0) \to 1$. The quantity $-c = \mathbb{E}_{\theta_0}[\log(f(X;\theta_0)/f(X;\theta_1))] > 0$ is called the **Kullback–Leibler divergence** between the two distributions, and it shows up everywhere in machine learning. The theorem says that the log-likelihood ratio grows like n times the KL divergence, which is why the likelihood is so good at telling distributions apart.

From the license, it's a short step to consistency.

Theorem (Consistency of the MLE). Under R0 through R3, with probability going to one there is a root $\hat{\theta}_n$ of the likelihood equation $\ell'(\theta) = 0$ with $\hat{\theta}_n \xrightarrow{p} \theta_0$.

Here's the idea of the proof. Fix a small $\delta > 0$ such that $[\theta_0 - \delta, \theta_0 + \delta] \subset \Theta$, which is possible because Θ is open (R1). By the license, with probability going to one, both $\ell(\theta_0 - \delta) < \ell(\theta_0)$ and $\ell(\theta_0 + \delta) < \ell(\theta_0)$. On that event, the continuous function ℓ is bigger in the middle of the interval than at both ends, so it has a local maximum strictly inside, and since ℓ is differentiable (R3), the derivative is zero there. That root is within δ of θ_0. Since δ was arbitrary, we have consistency.

For the Laplace, R3 fails, so this theorem doesn't directly apply. The sample median is still consistent, but we'd have to prove it another way.

13.3.1 Invariance

One more convenient property. If $\hat{\theta}_n$ is the MLE of θ, what's the MLE of some function $g(\theta)$? The answer is the obvious one.

Lemma (Invariance). If g is one-to-one and $\hat{\theta}_n$ is the MLE of θ, then $g(\hat{\theta}_n)$ is the MLE of $\delta = g(\theta)$.

To see this, reparametrize the family by δ. The likelihood in terms of δ is $\tilde{L}(\delta) = L(g^{-1}(\delta))$. For any δ,

$$\tilde{L}(\delta) = L(g^{-1}(\delta)) \leq L(\hat{\theta}_n) = \tilde{L}(g(\hat{\theta}_n)), \tag{13.14}$$

so $g(\hat{\theta}_n)$ maximizes $\tilde{L}$. (The result holds for functions that aren't one-to-one too, with a suitable definition.) So for the Poisson, the MLE of the standard deviation $\sqrt{\lambda}$ is $\sqrt{\bar{X}_n}$, and for the normal, the MLE of σ is $\hat{\sigma}_n$. Combined with the variance stabilizing transformation from the last chapter, $\sqrt{n}(\sqrt{\bar{X}_n} - \sqrt{\lambda}) \xrightarrow{d} \mathcal{N}(0, 1/4)$, we get an approximate confidence interval for the Poisson standard deviation, $\sqrt{\bar{X}_n} \pm z_{\alpha/2}/(2\sqrt{n})$.

13.4 Fisher Information

Some likelihoods have a sharp peak, and some are flat near the top. A sharp peak means the data strongly prefer one value of θ, and a flat top means lots of values fit almost equally well. The curvature of the log-likelihood at the peak is a measure of how much information the data carry about θ, which motivates the following definition. For a very regular family, the **Fisher information** in one observation is

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X;\theta) \right]. \tag{13.15}$$

Since the log-likelihood of n i.i.d. observations is a sum, the information in the whole sample is $I_n(\theta) = -\mathbb{E}[\ell''(\theta)] = nI(\theta)$. Information adds up over independent observations, which is exactly what you'd want from something called information. Before we do examples, let's establish two identities about the score of a single observation, $U(\theta) = \frac{\partial}{\partial \theta} \log f(X;\theta)$.

Identity 1. The score has mean zero. Start with $\int f(x;\theta)\,dx = 1$ and differentiate with respect to θ. By R3 we can move the derivative inside, and then we multiply and divide by f and use the fact that the derivative of a log is the derivative of the argument divided by the argument,

$$0 = \int \frac{\partial f(x;\theta)}{\partial \theta}\,dx = \int \frac{\partial \log f(x;\theta)}{\partial \theta} f(x;\theta)\,dx = \mathbb{E}_\theta\left[U(\theta)\right]. \tag{13.16}$$

(We already saw a version of this for the exponential family in Chapter 8, where it said $\mathbb{E}[Y] = b'(\theta)$.)

Identity 2. Information is the variance of the score. Differentiate the middle expression of Identity 1 again, using the product rule and the same trick on $\partial f/\partial\theta$,

$$\begin{aligned} 0 &= \int \frac{\partial^2 \log f(x;\theta)}{\partial \theta^2} f(x;\theta)\,dx + \int \frac{\partial \log f(x;\theta)}{\partial \theta} \frac{\partial f(x;\theta)}{\partial \theta}\,dx \\ &= \mathbb{E}_\theta\left[\frac{\partial^2}{\partial \theta^2} \log f(X;\theta) \right] + \mathbb{E}_\theta\left[U(\theta)^2\right]. \end{aligned} \tag{13.17}$$

So $I(\theta) = \mathbb{E}_\theta[U(\theta)^2]$, and since the score has mean zero, this is the same as

$$I(\theta) = \mathrm{Var}_\theta\left[\frac{\partial}{\partial \theta} \log f(X;\theta) \right] \geq 0. \tag{13.18}$$

So we have two formulas for the information, minus the expected curvature and the variance of the score. Use whichever one is easier.

Problem 2.

(1) Prove both identities for a discrete random variable.
(2) Show that $I(\theta) = 0$ for all $\theta \in \Theta$ is impossible for a very regular family. (Hint: if the score had variance zero, it would be the constant 0. What would that say about $f(x;\theta)$ as a function of θ, and which regularity condition does that break?)

Let's do some examples.

Poisson. $\log f(X;\lambda) = -\lambda + X\log\lambda - \log(X!)$, so the score is $U = X/\lambda - 1$, and

$$I(\lambda) = \operatorname{Var}\left[\frac{X}{\lambda} - 1\right] = \frac{\operatorname{Var}[X]}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}. \tag{13.19}$$

Bernoulli. $\log f(X;p) = X\log p + (1-X)\log(1-p)$, so

$$\frac{\partial}{\partial p}\log f = \frac{X}{p} - \frac{1-X}{1-p}, \qquad \frac{\partial^2}{\partial p^2}\log f = -\frac{X}{p^2} - \frac{1-X}{(1-p)^2}, \tag{13.20}$$

and taking minus the expectation,

$$I(p) = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}. \tag{13.21}$$

Look at how this behaves as p varies. The smallest value is $I(1/2) = 4$, and it blows up as $p \to 0$ or $p \to 1$. So a coin flip tells you the least about p when the coin is fair. That might seem backwards at first, but think about it this way. If $p = 0.001$, then the variance of $\hat{p}$ is tiny ($p(1-p)/n$), so a handful of flips pins down p very precisely in absolute terms. When $p = 1/2$, the outcomes are as unpredictable as possible, and each flip moves $\hat{p}$ around a lot.

Laplace. The score is $U = \operatorname{sgn}(X - \theta)$, which is ± 1, so $U^2 = 1$ and $I(\theta) = \mathbb{E}[U^2] = 1$. (The curvature formula doesn't work here, since $\log f$ has a corner, but the variance formula does.)

13.5 The Cramér–Rao Lower Bound

Between two unbiased estimators of θ, we'd prefer the one with the smaller variance. So how small can the variance of an unbiased estimator possibly be? The answer is given by the information.

Theorem (Cramér–Rao). Let $\mathcal{F}$ be very regular. If $T_n = T(X_1, \dots, X_n)$ is unbiased for

θ, then

$$\operatorname{Var}_\theta[T_n] \geq \frac{1}{nI(\theta)}. \tag{13.22}$$

The right side is called the **Cramér–Rao lower bound** (CRLB). To prove it, write unbiasedness as an integral against the likelihood, $\theta = \mathbb{E}_\theta[T_n] = \int T(\mathbf{x})L(\theta;\mathbf{x})\,d\mathbf{x}$, and differentiate both sides with respect to θ, using the same log trick,

$$\begin{aligned} 1 = \frac{d}{d\theta}\int T(\mathbf{x})L(\theta;\mathbf{x})\,d\mathbf{x} &= \int T(\mathbf{x})\frac{\partial L(\theta;\mathbf{x})}{\partial\theta}\,d\mathbf{x} \\ &= \int T(\mathbf{x})\frac{\partial \ell(\theta;\mathbf{x})}{\partial\theta}L(\theta;\mathbf{x})\,d\mathbf{x} \\ &= \mathbb{E}_\theta\left[T_n\,\ell'(\theta)\right]. \end{aligned} \tag{13.23}$$

The score $\ell'(\theta)$ has mean zero (it's a sum of single-observation scores), so this expectation is a covariance, $1 = \operatorname{Cov}_\theta[T_n, \ell'(\theta)]$. Now Cauchy–Schwarz (10.12) gives

$$1 = \operatorname{Cov}_\theta\left[T_n, \ell'(\theta)\right]^2 \leq \operatorname{Var}_\theta[T_n]\operatorname{Var}_\theta\left[\ell'(\theta)\right] = \operatorname{Var}_\theta[T_n]\cdot nI(\theta), \tag{13.24}$$

since the variance of a sum of n i.i.d. scores is $nI(\theta)$. Dividing through gives the bound. The bound makes sense of the name "information". The more information each observation carries, the smaller the variance an estimator can (potentially) achieve.

Definition. The **efficiency** of an unbiased estimator is

$$e_n(\theta) = \frac{\text{CRLB}}{\operatorname{Var}_\theta\left[\hat\theta_n\right]} = \frac{1}{nI(\theta)\operatorname{Var}_\theta\left[\hat\theta_n\right]} \leq 1. \tag{13.25}$$

An estimator is **efficient** if $e_n(\theta) = 1$ for all n and θ, and **asymptotically efficient** if $e_n(\theta) \to 1$.

For the Poisson, $\operatorname{Var}[\bar X_n] = \lambda/n$ and $I(\lambda) = 1/\lambda$, so $e_n = \frac{1}{n(1/\lambda)(\lambda/n)} = 1$. The sample mean is efficient, and no unbiased estimator of λ does better. For the Laplace (which isn't very regular, but let's compare against $1/(nI)$ anyway), the distribution is symmetric about θ, so $\bar X_n$ is unbiased. Its variance is $\operatorname{Var}[X]/n$, where

$$\operatorname{Var}[X] = \frac{1}{2}\int_{-\infty}^{\infty}(x-\theta)^2 e^{-|x-\theta|}\,dx = \int_0^\infty u^2 e^{-u}\,du = 2, \tag{13.26}$$

using $u = x - \theta$ and the symmetry to get rid of the $1/2$. The last integral is the second moment of an Exp(1), which we know is 2. So the efficiency of the sample mean is $e_n = 1/(n\cdot 1\cdot 2/n) = 1/2$. The sample mean wastes half of the information. The MLE, the median, has asymptotic variance $1/n$ (Problem 4), which matches the bound. This is a nice

example of the MLE beating the Golden Rule.

13.6 Asymptotic Normality Of The MLE

The MLE is more than just a convenient recipe. Here's the theorem that makes it, in some ways, the best estimator one could hope for.

Theorem. Let $\mathcal{F}$ be very regular (plus a mild condition on the third derivative), and let $\hat{\theta}_n$ be the consistent root of the likelihood equation. Then

$$\sqrt{n}\left(\hat{\theta}_n - \theta_0\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right). \tag{13.27}$$

So the MLE is CAN, and its asymptotic variance is exactly the Cramér–Rao bound. In other words, the MLE is asymptotically efficient. No (regular) unbiased estimator can do better in the limit.

The proof is a Taylor expansion of the score around the true θ_0. Since $\ell'(\hat{\theta}_n) = 0$,

$$0 = \ell'(\hat{\theta}_n) \approx \ell'(\theta_0) + \ell''(\theta_0)\left(\hat{\theta}_n - \theta_0\right), \tag{13.28}$$

which is a good approximation because $\hat{\theta}_n$ is consistent. Solving for the error and multiplying by $\sqrt{n}$, and splitting the $\sqrt{n}$ cleverly between the top and the bottom,

$$\sqrt{n}\left(\hat{\theta}_n - \theta_0\right) \approx -\frac{\ell'(\theta_0)/\sqrt{n}}{\ell''(\theta_0)/n} = \frac{A_n}{B_n}. \tag{13.29}$$

The numerator is $\sqrt{n}$ times the sample mean of the single-observation scores U_i, which are i.i.d. with mean 0 and variance $I(\theta_0)$ by our two identities. So by the CLT,

$$A_n = -\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} U_i - 0\right) \xrightarrow{d} \mathcal{N}\left(0, I(\theta_0)\right). \tag{13.30}$$

The denominator is minus a sample mean of the second derivatives, so by the law of large numbers,

$$B_n = -\frac{1}{n}\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta^2} \log f(X_i;\theta_0) \xrightarrow{p} I(\theta_0). \tag{13.31}$$

By Slutsky, the ratio converges in distribution to $\mathcal{N}(0, I(\theta_0))/I(\theta_0)$, which is $\mathcal{N}(0, 1/I(\theta_0))$. And we're done!

This theorem is what was secretly behind the Wald intervals in Part I. It says the MLE is

approximately $\mathcal{N}(\theta, 1/(nI(\theta)))$, and replacing $I(\theta)$ by $I(\hat{\theta}_n)$ (or by the observed curvature $-\ell''(\hat{\theta}_n)/n$), Slutsky gives the approximate $100(1-\alpha)\%$ confidence interval

$$\hat{\theta}_n \pm \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta}_n)}}. \tag{13.32}$$

For the Bernoulli, $1/(nI(\hat{p})) = \hat{p}(1-\hat{p})/n$, and this is the Wald interval for a proportion from Chapter 2. For logistic regression, the matrix version $(\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}$ is exactly the inverse of the Fisher information matrix, which is where the standard errors in Chapter 8 came from.

Problem 3.

Let $X_1, \ldots, X_n$ be i.i.d. Exp(λ) with density $\lambda e^{-\lambda x}$.

(1) Find $I(\lambda)$, and check that the asymptotic variance of the MLE $1/\bar{X}_n$ from the delta method in Chapter 12 equals $1/I(\lambda)$.

(2) Compute the efficiency of the unbiased estimator $\lambda'_n = (n-1)/\sum X_i$ for finite n. (You'll need $\mathbb{E}[1/Y^2]$ for $Y \sim \text{Gamma}(n, \lambda)$.) Is it efficient? Is it asymptotically efficient?

Problem 4.

For a continuous distribution with median m and density f with $f(m) > 0$, the sample median satisfies $\sqrt{n}(\widetilde{X}_n - m) \xrightarrow{d} \mathcal{N}(0, 1/(4f(m)^2))$. Use this to show the sample median is asymptotically efficient for the Laplace family. Then compute the asymptotic efficiency of the sample median relative to the sample mean for normal data. Which would you use for normal data? For Laplace data?

Problem 5.

Simulate 5,000 samples of size $n = 30$ from a Bernoulli(p) with $p = 0.1$. For each one, compute the Wald interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ and check whether it contains p. What is the actual coverage? Compare with the plus-four interval from Chapter 2, and explain the difference using the fact that $I(p)$ changes quickly near $p = 0$.

14

Sufficiency And The MVUE

> *"The statistic chosen should summarise the whole of the relevant information supplied by the sample."*
>
> – R. A. Fisher, *On the Mathematical Foundations of Theoretical Statistics*

The Cramér–Rao bound told us the smallest variance an unbiased estimator could possibly have, and the MLE gets there asymptotically. But two questions are left over. First, what about finite n? The MLE is often biased, and even when we fix the bias, the result might not hit the bound. Second, what about families like the uniform, which aren't regular and where Cramér–Rao doesn't even apply? In this chapter we'll find a completely different route to the best unbiased estimator, one that works for any n and doesn't need regularity. The idea is to throw away everything in the data that doesn't carry information about θ, and the tool for doing that is conditional expectation.

14.1 Conditional Expectation

We'll need two facts from probability. For random variables X and Y, the conditional expectation $\mathbb{E}[X \mid Y]$ is a function of Y (so it's itself a random variable), defined by $\mathbb{E}[X \mid Y = y] = \int x\, f(x \mid y)\, dx$ with $f(x \mid y) = f(x, y)/f_Y(y)$.

Law of total expectation. $\mathbb{E}\left[\mathbb{E}[X \mid Y]\right] = \mathbb{E}[X]$.

To see that this is true, write out the outer expectation as an integral against the density of

Y, and use the definition of the conditional density,

$$\begin{aligned}\mathbb{E}\left[\mathbb{E}[X \mid Y]\right] &= \int f_Y(y) \left(\int x \, \frac{f(x,y)}{f_Y(y)} \, dx \right) dy \\ &= \int \int x \, f(x,y) \, dx \, dy \\ &= \int x \, f_X(x) \, dx = \mathbb{E}[X], \end{aligned} \tag{14.1}$$

where in the last line we integrated y out of the joint density to get the marginal of X. In words, averaging the conditional averages gives you the overall average. If this gives such a neat result for the expectation, perhaps we wouldn't be wrong to expect something similar for the variance.

Law of total variance. $\mathrm{Var}[X] = \mathbb{E}\left[\mathrm{Var}[X \mid Y]\right] + \mathrm{Var}\left[\mathbb{E}[X \mid Y]\right]$.

To prove it, let $\mu = \mathbb{E}[X]$, and pivot (add and subtract $\mathbb{E}[X \mid Y]$) inside the square,

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}\Big[\big((X - \mathbb{E}[X \mid Y]) + (\mathbb{E}[X \mid Y] - \mu)\big)^2\Big] \\ &= \underbrace{\mathbb{E}\Big[(X - \mathbb{E}[X \mid Y])^2\Big]}_{\alpha} + \underbrace{\mathbb{E}\Big[(\mathbb{E}[X \mid Y] - \mu)^2\Big]}_{\beta} + 2\underbrace{\mathbb{E}\left[(\mathbb{E}[X \mid Y] - \mu)\,(X - \mathbb{E}[X \mid Y])\right]}_{\gamma}. \end{aligned} \tag{14.2}$$

By the law of total expectation, β is the variance of the random variable $\mathbb{E}[X \mid Y]$ (whose mean is μ), and α is $\mathbb{E}[\mathrm{Var}[X \mid Y]]$ after conditioning on Y inside. So the claim is that $\gamma = 0$. Condition on Y. Given $Y = y$, the factor $\mathbb{E}[X \mid Y = y] - \mu$ is just a constant, so it comes out of the inner expectation,

$$\gamma = \mathbb{E}\Big[\,(\mathbb{E}[X \mid Y] - \mu)\ \underbrace{\mathbb{E}\left[X - \mathbb{E}[X \mid Y] \mid Y\right]}_{=0}\Big] = 0, \tag{14.3}$$

because the average distance from the average is zero. And we're done. The important consequence for us is that both terms are nonnegative, so

$$\mathrm{Var}\left[\mathbb{E}[X \mid Y]\right] \leq \mathrm{Var}[X], \tag{14.4}$$

with equality only if $\mathrm{Var}[X \mid Y] = 0$, meaning X is already a function of Y. Conditioning can only reduce variance.

14.2 Rao–Blackwell

Now we can see where this is going. Suppose $\hat{\theta}_n$ is an unbiased estimator of θ, and Y is some statistic. Consider the new estimator

$$\theta'_n = \mathbb{E}_\theta \left[\hat{\theta}_n \mid Y \right]. \tag{14.5}$$

By the law of total expectation, $\mathbb{E}_\theta[\theta'_n] = \mathbb{E}_\theta[\hat{\theta}_n] = \theta$, so it's still unbiased. And by (14.4), $\mathrm{Var}_\theta[\theta'_n] \le \mathrm{Var}_\theta[\hat{\theta}_n]$. So conditioning made the estimator better for free. That seems too good to be true, and there is a catch. We put θ in the subscript of the expectation because the conditional expectation might depend on θ, and if it does, then θ'_n isn't an estimator at all, since we can't compute it without knowing θ.

Here's an example where it fails. Let $n = 2$ and X_1, X_2 i.i.d. Exp with mean β, so $\bar{X}_2$ is unbiased for β. Condition on $Y = X_2$,

$$\mathbb{E}_\beta \left[\frac{X_1 + X_2}{2} \;\middle|\; X_2 \right] = \frac{1}{2} \left(\mathbb{E}_\beta[X_1 \mid X_2] + X_2 \right) = \frac{1}{2} \left(\beta + X_2 \right), \tag{14.6}$$

where $\mathbb{E}[X_1 \mid X_2] = \mathbb{E}[X_1] = \beta$ by independence. This depends on β, so it isn't an estimator. The problem is that $Y = X_2$ throws away the information in X_1, and to fill the gap the conditional expectation has to use the true parameter. For conditioning to work, we need a Y that keeps *all* of the information about θ, so that once you know Y, the rest of the data has nothing left to say about θ.

14.2.1 Sufficient Statistics

Definition. A statistic $Y = q(X_1, \ldots, X_n)$ is **sufficient** for θ if the conditional distribution of $(X_1, \ldots, X_n)$ given Y does not depend on θ.

If Y is sufficient, then $\mathbb{E}_\theta[\hat{\theta}_n \mid Y]$ is computed from a θ-free conditional distribution, so it's θ-free, and we get the following.

Theorem (Rao–Blackwell). If $\hat{\theta}_n$ is unbiased for θ and Y is sufficient for θ, then $\theta'_n = \mathbb{E}[\hat{\theta}_n \mid Y]$ is an estimator, it is unbiased for θ, and $\mathrm{Var}_\theta[\theta'_n] \le \mathrm{Var}_\theta[\hat{\theta}_n]$ for every θ.

Let's check that the sum is sufficient for Bernoulli data. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(θ) and $Y = \sum X_i \sim \mathrm{Bin}(n, \theta)$. If $\sum x_i = y$, then the event $\{X_1 = x_1, \ldots, X_n = x_n\}$ is contained in the event $\{Y = y\}$, so the intersection is just the first event, and

$$\mathbb{P}_\theta \left(X_1 = x_1, \ldots, X_n = x_n \mid Y = y \right) = \frac{\prod_i \theta^{x_i}(1-\theta)^{1-x_i}}{\binom{n}{y}\theta^y(1-\theta)^{n-y}} = \frac{\theta^y(1-\theta)^{n-y}}{\binom{n}{y}\theta^y(1-\theta)^{n-y}} = \frac{1}{\binom{n}{y}}. \tag{14.7}$$

Everything dies! Given that there were y successes, every arrangement of them is equally likely, no matter what θ is. So the order of the successes carries no information about θ, and the count Y is sufficient. Now Rao–Blackwellize the silly unbiased estimator $\hat{\theta} = X_1$ (just use the first flip). By symmetry, $\mathbb{E}[X_1 \mid Y] = \mathbb{E}[X_i \mid Y]$ for every i, and these n things add up to $\mathbb{E}[Y \mid Y] = Y$, so

$$\mathbb{E}[X_1 \mid Y] = \frac{Y}{n} = \bar{X}_n. \tag{14.8}$$

Rao–Blackwell turned the worst reasonable estimator into the sample proportion, and cut the variance from $\theta(1-\theta)$ to $\theta(1-\theta)/n$.

14.3 The Factorization Theorem

If we step back, checking sufficiency from the definition is extremely tedious. Even when the data are as simple as Bernoulli, the calculation isn't trivial, and for continuous data conditioning on Y is worse. Fortunately, Neyman gave us a shortcut.

Theorem (Neyman's factorization). $Y = q(X_1, \dots, X_n)$ is sufficient for θ if and only if the likelihood can be factored as

$$L(\theta; x_1, \dots, x_n) = \alpha\big(\theta, q(x_1, \dots, x_n)\big)\, \beta(x_1, \dots, x_n) \tag{14.9}$$

for some nonnegative functions α and β.

In words, the part of the likelihood that involves θ can only see the data through Y, and the part that sees the rest of the data can't involve θ. We'll prove the discrete case. Write $\mathbf{x} = (x_1, \dots, x_n)$ and $y = q(\mathbf{x})$.

($\Rightarrow$) If Y is sufficient, then

$$L(\theta; \mathbf{x}) = \mathbb{P}_\theta(\mathbf{X} = \mathbf{x}) = \underbrace{\mathbb{P}(\mathbf{X} = \mathbf{x} \mid Y = y)}_{\beta(\mathbf{x})}\, \underbrace{\mathbb{P}_\theta(Y = y)}_{\alpha(\theta, y)}, \tag{14.10}$$

where the first factor doesn't depend on θ precisely because Y is sufficient.

($\Leftarrow$) Suppose the factorization holds. Then, using the fact that $\{\mathbf{X} = \mathbf{x}\} \subset \{Y = y\}$, and summing over all data sets $\mathbf{t}$ with $q(\mathbf{t}) = y$ to get the probability of $\{Y = y\}$,

$$\begin{aligned} \mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid Y = y) &= \frac{\mathbb{P}_\theta(\mathbf{X} = \mathbf{x})}{\mathbb{P}_\theta(Y = y)} = \frac{\alpha(\theta, y)\beta(\mathbf{x})}{\sum_{\mathbf{t}:\, q(\mathbf{t}) = y} \alpha(\theta, y)\beta(\mathbf{t})} \\ &= \frac{\beta(\mathbf{x})}{\sum_{\mathbf{t}:\, q(\mathbf{t}) = y} \beta(\mathbf{t})}, \end{aligned} \tag{14.11}$$

since $\alpha(\theta, y)$ is the same in every term of the sum and cancels. This is free of θ, so Y is sufficient.

Now sufficiency is just a matter of staring at the likelihood. Let's do some examples.

Poisson. The likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \underbrace{\lambda^{\sum x_i} e^{-n\lambda}}_{\alpha(\lambda, y)} \underbrace{\frac{1}{\prod x_i!}}_{\beta(\mathbf{x})}, \tag{14.12}$$

so $Y = \sum X_i$ is sufficient for λ.

Uniform. Here we have to keep track of the support with an indicator, $f(x;\theta) = \frac{1}{\theta}\mathbf{1}\{0 \le x \le \theta\}$. All n indicators equal 1 exactly when the smallest observation is at least 0 and the largest is at most θ, so

$$L(\theta) = \frac{1}{\theta^n} \prod_{i=1}^{n} \mathbf{1}\{0 \le x_i \le \theta\} = \underbrace{\theta^{-n} \mathbf{1}\left\{\max_i x_i \le \theta\right\}}_{\alpha(\theta, y)} \underbrace{\mathbf{1}\left\{\min_i x_i \ge 0\right\}}_{\beta(\mathbf{x})}, \tag{14.13}$$

and $Y = \max_i X_i$ is sufficient for θ.

The Hieu distribution. Now let's look at (undeniably) the most important distribution of all time, the shifted exponential $f(x;\theta) = e^{-(x-\theta)}\mathbf{1}\{x \ge \theta\}$. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i - \theta)} \mathbf{1}\{x_i \ge \theta\} = \underbrace{e^{n\theta}\, \mathbf{1}\left\{\min_i x_i \ge \theta\right\}}_{\alpha(\theta, y)} \underbrace{e^{-\sum x_i}}_{\beta(\mathbf{x})}, \tag{14.14}$$

so the sample minimum is sufficient for θ.

Problem 1.

Use the factorization theorem to find a sufficient statistic for each family.

(1) Exp(λ) and Geometric(p).

(2) $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Then with both unknown, where the sufficient statistic is a pair $(\sum X_i, \sum X_i^2)$.

(3) Any one-parameter exponential family from Chapter 8, $f(y;\theta) = \exp[(y\theta - b(\theta))/a(\phi) + c(y, \phi)]$ with ϕ known.

Explain why the order statistics $(X_{(1)}, \ldots, X_{(n)})$ are always sufficient, for any family.

14.3.1 MLE And Sufficiency

If we look at the examples, the sufficient statistics were $\sum X_i$, $\max X_i$ and $\min X_i$, and the MLEs were $\bar{X}_n$, $\max X_i$ and (as you found in Problem 1 of Chapter 13) $\min X_i$. Is it a coincidence that the MLE always seems to be a function of the sufficient statistic? No.

Lemma 1. If Y is sufficient for θ and δ is one-to-one, then $\delta(Y)$ is also sufficient.

Lemma 2. If Y is sufficient and the MLE is unique, then the MLE is a function of Y.

For Lemma 1, let $T = \delta(Y)$, so $Y = \delta^{-1}(T)$, and substitute into the factorization, $L = \alpha(\theta, \delta^{-1}(t))\beta(\mathbf{x}) = \tilde{\alpha}(\theta, t)\beta(\mathbf{x})$, which is a factorization through T. For Lemma 2, write $L(\theta) = \alpha(\theta, Y)\beta(\mathbf{X})$. The factor $\beta(\mathbf{X})$ doesn't depend on θ and is nonnegative, so it scales the likelihood without moving the location of the maximum (when $\beta > 0$, which it is for any data we could actually observe). So

$$\hat{\theta}_n = \arg\max_{\theta} \alpha(\theta, Y), \tag{14.15}$$

and this depends on the data only through Y. For example, for Exp(λ) data, $\hat{\lambda}_n = n/\sum X_i$ is a one-to-one function of $\sum X_i$, so by Lemma 1 the MLE is itself sufficient.

14.4 Completeness And The MVUE

Now we can state what we're after, which is one of the climaxes of this book.

Definition. An unbiased estimator $\hat{\theta}_n$ is a **minimum variance unbiased estimator** (MVUE) for θ if for every other unbiased estimator $\tilde{\theta}_n$,

$$\mathrm{Var}_\theta\left[\hat{\theta}_n\right] \le \mathrm{Var}_\theta\left[\tilde{\theta}_n\right] \quad \text{for every } \theta \in \Theta. \tag{14.16}$$

If the family is very regular and $\hat{\theta}_n$ is efficient (it hits the Cramér–Rao bound), then it's automatically an MVUE. But the converse is false. Lots of families have an MVUE whose variance is strictly above the bound for every finite n, and families like the uniform don't have a Cramér–Rao bound at all. So we need another way to certify that an estimator is the best, and it comes from one more property.

Rao–Blackwell says that conditioning on a sufficient statistic improves any unbiased estimator. But different starting estimators could, in principle, Rao–Blackwellize to different functions of Y. What we need is a guarantee that there's only one unbiased function of Y to land on.

Definition. A statistic Y is **complete** for θ if for every function δ,

$$\mathbb{E}_\theta[\delta(Y)] = 0 \text{ for all } \theta \in \Theta \quad \Longrightarrow \quad \mathbb{P}_\theta(\delta(Y) = 0) = 1 \text{ for all } \theta. \tag{14.17}$$

In words, the only function of Y that has mean zero no matter what θ is, is the zero function ("almost surely", meaning it can be nonzero on a set of probability zero). To check completeness, we need the distribution of Y.

Poisson. $Y = \sum X_i \sim \text{Poisson}(n\lambda)$. Suppose $\mathbb{E}_\lambda[\delta(Y)] = 0$ for every $\lambda > 0$. Then

$$0 = \sum_{k=0}^{\infty} \delta(k) e^{-n\lambda} \frac{(n\lambda)^k}{k!} \quad \Longrightarrow \quad 0 = \sum_{k=0}^{\infty} \frac{n^k \delta(k)}{k!} \lambda^k \quad \text{for all } \lambda > 0, \tag{14.18}$$

after multiplying by $e^{n\lambda}$. The right side is a power series in λ that is identically zero, so by the uniqueness of power series every coefficient is zero. So $\delta(k) = 0$ for every k, and Y is complete.

Uniform. $Y = \max X_i$ has density $f_Y(y) = n y^{n-1}/\theta^n$ on $[0, \theta]$ (the derivative of $(y/\theta)^n$). Suppose $\mathbb{E}_\theta[\delta(Y)] = 0$ for every $\theta > 0$. Then

$$0 = \int_0^\theta \delta(y) \, \frac{n y^{n-1}}{\theta^n} \, dy \quad \Longrightarrow \quad 0 = \int_0^\theta \delta(y) y^{n-1} \, dy \quad \text{for all } \theta > 0. \tag{14.19}$$

We don't know what δ is, but we can still use the fundamental theorem of calculus. Differentiating both sides with respect to θ gives $0 = \delta(\theta)\theta^{n-1}$, so $\delta(\theta) = 0$ for every $\theta > 0$ (at least wherever δ is continuous, and a bit of measure theory handles the rest). So the max is complete.

> Problem 2.
>
> Show that $Y = \sum X_i$ is complete for i.i.d. Bernoulli(θ) data. (Hint: $\mathbb{E}_\theta[\delta(Y)]$ is $(1-\theta)^n$ times a polynomial in $\theta/(1-\theta)$.) Then give an example of a sufficient statistic that is *not* complete, for instance $Y = (X_1, \sum X_i)$, by finding a nonzero δ with mean zero.

14.4.1 The Lehmann–Scheffé Theorem

Theorem (Lehmann–Scheffé). Suppose Y is sufficient and complete for θ.

(1) If $h_1(Y)$ and $h_2(Y)$ are both unbiased for θ, then $h_1(Y) = h_2(Y)$ almost surely.
(2) If $\hat{\theta}_n$ is any unbiased estimator of θ, then $\theta'_n = \mathbb{E}[\hat{\theta}_n \mid Y]$ is the (essentially unique)

MVUE of θ.

The first part is terribly easy. Let $\delta = h_1 - h_2$. By linearity, $\mathbb{E}_\theta[\delta(Y)] = \theta - \theta = 0$ for every θ, and completeness says $\delta(Y) = 0$ almost surely. For the second part, $\theta'_n = \Psi(Y)$ is an unbiased function of Y (by Rao–Blackwell, using sufficiency). Let θ''_n be any competing unbiased estimator. Rao–Blackwellize it too, $\theta'''_n = \mathbb{E}[\theta''_n \mid Y]$, which is another unbiased function of Y. By part 1 there's only one unbiased function of Y, so $\theta'''_n = \theta'_n$, and then by (14.4),

$$\operatorname{Var}_\theta[\theta'_n] = \operatorname{Var}_\theta\left[\mathbb{E}[\theta''_n \mid Y]\right] \le \operatorname{Var}_\theta\left[\theta''_n\right]. \tag{14.20}$$

So θ'_n beats every competitor, and we're done.

The theorem gives a really simple strategy for finding an MVUE, which we'll call the L–S strategy. Find a complete sufficient statistic Y, then just fool around until you find *any* function of Y that's unbiased. That's it, it's the MVUE. You don't even need to compute a conditional expectation. Let's use it.

Uniform. $Y = \max X_i$ is complete and sufficient, and we know $\mathbb{E}_\theta[Y] = \frac{n}{n+1}\theta$. So

$$\theta'_n = \frac{n+1}{n} \max_i X_i \tag{14.21}$$

is unbiased, and it's a function of Y, so it's the MVUE for θ. No regularity needed.

Exponential. For $\text{Exp}(\lambda)$ data, $Y = \sum X_i$ is sufficient (and complete, which follows from a Laplace transform argument much like the Poisson one). In Chapter 12 we found, using Jensen and the Gamma distribution, that $\frac{n-1}{\sum X_i}$ is unbiased for λ. It's a function of Y, so it's the MVUE. Its variance is $\lambda^2/(n-2)$, which is strictly bigger than the Cramér–Rao bound λ^2/n for every n (you computed this in Problem 3 of Chapter 13). So here is an MVUE that is not efficient. No unbiased estimator of λ hits the bound, and this one is the best you can do.

Poisson. $\bar{X}_n$ is an unbiased function of the complete sufficient statistic $\sum X_i$, so it's the MVUE for λ (which we already knew, since it's efficient). The L–S strategy is more interesting for functions of λ. Say we want to estimate $\mathbb{P}(X = 0) = e^{-\lambda}$, the probability of a quiet day. The crude estimator $\mathbf{1}\{X_1 = 0\}$ is unbiased, and Rao–Blackwellizing it gives the MVUE (Problem 3).

Problem 3.

Let $X_1, \ldots, X_n$ be i.i.d. Poisson(λ) and $Y = \sum X_i$.

(1) Show that given $Y = y$, $X_1 \sim \mathrm{Bin}(y, 1/n)$.
(2) Conclude that the MVUE of $e^{-\lambda}$ is $\left(1 - \frac{1}{n}\right)^Y$.
(3) Compare it with the MLE $e^{-\bar{X}_n}$ (by invariance). Show that the two agree in the limit $n \to \infty$.

Problem 4.

For the Hieu distribution $f(x;\theta) = e^{-(x-\theta)}\mathbf{1}\{x \geq \theta\}$, let $Y = \min_i X_i$.
(1) Show that $Y - \theta \sim \mathrm{Exp}(n)$ (rate n), so $\mathbb{E}[Y] = \theta + 1/n$.
(2) Show that Y is complete, by imitating the uniform argument.
(3) Find the MVUE of θ, and compare its variance with that of the unbiased estimator $\bar{X}_n - 1$.

15

Hypothesis Testing, Revisited

"Without hoping to know whether each separate hypothesis is true or false, we may search for rules to govern our behaviour with regard to them."

– Jerzy Neyman and Egon Pearson, 1933

Now that we've spent a lot of time on estimation theory, we turn our heads back to hypothesis testing. This is no doubt something you've seen before, since we spent all of Chapter 3 on it. But there, every test was handed to us. Someone told us to use the z statistic, or the t statistic, or the χ^2 statistic, and we used it. Here we ask a deeper question. Given a family and a pair of hypotheses, what is the *best* test, and how do we find it? The answer is going to be the likelihood again.

15.1 The Setup

Let $\mathcal{F} = \{f(\cdot\,;\theta) : \theta \in \Theta\}$ be a parametric family, and **decompose** the parameter space into two disjoint, nonempty pieces $\Theta = \Theta_0 \cup \Theta_a$. These are associated with the **null** and **alternative** hypotheses,

$$H_0 : \theta \in \Theta_0 \quad \text{vs.} \quad H_a : \theta \in \Theta_a. \tag{15.1}$$

Based on the data, we either reject H_0 in favor of H_a, or we fail to reject H_0. (If you're super old, like me, you NEVERRRRR say "accept the null." In more modern writing you might see "accept" thrown around, and that's fine, it's just a product of how we were raised.) Since the data are random, we can make mistakes. As in Chapter 3,

- a **Type I error** is rejecting H_0 when it's true, and
- a **Type II error** is failing to reject H_0 when H_a is true.

Formally, a **test** is a **critical region** $C \subset \mathbb{R}^n$. If the data $(X_1, \ldots, X_n)$ land in C, we reject H_0, and otherwise we don't. The **power function** of the test is the probability of rejecting as a function of the true parameter,

$$\gamma(\theta) = \mathbb{P}_\theta \left((X_1, \ldots, X_n) \in C \right). \tag{15.2}$$

For $\theta \in \Theta_0$ this is the probability of a Type I error, and the **size** (or significance level) of the test is the worst case,

$$\alpha = \max_{\theta \in \Theta_0} \gamma(\theta). \tag{15.3}$$

For $\theta \in \Theta_a$, $\gamma(\theta)$ is the **power**, the probability of correctly rejecting, and $\beta(\theta) = 1 - \gamma(\theta)$ is the probability of a Type II error. So γ isn't a Type II error probability, but it contains it.

The trouble is that we can't make both errors small at the same time. The critical region $C = \varnothing$ (never reject) has no Type I errors and no power, and $C = \mathbb{R}^n$ (always reject) has full power and $\alpha = 1$. The Neyman–Pearson approach is to treat the two errors asymmetrically. We fix α ahead of time, and then, among all tests with that size, look for the one with the most power.

15.2 The Neyman–Pearson Lemma

We start with the simplest case, a **simple versus simple** test, where each hypothesis is a single point,

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_a : \theta = \theta_a. \tag{15.4}$$

Definition. A test C of size α is a **most powerful test** (MPT) of size α if $\gamma_C(\theta_a) \geq \gamma_D(\theta_a)$ for every other test D of size at most α.

Here's the natural candidate. The license to practice MLE told us that the likelihood is great at telling distributions apart, so let's reject H_0 when the data are much less likely under θ_0 than under θ_a. The **likelihood ratio** is

$$\mathrm{LR} = \frac{L(\theta_0)}{L(\theta_a)}, \tag{15.5}$$

and the **likelihood ratio test** (LRT) with cutoff $k > 0$ has critical region

$$C_k = \{\mathbf{x} \in \mathbb{R}^n : \mathrm{LR}(\mathbf{x}) < k\}, \tag{15.6}$$

where k is chosen so that $\mathbb{P}_{\theta_0}(\mathrm{LR} < k) = \alpha$. (If $\mathrm{LR} = k$ has positive probability, which can happen for discrete data, we'll need to randomize on that set. We'll get to that.)

Theorem (Neyman–Pearson lemma). The likelihood ratio test C_k of size α is a most powerful test of size α.

We'll prove the continuous case. Let D be any other test of size at most α. We want $\gamma_C - \gamma_D \geq 0$, where $\gamma_C = \int_C L(\theta_a)$ and $\gamma_D = \int_D L(\theta_a)$ (integrals over $\mathbf{x} \in \mathbb{R}^n$). Split both regions along $C \cap D$, which cancels,

$$\gamma_C - \gamma_D = \int_{C \cap D^c} L(\theta_a) - \int_{C^c \cap D} L(\theta_a). \tag{15.7}$$

On C, the likelihood ratio is less than k, so $L(\theta_a) > L(\theta_0)/k$. Off C, the inequality flips, $L(\theta_a) \leq L(\theta_0)/k$. Using these on the two integrals,

$$\begin{aligned} \gamma_C - \gamma_D &\geq \frac{1}{k} \int_{C \cap D^c} L(\theta_0) - \frac{1}{k} \int_{C^c \cap D} L(\theta_0) \\ &= \frac{1}{k} \left(\int_C L(\theta_0) - \int_D L(\theta_0) \right) \\ &= \frac{1}{k} \left(\alpha - \alpha_D \right) \geq 0, \end{aligned} \tag{15.8}$$

where in the middle line we added and subtracted $\int_{C \cap D} L(\theta_0)$, and in the last line we used that the integral of the null likelihood over a critical region is the size of the test, with $\alpha_D \leq \alpha$. And we're done!

The intuition is a knapsack problem. Each little piece of $\mathbb{R}^n$ we put in the critical region "costs" some null probability (we have a budget of α) and "buys" some alternative probability (which we want to maximize). The best deal is to buy the pieces with the best ratio of alternative to null probability first, which is exactly what the LRT does.

15.2.1 The Normal Mean

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\theta, 1)$, and test $H_0 : \theta = 0$ versus $H_a : \theta = 1$ at $\alpha = 0.05$ with $n = 16$. The likelihood ratio is

$$\begin{aligned} \mathrm{LR} = \frac{L(0)}{L(1)} &= \frac{\prod_i \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x_i^2}}{\prod_i \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x_i - 1)^2}} \\ &= \exp\left[-\frac{1}{2} \sum_{i=1}^{n} \left(x_i^2 - (x_i - 1)^2 \right) \right] \end{aligned}$$

$$= \exp\left[\frac{n}{2} - \sum_{i=1}^{n} x_i\right]. \tag{15.9}$$

So the LR depends on the data only through $T = \sum X_i$, which is the sufficient statistic for this family. That's not an accident (Problem 1). And since the LR is a decreasing function of T, "LR is small" is the same as "T is big", so

$$\text{LR} < k \iff T > \frac{n}{2} - \log k \equiv c_n. \tag{15.10}$$

This is a nice feature of the Neyman–Pearson approach. We never actually need k. We just need the direction of the inequality, and then we choose c_n to get the right size. Under H_0, $T \sim \mathcal{N}(0, n)$, so

$$\alpha = \mathbb{P}_0\left(T > c_n\right) = 1 - \Phi\left(\frac{c_n}{\sqrt{n}}\right) \quad \Longrightarrow \quad c_n = \sqrt{n}\, z_\alpha, \tag{15.11}$$

with z_α the upper-tail critical value. So the most powerful test rejects H_0 when

$$T = \sum_{i=1}^{16} X_i > 1.645\sqrt{16} = 6.58. \tag{15.12}$$

For the power, under H_a we have $T \sim \mathcal{N}(n, n)$, so we can write $T = n + \sqrt{n}Z$ and

$$\gamma = \mathbb{P}_1\left(n + \sqrt{n}Z > z_\alpha\sqrt{n}\right) = \mathbb{P}\left(Z > z_\alpha - \sqrt{n}\right) = \Phi\left(\sqrt{n} - z_\alpha\right), \tag{15.13}$$

where the last step is the symmetry of the normal. With $n = 16$, $\gamma = \Phi(4 - 1.645) = 0.9907$. The two distributions of T and the test are shown in Figure 15.1. No other test of size 0.05 based on 16 observations has more power than this.

15.2.2 How Fast Does Power Go To 1?

The formula $\gamma_n = \Phi(\sqrt{n} - z_\alpha)$ says the power goes to 1 as $n \to \infty$, but how fast? The Type II error probability is

$$\beta_n = 1 - \gamma_n = \mathbb{P}\left(Z > \sqrt{n} - z_\alpha\right). \tag{15.14}$$

In Problem 2 of Chapter 10 we used a Chernoff bound to show that $\mathbb{P}(Z \geq t) \leq e^{-t^2/2}$ for $t > 0$. So for $n > z_\alpha^2$,

$$\beta_n \leq \exp\left[-\frac{(\sqrt{n} - z_\alpha)^2}{2}\right] = \exp\left[-\frac{n}{2} + z_\alpha\sqrt{n} - \frac{z_\alpha^2}{2}\right]. \tag{15.15}$$

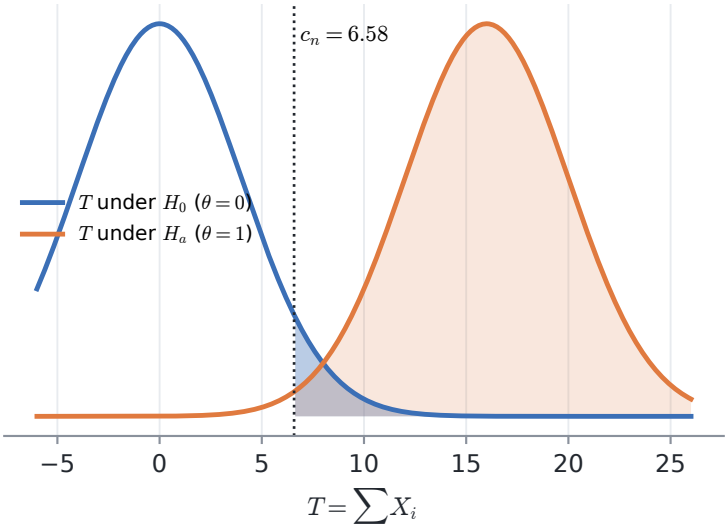


Figure 15.1: The distribution of $T = \sum X_i$ under H_0 and H_a for the normal mean example with $n = 16$. The darker shaded area is the size $\alpha = 0.05$, and the lighter area is the power $\gamma \approx 0.99$.

As $n \to \infty$, the leading $-n/2$ dominates the exponent, so for any $0 < \delta < 1/2$ there is an N with $\beta_n \leq e^{-n\delta}$ for all $n > N$. The Type II error goes to zero *exponentially fast*. Compare this with estimation, where errors shrink like $1/\sqrt{n}$. Telling two fixed hypotheses apart is a much easier problem than pinning down a parameter. In general, the exponential rate is governed by the KL divergence between $f(\cdot\,; \theta_0)$ and $f(\cdot\,; \theta_a)$, which ties this right back to the license to practice MLE.

Problem 1.

Use the factorization theorem to show that if Y is sufficient for θ, then the likelihood ratio $L(\theta_0)/L(\theta_a)$ depends on the data only through Y. Conclude that the most powerful test can always be written in terms of a sufficient statistic.

Problem 2.

In the normal mean example, show that the same critical region $T > \sqrt{n} z_\alpha$ is most powerful for testing $H_0 : \theta = 0$ against $H_a : \theta = \theta_a$ for *every* $\theta_a > 0$. A test like this, which is most powerful against every alternative in a composite $H_a : \theta > 0$ at once, is

called **uniformly most powerful**. Is there a UMP test for $H_a : \theta \neq 0$? Why not?

15.3 More Examples

The recipe is always the same. Write down the LR, find the sufficient statistic T it depends on, figure out which direction of T makes the LR small, and use the null distribution of T to set the cutoff.

Normal variance. Let $X_i \sim \mathcal{N}(0, \theta)$ and test $H_0 : \theta = 1$ versus $H_a : \theta = 4$ with $\alpha = 0.1$ and $n = 9$. The likelihood is $L(\theta) = (2\pi\theta)^{-n/2} \exp\left[-\frac{1}{2\theta}\sum x_i^2\right]$, so

$$\text{LR} = \frac{L(1)}{L(4)} = \frac{(2\pi)^{-n/2} e^{-\frac{1}{2}\sum x_i^2}}{(8\pi)^{-n/2} e^{-\frac{1}{8}\sum x_i^2}} = 2^n \exp\left[-\frac{3}{8}\sum_{i=1}^{n} x_i^2\right]. \tag{15.16}$$

The test statistic is the sum of squares $T = \sum X_i^2$, and the LR is small when T is big. Under H_0, T is a sum of nine squared standard normals, so $T \sim \chi^2_9$, and

$$\alpha = \mathbb{P}_1(T > c) \quad \Longrightarrow \quad c = \chi^2_{0.1,9} = 14.684. \tag{15.17}$$

So we reject H_0 if $\sum_{i=1}^{9} X_i^2 > 14.684$. Under H_a, $X_i/2$ is standard normal, so $T/4 \sim \chi^2_9$ and the power is

$$\gamma = \mathbb{P}_4(T > 14.684) = \mathbb{P}\left(\chi^2_9 > \frac{14.684}{4}\right) = \mathbb{P}\left(\chi^2_9 > 3.671\right) \approx 0.93. \tag{15.18}$$

Et voilà!

Exponential. Let $X_i \sim \text{Exp}(\lambda)$ and test $H_0 : \lambda = 1$ versus $H_a : \lambda = 3$ with $\alpha = 0.01$ and $n = 25$. Then

$$\text{LR} = \frac{e^{-\sum x_i}}{3^n e^{-3\sum x_i}} = 3^{-n} e^{2\sum x_i}, \tag{15.19}$$

which is small when $T = \sum X_i$ is *small*. That makes sense, since a bigger rate means shorter lifetimes. Under H_0, $T \sim \text{Gamma}(25, 1)$, so with G_{25} the Gamma$(25, 1)$ CDF,

$$\alpha = \mathbb{P}_1(T < c) = G_{25}(c) \quad \Longrightarrow \quad c = G_{25}^{-1}(0.01) = 14.85. \tag{15.20}$$

For the power, we use the fact that if $H \sim \text{Exp}(\lambda)$ then $\lambda H \sim \text{Exp}(1)$. Under H_a, $3T \sim \text{Gamma}(25, 1)$, so

$$\gamma = \mathbb{P}_3(T < 14.85) = \mathbb{P}(3T < 44.55) = G_{25}(44.55) \approx 0.9994. \tag{15.21}$$

Poisson. Let $X_i \sim \text{Poisson}(\lambda)$ and test $H_0 : \lambda = 1$ versus $H_a : \lambda = 5$. Then

$$\text{LR} = \frac{L(1)}{L(5)} = \frac{e^{-n}}{5^{\sum x_i} e^{-5n}} = e^{4n} 5^{-\sum x_i}, \tag{15.22}$$

which is small when $T = \sum X_i$ is big. Under H_0, $T \sim \text{Poisson}(n)$, and under H_a, $T \sim \text{Poisson}(5n)$. If F_μ is the Poisson(μ) CDF, the test rejects when $T > c$, with c the smallest integer such that $1 - F_n(c) \le \alpha$, and the power is $1 - F_{5n}(c)$. For example, with $n = 10$ and $\alpha = 0.05$, we get $c = 15$, so we reject if $T \ge 16$. The size is $\mathbb{P}(\text{Poisson}(10) \ge 16) = 0.0487$, a little under 0.05, because a discrete T can't hit α exactly. The power is essentially 1, since under H_a we expect T to be around 50.

15.3.1 Randomized Tests

That last example brings up a real issue with discrete data. Let $X_i \sim \text{Bernoulli}(\theta)$ and test the fair coin $H_0 : \theta = 1/2$ against $H_a : \theta = 1/3$. Writing the likelihood as $L(\theta) = \theta^T (1-\theta)^{n-T}$ with $T = \sum X_i$,

$$\text{LR} = \frac{L(1/2)}{L(1/3)} = \frac{\left(\frac{1}{2}\right)^n}{\left(\frac{1}{3}\right)^T \left(\frac{2}{3}\right)^{n-T}} = \left(\frac{3}{4}\right)^n 2^T, \tag{15.23}$$

which is small when T is small (fewer heads favors the coin biased toward tails). Under H_0, $T \sim \text{Bin}(n, 1/2)$. Take $n = 20$ and $\alpha = 0.05$. The binomial CDF jumps, so no cutoff gives exactly 5%,

$$\mathbb{P}_{1/2}(T \le 5) = 0.0207, \qquad \mathbb{P}_{1/2}(T \le 6) = 0.0577. \tag{15.24}$$

Rejecting for $T \le 5$ wastes some of the budget, and $T \le 6$ goes over it. The Neyman–Pearson fix is to **randomize** on the boundary. Reject for sure if $T \le 5$, and if $T = 6$, flip a biased coin and reject with probability r, where r makes the size exactly α,

$$0.0207 + r\,\mathbb{P}_{1/2}(T = 6) = 0.05 \quad \Longrightarrow \quad r = \frac{0.05 - 0.0207}{0.0370} = 0.793. \tag{15.25}$$

The power is then $\mathbb{P}_{1/3}(T \le 5) + r\,\mathbb{P}_{1/3}(T = 6) = 0.297 + 0.793(0.182) = 0.442$. With the randomization, the Neyman–Pearson lemma holds for discrete data too. In practice, almost nobody flips a coin to decide a hypothesis test. People just report the p value, or use the conservative cutoff. But the randomized test is the one the theory says is optimal.

Uniform. Finally, our favorite family. Let $X_i \sim \text{Uniform}(0, \theta)$ and test $H_0 : \theta = 1$ versus $H_a : \theta = 4$. With $T = \max X_i$,

$$\text{LR} = \frac{L(1)}{L(4)} = \frac{\mathbf{1}\{T \le 1\}}{4^{-n}\mathbf{1}\{T \le 4\}} = \begin{cases} 4^n & T \le 1 \\ 0 & 1 < T \le 4. \end{cases} \tag{15.26}$$

If $T > 1$, the data are impossible under H_0, and we should obviously reject. If $T \le 1$, the LR is the constant 4^n, so it can't tell the hypotheses apart at all, and we have to randomize. Rejecting with probability α when $T \le 1$ gives size $\mathbb{P}_1(T > 1) + \alpha\,\mathbb{P}_1(T \le 1) = 0 + \alpha = \alpha$, and power

$$\gamma = \mathbb{P}_4(T > 1) + \alpha\,\mathbb{P}_4(T \le 1) = 1 - \left(\frac{1}{4}\right)^n (1 - \alpha), \tag{15.27}$$

using $F_{\max} = F^n$ again. The power goes to 1 exponentially fast, as expected.

Problem 3.

Prove the Neyman–Pearson lemma for discrete data, allowing for randomization on the set $\{\text{LR} = k\}$. (Hint: replace the critical region by a function $\phi(\mathbf{x}) \in [0, 1]$, the probability of rejecting, and redo the proof with sums.)

Problem 4.

For the Poisson test with $H_0 : \lambda = 1$ versus $H_a : \lambda = 1.5$, find the smallest n for which the (nonrandomized) most powerful test of size at most 0.05 has power at least 0.9. Then estimate n using the normal approximation to T and compare.

15.4 The Generalized Likelihood Ratio Test

Simple versus simple tests are clean, but almost no real question looks like that. What we usually want is something like

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_a : \theta \neq \theta_0, \tag{15.28}$$

and there's no single θ_a to put in the denominator of the LR. The natural fix is to put the best one, the MLE. The **generalized likelihood ratio** is

$$\Lambda = \frac{L(\theta_0)}{L(\hat{\theta}_n)} \le 1, \tag{15.29}$$

since the MLE maximizes the likelihood. If H_0 is true, $\hat{\theta}_n$ should be close to θ_0 and Λ should be close to 1. If H_a is true, then by the license to practice MLE, $L(\theta_0)$ will be much smaller than the likelihood near the truth, and Λ will be small. So we reject when Λ is small, or equivalently when $-2\log\Lambda$ is big. The question is how big is big, and for that we need the

null distribution of $-2\log\Lambda$. Usually it's impossible to find exactly, but for a very regular family, there's a beautiful approximation.

Theorem (Wilks). If $\mathcal{F}$ is very regular and $H_0 : \theta = \theta_0$ is true, then

$$-2\log\Lambda \xrightarrow{d} \chi^2_1. \tag{15.30}$$

To prove it, Taylor expand the log-likelihood around the MLE and evaluate it at θ_0. Since $\ell'(\hat{\theta}_n) = 0$, the first-order term vanishes, and

$$\log\Lambda = \ell(\theta_0) - \ell(\hat{\theta}_n) \approx \frac{1}{2}\ell''(\hat{\theta}_n)\left(\theta_0 - \hat{\theta}_n\right)^2. \tag{15.31}$$

Multiplying by -2 and splitting up the n the same way we did for asymptotic normality of the MLE,

$$-2\log\Lambda \approx \underbrace{\left(-\frac{\ell''(\hat{\theta}_n)}{n}\right)}_{\xrightarrow{p} I(\theta_0)} \underbrace{\left(\sqrt{n}(\hat{\theta}_n - \theta_0)\right)^2}_{\xrightarrow{d} Z^2/I(\theta_0)}. \tag{15.32}$$

The first factor converges to $I(\theta_0)$ by the law of large numbers (plus consistency of $\hat{\theta}_n$), and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} Z/\sqrt{I(\theta_0)}$ with $Z \sim \mathcal{N}(0,1)$ by the asymptotic normality of the MLE, so by continuous mapping its square converges to $Z^2/I(\theta_0)$. By Slutsky, the product converges to $I(\theta_0)\cdot Z^2/I(\theta_0) = Z^2$, which is χ^2_1. And we're done!

The Fisher information cancelled out, which is the magic of the theorem. The limit doesn't depend on the family or on θ_0 at all. So the GLRT at level α rejects H_0 when $-2\log\Lambda > \chi^2_{\alpha,1}$, whatever the family is. With k parameters being tested, the same argument with a multivariate Taylor expansion gives χ^2_k. That's the result we used without proof for the deviance tests in logistic regression in Chapter 8. And Pearson's χ^2 test for contingency tables in Chapter 7 turns out to be a second-order approximation to a GLRT too. So a lot of Part I was secretly this one theorem.

Problem 5.

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\theta, 1)$ and test $H_0 : \theta = \theta_0$ against $H_a : \theta \neq \theta_0$. Show that $-2\log\Lambda = n(\bar{X}_n - \theta_0)^2$ exactly, so Wilks' theorem holds with no approximation, and that the GLRT is the two-sided z test from Chapter 3.

Problem 6.

Let $X_1, \ldots, X_n$ be i.i.d. Poisson(λ), and test $H_0 : \lambda = \lambda_0$ against $H_a : \lambda \neq \lambda_0$.

(1) Show that $-2\log\Lambda = 2n\left[\bar{X}_n \log(\bar{X}_n/\lambda_0) - (\bar{X}_n - \lambda_0)\right]$.

(2) Simulate 10,000 data sets with $\lambda_0 = 2$ and $n = 15$, and compare the histogram of $-2\log\Lambda$ with the χ^2_1 density. What fraction of the time does the test reject at $\alpha = 0.05$?

(3) Compare with the Wald test that rejects when $|\bar{X}_n - \lambda_0|/\sqrt{\bar{X}_n/n} > 1.96$.

Bibliography

[1] Alan Agresti. *Categorical Data Analysis*. Wiley, 3rd edition, 2013.

[2] George Casella and Roger L. Berger. *Statistical Inference*. Duxbury, 2nd edition, 2002.

[3] Siddhartha R. Dalal, Edward B. Fowlkes, and Bruce Hoadley. Risk analysis of the space shuttle: Pre-challenger prediction of failure. *Journal of the American Statistical Association*, 84(408):945–957, 1989.

[4] Richard J. Larsen and Morris L. Marx. *An Introduction to Mathematical Statistics and Its Applications*. Pearson, 6th edition, 2018.

[5] Student. The probable error of a mean. *Biometrika*, 6(1):1–25, 1908.

[6] Larry Wasserman. *All of Statistics: A Concise Course in Statistical Inference*. Springer, 2004.